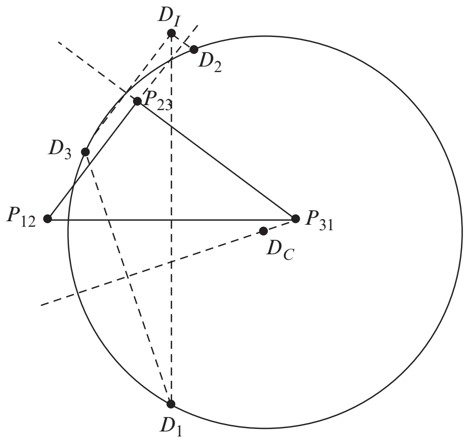

Figure 9.10 Location of point D in the three positions (D_1, D_2, and D_3) .

A straightforward approach for finding the locations of points D_2 and D_3 is to use the so-called *image point* (also referred to as the *image pole*). The image point, denoted here D_I, is defined as that point that will give $D_i (i = 1, 2, 3)$ when reflected across the side (or the extended side) of the pole triangle with the common subscript i. In other words, D_1 is the reflection of D_I about side 1 ($P_{31} P_{12}$), D_2 is the reflection of D_I across side 2 ($P_{12}P_{23}$), and D_3 is the reflection of D_I across side 3 ($P_{23} P_{31}$). Points D_1, D_2, and D_3 are shown in Fig. 9.10.

The three points D_1, D_2, and D_3 must also lie on the circumference of a circle, that is, we know that it is always possible to find a circle that passes through the three locations defined by any arbitrary point fixed in the body as the body travels through three finitely separated postures. This makes it possible to design a four-bar linkage to guide a rigid body through three finitely separated postures, since pins A and B on the coupler link must be located on circular arcs.

The center of the circle that passes through the three points D_1, D_2, and D_3 is called a *center point* and is denoted D_C. All points such as D_C make up the center point system (or center system), which will be explained in more detail later. The center point, D_C, is the point of intersection of the perpendicular bisectors of lines D_1D_3 and D_2D_3. Since the perpendicular bisector of line D_1D_2 must also pass through the center point, D_C, this can be used to check the accuracy of the previous constructions. Also note that, consistent with the definition of a pole, the perpendicular bisector of line D_iD_j must pass through pole P_{ij}.

The geometric relationships can be written as follows:

$$\begin{aligned}
\angle D_1P_{12}D_C &= \angle D_CP_{12}D_2 = \phi_{12},\\
\angle D_2P_{23}D_C &= \angle D_CP_{23}D_3 = \phi_{23},\\
\angle D_3P_{31}D_C &= \angle D_CP_{31}D_1 = \phi_{31}.
\end{aligned} \tag{9.18}$$

EXAMPLE 9.2

Assume we are given the pole triangle and center point E_C, as shown in Fig. 9.11. The length of the side $P_{31}P_{12}$ is 2 in, the interior angles $\phi_{12} = \theta_{12}/2 = 25°$ ccw and $\phi_{31} = \theta_{31}/2 = 40°$ ccw, the distance from P_{12} to center point E_C is 1 in, and the angle $\angle P_{31}P_{12}E_C = 45°$cw, as shown in Fig. 9.11. Find the locations of point E when the body is in postures 1, 2, and 3 (that is, find E_1, E_2, and E_3), and determine the radius of the circle that passes through these three points.

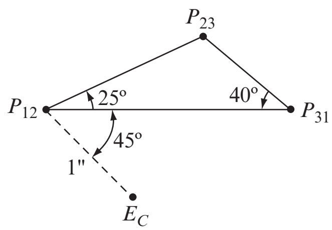

Figure 9.11 Pole triangle and the center point, E_C.

SOLUTION

The interior angle from posture 1 to posture 2 can be written as

$$\phi_{12} = \angle P_{31}P_{12}P_{23} = 25° \text{ ccw}, \tag{1a}$$

and the interior angle from posture 3 to posture 1 can be written as

$$\phi_{31} = \angle P_{23}P_{31}P_{12} = 40° \text{ ccw}. \tag{1b}$$

Therefore, the interior angle from posture 2 to posture 3 is

$$\phi_{23} = \angle P_{12}P_{23}P_{31} = 115° \text{ ccw}. \tag{1c}$$

The geometric relationships are

$$\begin{aligned} \angle E_1P_{12}E_C &= \angle E_CP_{12}E_2 = \phi_{12}, \\ \angle E_2P_{23}E_C &= \angle E_CP_{23}E_3 = \phi_{23}, \\ \angle E_3P_{31}E_C &= \angle E_CP_{31}E_1 = \phi_{31}. \end{aligned} \tag{2}$$

Two of these three equations can be used to locate E_1, that is, the first and the third equations can be written as

$$\begin{aligned} \angle E_CP_{12}E_1 &= -\phi_{12} = 25° \text{ cw}, \\ \angle E_CP_{31}E_1 &= \phi_{31} = 40° \text{ ccw}. \end{aligned} \tag{3}$$

Therefore, the point of intersection of these two lines is E_1, as shown in Fig. 9.12. The location of point E in positions 2 and 3 can be obtained from a similar procedure; that is, to locate E_2, use the first two of Eqs. (2), and to locate E_3 use the second and third equations. An alternative procedure is to find the image point E_I and then reflect this point across the common side 2 and the common side 3 of the pole triangle to find E_2 and E_3, respectively.

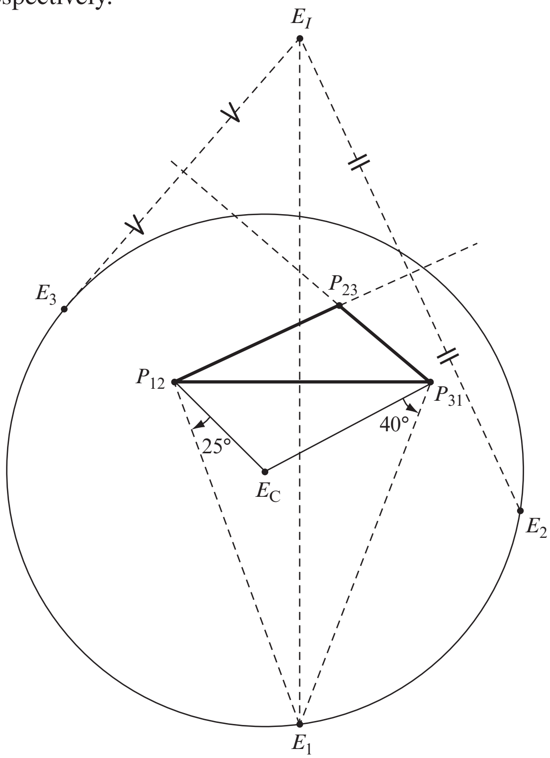

Figure 9.12 Pole triangle and points E_1, E_2, and E_3.

The circle can be drawn with center E_C and passing through points E_1, E_2, and E_3. The radius of this circle is measured as

$$E_C E_1 = E_C E_2 = E_C E_3 = 1.9 \text{ in.}$$

A special case, which is important in synthesis, is when the radius of the circle is infinite, that is, when the circle degenerates to a straight line. This implies that point E is moving on a straight line through the three finitely separated positions. In such a case, point E would be suitable for a prismatic joint. An example of this special case is presented next.

EXAMPLE 9.3

We are given the pole triangle and point D_1, as shown in Fig. 9.13. The length of side $P_{31}P_{12}$ is 2 in, and the interior angles are $\phi_{12} = 30°$ ccw, $\phi_{23} = 90°$ ccw, and $\phi_{31} = 60°$ ccw. The location of point D_1 is midway between poles P_{31} and P_{12} and 1 in below this line, as shown in Fig. 9.13. Find the location of point D when the rigid body is in postures 2 and 3; that is, find the locations of D_2 and D_3.

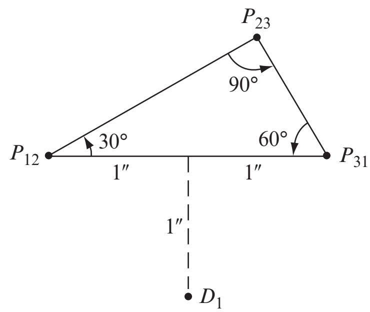

Figure 9.13 Pole triangle and point D_1.

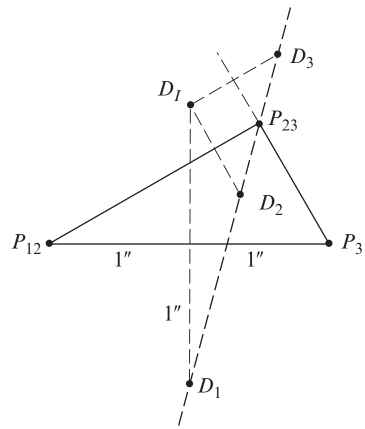

Figure 9.14 Locations of points D_1, D_2, and D_3.

SOLUTION

The pole triangle is a 1:2:$\sqrt{3}$ right-angled triangle; therefore, the lengths of the remaining two sides are $P_{12}P_{23} = \sqrt{3}$ in and $P_{23}P_{31} = 1$ in. Reflecting point D_1 across the side $P_{31}P_{12}$ gives the image point D_I. Then, reflecting D_I across the side $P_{12}P_{23}$ gives point D_2, and, finally, reflecting D_I across the side $P_{23}P_{31}$ gives point D_3. Points D_1, D_2 and D_3 are shown in Fig. 9.14.

Note that the three points D_1, D_2, and D_3 lie on a straight line. Also note that this line passes through pole P_{23}, that is, the rotation angle

$$\angle D_2P_{23}D_3 = 2\phi_{23} = 2\angle P_{12}P_{23}P_{31} = 180° \text{ ccw}.$$

The following will clarify why the path of point D, in this example, is a straight line.

Circumscribing Circle

An important geometric property of a triangle is its circumscribing circle. A circle can always be drawn such that the three vertices of the triangle lie on the circumference of this circle, referred to as the *circumscribing circle*. The center of the circumscribing circle is the intersection of the perpendicular bisectors of the three sides of the triangle. The center of the circumscribing circle is denoted point O, as shown in Fig. 9.15.

The reflection of point O across side 1 ($P_{31}P_{12}$) is denoted O_1, that across side 2 ($P_{12}P_{23}$) is denoted O_2, and that across side 3 ($P_{23}P_{31}$) is denoted O_3. Poles P_{31} and P_{12} lie on the circumference of a circle with the same radius as the circumscribing circle and with center O_1, poles P_{12} and P_{23} lie on the circumference of a circle with the same radius as the circumscribing circle and with center O_2, and poles P_{23} and P_{31} lie on the circumference of a circle with the same radius as the circumscribing circle and with center O_3. These three circles also intersect at a unique point that is referred to as the *orthocenter* and is denoted as point H, as shown in Fig. 9.16.

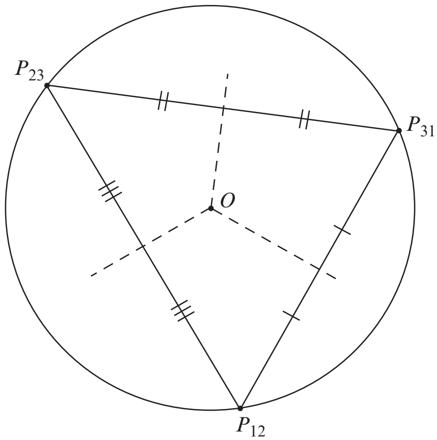

Figure 9.15 Circumscribing circle of a pole triangle.

The orthocenter of a triangle is defined as the point of intersection of the lines drawn through the vertices of the triangle and perpendicular to the opposite sides. Therefore, if the pole triangle is an acute-angled triangle, then the orthocenter lies inside the triangle. If the pole triangle is a right-angled triangle, then the orthocenter is coincident with a pole

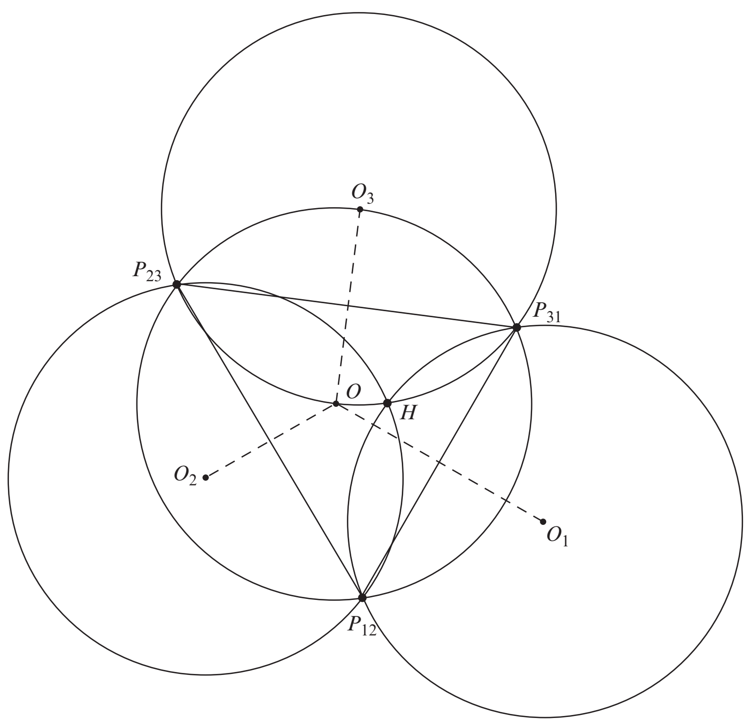

Figure 9.16 Orthocenter.

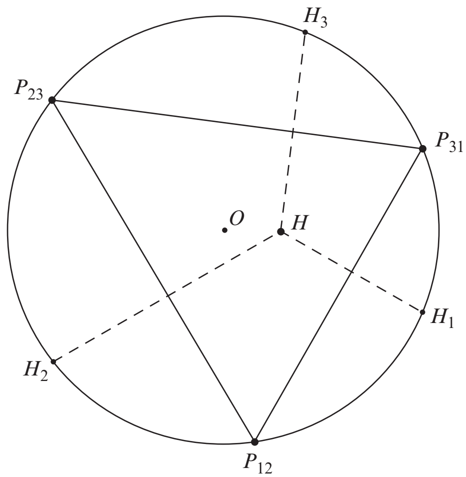

Figure 9.17 Reflections of the orthocenter.

and is located at the apex of the right angle. If the pole triangle has an obtuse angle, then the orthocenter lies outside the triangle.

The reflection of the orthocenter, H, across side 1 ($P_{31}P_{12}$) is denoted H_1, the reflection of point H across side 2 ($P_{12}P_{23}$) is denoted H_2, and the reflection of point H across side 3 ($P_{23}P_{31}$) is denoted H_3. Note that the reflections of the orthocenter lie on the circumscribing circle of the pole triangle, as shown in Fig. 9.17.

EXAMPLE 9.4

For the pole triangle given in Example 9.3, find the center of the circumscribing circle, point O, and draw the circumscribing circle. Then, locate points O_1, O_2, and O_3, and draw the circles with the same radii as the circumscribing circle and with centers at O_1, O_2, and O_3. Finally, determine the locus of all points in the body having three positions on straight lines.

SOLUTION

Since the pole triangle from Example 9.3 is a right-angled triangle, the center of the circumscribing circle, point O, must lie at the midpoint of the hypotenuse. Note that the hypotenuse is side 1 of the pole triangle, with common subscript 1; therefore, point O_1 is coincident with O. Reflecting point O across side 2 of the pole triangle with the common 2 subscript gives O_2, and reflecting point O across side 3 of the pole triangle with the common 3 subscript gives O_3, as shown in Fig. 9.18.

Since point O_1 is coincident with O, then the circle with center at O_1 and the same radius as the circumscribing circle is coincident with the circumscribing circle. Also note that the given point D_1 in Example 9.3 lies on this circle; therefore, points D_2 and D_3 must lie on the circles with centers at O_2 and O_3. The conclusion is that the locus of points having three positions on a straight line are the circles with the same radii as the circumscribing circle and centers at O_1, O_2, and O_3. These are the only points in the body that can travel on straight lines through the three finitely separated postures.

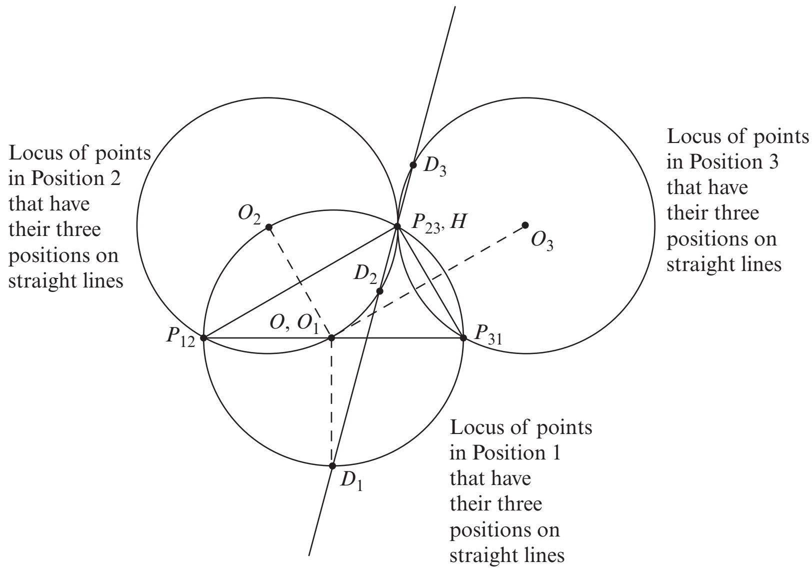

Figure 9.18 Locus of points having three positions on a straight line.

This explains why point D in Example 9.3 travels on a straight line.

It is interesting to note that, for three infinitesimally separated postures of the body, the three circles coalesce and become the inflection circle (Sec. 4.12). Recall that the inflection circle is the locus of all inflection points in the body, that is, points that instantaneously travel on straight lines.

In designing a four-bar linkage to guide the coupler link through three finitely separated postures there are ∞^4 possible solutions, that is, ∞^2 choices for the crankpin A in the coupler link and ∞^2 choices for the crankpin B in the coupler link. Based on the choices of A and B, the locations of the ground pivots O_A and O_B are uniquely determined.

9.5 FOUR FINITELY SEPARATED POSTURES OF A RIGID BODY ($N = 4$)

For four finitely separated postures of a rigid body, there are six poles: (*a*) pole P_{12} for the finite displacement from posture 1 to posture 2, (*b*) pole P_{13} for the finite displacement from posture 1 to posture 3, (*c*) pole P_{14} for the finite displacement from posture 1 to posture 4, (*d*) pole P_{23} for the finite displacement from posture 2 to posture 3, (*e*) pole P_{24} for the finite displacement from posture 2 to posture 4, and (*f*) pole P_{34} for the finite displacement from posture 3 to posture 4.

The four finitely separated positions of an arbitrary point fixed in the moving body do not, in general, lie on the circumference of a circle. However, there are some points of the moving body whose four positions do lie on the circumference of a circle; these points are important in kinematic synthesis of a mechanism to guide the body through four given postures and are referred to as *circle points*. Circle points are suitable for the crankpins of a four-bar linkage with the ground pivots at the centers of the circles; these are called *center points*. For four finitely separated postures of a plane, all circle points lie on a curve that is referred to as the *circle point curve,* and all center points lie on a curve that is referred to as the *center point curve*. These two curves are each of third degree [1] or cubic curves, that is, the curves can be described by third-order polynomials.

To design a four-bar linkage to guide a rigid body through four finitely separated postures, we only need to focus on four of the six poles. In other words, the circle point curve and the center point curve can be obtained by considering only four of the six poles. If the solution is not satisfactory, then we can choose a different combination of four poles and repeat the synthesis procedure. The geometry involved is that of a pole quadrilateral referred to as an *opposite pole quadrilateral*.

Two poles that do not contain a common numeric subscript are referred to as *opposite poles*. There are three pairs of *opposite poles*, namely, (P_{12}, P_{34}), (P_{13}, P_{24}), and (P_{14}, P_{23}). Two poles that do contain a common numeric subscript are referred to as *adjacent poles*. There are twelve pairs of adjacent poles, namely, (P_{12}, P_{13}), (P_{12}, P_{14}), (P_{12}, P_{23}), (P_{12}, P_{24}), (P_{13}, P_{14}), (P_{13}, P_{23}), (P_{13}, P_{34}), (P_{14}, P_{24}), (P_{14}, P_{34}), (P_{23}, P_{24}), (P_{23}, P_{34}), and (P_{24}, P_{34}). There a total of three opposite pole quadrilaterals; they have diagonals that connect opposite poles. Therefore, the sides of an opposite pole quadrilateral are lines connecting adjacent poles. The three possible opposite pole quadrilaterals are: (a) (P_{13}, P_{12}), (P_{12}, P_{24}), (P_{24}, P_{34}), and (P_{34}, P_{13}); (b) (P_{14}, P_{12}), (P_{12}, P_{23}), (P_{23}, P_{34}), and (P_{34}, P_{14}); and (c) (P_{14}, P_{13}), (P_{13}, P_{23}), (P_{23}, P_{24}), and (P_{24}, P_{14}). These are shown in Fig. 9.19.

Theorem *If four positions of a point fixed in a rigid body lie on the circumference of a circle, then the center of that circle (that is, the center point) views opposite sides of an opposite pole quadrilateral under angles that are equal or differ by 180°.*

The converse is also true, namely: *If a point (say, E_C) views opposite sides of an opposite pole quadrilateral under angles that are equal or differ by 180°, then that point is the center of a circle that passes through the four positions of point E of the body.*

The procedure to find circle points is shown in Fig. 9.20; it is as follows. Consider one of the three opposite pole quadrilaterals, say the opposite pole quadrilateral (P_{13}, P_{12}), (P_{12}, P_{24}), (P_{24}, P_{34}), and (P_{34}, P_{13}). Then, choose one pair of opposite sides of this quadrilateral, say the opposite sides (P_{12}, P_{24}) and (P_{13}, P_{34}) Draw a circle with the side (P_{12}, P_{24}) as a chord; that is, perpendicularly bisect the side (P_{12}, P_{24}), and choose the

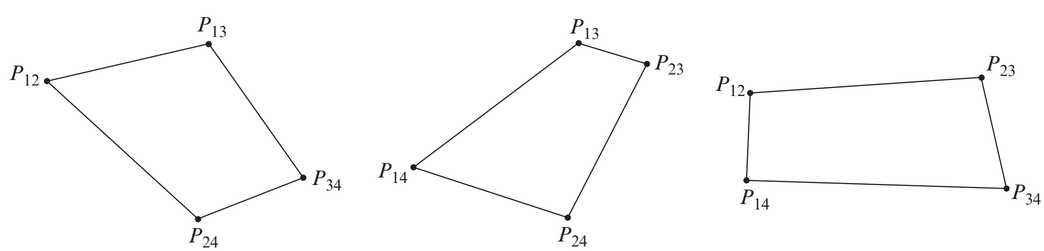

Figure 9.19 Three possible opposite pole quadrilaterals.

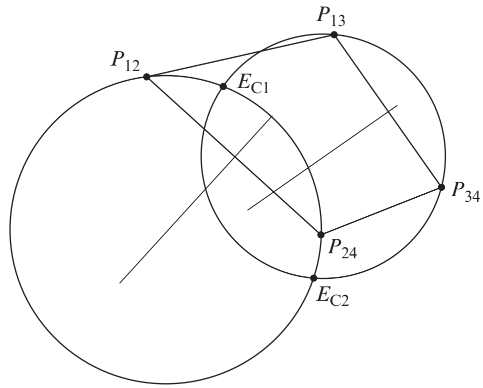

Figure 9.20 Circle with side (P_{12}, P_{24}) as a chord.

center of the circle as a point on this bisector. Choose a convenient radius R, and draw the circle as shown in Fig. 9.20. The center of this circle can be chosen on the left side of line (P_{12}, P_{24}) or on the right side of the line. Left and right are defined as follows: If we stand on the pole P_{12}, for example, and look towards the pole P_{24}, that is, we are looking from 1 to 4, as we ignore the common 2 subscript. So, as shown in Fig. 9.20, the center of the circle has been chosen on the right side of the line.

Now we draw a circle with the opposite side (P_{13}, P_{34}) as a chord. The radius, r, of this circle must be chosen in the ratio

$$r = \frac{P_{13}P_{34}}{P_{12}P_{24}} R. \tag{9.19}$$

The center of this circle must be chosen on the same side of line (P_{13}, P_{34}) as the center of the first circle was chosen relative to line (P_{12}, P_{24}). So, we stand on pole (P_{13}) and look towards the pole (P_{34}), that is, looking from 1 to 4, as we ignore the common 3 subscript, consistent with the previous procedure. The center must again be chosen on the right of the line, as shown in Fig. 9.20.

In general, these two circles will intersect in two points, denoted here as E_{C_1} and E_{C_2}, and both are possible center points. These points both satisfy the theorem, that is, they both view the opposite sides of the opposite pole quadrilateral under angles that are equal or differ by 180°. The relationships can be written as

$$\begin{aligned} \angle P_{12}E_{C_1}P_{24} &= \angle P_{13}E_{C_1}P_{34}, \\ \angle P_{12}E_{C_2}P_{24} &= \angle P_{13}E_{C_2}P_{34}, \end{aligned} \tag{9.20}$$

or

$$\begin{aligned} \angle P_{12}E_{C_1}P_{24} &= \angle P_{13}E_{C_1}P_{34} \pm 180^\circ, \\ \angle P_{12}E_{C_2}P_{24} &= \angle P_{13}E_{C_2}P_{34} \pm 180^\circ. \end{aligned} \tag{9.21}$$

The second of Eqs. (9.20) and the first of Eqs. (9.21) can be verified to be true for the example shown in Fig. 9.20.

By choosing different values for R (and r) and repeating the above procedure, a set of center points can be obtained. A curve can then be drawn through these points and is called the *center point curve*. Points on this curve are all suitable candidates for fixed pivots of the four-bar linkage. Note that all six poles, by definition, must lie on the center point curve.

EXAMPLE 9.5

The locations of five of the six poles for four finitely separated postures of the coupler link of a planar four-bar linkage are as shown in Fig. 9.21. Draw an opposite pole quadrilateral. Draw the center point curve, which can be used in the synthesis of a four-bar linkage. For the specified fixed pivots of the four-bar linkage, O_A and O_B, shown in Fig. 9.21, locate the corresponding circle points A and B in the first three postures of the linkage (that is, A_1B_1, A_2B_2, and A_3B_3). Then, specify the lengths of the two links O_AA and O_BB and the length of the coupler link AB of the synthesized four-bar linkage O_A, A, B, and O_B. Is the synthesized four-bar linkage a Grashof four-bar linkage or a non-Grashof four-bar linkage?

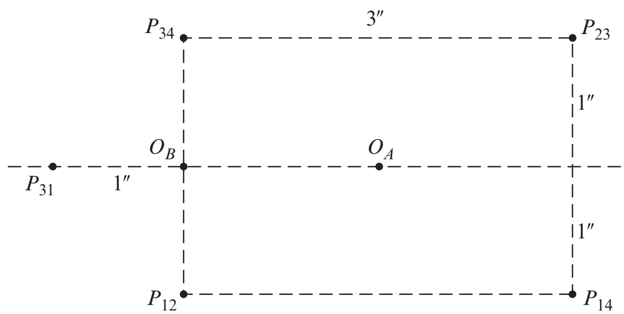

Figure 9.21 Four finitely separated postures of a coupler link.

SOLUTION

First, an opposite pole quadrilateral is drawn. The four sides of the opposite pole quadrilateral are (P_{12}, P_{14}), (P_{14}, P_{34}), (P_{34}, P_{23}), and (P_{23}, P_{12}), as shown in Fig. 9.22. Note that this is only one of three possible opposite pole quadrilaterals. However, it is the only one that can be drawn from the five poles given. Pole P_{24} is not known; therefore, pole P_{31} cannot be used to draw an opposite pole quadrilateral.

The procedure to draw the center point curve is as follows:

1. Choose a side (P_{12}, P_{14}) and the opposite side (P_{34}, P_{23}) of the opposite pole quadrilateral. From the property of similar triangles,

$$\frac{R}{r} = \frac{P_{12}P_{14}}{P_{23}P_{34}} = \frac{3\text{ in}}{3\text{ in}} = 1, \quad \text{that is,} \quad R = r, \tag{1}$$

where R is the radius of the circle with center on the perpendicular bisector of $P_{12}P_{14}$, and r is the radius of the circle with center on the perpendicular bisector of $P_{34}P_{23}$.

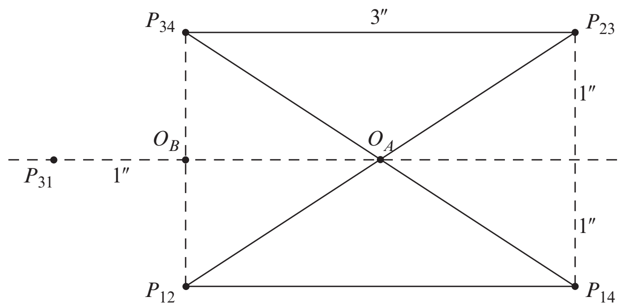

Figure 9.22 An opposite pole quadrilateral.

2. Consider the side (P_{12}, P_{14}) of the opposite pole quadrilateral. Standing at pole P_{12}, we look at pole P_{14}; we are looking from 2 to 4, ignoring the common subscript 1. We draw a circle with radius R and with center, say, to our right on the perpendicular bisector of (P_{12}, P_{14}).
3. Now we consider the opposite side (P_{23}, P_{34}) of the opposite pole quadrilateral. Standing at pole P_{23}, we look at pole P_{34}, looking from 2 to 4, ignoring the common subscript 3. We draw a circle with radius $r = R$ and with the center again to our right on the perpendicular bisector of (P_{23}, P_{34}).
4. Identify the two points of intersection of the two circles in steps 2 and 3 as two center points.
5. Choose different values for R (and r) and follow the above procedure (steps 2, 3, and 4) to obtain another set of center points.
6. Repeating steps 2 through 5 several times, we draw a curve through these center points.

This results in the center point curve shown in Fig. 9.23.

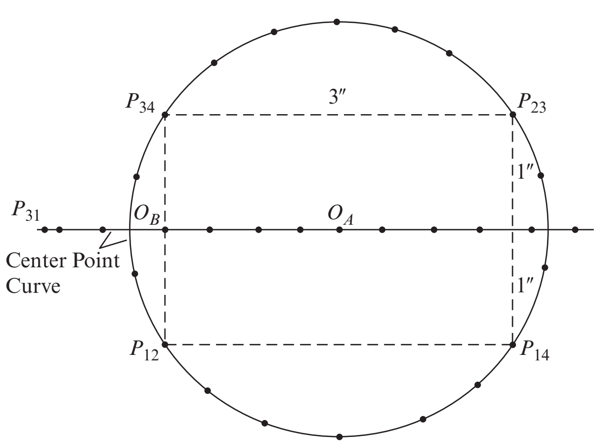

Figure 9.23 Center point curve.

Note that any two points on this curve are suitable as fixed pivots of the four-bar linkage. Note, also, that the center point curve for this example is the straight horizontal line passing through O_A and O_B and the circle circumscribing the opposite pole quadrilateral. This is an overly simplified and a degenerate case; that is, the cubic curve here consists of a straight line and a circle. Still, we note that the center point curve passes through the four poles of the opposite pole quadrilateral (and the given fifth pole). As mentioned previously, the poles (by definition) must always lie on the center point curve; this helps to confirm the graphic construction.

To locate circle points A and B corresponding to the given center points O_A and O_B, consider the pole triangle formed by poles P_{12}, P_{23}, and P_{31}; the interior angles are

$$\phi_{12} = 101.3° \text{ cw}, \quad \phi_{23} = 19.6° \text{ cw}, \quad \text{and} \quad \phi_{31} = 59.1° \text{ cw}. \tag{2}$$

Consider the center point, that is, the fixed pivot O_A; then, we find the angles

$$\angle O_A P_{12} A_1 = -\phi_{12} = 101.3° \text{ ccw} \quad \text{and} \quad \angle O_A P_{31} A_1 = +\phi_{31} = 59.1° \text{ cw}. \tag{3}$$

We draw a line through pole P_{12} such that it makes an angle of 101.3° ccw from line $O_A P_{12}$. Then, we draw a line through pole P_{31} such that it makes an angle of 59.1° cw from line $O_A P_{31}$. The intersection of these two lines gives point A_1. Note that point A_1 is coincident with pole P_{31}. Next, we reflect A_1 across the side 1 ($P_{12}P_{31}$) to find image point A_I. Then, we reflect image point A_I across side 2 ($P_{12}P_{23}$) and side 3 ($P_{23}P_{31}$) to give points A_2 and A_3, respectively. The locations of points A_1, A_2, and A_3 are as shown in Fig. 9.24. Note that the image point A_I is coincident with point A_1.

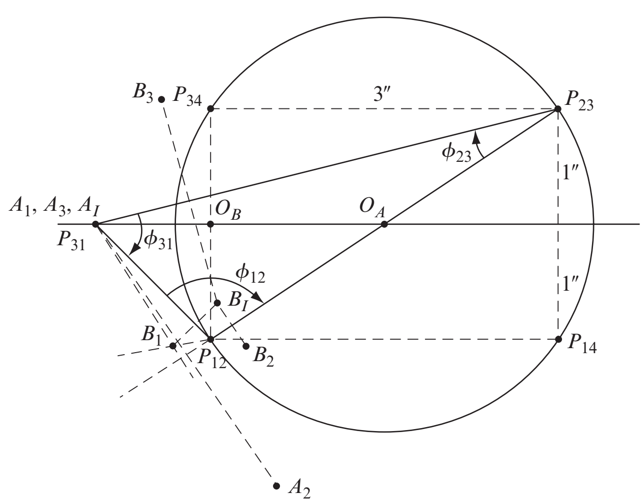

Figure 9.24 Four-bar linkage.

Similarly, we consider the center point, that is, the fixed pivot O_B, where the angles are

$$\angle O_B P_{12} B_1 = -\phi_{12} = 101.3^\circ \text{ ccw} \quad \text{and} \quad \angle O_B P_{31} B_1 = +\phi_{31} = 59.1^\circ \text{ cw.} \tag{4}$$

Then, we follow the same procedure as above to obtain points B_1, B_2, and B_3. The locations of points B_1, B_2, and B_3. are as shown in Fig. 9.24.

The lengths of the links of the synthesized four-bar linkage $O_A ABO_B$ are measured and found to be

$$AB = 1.25 \text{ in} = p, \quad O_A O_B = 1.5 \text{ in} = q, \quad O_A A = 2.5 \text{ in} = l, \quad \text{and} \quad O_B B = 1.2 \text{ in} = s.$$

From the Grashof criterion, Eq. (1.6), a planar four-bar linkage has a crank if and only if

$$s + l \leq p + q,$$

where $l =$ the length of the longest link, $s =$ the length of the shortest link, and p and q are the lengths of the remaining two links. From the measurements

$$s + l = 1.2 \text{ in} + 2.5 \text{ in} = 3.7 \text{ in} \quad \text{and} \quad p + q = 1.25 \text{ in} + 1.5 \text{ in} = 2.75 \text{ in.}$$

Therefore, the synthesized planar four-bar linkage is a non-Grashof four-bar linkage and does not have a continuously rotating crank.

EXAMPLE 9.6

For a rigid body in plane motion, posture 2 coincides with posture 1, and posture 4 coincides with posture 3. The locations of the six poles for four finitely separated postures of the moving body are as shown in Fig. 9.25. Draw the center point curve that can be used in the synthesis of a four-bar linkage to guide the rigid body through the four finitely separated postures.

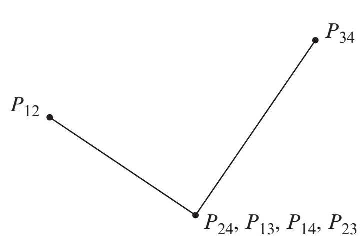

Figure 9.25 Four finitely separated postures of a rigid body.

SOLUTION

Following the procedure that was outlined in the previous example, the center point curve is as shown in Fig. 9.26.

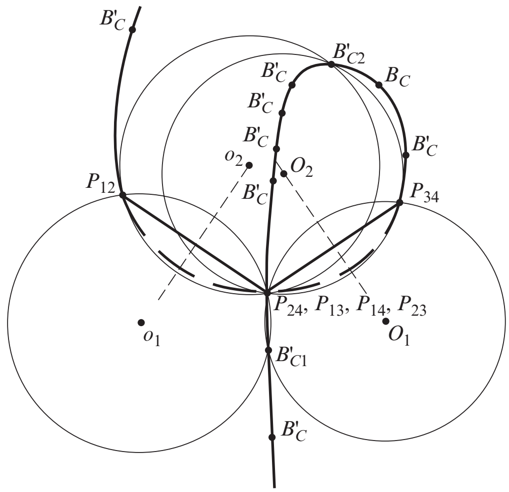

Figure 9.26 Center point curve.

9.6 FIVE FINITELY SEPARATED POSTURES OF A RIGID BODY ($N = 5$)

It may be possible to guide a rigid body through five specified finitely separated postures using a four-bar linkage. However, to do this, we must find two points belonging to the rigid body and having their five positions on the circumferences of two circles. If such circle points exist, they must lie at the intersection of two circle point curves. For example, one could plot the circle point curve that is associated with postures 1, 2, 3, and 4 and then plot the circle point curve that is associated with postures 1, 2, 3, and 5. The points of intersection of these two circle point curves are called *Burmester points* [8], and they are possible locations for the coupler pivots of the four-bar linkage. If the two center points corresponding to the two chosen coupler pivots lie in locations where there are no practical obstructions to establishing these center points as ground pivots for the linkage, then a workable design may be possible.

We probably agree that it is not likely that these particular points will also happen to fall on the circle point curve of yet another group of four postures of the body. Therefore, it is generally *not* possible to synthesize a planar four-bar linkage that passes through more than five arbitrarily prescribed postures.

It is worth noting that a four-bar linkage designed using the methods presented in this section may not be workable even though the theory has been applied correctly. Since we are dealing with finitely separated postures of the coupler link, and we are not exerting any control over intermediate postures, it is possible that the designed four-bar linkage may not pass through the desired motion because of intermediate limiting postures.

9.7 PRECISION POSTURES; STRUCTURAL ERROR; CHEBYSHEV SPACING

The synthesis examples in the preceding sections are of the body guidance type. However, it was pointed out that linkages of the function generation type can also be synthesized by kinematic inversion if we consider the motion of the input with respect to the output. That is, if x is the orientation of the input, and y is the orientation of the output, then, for function generation, we are trying to find the dimensions of a linkage for which the input/output relationship fits a given functional relationship:

$$y = f(x). \tag{a}$$

In general, however, a mechanism has only a limited number of design parameters, a few link lengths, starting angles for the input and output, and a few more. Therefore, except for very special cases, a linkage synthesis problem usually has no exact solution over its entire range of travel.

In the preceding sections, we have chosen to work with two, three, four, or even five postures of the linkage, called *precision postures*, and to seek a linkage that exactly satisfies the desired requirements at these few chosen postures. Our implicit assumption has been that if the design fits the specifications at these few postures, then it will probably deviate only slightly from the desired motion between the precision postures, and that the deviation will probably be acceptably small. *Structural error* is defined as the theoretical difference between the function produced by the synthesized linkage and the function originally prescribed. For many function generation problems, the structural error in a four-bar linkage solution can be held to less than 4% [3]. We should note, however, that structural error usually exists even with no *graphic error* resulting from a graphic solution process and even with no *mechanical error*, which stems from imperfect manufacturing tolerances.

Of course, the amount of structural error in the solution can be affected by the choice of the precision postures. One objective of linkage design is to select a set of precision postures for use in the synthesis procedure that minimizes this structural error.

Although it is not perfect, a very good trial for the distribution of these precision postures is called *Chebyshev* spacing [4]. For N precision postures in the range $x_0 \le x \le x_{N+1}$, Chebyshev spacing is given by

$$x_j = \frac{1}{2}(x_{N+1} + x_0) - \frac{1}{2}(x_{N+1} - x_0)\cos\frac{(2j-1)\pi}{2N} \qquad j = 1, 2, \ldots, N. \tag{9.22}$$

As an example, suppose we wish to design a linkage to generate the function

$$y = x^{0.8} \tag{b}$$

over the range $1 \le x \le 3$ using three precision postures.

From Eq. (9.22), the three values of x_j are

$$x_1 = \frac{1}{2}(3+1) - \frac{1}{2}(3-1)\cos\frac{(2-1)\pi}{2(3)} = 2 - \cos\frac{\pi}{6} = 1.134,$$

$$x_2 = 2 - \cos\frac{3\pi}{6} = 2.000,$$

$$x_3 = 2 - \cos\frac{5\pi}{6} = 2.866.$$

From Eq. (b), we find the corresponding values of y to be

$$y_1 = 1.106, \quad y_2 = 1.741, \quad \text{and} \quad y_3 = 2.322.$$

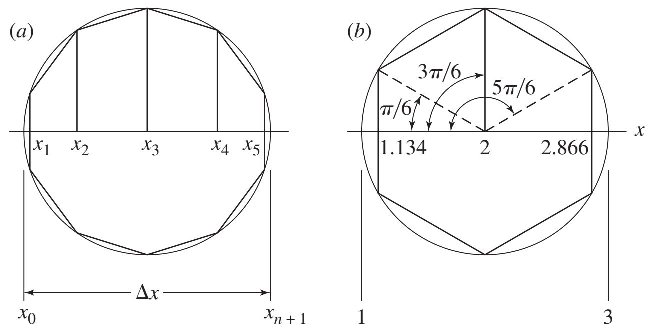

Figure 9.27 Graphic construction for Eq. (9.22): (*a*) $N = 5$; (*b*) $N = 3$.

Chebyshev spacing of the precision postures is also easily determined using the graphic approach shown in Fig. 9.27. As shown in Fig. 9.27*a*, a circle is first constructed whose diameter is equal to the range Δx, given by

$$\Delta x = x_{N+1} - x_0. \tag{c}$$

Next, we inscribe a regular polygon having $2N$ sides in this circle, with its first side placed perpendicular to the x axis. Perpendiculars dropped from each j^{th} vertex now intersect diameter Δx at the precision posture value of x_j. Figure 9.27*b* shows the construction for this numeric example of Eq. (9.22).

It should be noted that Chebyshev spacing is a good approximation of precision postures that usually reduce structural error in a design. Depending on the accuracy requirements of the problem, it may be satisfactory. If additional accuracy is required, then by plotting a curve of structural error versus x, we can usually determine visually the adjustments to be made in the choice of precision postures for another trial.

Before closing this section, however, we should note two more problems that can arise to confound the designer in choosing precision postures for synthesis. These are called *branch defect* and *order defect* [19]. Branch defect refers to a possible completed design that meets all of the prescribed requirements at each of the precision postures but that cannot be moved continuously between these postures without being taken apart and reassembled. Order defect refers to a completed linkage design that can reach all of the precision postures, but not in the desired order.

9.8 OVERLAY METHOD

Synthesis of a function generator mechanism, say, using the overlay method, is one of the easiest and quickest of all methods of synthesis in use. It is not always possible to obtain a solution, and sometimes the accuracy may be rather poor. Theoretically, however, one can employ as many precision postures as are desired in the process.

Let us design a function generator linkage to generate the function

$$y = x^{0.8}, \quad 1 \le x \le 3. \tag{a}$$

Suppose we choose six precision postures of the linkage for this example and use uniform spacing of the output. Table 9.1 indicates the rounded values of x and y and the corresponding angles selected for the input, ψ, and output, ϕ.

The first step in the synthesis is shown in Fig. 9.28*a*. We use a sheet of tracing paper and construct the input, O_2A, in all six postures. This requires an arbitrary choice for the length of O_2A. Also, on this sheet, we choose another arbitrary length for coupler AB and draw arcs numbered 1 to 6 using points A_1 to A_6, respectively, as centers.

Now, on another sheet of paper, we construct the output, whose length is unknown, in all six postures, as shown in Fig. 9.28*b*. Through point O_4 we draw a number of arbitrarily spaced arcs intersecting lines O_41, O_42, and so on; these represent possible lengths of the output rocker.

As the final step, we lay the tracing over the drawing and manipulate it in an effort to find a fit. In this case, a fit is found, and the result is shown in Fig. 9.29.

Table 9.1

Position	x	ψ, deg	y	ϕ, deg
1	1	0	1	0
2	1.366	22.0	1.284	14.2
3	1.756	45.4	1.568	28.4
4	2.16	69.5	1.852	42.6
5	2.58	94.8	2.136	56.8
6	3.02	121.0	2.420	71.0

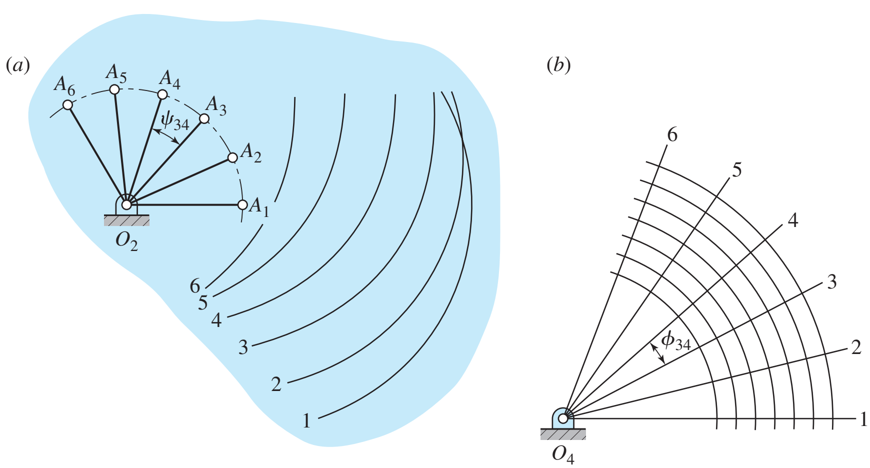

Figure 9.28

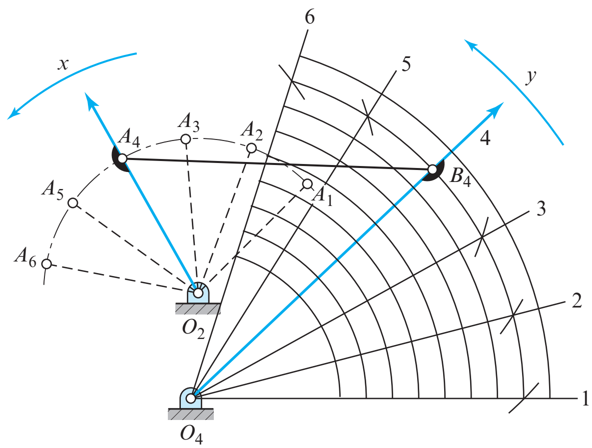

Figure 9.29

9.9 COUPLER-CURVE SYNTHESIS*

In this section, we synthesize a four-bar linkage so that a tracing point on the coupler traces a specified path when the linkage is moved. Then, in Secs. 9.10 through 9.14, we discover that paths having certain characteristics are particularly useful, for example, in synthesizing linkages having dwells of the output for certain periods of rotation of the input.

In synthesizing a linkage to generate a specified path, we can choose up to six precision points along the path. If the synthesis is successful, the tracing point will pass through each precision position. Because of the branch or order defects, the final result may or may not approximate the desired path.

Two postures of a four-bar linkage are shown in Fig. 9.30. Link 2 is the input; it is connected at A to coupler 3, containing tracing point C, and connected to output link 4 at B. Two postures of the linkage are shown by subscripts 1 and 3. Points C_1 and C_3 are two positions of the tracing point on the path to be generated. In this example, C_1 and C_3 have been especially selected so that the perpendicular bisector c_{13} passes through O_4. Note, for this selection of points, that the angle $\angle C_1O_4C_3$ is equal to the angle $\angle A_1O_4A_3$, as indicated in Fig. 9.30.

The advantage of making these two angles equal is that when the linkage is finally synthesized, triangles $C_3A_3O_4$ and $C_1A_1O_4$ are congruent. Thus, if the tracing point is made to pass through C_1 on the path, it will also pass through C_3.

To synthesize a linkage so that the coupler point will pass through four precision positions, we locate any four points, C_1, C_2, C_3, and C_4, on the desired path (Fig. 9.31). Choosing C_1 and C_3, say, we first locate O_4 anywhere on the perpendicular bisector, c_{13}. Then, with O_4 as a center and using any radius R, we construct a circular arc. Next, with centers at C_1 and C_3, and any other radius, r, we strike arcs to intersect the arc of radius R. These two intersections define points A_1 and A_3 on the input link. We construct

*The methods presented here were devised and presented by Hain in [8, Chap. 17].

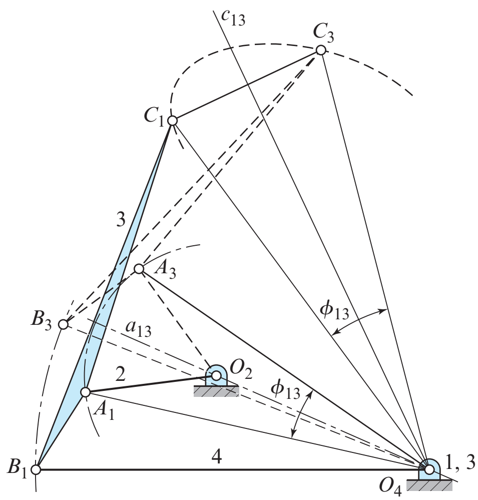

Figure 9.30

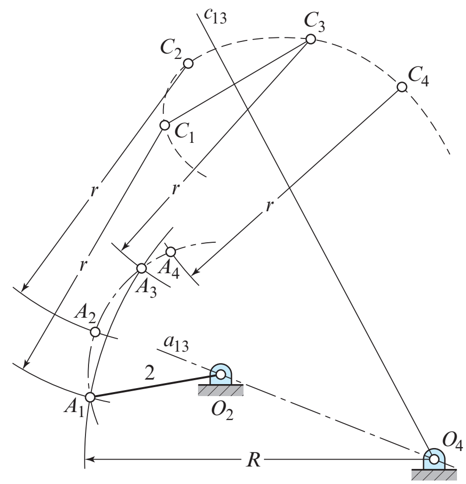

Figure 9.31

the perpendicular bisector, a_{13}, to A_1A_3 and note that it passes through O_4. We locate O_2 anywhere on a_{13}. This provides an opportunity to choose a convenient length for the input link. Now, we use O_2 as a center and draw the crank circle through A_1 and A_3. Points A_2 and A_4 on this circle are obtained by striking arcs of radius r again about C_2 and C_4. This completes the first phase of the synthesis; we have located O_2 and O_4 relative to the desired path and hence defined distance O_2O_4. We have also defined the length of the input link and located its positions relative to the four precision points on the path.

Our next task is to locate point B, the point of attachment of the coupler and output link. Any one of the four locations of B can be used; in this example we use the B_1 position.

Before beginning the final step, we note that the linkage is now defined. Four arbitrary decisions were made: the location of O_4, the radii R and r, and the location of O_2. Thus, ∞^4 solutions are possible.

Referring to Fig. 9.32, we locate point 2 by making triangles $C_2A_2O_4$ and C_1A_12 congruent. We locate point 4 by making $C_4A_1O_4$ and C_1A_14 congruent. Points 4, 2, and O_4 lie on a circle whose center is B_1. So, we find B_1 at the intersection of the perpendicular bisectors of O_42 and O_44. Note that the procedure used causes points 1 and 3 to coincide with O_4. With B_1 located, the links can be drawn in place and the mechanism tested to see how well it traces the desired path.

To synthesize a linkage to generate a path through five precision points, we can make two point reductions. We begin by choosing five points, C_1 to C_5, on the path to be traced. We choose two pairs of these for reduction purposes. In Fig. 9.33, we choose the pairs C_1C_5 and C_2C_3. Other pairs that could have been chosen are

$$C_1C_5, C_2C_4; \quad C_1C_5, C_3C_4; \quad C_1C_4, C_2C_3; \quad C_2C_5, C_3C_4.$$

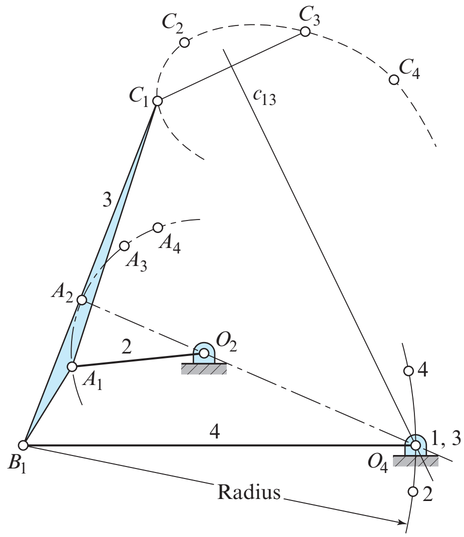

Figure 9.32

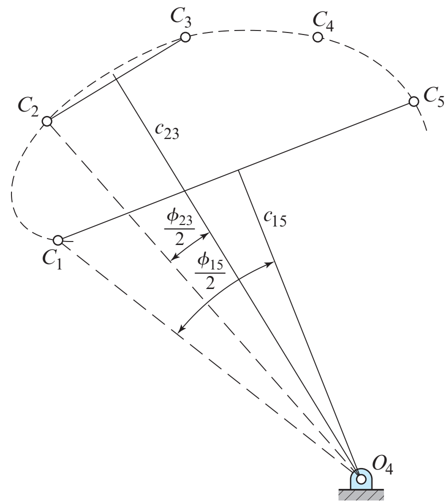

Figure 9.33

We construct the perpendicular bisectors c_{23} and c_{15} of the lines connecting each pair. These intersect at point O_4. Note that O_4 can, therefore, be located conveniently by a judicious choice of the pairs to be used as well as by the choice of the positions of the points C_i on the path.

The next step is best performed using a sheet of tracing paper as an overlay. We secure the tracing paper to the drawing and mark upon it center O_4, perpendicular bisector c_{23}, and another line from O_4 to C_2. Such an overlay is shown in Fig. 9.34*a* with line O_4C_2 designated O_4C_2'. This defines the angle $\phi_{23}/2$. Now, we rotate the overlay about point O_4 until the perpendicular bisector coincides with c_{15} and repeat for point C_1. This defines the angle $\phi_{15}/2$ and the corresponding line, O_4C_1'.

Now, we pin the overlay at point O_4, using a thumbtack, and rotate it until a good position is found. It is helpful to set the compass for some convenient radius, r, and draw circles about each point C_i. The intersection of these circles with lines O_4C_1' and O_4C_2' on the overlay, and with each other, will reveal which areas will be profitable to investigate (Fig. 9.34*b*).

The final steps in the solution are shown in Fig. 9.35. Having located a good position for the overlay, we transfer the three lines to the drawing and remove the overlay. Now, we draw a circle of radius r to intersect O_4C_1' and locate point A_1. Another arc of the same radius r from point C_2 intersects O_4C_2' at point A_2. With A_1 and A_2 located, we draw the perpendicular bisector a_{12}; it intersects the perpendicular bisector a_{23} at O_2, giving us the length of the input rocker. A circle through A_1 about O_2 contains all the design positions of point A; we use the same radius r and locate A_3, A_4, and A_5 on arcs about C_3, C_4, and C_5.

We have now located everything except point B_1, and this is determined as before. Note that points 2 and 3 are coincident (called a *double point*) because of the choice of point O_4 on the perpendicular bisector c_{23}. To locate this double point, we strike an arc from C_1 of radius C_2O_4. Then, we strike another arc from A_1 of radius A_2O_4. These

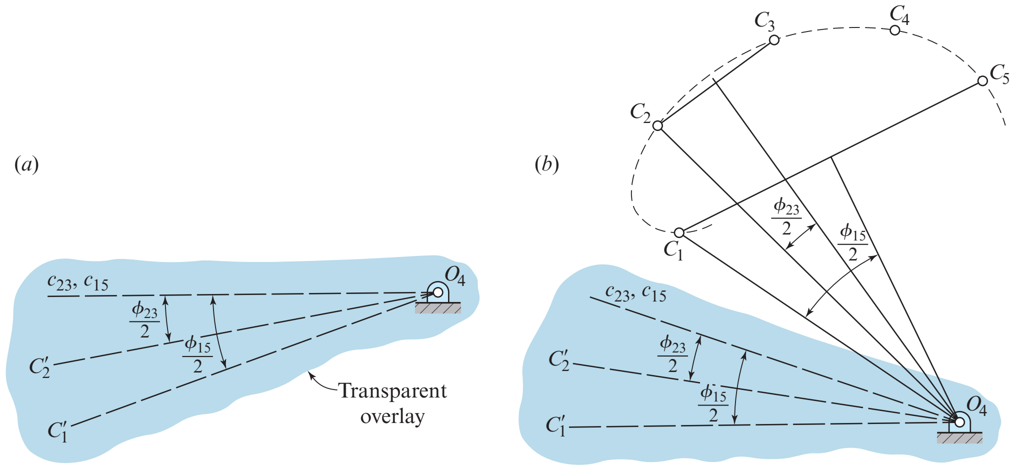

Figure 9.34

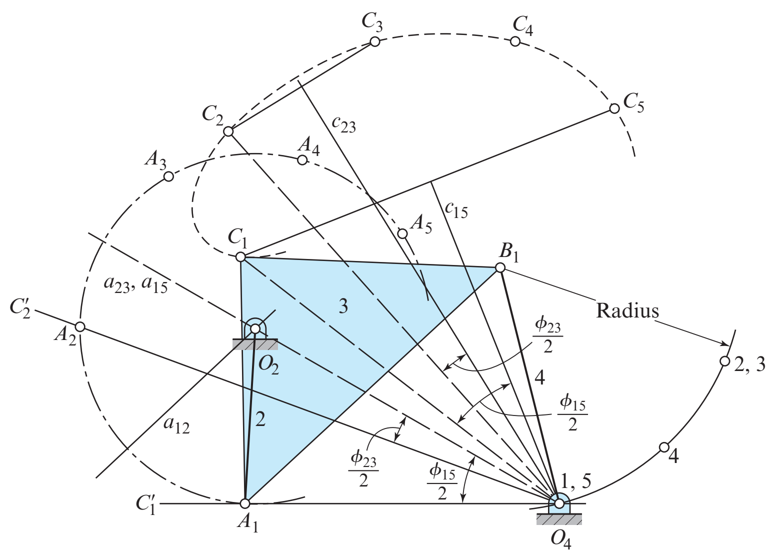

Figure 9.35

intersect at the double point 2, 3. To locate point 4, we strike an arc from C_1 of radius C_4O_4, and another from A_1 of radius A_4O_4. Note that point O_4 and the double point 1, 5 are coincident, since the synthesis is based on inversion on the O_4B_1 position. Points O_4, 4, and double point 2, 3 lie on a circle whose center is B_1, as shown in Fig. 9.35. The linkage is completed by drawing the coupler link and the follower link in the first design posture.

9.10 COGNATE LINKAGES; ROBERTS-CHEBYSHEV THEOREM

One of the remarkable properties of the planar four-bar linkage is that there is not just one but three four-bar linkages that generate the same coupler curve. This was discovered by Roberts* in 1875 and by Chebyshev in 1878 and hence is known as the Roberts-Chebyshev theorem. Although mentioned in an English publication in 1954 [5], it did not appear in the American literature until 1958 when it was presented, independently and almost simultaneously, by Hartenberg and Denavit of Northwestern University [10] and by Hinkle of Michigan State University [12].

In Fig. 9.36, let O_1ABO_2 be the original four-bar linkage with coupler point P attached to AB. The remaining two linkages defined by the Roberts-Chebyshev theorem were termed *cognate linkages* by Hartenberg and Denavit. Each of the cognate linkages is shown in Fig. 9.36; one is $O_1A_1C_1O_3$ and uses short dashes for showing the links, and the other is $O_2B_2C_2O_3$ and uses long dashes. The construction is evident by observing that there are four similar triangles, each containing the angles α, β, and γ, and three different parallelograms.

A good way to obtain the dimensions of the two cognate linkages is to imagine that the frame connections O_1, O_2, and O_3 can be unfastened. Then, imagine that O_1, O_2, and O_3 are "pulled" away from each other until a straight line is formed by the crank, coupler, and follower of each linkage. If we were to do this for Fig. 9.36, then we would obtain Fig. 9.37. Note that the frame distances are now incorrect, but all the movable links are of the correct lengths. Given any four-bar linkage and its coupler point, one can create a

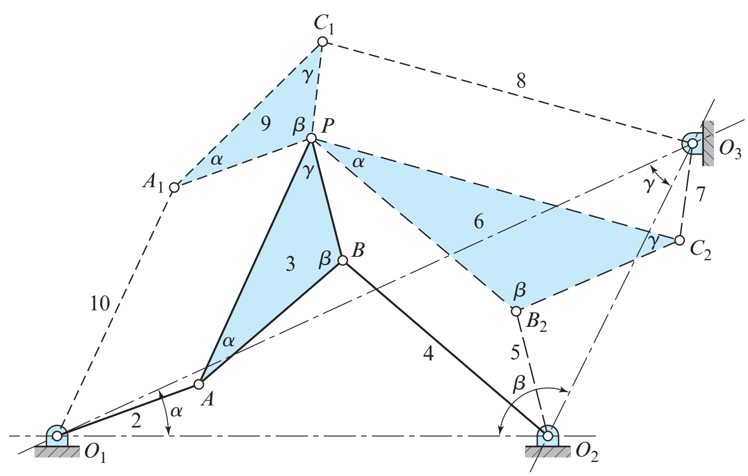

Figure 9.36

* Samuel Roberts (1827–1913), a mathematician; this was not the same Roberts of the approximate straight-line generator (Fig. 1.24*b*).

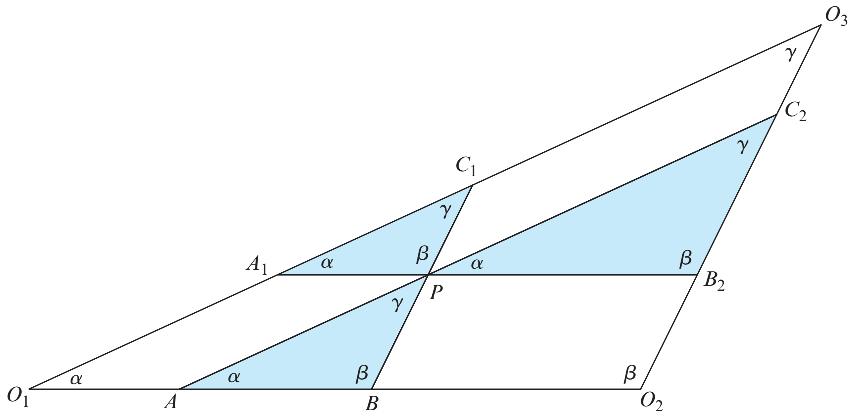

Figure 9.37 The Cayley diagram.

drawing similar to Fig. 9.37 and obtain the dimensions of the other two cognate linkages. This approach was discovered by A. Cayley and is called the *Cayley diagram*.*

If the tracing point, P is on the straight line AB or its extensions, a figure like Fig. 9.37 is of little help, since all three linkages are compressed into a single straight line. An example is shown in Fig. 9.38, where O_1ABO_2 is the original linkage having a coupler point, P on an extension of AB. To find the cognate linkages, locate O_3 on an extension of O_1O_2 in the same ratio as AB is to BP. Then, construct, in order, the parallelograms O_1A_1PA, O_2B_2PB, and $O_3C_1PC_2$.

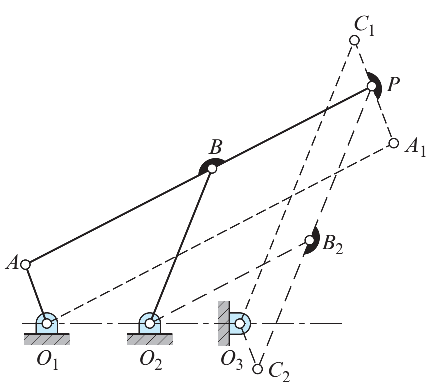

Figure 9.38

* Arthur Cayley (1821–1895), 1876. On three-bar motion, *Proc. Lond. Math. Soc.* **7**:136–66. In Cayley's time, a four-bar linkage was described as a three-bar mechanism, since the idea of a kinematic chain had not yet been conceived.

Hartenberg and Denavit demonstrated that the angular-velocity relations between the links in Fig. 9.36 are

$$\omega_9 = \omega_2 = \omega_7, \quad \omega_{10} = \omega_3 = \omega_5, \quad \omega_8 = \omega_4 = \omega_6. \tag{9.23}$$

They also observed that if crank 2 is driven at a constant angular velocity and if the velocity relationships are to be preserved during generation of the coupler curve, the cognate mechanisms must be driven at variable angular velocities.

9.11 FREUDENSTEIN'S EQUATION

In Fig. 9.39, we replace the links of a four-bar linkage by position vectors and write the loop-closure equation

$$\overset{\surd\surd}{\mathbf{r}_1} + \overset{\surd I}{\mathbf{r}_2} + \overset{\surd ?}{\mathbf{r}_3} + \overset{\surd ?}{\mathbf{r}_4} = \mathbf{0} \tag{a}$$

In complex polar notation, Eq. (a) is written as

$$r_1 e^{j\theta_1} + r_2 e^{j\theta_2} + r_3 e^{j\theta_3} + r_4 e^{j\theta_4} = 0. \tag{b}$$

From Fig. 9.39, we see that $\theta_1 = 180° = \pi$ radians, from which $e^{j\theta_1} = -1$. Therefore, if Eq. (b) is transformed into complex rectangular form, and if the real and the imaginary components are separated, we obtain the two algebraic equations

$$-r_1 + r_2 \cos\theta_2 + r_3 \cos\theta_3 + r_4 \cos\theta_4 = 0, \tag{c}$$

$$r_2 \sin\theta_2 + r_3 \sin\theta_3 + r_4 \sin\theta_4 = 0. \tag{d}$$

The coupler angle θ_3 can be deleted, and the output angle θ_4 can be expressed in terms of the input angle θ_2 by the following procedure. Moving all terms except those involving θ_3 to the right-hand side and squaring both sides gives

$$r_3^2 \cos^2\theta_3 = (r_1 - r_2\cos\theta_2 - r_4\cos\theta_4)^2 \tag{e}$$

$$r_3^2 \sin^2\theta_3 = (-r_2\sin\theta_2 - r_4\sin\theta_4)^2. \tag{f}$$

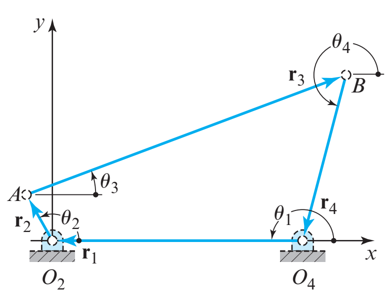

Figure 9.39

Now, expanding the right-hand sides and adding the two equations gives

$$r_3^2 = r_1^2 + r_2^2 + r_4^2 - 2r_1 r_2 \cos\theta_2 - 2r_1 r_4 \cos\theta_4 + 2r_2 r_4 (\cos\theta_2 \cos\theta_4 + \sin\theta_2 \sin\theta_4). \quad (g)$$

Making the substitution $(\cos\theta_2 \cos\theta_4 + \sin\theta_2 \sin\theta_4) = \cos(\theta_2 - \theta_4)$, and then dividing by the factor $2r_2r_4$, and rearranging again gives

$$\frac{r_3^2 - r_1^2 - r_2^2 - r_4^2}{2r_2r_4} + \frac{r_1}{r_4}\cos\theta_2 + \frac{r_1}{r_2}\cos\theta_4 = \cos(\theta_2 - \theta_4). \quad (h)$$

Freudenstein [3] writes this equation in the form

$$K_1 \cos\theta_2 + K_2 \cos\theta_4 + K_3 = \cos(\theta_2 - \theta_4), \quad (9.24)$$

where

$$K_1 = \frac{r_1}{r_4}, \quad (9.25)$$

$$K_2 = \frac{r_1}{r_2}, \quad (9.26)$$

$$K_3 = \frac{r_3^2 - r_1^2 - r_2^2 - r_4^2}{2r_2r_4}. \quad (9.27)$$

We have already learned graphic methods for synthesizing a four-bar linkage so that the motion of the output is coordinated with that of the input. Freudenstein's equation, Eq. (9.24), enables us to perform this same task by analytic means. Thus, suppose we wish the output of a four-bar linkage to occupy the postures ψ_1, ψ_2, and ψ_3 corresponding to the postures ϕ_1, ϕ_2, and ϕ_3 of the input. In Eq. (9.24), we simply replace θ_2 with ϕ_i, and θ_4 with ψ_i, and write the equation three times, once for each posture. This gives

$$\begin{aligned} K_1 \cos\phi_1 + K_2 \cos\psi_1 + K_3 &= \cos(\phi_1 - \psi_1), \\ K_1 \cos\phi_2 + K_2 \cos\psi_2 + K_3 &= \cos(\phi_2 - \psi_2), \\ K_1 \cos\phi_3 + K_2 \cos\psi_3 + K_3 &= \cos(\phi_3 - \psi_3). \end{aligned} \quad (i)$$

Equations (i) are then solved simultaneously for the three unknowns, K_1, K_2, and K_3. Then, a length, say r_1, is selected for one of the links, and Eqs. (9.25) through (9.27) are solved for the dimensions of the other three links. The method is best illustrated by an example.

EXAMPLE 9.7

Synthesize a function generator to follow the equation

$$y = \frac{1}{x} \quad \text{over the range} \quad 1 \le x \le 2$$

using three precision postures.

SOLUTION

Choosing Chebyshev spacing, we find, from Eq. (9.22), the values of x and the corresponding values of y to be

$$x_1 = 1.067, \quad y_1 = 0.937,$$
$$x_2 = 1.500, \quad y_2 = 0.667,$$
$$x_3 = 1.933, \quad y_3 = 0.517.$$

We must now choose starting angles for the input and output and also total displacement angles for each. These are arbitrary decisions and may or may not result in a good linkage design in the sense that the structural errors between the precision postures may be large or the transmission angles may be poor. Sometimes, in such a synthesis, it is found that one of the pivots must be disconnected to move from one precision posture to another. Generally, some trial-and-error work may be necessary to discover the best choices of starting angles and total displacement angles.

Here, for the input, we choose a starting angle of $\phi_{\min} = 30°$ and a total displacement angle of $\Delta\phi = \phi_{\max} - \phi_{\min} = 90°$. For the output, we choose a starting angle of $\psi_{\min} = 240°$ and again choose a displacement of $\Delta\psi = \psi_{\max} - \psi_{\min} = 90°$ total travel. With these choices made, the first and last rows of Table 9.2 can be completed.

Next, to obtain the values of ϕ and ψ corresponding to the precision postures, we write

$$\phi = ax + b, \quad \psi = cy + d, \tag{1}$$

and use the data in the first and last rows of Table 9.2 to evaluate constants a, b, c, and d. When this is done, we find Eqs. (1) are

$$\phi = 90°x - 60°, \quad \psi = -180°y + 420°. \tag{2}$$

These equations can now be used to compute the data for the remaining rows in Table 9.2 and to determine the scales of the input and output links of the synthesized linkage.

Now, we take the values of ϕ and ψ from the second line of Table 9.2 and substitute them for θ_2 and θ_4 in Eq. (9.24). Then, if we repeat this for the third and fourth lines, we have the three equations

$$\begin{aligned} K_1 \cos 36.03° + K_2 \cos 251.34° + K_3 &= \cos\left(36.03° - 251.34°\right), \\ K_1 \cos 75.00° + K_2 \cos 300.00° + K_3 &= \cos\left(75.00° - 300.00°\right), \\ K_1 \cos 113.97° + K_2 \cos 326.94° + K_3 &= \cos\left(113.97° - 326.94°\right). \end{aligned} \tag{3}$$

When the trigonometric operations are carried out, we have

$$\begin{aligned} 0.8087K_1 - 0.3200K_2 + K_3 &= -0.8160, \\ 0.2588K_1 + 0.5000K_2 + K_3 &= -0.7071, \\ -0.4062K_1 + 0.8381K_2 + K_3 &= -0.8389. \end{aligned} \tag{4}$$

Table 9.2 Precision Postures

Position	x	ψ, deg	y	ϕ, deg
—	1.000	30.00	1.000	240.00
1	1.067	36.03	0.937	251.34
2	1.500	75.00	0.667	300.00
3	1.933	113.97	0.517	326.94
—	2.000	120.00	0.500	330.00

Upon solving Eqs. (4), we obtain

$$K_1 = 0.4032, \quad K_2 = 0.4032, \quad K_3 = -1.0130.$$

Using $r_1 = 1.000$ units, we obtain, from Eq. (9.25),

$$r_4 = \frac{r_1}{K_1} = \frac{1.000}{0.4032} = 2.480 \text{ units.} \qquad \textit{Ans.}$$

Similarly, from Eqs. (9.26) and (9.27), we learn that

$$r_2 = 2.480 \text{ units} \quad \text{and} \quad r_3 = 0.917 \text{ units.} \qquad \textit{Ans.}$$

The result is the crossed four-bar linkage shown in Fig. 9.40.

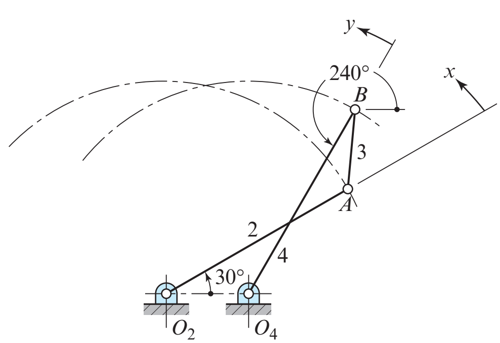

Figure 9.40 The crossed four-bar linkage.

Freudenstein offers the following suggestions, which are helpful in synthesizing such function generators.

1. The total displacement angles of the input and output links should be less than 120°.
2. Avoid the generation of symmetric functions, such as $y = x^2$, over a symmetric range, such as $-1 \leq x \leq 1$.
3. Avoid the generation of functions having abrupt changes in slope.

9.12 ANALYTIC SYNTHESIS USING COMPLEX ALGEBRA

Another very powerful approach to the synthesis of planar linkages takes advantage of the concept of precision postures and the operations available through the use of complex algebra. Basically, as with Freudenstein's equation in the previous section, the idea is to write complex-algebraic equations describing the final linkage in each of its precision postures.

Since links do not change lengths during the motion, the magnitudes of these complex vectors do not change from one posture to the next, but their angles vary. By writing equations at several precision postures, we obtain a set of simultaneous equations that may be solved for the unknown magnitudes and angles.

The method is very flexible and much more general than is illustrated here. More complete coverage is given in texts such as that by Erdman, Sandor, and Kota [2, Chap. 8]. However, the fundamental ideas and some of the operations are illustrated here by an example.

EXAMPLE 9.8

In this example, we wish to design a mechanical strip-chart recorder. The concept of the final design is shown in Fig. 9.41. We assume that the signal to be recorded is available as a shaft rotation having a range of $0 \leq \phi \leq 90°$ clockwise. This rotation is to be converted into a straight-line motion of the recorder pen over a range of $0 \leq s \leq 4$ in to the right, with a linear relationship between changes of ϕ and s.*

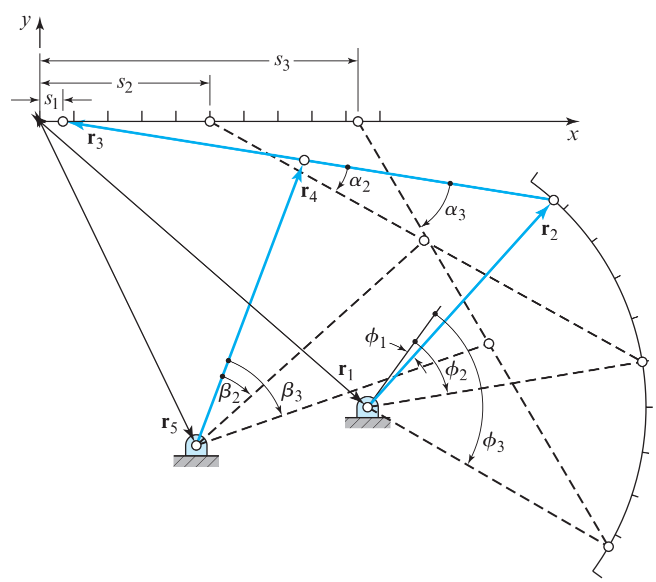

Figure 9.41 Three-posture synthesis of chart-recorder linkage using complex algebra.

* A similar problem is solved graphically in [11, pp. 244–8 and 274–8].

SOLUTION

We choose three accuracy postures for our design approach. Using Chebyshev spacing over the range to reduce structural error and taking counterclockwise angles as positive, the three accuracy postures are given by Eq. (9.22):

$$\begin{aligned} \phi_1 &= -6^\circ = -0.104\,72 \text{ rad}, & s_1 &= 0.267\,95 \text{ in}, \\ \phi_2 &= -45^\circ = -0.785\,40 \text{ rad}, & s_2 &= 2.000\,00 \text{ in}, \\ \phi_3 &= -84^\circ = -1.466\,08 \text{ rad}, & s_3 &= 3.732\,05 \text{ in}. \end{aligned} \tag{1}$$

First we tackle the design of the dyad consisting of the input crank, $\mathbf{r}_2$, and the coupler link, $\mathbf{r}_3$. Taking these to be complex vectors for the mechanism in its first accuracy posture, and designating the unknown location of the fixed pivot by the complex vector $\mathbf{r}_1$, we write a loop-closure equation at each of the three precision postures:

$$\begin{aligned} \mathbf{r}_1 + \mathbf{r}_2 \;\;+ \mathbf{r}_3 \;\;&= s_1, \\ \mathbf{r}_1 + \mathbf{r}_2 e^{j(\phi_2-\phi_1)} + \mathbf{r}_3 e^{j\alpha_2} &= s_2, \\ \mathbf{r}_1 + \mathbf{r}_2 e^{j(\phi_3-\phi_1)} + \mathbf{r}_3 e^{j\alpha_3} &= s_3, \end{aligned} \tag{2}$$

where the angles, α_j, represent the angular displacements of the coupler link from its first posture. Next, by subtracting the first of these equations from each of the others and rearranging, we obtain

$$\begin{aligned} \left[e^{j(\phi_2-\phi_1)} - 1\right]\mathbf{r}_2 + \left[e^{j\alpha_2} - 1\right]\mathbf{r}_3 &= s_2 - s_1, \\ \left[e^{j(\phi_3-\phi_1)} - 1\right]\mathbf{r}_2 + \left[e^{j\alpha_3} - 1\right]\mathbf{r}_3 &= s_3 - s_1. \end{aligned} \tag{3}$$

Here, we note that we have two complex equations in two complex unknowns, $\mathbf{r}_2$ and $\mathbf{r}_3$, except we note that the coupler displacement angles, α_j, which appear in the coefficients, are also unknowns. Thus, we have more unknowns than equations and are free to specify additional conditions or additional data for the problem. Making estimates based on crude sketches of our contemplated design, therefore, we make the following arbitrary decisions:

$$\begin{aligned} \alpha_2 &= -20^\circ = -0.349\,07 \text{ rad}, \\ \alpha_3 &= -50^\circ = -0.872\,66 \text{ rad}. \end{aligned} \tag{4}$$

Collecting the data from Eqs. (1) and (4), substituting into Eqs. (3), and evaluating, we find

$$\begin{aligned} -(0.222\,85 + j0.629\,32)\mathbf{r}_2 - (0.060\,31 + j0.342\,02)\mathbf{r}_3 &= 1.732\,05, \\ -(0.792\,09 + j0.978\,15)\mathbf{r}_2 - (0.357\,21 + j0.766\,04)\mathbf{r}_3 &= 3.464\,10, \end{aligned}$$

which we can now solve for the two unknowns:

$$\mathbf{r}_2 = 2.153\,26 + j2.448\,60 = 3.261 \text{ in}\angle 48.67^\circ, \qquad \textit{Ans.}$$

$$\mathbf{r}_3 = -5.725\,48 + j0.952\,04 = 5.804 \text{ in}\angle 170.56^\circ. \qquad \textit{Ans.}$$

Then, using the first of Eqs. (2), we solve for the location of the fixed pivot:

$$\begin{aligned}\mathbf{r}_1 &= s_1 - \mathbf{r}_2 - \mathbf{r}_3 \\ &= 3.840\,17 - j3.400\,64 = 5.129 \text{ in}\angle -41.53^\circ.\end{aligned} \qquad \textit{Ans.}$$

Thus far, we have completed the design of the dyad, which includes the input crank. Before proceeding, we should note that an identical procedure could have been used for the design of a slider-crank linkage, for one dyad of a four-bar linkage used for any path generation or motion generation problem, or for a variety of other applications. Our total design is not yet completed, but we should note the general applicability of the procedures covered to other linkage synthesis problems.

Continuing with our design of the recording instrument, however, we now must find the postures and dimensions of the dyad, $\mathbf{r}_4$ and $\mathbf{r}_5$ of Fig. 9.41. As shown, we choose to connect the moving pivot of the output crank at the midpoint of the coupler link to minimize its mass and to keep dynamic forces low. Thus, we can write another loop-closure equation including the rocker at each of the three precision postures:

$$\begin{aligned}\mathbf{r}_5 + \mathbf{r}_4 \quad + 0.5\mathbf{r}_3 \quad &= s_1, \\ \mathbf{r}_5 + \mathbf{r}_4 e^{j\beta_2} + 0.5\mathbf{r}_3 e^{j\alpha_2} &= s_2, \\ \mathbf{r}_5 + \mathbf{r}_4 e^{j\beta_3} + 0.5\mathbf{r}_3 e^{j\alpha_3} &= s_3.\end{aligned} \tag{5}$$

Substituting the known data into these equations and rearranging, we obtain

$$\begin{aligned}\mathbf{r}_5 + \mathbf{r}_4 + (-3.130\,69 + j0.476\,02) &= \mathbf{0}, \\ \mathbf{r}_5 + \left(e^{j\beta_2}\right)\mathbf{r}_4 + (-4.527\,25 + j1.426\,52) &= \mathbf{0}, \\ \mathbf{r}_5 + \left(e^{j\beta_3}\right)\mathbf{r}_4 + (-5.207\,45 + j2.499\,03) &= \mathbf{0}.\end{aligned} \tag{6}$$

These are three simultaneous complex equations in only two vector unknowns, $\mathbf{r}_4$ and $\mathbf{r}_5$, and thus we are not free to choose the rotation angles, β_2 and β_3, arbitrarily. For Eqs. (6) to have consistent nontrivial solutions, it is necessary that the determinant of the matrix of coefficients be zero. Thus, β_2 and β_3 must be chosen such that

$$\begin{vmatrix} 1 & 1 & (-3.130\,69 + j0.476\,02) \\ 1 & e^{j\beta_2} & (-4.527\,25 + j1.426\,52) \\ 1 & e^{j\beta_3} & (-5.207\,45 + j2.499\,03) \end{vmatrix} = 0, \tag{7}$$

which expands to

$$(-2.076\,76 + j2.023\,01)\,e^{j\beta_2} + (1.396\,56 - j0.950\,50)\,e^{j\beta_3} + (0.680\,20 - j1.072\,51) = 0.$$

Solving this for $e^{j\beta_2}$ gives

$$e^{j\beta_2} = (0.573\,82 + j0.101\,28)\,e^{j\beta_3} + (0.426\,19 - j0.101\,28),$$

and equating the real and imaginary parts, respectively, gives

$$\begin{aligned}\cos\beta_2 &= 0.573\,82\cos\beta_3 - 0.101\,28\sin\beta_3 + 0.426\,19, \\ \sin\beta_2 &= 0.101\,28\cos\beta_3 + 0.573\,82\sin\beta_3 - 0.101\,28.\end{aligned} \tag{8}$$

These equations can now be squared and added to eliminate the unknown β_2. The result, after rearrangement, is a single equation in β_3:

$$0.468\,59\cos\beta_3 - 0.202\,56\sin\beta_3 - 0.468\,57 = 0. \tag{9}$$

This equation can be solved by substituting the tangent of the half-angle identities

$$x = \tan\frac{\beta_3}{2}, \quad \cos\beta_3 = \frac{1-x^2}{1+x^2}, \quad \sin\beta_3 = \frac{2x}{1+x^2}, \tag{10}$$

$$0.468\,59\left(1-x^2\right) - 0.202\,56\,(2x) - 0.468\,57\left(1+x^2\right) = 0.$$

This reduces to the quadratic equation

$$-0.937\,16x^2 - 0.405\,12x + 0.000\,02 = 0,$$

for which the roots are

$$x = -0.432\,66 \quad \text{or} \quad x = 0.000\,27,$$

and, from Eq. (10),

$$\beta_3 = -46.79^\circ \quad \text{or} \quad \beta_3 = 0.03^\circ.$$

Guided by our sketch of the desired design, we choose the first of these roots, $\beta_3 = -46.79^\circ$. Then, returning to Eqs. (8), we find the value $\beta_2 = -26.76^\circ$, and finally, from Eqs. (6), we find the final solution:

$$\mathbf{r}_4 = 1.299\,71 + j3.410\,92 = 3.650\text{ in}\angle 69.14^\circ, \qquad \textit{Ans.}$$

$$\mathbf{r}_5 = 1.830\,98 - j3.886\,94 = 4.297\text{ in}\angle -64.78^\circ. \qquad \textit{Ans.}$$

Note that this second part of the solution, solving for the rocker $\mathbf{r}_4$, is also a very general approach that could be used to design a crank to go through three given precision postures in a variety of other problems. Although only a specific case is presented here, the approach arises repeatedly in linkage design.

Of course, before we finish, we should evaluate the quality of our solution by analysis of the linkage we have designed. This was performed here using the equations of Chap. 2 to find the locations of the coupler point for 20 equally spaced crank increments spanning the given range of motion. As expected, there is structural error; the coupler curve of the recording pen tip is not exactly straight, and the displacement increments are not perfectly linear over the range of travel of the pen. However, the solution is quite good; the deviation from a straight line is less than 0.020 in, or 0.5% of the travel, and the linearity between the input crank rotation and coupler point travel is better than 1% of the travel. As expected, the structural error follows a regular pattern and vanishes at the three precision postures. The transmission angle remains larger than 70° throughout the range; thus, no problems with force transmission are expected. Although the design might be improved slightly using additional precision postures, the present solution seems excellent, and the additional effort does not appear necessary.

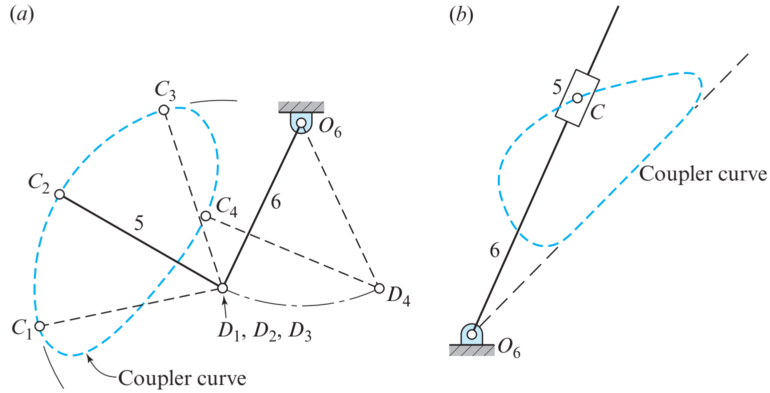

Figure 9.42 Link 6 dwells as point C travels: (*a*) the circular-arc path $C_1C_2C_3$; (*b*) on the straight portion of the coupler curve.

9.13 SYNTHESIS OF DWELL LINKAGES

One of the most interesting uses of coupler curves having straight-line or circular-arc segments is in the synthesis of linkages having a substantial dwell during a portion of their operating cycle. Using segments of coupler curves, it is not difficult to synthesize linkages having a dwell at either or both of the extremes of their motion or at an intermediate posture.

In Fig. 9.42*a*, a coupler curve having an approximately elliptic shape is selected from the Hrones and Nelson atlas [14] so that a substantial portion of the curve approximates a circular arc. The four-bar linkage that generates this coupler curve is not shown in the figure. Connecting link 5 is given a length equal to the radius of this arc. Thus, in Fig. 9.42*a*, points D_1, D_2, and D_3 are stationary as coupler point C moves through positions C_1, C_2, and C_3. Also, the same purpose is accomplished with a slider, as shown in Fig. 9.42*b*; link 6 dwells as point C travels along the straight portion of the coupler curve. The length of output link 6 and the location of the frame pivot, O_6, depend upon the desired angle of oscillation of this link. The frame pivot should also be positioned for a desirable transmission angle.

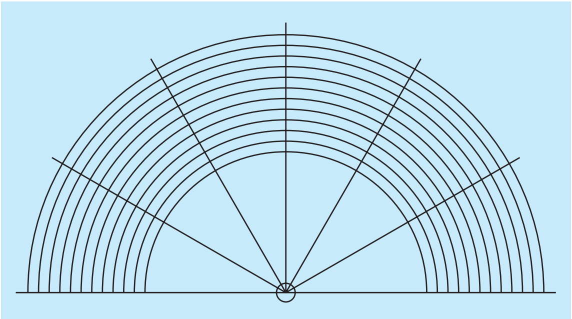

Figure 9.43 Overlay for use with the Hrones and Nelson atlas.

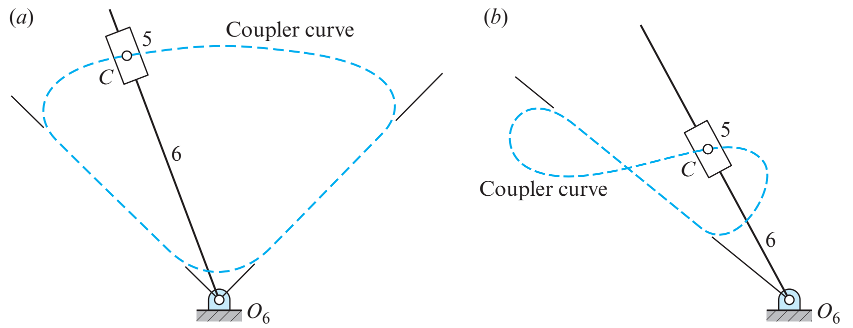

Figure 9.44 Link 6 dwells: (*a*) at each end of its swing; (*b*) in the central portion of its swing.

When segments of circular arcs are desired for a coupler curve, an organized method of searching the Hrones and Nelson atlas can be employed. The overlay, shown in Fig. 9.43, is made on a sheet of tracing paper and can be fitted over the paths in the atlas very quickly. It reveals immediately the radius of curvature of the segment, the location of pivot point D, and the displacement angle of the connecting link.

Figure 9.44 shows two ideas for dwell mechanisms employing a slider. A coupler curve having a straight-line segment is used in each, and the pivot point, O_6, is placed on an extension of this line.

The arrangement shown in Fig. 9.44*a* has a dwell at both extremes of the motion of link 6. A practical design of this mechanism may be difficult to achieve, however, since link 6 has a high velocity when the slider is near the pivot, O_6.

The slider mechanism of Fig. 9.44*b* uses a figure-eight coupler curve having a straight-line segment to produce an intermediate dwell linkage. Pivot O_6 must be located on an extension of the straight-line segment, as shown.

9.14 INTERMITTENT ROTARY MOTION

The *Geneva wheel*, or *Maltese cross*, is a cam-like mechanism that provides intermittent rotary motion and is widely used in both low-speed and high-speed machinery. Although originally developed as a stop to prevent overwinding of watches, it is now used extensively in automatic machinery, for example, where a spindle, turret, or worktable must be indexed. It is also used in motion-picture projectors to provide the intermittent advance of the film.

A drawing of a six-slot Geneva mechanism is shown in Fig. 9.45. Note that the centerlines of the slot and crank are mutually perpendicular at engagement and at disengagement. The crank, which usually rotates at a uniform angular velocity, carries a roller to engage with the slots. During one revolution of the crank, the Geneva wheel rotates a fractional part of a revolution, the amount of which is dependent upon the number of slots. The circular segment attached to the crank effectively locks the wheel against rotation when the roller is not in engagement and also locates the wheel for correct engagement of the roller with the next slot.

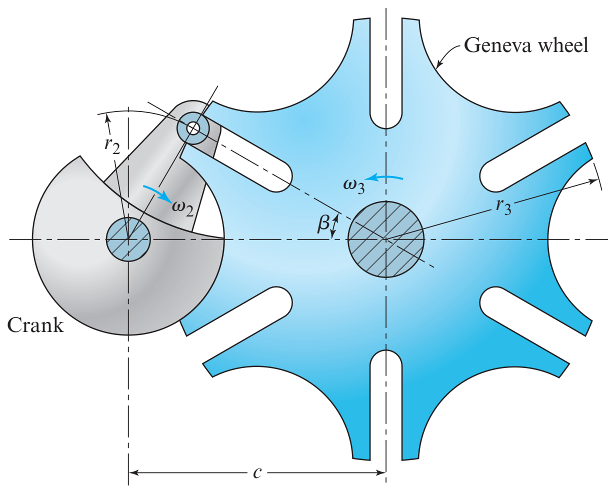

Figure 9.45 Geneva mechanism.

The design of a Geneva mechanism is initiated by specifying the crank radius, the roller diameter, and the number of slots. At least three slots are necessary, but most problems can be solved with wheels having from 4 to 12 slots. The design procedure is shown in Fig. 9.46. The angle β is half the angle subtended by adjacent slots; that is,

$$\beta = \frac{360^\circ}{2n}, \tag{a}$$

where n is the number of slots in the wheel. Then, defining r_2 as the crank radius, we have

$$c = \frac{r_2}{\sin\beta}, \tag{b}$$

where c is the center distance. Note, too, from Fig. 9.46, that the actual Geneva-wheel radius is more than that which would be obtained by a zero-diameter roller. This is because of the difference between the sine and the tangent of the angle subtended by the roller, measured from the wheel center.

After the roller has entered the slot and is driving the wheel, the geometry is that of Fig. 9.47. Here, θ_2 is the crank angle, and θ_3 is the wheel angle. They are related trigonometrically by

$$\tan\theta_3 = \frac{\sin\theta_2}{(c/r_2) - \cos\theta_2}. \tag{c}$$

We can determine the angular velocity of the wheel for any value of θ_2 by differentiating Eq. (c) with respect to time. This produces

$$\omega_3 = \frac{(c/r_2)\cos\theta_2 - 1}{1 + \left(c^2/r_2^2\right) - 2\,(c/r_2)\cos\theta_2}\,\omega_2. \tag{9.28}$$

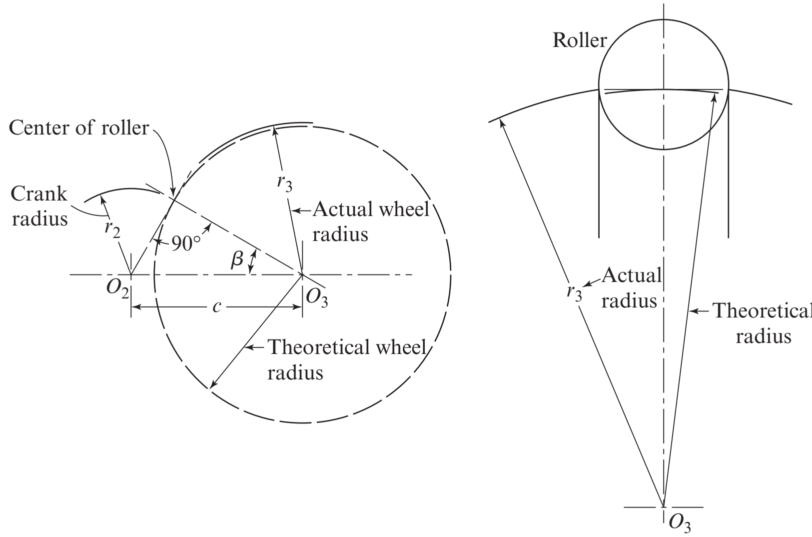

Figure 9.46 Design of a Geneva wheel.

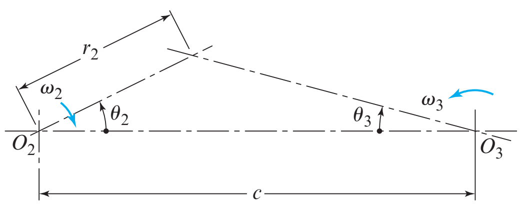

Figure 9.47

The maximum wheel velocity occurs when the crank angle is zero. Substituting $\theta_2 = 0$, therefore, gives

$$(\omega_3)_{\max} = \frac{r_2}{c - r_2}\omega_2. \tag{9.29}$$

The angular acceleration of the wheel with constant input crank speed, obtained by differentiating Eq. (9.28) with respect to time, is

$$\alpha_3 = \frac{(c/r_2)\sin\theta_2\left(1 - c^2/r_2^2\right)}{\left[1 + (c/r_2)^2 - 2(c/r_2)\cos\theta_2\right]^2}\omega_2^2. \tag{9.30}$$

The angular acceleration reaches a maximum when the crank angle is

$$\theta_2 = \cos^{-1}\left\{\pm\sqrt{\left[\frac{1 + \left(c^2/r_2^2\right)}{4(c/r_2)}\right]^2 + 2} - \frac{1 + (c/r_2)^2}{4(c/r_2)}\right\}. \tag{9.31}$$

This occurs when the roller has advanced about 30% into the slot.

Several methods have been employed to reduce the wheel angular acceleration that reduces inertia forces and the consequent wear on the sides of the slot. Among these is the idea of using a curved slot. This can reduce the acceleration but also increases the deceleration and consequently the wear on the other side of the slot.

Another method uses the Hrones and Nelson atlas [14] for synthesis. The idea is to place the roller on the coupler of a four-bar linkage. During the period in which it drives the wheel, the path of the roller should be curved, and the roller should have a low value of angular acceleration. Figure 9.48 shows one solution and includes the path taken by the roller. This is the type of path that is sought while leafing through the atlas.

The inverse Geneva mechanism of Fig. 9.49 enables the wheel to rotate in the same direction as the crank and requires less radial space. The locking device is not shown, but this can be a circular segment attached to the crank, as before, which locks by wiping against a built-up rim on the periphery of the wheel.

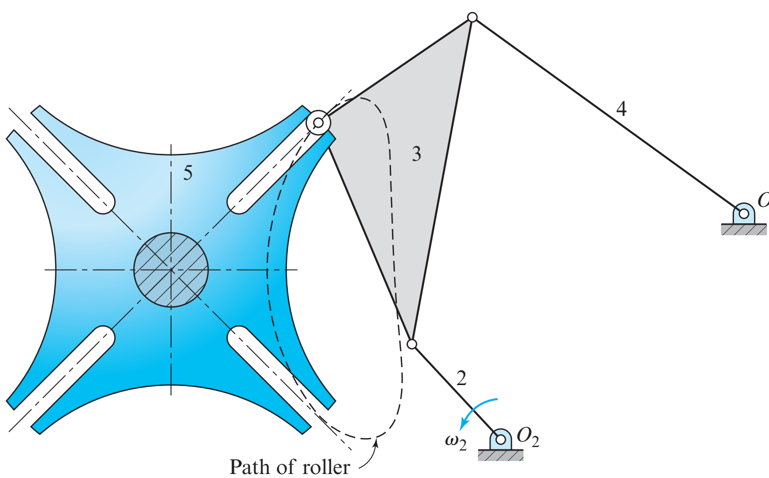

Figure 9.48 Geneva wheel activated by a four-bar linkage with link 2 as the crank.

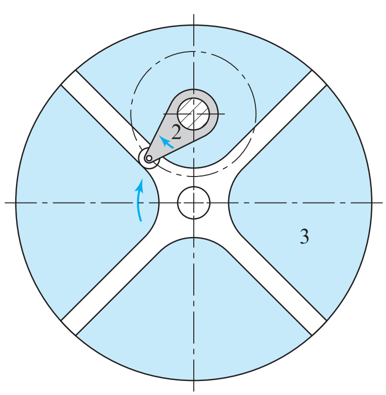

Figure 9.49 Inverse Geneva mechanism.

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PROBLEMS

9.1 A function varies from 0 to 10. Find the Chebyshev spacing for two, three, four, five, and six precision positions.

9.2 Determine the link lengths of a slider-crank linkage to have a stroke of 600 mm and an advance-to-return ratio of 1.20.

9.3 Determine a set of link lengths for a slider-crank linkage such that the stroke is 16 in and the advance-to-return ratio is 1.25.

9.4 The rocker of a crank-rocker linkage is to have a length of 500 mm and swing through a total angle of 45° with an advance-to-return ratio of 1.25. Determine a suitable set of dimensions for r_1, r_2, and r_3.

9.5 A crank-rocker linkage is to have a rocker of 6 ft length and a rocking angle of 75°. If the advance-to-return ratio is to be 1.32, what is a suitable set of link lengths for the remaining three links?

9.6 Design a crank and coupler to drive rocker 4 such that slider 6 reciprocates through a distance of 16 in with an advance-to-return ratio of 1.20. Use $a = r_4 = 16$ in and $r_5 = 24$ in with $\mathbf{r}_4$ vertical at midstroke. Record the location of O_2 and dimensions r_2 and r_3.

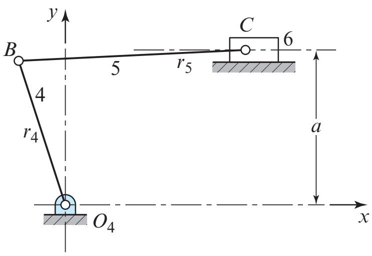

Figure P9.6

9.7 Design a crank and rocker for a six-bar linkage such that the slider in Fig. P9.6 reciprocates a distance of 800 mm with an advance-to-return ratio of 1.12; use $a = r_4 = 1\,200$ mm and $r_5 = 1\,800$ mm. Locate O_4 such that rocker 4 is vertical when the slider is at midstroke. Find suitable coordinates for O_2 and lengths for r_2 and r_3.

9.8 Two postures of a folding seat used in the aisles of buses to accommodate extra passengers are shown. Design a four-bar linkage to support the seat so that it will lock in the open posture and fold to a stable closing posture along the side of the aisle.

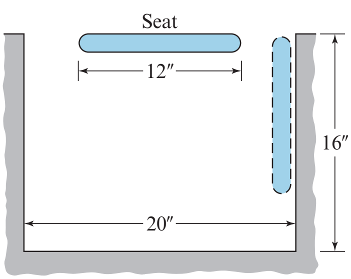

Figure P9.8

9.9 Design a spring-operated four-bar linkage to support a heavy lid, like the trunk lid of an automobile. The lid is to swing through an angle of 80° from the closed to the open posture. The springs are to be mounted so that the lid will be held closed against a stop, and they should also hold the lid in a stable open posture without the use of a stop.

9.10 Synthesize a linkage to move AB from posture 1 to posture 2 and return.

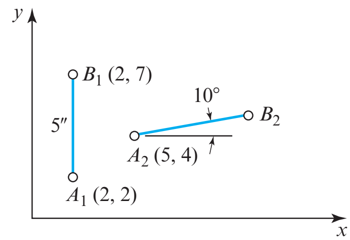

Figure P9.10

9.11 Synthesize a linkage to move AB successively through postures 1, 2, and 3.

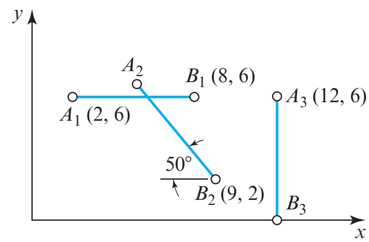

Figure P9.11

9.12 to 9.21* The figure shows a function-generator linkage in which the motion of rocker 2 corresponds to x and the motion of rocker 4 to the function $y = f(x)$. Use four precision points with Chebyshev spacing, and synthesize a linkage to generate the functions in the table. Plot a curve of the desired function and a curve of the actual function that the linkage generates. Compute the maximum error between them in percent.

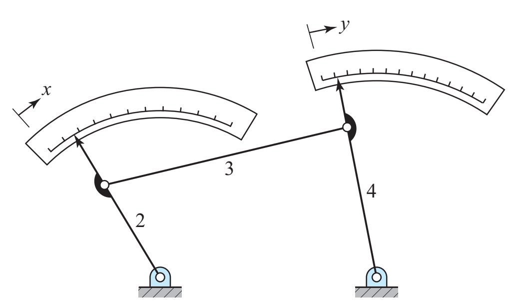

Figure P9.12

* Solutions for these problems were among the earliest computer work in kinematic synthesis and results are reported in F. Freudenstein, 1958. Four-bar function generators, *Mach. Design* **30**(24): 119–23.

Table P9.12 to P9.31.

Problem number	Function, y=	Range
P9.12, P9.22	$\log_{10} x$	$1 \leq x \leq 2$
P9.13, P9.23	$\sin x$	$0 \leq x \leq \pi/2$
P9.14, P9.24	$\tan x$	$0 \leq x \leq \pi/4$
P9.15, P9.25	e^x	$0 \leq x \leq 1$
P9.16, P9.26	$1/x$	$1 \leq x \leq 2$
P9.17, P9.27	$x^{1.5}$	$0 \leq x \leq 1$
P9.18, P9.28	x^2	$0 \leq x \leq 1$
P9.19, P9.29	$x^{2.5}$	$0 \leq x \leq 1$
P9.20, P9.30	x^3	$0 \leq x \leq 1$
P9.21, P9.31	x^2	$-1 \leq x \leq 1$

9.22 to 9.31 Repeat Problems 9.12 to 9.21 using the overlay method.

9.32 The figure shows a coupler curve generated by a four-bar linkage (not shown). Link 5 is to be attached to the coupler point, and link 6 is to be a rotating member with O_6 as the frame connection. In this problem, we wish to find a coupler curve from the Hrones and Nelson atlas or by precision postures, such that, for an appreciable distance, point C moves through an arc of a circle. Link 5 is then proportioned so that D lies at the center of curvature of this arc. The result is then called a *hesitation motion*, since link 6 hesitates in its rotation for the period during which point C transverses the approximate circular arc. Make a drawing of the complete linkage, and plot the first-order kinematic coefficient, ϕ'_6, of link 6 for 360° of displacement of the input link.

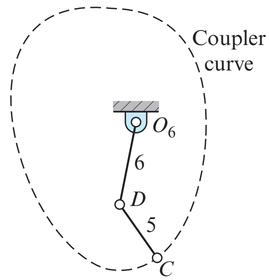

Figure P9.32

9.33 Synthesize a four-bar linkage to obtain a coupler curve having an approximate straight-line segment. Then, using the suggestion included in Fig. 9.42*b* or Fig. 9.44*b*, synthesize a dwell motion. Using an input crank angular velocity of unity, plot the first-order kinematic coefficient ϕ'_6 of rocker 6 versus the input crank displacement.

9.34 Synthesize a dwell mechanism using the idea suggested in Fig. 9.42*a* and the Hrones and Nelson atlas. Rocker 6 is to have a total angular displacement of 60°. Using this displacement as the abscissa, plot the first-order kinematic coefficient, ϕ'_6, of the motion of the rocker to show the dwell motion.

10 Spatial Mechanisms and Robotics

10.1 INTRODUCTION

The large majority of mechanisms in use today have *planar* motion, that is, the motions of all points produce paths that lie in a single plane or in parallel planes. This means that all motions can be seen in true size and shape from a single viewing direction and that graphic methods of analysis require only a single view. If the coordinate system is chosen with the x and y axes parallel to the plane(s) of motion, then all z values remain constant, and the problem can be solved, either graphically or analytically, with only two-dimensional methods. Although this is usually the case, it is not a necessity. Mechanisms having three-dimensional point paths do exist and are called *spatial mechanisms*. Another special category, called *spherical mechanisms*, have point paths that lie on concentric spherical surfaces.

Recall that these definitions were raised in Chap. 1; however, almost all of the examples presented in previous chapters have dealt only with planar mechanisms. This is justified because of their very extensive use in practical applications. Although a few nonplanar mechanisms, such as universal shaft couplings and bevel gears, have been known for centuries, it is only relatively recently that kinematicians have made substantial progress in developing design procedures for spatial mechanisms. It is probably not a coincidence, given the greater difficulty of the mathematic manipulations, that the emergence of such tools has awaited the development and availability of computers.

Although we have concentrated so far on mechanisms with planar motion, a brief review demonstrates that most of the previous theory has been derived in sufficient generality for both planar and spatial motion. The focus has been planar, since planar motion can be more easily visualized and requires less tedious computations than three-dimensional applications. However, most of the theory extends directly to spatial mechanisms. This chapter reviews some of the previous techniques, including examples with spatial motion. The chapter also introduces new mathematical tools and solution techniques that were not required for planar motion.

In Sec. 1.6, we learned that the mobility of a mechanism can be obtained from the Kutzbach criterion. The three-dimensional form of this criterion was given in Eq. (1.3); that is,

$$m = 6(n-1) - 5j_1 - 4j_2 - 3j_3 - 2j_4 - 1j_5, \tag{10.1}$$

where n is the number of links, and each j_k is the number of joints having k degrees of freedom. There are many possible combinations of n and j_k that produce a mobility of $m = 1$. One possibility has $n = 7$, $j_1 = 7$, and $j_2 = j_3 = j_4 = j_5 = 0$. Harrisberger [10] called such a combination a mechanism type; in particular, he called this the $7j_1$ type. Other combinations of j_k's, of course, produce different types of mechanisms. For example, the $3j_1 + 2j_2$ type has $n = 5$ links, and the $1j_1 + 2j_3$ type has $n = 3$ links.

Each mechanism type contains a finite number of *kinds* of mechanisms; there are as many kinds of mechanisms of each type as there are ways of arranging the different types of joints between the links. In Table 1.1, we saw that three of the six lower pairs have one degree of freedom, the revolute, R, the prismatic, P, and the helical, H. Thus, using any seven of these lower pairs, we obtain 36 kinds of type $7j_1$ mechanisms. Altogether,

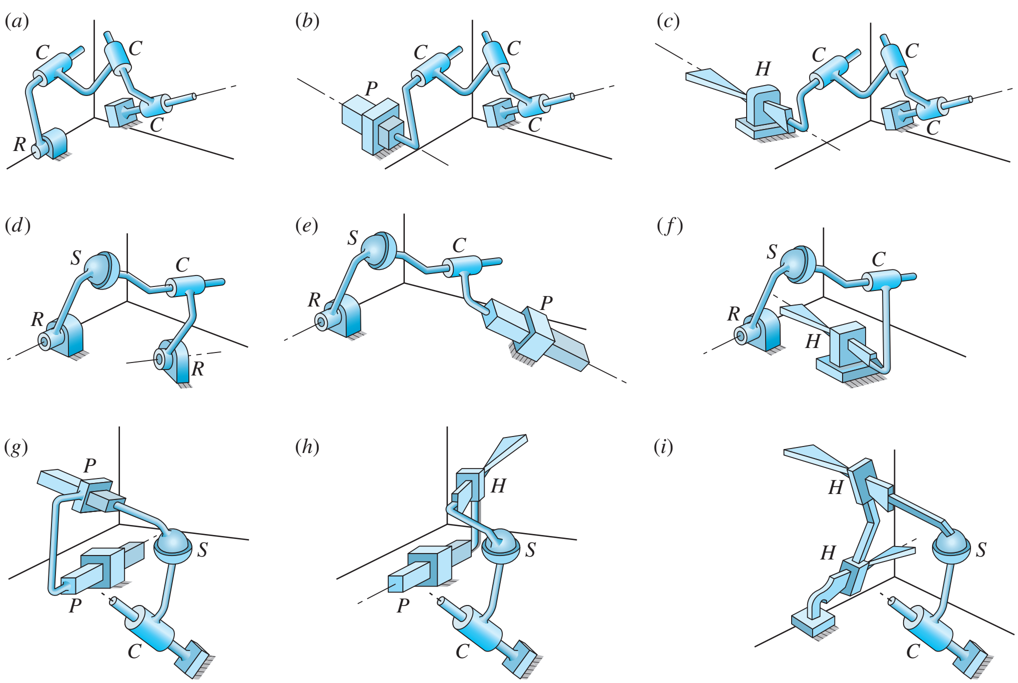

Figure 10.1 (*a*) *RCCC*; (*b*) *PCCC*; (*c*) *HCCC*; (*d*) *RSCR*; (*e*) *RSCP*; (*f*) *RSCH*; (*g*) *PPSC*; (*h*) *PHSC*; (*i*) *HHSC*.

Harrisberger lists 435 kinds of mechanisms that satisfy the Kutzbach criterion with a mobility of $m = 1$. Not all of these types, or kinds, however, are likely to have much practical value. Consider, for example, the $7j_1$ type with all revolute joints, each connected in series in a single loop.*

For mechanisms having mobility $m = 1$, Harrisberger selected nine kinds from two types that appeared to him to be useful; these are shown in Fig. 10.1. They are all spatial four-bar linkages having four joints with either rotating or sliding input and output. The designations in the legend, such as the *RSCH* four-bar linkage in Fig. 10.1*f*, identify the kinematic pair types (Table 1.1) beginning with the input and proceeding through the coupler and the output back to the frame. Thus, for this linkage, the input is connected to the frame by a revolute pair, *R*, and to the coupler by a spheric pair, *S*, the coupler is connected to the output by a cylindric pair, *C*, and the output is connected to the frame by a helical pair, *H*. The freedoms of these pairs, from Table 1.1 are $R = 1$, $S = 3$, $C = 2$, and $H = 1$.

The linkages of Fig. 10.1*a* through Fig. 10.1*c* are described by Harrisberger as type 1, or of the $1j_1 + 3j_2$ type. The remaining linkages of Fig. 10.1 are described as type 2, or of the $2j_1 + 1j_2 + 1j_3$ type. All have $n = 4$ links and have a mobility of $m = 1$.

10.2 EXCEPTIONS TO THE MOBILITY CRITERION

Curiously, the most common and most useful spatial mechanisms that have been discovered date back many years and are, in fact, exceptions to the Kutzbach criterion. As shown by the example in Fig. 1.6, geometric conditions sometimes occur that are not accounted for in the Kutzbach criterion and lead to apparent exceptions. As a case in point, consider that every planar mechanism, once constructed, truly exists in three dimensions. Yet, a planar four-bar linkage has $n = 4$ and is of type $4j_1$; thus Eq. (10.1) predicts a mobility of $m = 6(4 - 1) - 5(4) = -2$ (implying redundant constraints). However, we know that the mobility is, in fact, $m = 1$. The special geometric conditions in this case lie in the fact that all revolute axes remain parallel and are all perpendicular to the plane of motion. The Kutzbach criterion does not consider such geometric conditions and can produce false predictions.

Three more *RRRR* four-bar linkages are exceptions to the Kutzbach criterion, namely: the spheric four-bar linkage; the wobble-plate mechanism; and the Bennett linkage. As with the planar four-bar linkage, the Kutzbach criterion for these mechanisms predicts $m = -2$, and yet they are truly of mobility $m = 1$. The spheric four-bar linkage is shown in Fig. 10.2; the axes of all four revolute joints intersect at the center of a sphere. The links may be regarded as great-circle arcs existing on the surface of the sphere; what had been link lengths are now spheric angles. By properly proportioning these angles, it is possible to design all of the spheric counterparts of the planar four-bar linkage such as the spheric crank-rocker linkage and the spheric drag-link linkage. The spheric four-bar linkage is easy to design and manufacture and hence is one of the most useful of all spatial linkages. The

* The only application known to the authors for this type is its use in the front landing gear of the Boeing 727 aircraft.

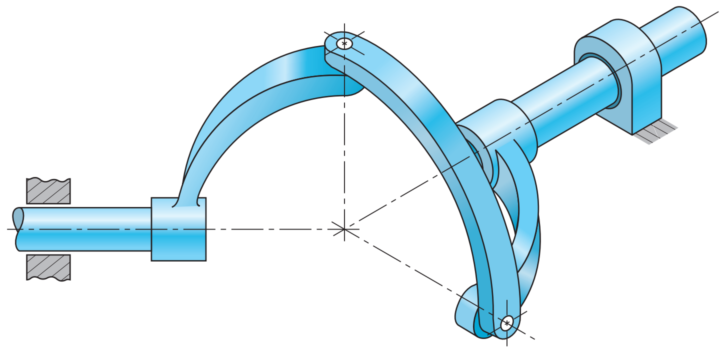

Figure 10.2 Spheric four-bar linkage.

Hooke or Cardan joint, which is the basis of a universal shaft coupling, is a special case of a spheric four-bar linkage with input and output cranks that subtend equal spheric angles.

The wobble-plate mechanism is shown in Fig. 10.3. Note that all of the revolute axes intersect at the origin; thus, it is also a spheric mechanism. Note also that input crank 2 rotates and output shaft 4 oscillates; also, when $\delta = 90°$, the mechanism is called a *spheric-slide oscillator*, and if $\gamma > \delta$, the output shaft rotates.

The Bennett linkage [2], shown in Fig. 10.4, is probably one of the less practical of the known spatial four-bar linkages. However, it has stimulated development of kinematic

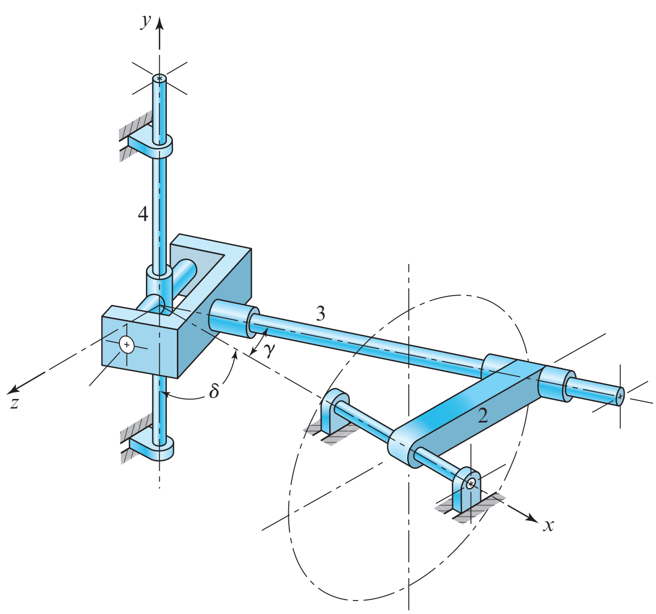

Figure 10.3 Wobble-plate mechanism.

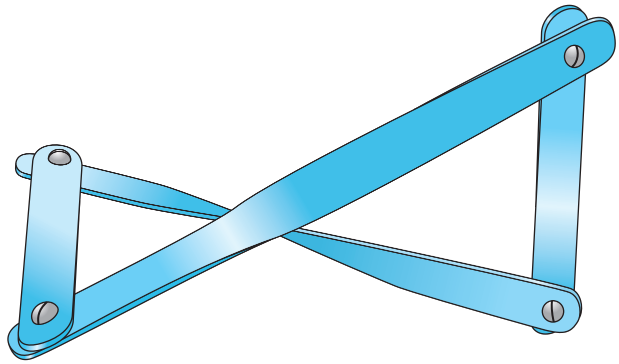

Figure 10.4 Bennett linkage.

theory for spatial mechanisms [1]. In this linkage, opposite links are of equal lengths and are twisted by equal amounts. The sines of the twist angles, α_1 and α_2, are proportional to the link lengths, a_1 and a_2, according to the equation

$$\frac{\sin\alpha_1}{a_1} = \pm\frac{\sin\alpha_2}{a_2}.$$

Two more exceptions to the Kutzbach mobility criterion are the Goldberg (Michael, not Rube!) five-bar *RRRRR* linkage and the Bricard six-bar *RRRRRR* linkage [8].* Again, it is doubtful whether these linkages have much practical value.

Harrisberger and Soni [11] identified all spatial linkages having one general constraint, that is, that have mobility of $m = 1$, but for which the Kutzbach criterion predicts $m = 0$. They identified eight types and 212 kinds and found seven new mechanisms that may have useful applications.

The spatial four-bar *RSSR* linkage, shown in Fig. 10.5, is an important and useful linkage. Since $n = 4$, $j_1 = 2$, and $j_3 = 2$, the Kutzbach criterion of Eq. (10.1) predicts a mobility of $m = 2$. Although this might appear at first glance to be another exception, upon closer examination we find that the second degree of freedom actually exists; it is the freedom of the coupler to spin about the axis between the two spheric joints. Since this degree of freedom does not affect the input-output kinematic relationship, it is called an *idle* degree of freedom. This extra freedom does no harm if the mass of the coupler is truly distributed along its axis; in fact, it may be an advantage, since the spin of the coupler about its axis may equalize wear on the two ball-and-socket joints. If the mass center of the coupler lies off axis, however, then this second freedom is not idle dynamically and can cause quite erratic performance of the linkage at high speed.

Note that all mechanisms that defy the Kutzbach criterion always predict a mobility less than the actual mobility. This is always the case; the Kutzbach criterion always predicts a lower limit on the mobility; the reason for this is mentioned in Sec. 1.6. The

* For pictures of these, see [12].

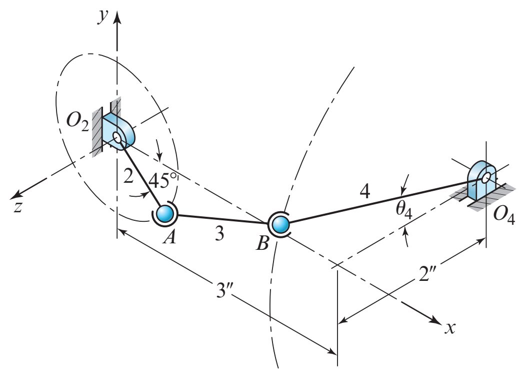

Figure 10.5 *RSSR* four-bar linkage.

argument for the development of the Kutzbach equation, Eq. (10.1), came from counting the freedoms for motion of all links before any connections are made, less the numbers of these presumably eliminated by connecting various types of joints. Yet, when there are special geometric conditions, such as intersections or parallelism between joint axes, the criterion counts each joint as eliminating its own share of freedoms although two (or more) of them may eliminate the same freedom(s). Thus, the exceptions arise from the false assumption of independence among the constraints of the joints.

Let us carry this thought one step further. When two (or more) of the constraint conditions eliminate the same motion freedom, the problem is said to have *redundant constraints*. Under these conditions, the same redundant constraints also determine how the forces are shared where the motion freedom is eliminated. Thus, when we come to analyzing forces, we find that there are too many constraints (equations) relating the number of unknown forces. The force analysis problem is then said to be *overconstrained*, and we find that there are statically indeterminate forces in the same number as the error in the predicted mobility.

There is an important lesson buried in this argument:* whenever there are redundant constraints on the motion, there are an equal number of statically indeterminate force components in the mechanism. Despite the higher simplicity in the design equations of planar linkages, for example, we should consider the force effects of these redundant constraints. All of the out-of-plane force and moment components become statically indeterminate. Slight machining tolerance errors or misalignments of axes can cause indeterminate stresses with cyclic loading as the mechanism is operated. The engineer needs to address the question: What effects will these stresses have on the fatigue life of the members of the mechanism?

On the other hand, as pointed out by Phillips [18], when motion is only occasional and loads are not high, this might be an ideal design decision.† If errors are small, the additional indeterminate forces may be small and such designs are tolerated although they may seem

* A detailed discussion of this entire topic forms one main theme of an excellent two volume set by Phillips [18].

† Ibid, Sec. 20.16, The advantages of overconstraint, p.151.

to exhibit friction effects and wear in the joints. As errors become larger, however, we may find that binding becomes a major issue.

The very existence of the large number of planar linkages in the world testify that such effects can be tolerated with appropriate tolerances on manufacturing errors, good lubrication, and proper fits between mating joint elements. Still, too few mechanical designers truly understand that the root of the problem can be eliminated by removing the redundant constraints in the first place. Thus, even in planar motion mechanisms, for example, we can make use of ball-and-socket or cylindric joints that, when properly located, can help to relieve statically indeterminate force and moment components.

10.3 SPATIAL POSTURE-ANALYSIS PROBLEM

Like planar mechanisms, a spatial mechanism is often connected to form one or more closed loops. Thus, employing methods similar to those of Sec. 2.7, loop-closure equations can be written that define the kinematic relationships of the mechanism. A variety of different mathematical tools can be used, including vectors [3], dual numbers [17], quaternions [23], and transformation matrices [22]. In vector notation, the closure of a spatial mechanism, such as the four-bar *RSSR* linkage shown in Fig. 10.5, can be defined by a loop-closure equation of the form

$$\mathbf{r} + \mathbf{s} + \mathbf{t} + \mathbf{C} = \mathbf{0}. \tag{10.2}$$

This equation is called the *vector tetrahedron equation*, since the individual vectors can be thought of as defining four of the six edges of a tetrahedron.

The vector tetrahedron equation is three dimensional and hence can be solved for three scalar unknowns. These can be either distances or angles and can exist in any combination in vectors $\mathbf{r}$, $\mathbf{s}$, and $\mathbf{t}$. Vector $\mathbf{C}$ is the sum of all known vectors in the loop. Using spheric coordinates, each of vectors $\mathbf{r}$, $\mathbf{s}$, and $\mathbf{t}$ can be expressed as a magnitude and two angles. Vector $\mathbf{r}$, for example, is defined once its magnitude, r, and two angles, θ_r and ϕ_r, are known. Thus, in Eq. (10.2), any three of the nine quantities r, θ_r, ϕ_r, s, θ_s, ϕ_s, t, θ_t, and ϕ_t can be unknowns that must be found from the vector equation. Chace has solved these nine cases by first reducing each to a polynomial [3]. He classifies the solutions depending upon whether the three unknowns occur in one, two, or three separate vectors, and he tabulates the forms of the solutions as shown in Table 10.1.

In Table 10.1, the unit vectors $\hat{\boldsymbol{\omega}}_r$, $\hat{\boldsymbol{\omega}}_s$, and $\hat{\boldsymbol{\omega}}_t$ are axes about which the angles ϕ_r, ϕ_s, and ϕ_t are measured. In case 1, vectors $\mathbf{s}$ and $\mathbf{t}$ are not needed; therefore, they are dropped from the equation. The three unknowns are all in vector $\mathbf{r}$. In cases 2a, 2b, 2c, and 2d, vector $\mathbf{t}$ is not needed and dropped; the three unknowns are shared by vectors $\mathbf{r}$ and $\mathbf{s}$. Cases 3a, 3b, 3c, and 3d have single unknowns in each of vectors $\mathbf{r}$, $\mathbf{s}$, and $\mathbf{t}$.

One advantage of the Chace vector tetrahedron solutions is that, since they provide known forms for the solutions of the nine cases, we can write a set of nine program modules for numeric evaluation. Eight of the nine cases have been reduced to explicit closed-form solutions for the unknowns and therefore can be quickly evaluated. Only case 3d, involving the solution of an eighth-order polynomial, must be solved by numeric iteration.

Although the vector tetrahedron equation and its nine case solutions can be used to solve most practical spatial problems, we recall from Sec. 10.1 that the Kutzbach criterion

Table 10.1 Classification of solutions of the vector tetrahedron equation

Case number	Unknowns	Known quantities: Vectors	Known quantities: Scalars	Degree of polynomial
1	r, θ_r, ϕ_r	$\mathbf{C}$		1
2*a*	r, θ_r, s	$\mathbf{C}, \hat{\boldsymbol{\omega}}_r, \hat{\mathbf{s}}$	ϕ_r	2
2*b*	r, θ_r, θ_s	$\mathbf{C}, \hat{\boldsymbol{\omega}}_r, \hat{\boldsymbol{\omega}}_s$	ϕ_r, s, ϕ_s	4
2*c*	θ_r, ϕ_r, s	$\mathbf{C}, \hat{\mathbf{s}}$	r	2
2*d*	$\theta_r, \phi_r, \theta_s$	$\mathbf{C}, \hat{\boldsymbol{\omega}}_s$	r, s, ϕ_s	2
3*a*	r, s, t	$\mathbf{C}, \hat{\mathbf{r}}, \hat{\mathbf{s}}, \hat{\mathbf{t}}$		1
3*b*	r, s, θ_t	$\mathbf{C}, \hat{\mathbf{r}}, \hat{\mathbf{s}}, \hat{\boldsymbol{\omega}}_t$	t, ϕ_t	2
3*c*	r, θ_s, θ_t	$\mathbf{C}, \hat{\mathbf{r}}, \hat{\boldsymbol{\omega}}_s, \hat{\boldsymbol{\omega}}_t$	s, ϕ_s, t, ϕ_t	4
3*d*	$\theta_r, \theta_s, \theta_t$	$\mathbf{C}, \hat{\boldsymbol{\omega}}_r, \hat{\boldsymbol{\omega}}_s, \hat{\boldsymbol{\omega}}_t$	$r, \phi_r, s, \phi_s, t, \phi_t$	8

predicts the existence of up to seven j_1 joints in a single-loop, one-degree-of-freedom mechanism. For example, the seven-link 7*R* linkage has one input and six unknown joint variables [6]. Using the vector form of the loop-closure equation, it is not possible to determine the values for the six unknowns, since the vector equation is equivalent to only three scalar equations and does not fully describe three-dimensional rotation. Therefore, other mathematical tools, such as dual numbers, quaternions, or transformation matrices, must be used. In general, such problems often require numeric techniques for their evaluation [6]. Anyone attempting the solution of such problems by hand-algebraic techniques quickly appreciates that *posture* analysis, not velocity or acceleration analysis, is the most difficult problem in kinematics.

Solving the polynomials of the Chace vector tetrahedron equation turns out to be equivalent to finding the intersections of straight lines or circles with various surfaces of revolution. In general, such problems can be solved easily and quickly by the methods of descriptive geometry. The graphic approach has the added advantage that the geometric nature of the problem is not concealed in a multiplicity of mathematic operations. Graphic methods performed by hand are not nearly as accurate as an analytic or a numeric approach. However, graphic methods performed by computer software can give the same accuracy as an analytic or numeric approach.

EXAMPLE 10.1

The dimensions of the links of the spatial *RSSR* four-bar linkage shown in Fig. 10.5 are $R_{AO_2} = 1$ in, $R_{BA} = 3.5$ in, $R_{BO_4} = 4$ in. Note that the planes of rotation of the input and output are known. For the input angle $\theta_2 = -45°$, determine the postures of the coupler 3 and output link 4.

GRAPHIC SOLUTION

If we replace link 4 by vector $\mathbf{R}_{BO_4}$, then the only unknown is the angle θ_4, since the magnitude of the vector and the plane of rotation are given. Similarly, if link 3 is replaced by vector $\mathbf{R}_{BA}$, then the magnitude is known, and there are only two unknown spheric coordinate angular direction variables for this vector. The loop-closure equation is

$$(\mathbf{R}_{BA}) + (-\mathbf{R}_{BO_4}) + \left(\mathbf{R}_{AO_2} - \mathbf{R}_{O_4O_2}\right) = \mathbf{0}. \qquad (1)$$

We identify this as case $2d$ in Table 10.1, requiring the solution of a second-degree (quadratic) polynomial, and hence it yields two solutions.

This problem is solved graphically using two orthographic views, the frontal and profile views. If we imagine, in Fig. 10.5, that the coupler is disconnected from the output crank at B and allowed to occupy any possible position with respect to A, then B of link 3 must lie on the surface of a sphere of known radius with its center at A. With joint B still disconnected, the locus of B of link 4 is a circle of known radius about O_4 in a plane parallel to the yz plane. Therefore, to solve this problem, we need to find the two points of intersection of a circle and a sphere.

The solution is shown in Fig. 10.6, where the subscripts F and P denote projections on the frontal and profile planes, respectively. First, we locate points O_2, A, and O_4 in both views. In the profile view, we draw a circle of radius $R_{BO_4} = 4$ in about center O_{4P}; this is the locus of point B_4. This circle appears in the frontal view as the vertical line $M_F O_{4F} N_F$.

Next, in the frontal view, we construct the outline of a sphere with A_F as center and coupler length $R_{BA} = 3.5$ in as its radius. If $M_F O_{4F} N_F$ is regarded as the trace of a plane normal to the frontal view plane, the intersection of this plane with the sphere appears as the full circle in the profile view, having diameter $M_P N_P = M_F N_F$. The circular arc of radius R_{BO_4} intersects the circle at two points, yielding two solutions. One of the points

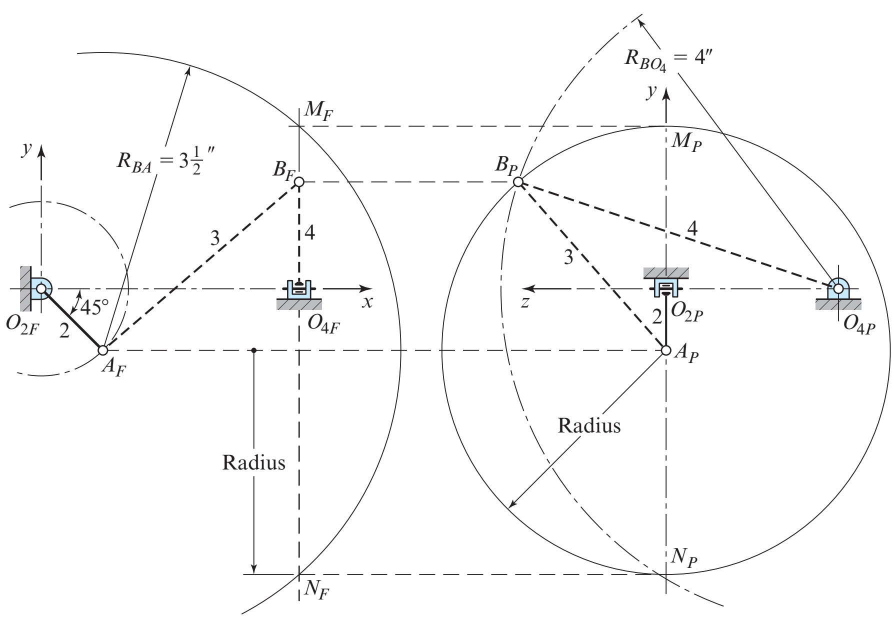

Figure 10.6 Graphic posture analysis.

is chosen as B_P and is projected back to the frontal view to locate B_F. Links 3 and 4 are drawn next as the dashed lines shown in the profile and frontal views of Fig. 10.6.

Measuring the x, y, and z projections from the graphic solution, we can write the vectors for each link:

$$\mathbf{R}_{O_4O_2} = 3.00\hat{\mathbf{i}} - 2.00\hat{\mathbf{k}} \text{ in}, \qquad \textit{Ans.}$$

$$\mathbf{R}_{AO_2} = 0.71\hat{\mathbf{i}} - 0.71\hat{\mathbf{j}} \text{ in}, \qquad \textit{Ans.}$$

$$\mathbf{R}_{BA} = 2.30\hat{\mathbf{i}} + 1.95\hat{\mathbf{j}} + 1.77\hat{\mathbf{k}} \text{ in}, \qquad \textit{Ans.}$$

$$\mathbf{R}_{BO_4} = 1.22\hat{\mathbf{j}} + 3.81\hat{\mathbf{k}} \text{ in}. \qquad \textit{Ans.}$$

These components were obtained from a true-size graphic solution; better accuracy is possible, of course, by making the drawings two or four times actual size.

ANALYTIC SOLUTION

The given information establishes that

$$\begin{aligned}
\mathbf{R}_{O_4O_2} &= 3.000\hat{\mathbf{i}} - 2.000\hat{\mathbf{k}} \text{ in}, \\
\mathbf{R}_{AO_2} &= \cos\left(-45^\circ\right) + \sin\left(-45^\circ\right) = 0.707\hat{\mathbf{i}} - 0.707\hat{\mathbf{j}} \text{ in}, \\
\mathbf{R}_{BA} &= R_{BA}^x\hat{\mathbf{i}} + R_{BA}^y\hat{\mathbf{j}} + R_{BA}^z\hat{\mathbf{k}}, \\
\mathbf{R}_{BO_4} &= -4.000\sin\theta_4\hat{\mathbf{j}} + 4.000\cos\theta_4\hat{\mathbf{k}} \text{ in}.
\end{aligned}$$

The same loop-closure equation as in Eq. (1) above must be solved. Substituting the given information gives

$$\begin{aligned}
&\left(R_{BA}^x\hat{\mathbf{i}} + R_{BA}^y\hat{\mathbf{j}} + R_{BA}^z\hat{\mathbf{k}}\right) + \left(4.000\sin\theta_4\hat{\mathbf{j}} - 4.000\cos\theta_4\hat{\mathbf{k}}\right) \\
&\quad + \left(0.707\hat{\mathbf{i}} - 0.707\hat{\mathbf{j}} - 3.000\hat{\mathbf{i}} + 2.000\hat{\mathbf{k}}\right) = \mathbf{0}.
\end{aligned}$$

Then, separating this equation into its $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ components and rearranging gives the following three equations:

$$\begin{aligned}
R_{BA}^x &= 2.293 \text{ in}, \\
R_{BA}^y &= -4.000\sin\theta_4 + 0.707 \text{ in}, \\
R_{BA}^z &= 4.000\cos\theta_4 - 2.000 \text{ in}.
\end{aligned}$$

Next, we square and add these equations, and, remembering that $R_{BA} = 3.500$ in, we obtain

$$(2.293 \text{ in})^2 + (-4.000\sin\theta_4 + 0.707 \text{ in})^2 + (4.000\cos\theta_4 - 2.000 \text{ in})^2 = (3.500 \text{ in})^2.$$

Expanding this equation and rearranging gives

$$-5.656\sin\theta_4 - 16.000\cos\theta_4 + 13.508 \text{ in}^2 = 0.$$

We now have a single equation with only one unknown, namely, θ_4. However, it contains both sine and cosine terms (a transcendental equation). A standard technique for dealing with this situation is to convert the equation into a second-order (quadratic) polynomial by using the tangent of half-angle identities:

$$\sin\theta_4 = \frac{2\tan(\theta_4/2)}{1+\tan^2(\theta_4/2)} \quad \text{and} \quad \cos\theta_4 = \frac{1-\tan^2(\theta_4/2)}{1+\tan^2(\theta_4/2)}.$$

Substituting these identities into the above equation and multiplying by the common denominator gives a single quadratic equation in one unknown, tan($\theta_4/2$), that is,

$$-11.312\tan(\theta_4/2) - 16.000\left[1-\tan^2(\theta_4/2)\right] + 13.508\left[1+\tan^2(\theta_4/2)\right] = 0,$$

or

$$29.508\tan^2(\theta_4/2) - 11.312\tan(\theta_4/2) - 2.492 = 0.$$

This quadratic equation can now be solved for two roots, which are

$$\tan\frac{\theta_4}{2} = \begin{cases} 0.5398; & \theta_4 = -123.3^\circ \\ -0.1565; & \theta_4 = 162.2^\circ \end{cases}.$$

Of these two roots, we choose the second as the solution of interest. Back-substituting into the previous conditions, we find numeric values for the four vectors, namely:

$$\mathbf{R}_{O_4O_2} = 3.000\hat{\mathbf{i}} - 2.000\hat{\mathbf{k}} \text{ in}, \qquad \textit{Ans.}$$

$$\mathbf{R}_{AO_2} = 0.707\hat{\mathbf{i}} - 0.707\hat{\mathbf{j}} \text{ in}, \qquad \textit{Ans.}$$

$$\mathbf{R}_{BA} = 2.293\hat{\mathbf{i}} + 1.930\hat{\mathbf{j}} + 1.809\hat{\mathbf{k}} \text{ in}, \qquad \textit{Ans.}$$

$$\mathbf{R}_{BO_4} = 1.223\hat{\mathbf{j}} + 3.809\hat{\mathbf{k}} \text{ in}. \qquad \textit{Ans.}$$

These results match those of the graphic solution within graphic error.

We note from this example that we have solved for the output angle θ_4 and the current values of the four vectors for the given position of the input crank angle θ_2. However, have we really solved for the postures of all links? The answer is no! As indicated above, we have not determined how the coupler link may have rotated about its axis $\mathbf{R}_{BA}$ between the two spheric joints. This is still unknown and explains how we were able to solve this two-degree-of-freedom system without specifying another input angle for the idle freedom. Depending upon our motivation, the above solution may be sufficient; however, it is important to note that the above vectors are not sufficient to solve for this additional unknown variable.

The spheric four-bar *RRRR* linkage shown in Fig. 10.2 is also case 2*d* of the vector tetrahedron equation and, for a specified posture of the input link, can be solved in the same manner as the example shown.

10.4 SPATIAL VELOCITY AND ACCELERATION ANALYSES

Once the postures of all links of a spatial mechanism are known, the velocities and accelerations can then be determined using the methods of Chaps. 3 and 4. In planar mechanisms, the angular velocity and acceleration vectors are always perpendicular to the plane of motion. This considerably reduces the effort required in the solution process for both graphic and analytic approaches. In spatial problems, however, these vectors may be skewed in space. Otherwise, the equations and the methods of analysis are the same. The following example illustrates the differences.

EXAMPLE 10.2

For the spatial *RSSR* four-bar linkage in the posture shown in Fig. 10.7, the constant angular velocity of link 2 is $\omega_2 = 40\hat{\mathbf{k}}$ rad/s. Find the angular velocities and angular accelerations of links 3 and 4 and the velocity and acceleration of point *B* for the specified posture.

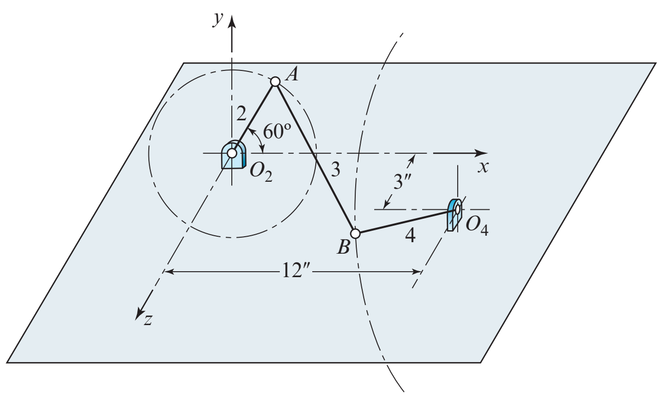

Figure 10.7 $R_{AO_2} = 4\ in$, $R_{BA} = 15$ in, and $R_{BO_4} = 10$ in.

GRAPHIC POSTURE SOLUTION

The graphic posture analysis follows exactly the procedure given in Sec. 10.3 and Example 10.1. The solution is shown in Fig. 10.8

ANALYTIC POSTURE SOLUTION

The analytic posture solution also follows the procedure given in Sec. 10.3 and Example 10.1. For the given input angle, the results are:

$$\mathbf{R}_{O_4O_2} = 12.000\hat{\mathbf{i}} + 3.000\hat{\mathbf{k}} \text{ in},$$

$$\mathbf{R}_{AO_2} = 2.000\hat{\mathbf{i}} + 3.464\hat{\mathbf{j}} \text{ in},$$

$$\mathbf{R}_{BA} = 10.000\hat{\mathbf{i}} + 2.746\hat{\mathbf{j}} + 10.838\hat{\mathbf{k}} \text{ in},$$

$$\mathbf{R}_{BO_4} = 6.210\hat{\mathbf{j}} + 7.838\hat{\mathbf{k}} \text{ in}.$$

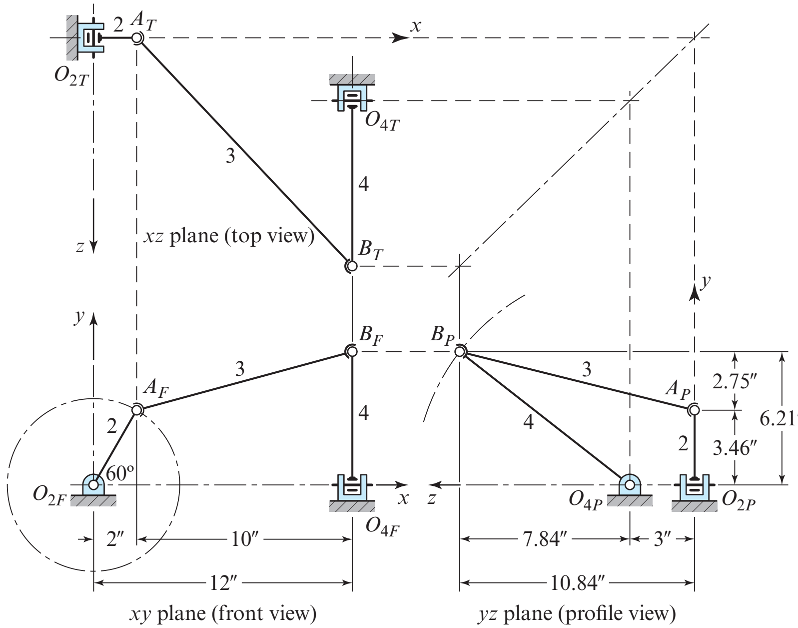

Figure 10.8 Graphic posture analysis.

GRAPHIC VELOCITY AND ACCELERATION SOLUTIONS

The determination of the velocities and accelerations of a spatial mechanism by graphic means can be conducted in the same manner as for a planar mechanism. However, the posture information as well as the velocity and acceleration vectors often do not appear in their true lengths in the front, top, and profile views, but are foreshortened. This means that spatial motion problems usually require the use of auxiliary views where the vectors appear in true lengths.

The velocity solution for this example is shown in Fig. 10.9 with notation corresponding to that used in many texts on descriptive geometry. The subscripts F, T, and P designate the front, top, and profile views, and the subscripts 1 and 2 indicate the first and second auxiliary views, respectively.

The steps to obtain the solution to the velocity problem are as follows:

1. We first construct to scale the front, top, and profile views of the linkage and designate each point.
2. Next, we calculate $\mathbf{V}_A$ as above and place this vector in position with its origin at A in the three views. The velocity $\mathbf{V}_A$ shows in true length in the frontal view. We designate its terminus as point a_F and project this point into the top and profile views to find a_T and a_P.
3. The magnitude of velocity $\mathbf{V}_B$ is unknown, but its direction is perpendicular to $\mathbf{R}_{BO_4}$, and, once the problem is solved, it will show in true size in the profile view. Therefore, we construct a line in the profile view that originates at point A_P (which becomes the origin of our velocity polygon) and corresponds in direction

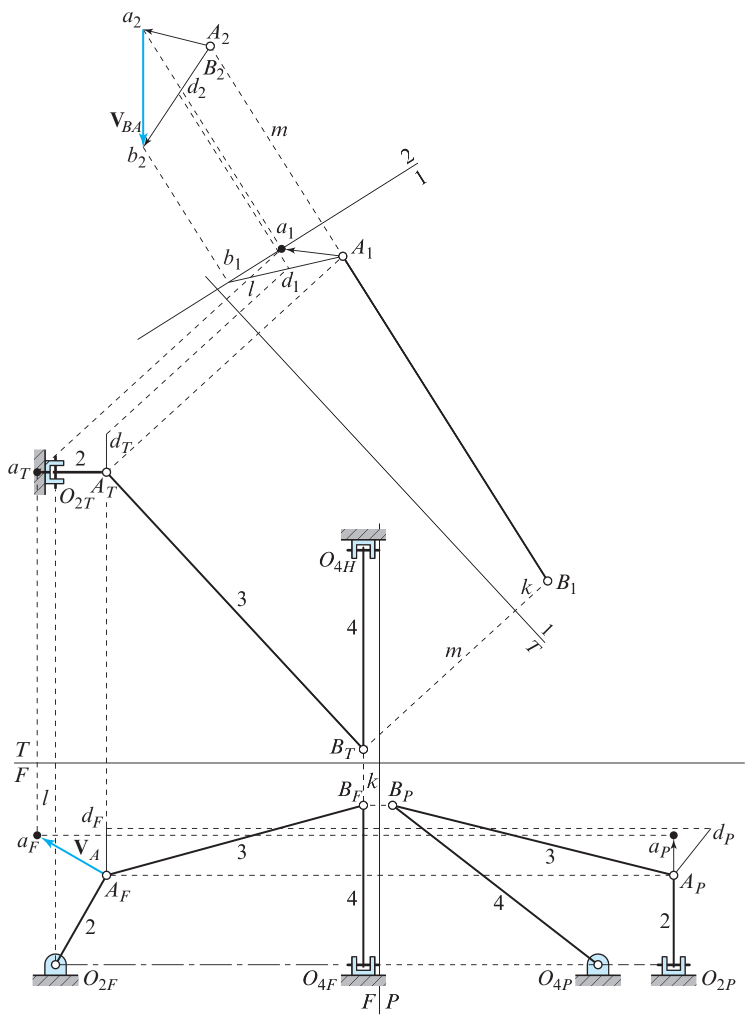

Figure 10.9 Graphic velocity analysis.

to that of $\mathbf{V}_B$ (perpendicular to $R_{B_P O_{4P}}$). We then choose any point d_P on this line and project it into the front view (d_F) and top view (d_T) to establish the line of action of $\mathbf{V}_B$ in those views.

4. The equation to be solved is

$$\mathbf{V}_B = \mathbf{V}_A + \mathbf{V}_{BA}, \tag{1}$$

where $\mathbf{V}_A$ and the directions of $\mathbf{V}_B$ and $\mathbf{V}_{BA}$ are known. We note that $\mathbf{V}_{BA}$ must lie in a plane perpendicular (in space) to $\mathbf{R}_{BA}$, but its magnitude and that of $\mathbf{V}_B$ are unknown.

Vector $\mathbf{V}_{BA}$ must originate at the terminus of $\mathbf{V}_A$ and must lie in a plane perpendicular to $\mathbf{R}_{BA}$; it terminates by intersecting line Ad or its extension. To find the plane perpendicular to $\mathbf{R}_{BA}$, we start by constructing the first auxiliary view, which shows vector $\mathbf{R}_{BA}$ in true length, so we construct the edge view of plane 1 parallel to $A_T B_T$ and project $\mathbf{R}_{BA}$ to this plane. In this projection, we note that distances k and l are the same in this first auxiliary view as in the frontal view. The first auxiliary view of AB is $A_1 B_1$, which is true length. We also project points a and d into this view, but the remaining links need not be projected.

5. In this step, we construct a second auxiliary view, plane 2, such that the projection of AB upon it is a point. Then, all lines drawn parallel to this plane are perpendicular to $\mathbf{R}_{BA}$. The edge view of such a plane must be perpendicular to $A_1 B_1$ extended. In this example, it is convenient to choose this plane so that it contains point a; therefore, we construct the edge view of plane 2 through point a_1 perpendicular to $A_1 B_1$ extended. Now we project points A, B, a, and d into this plane. Note that the distances—m, for example—of points from plane 1 must be the same in the top view and in the second auxiliary view.
6. We now extend line $A_1 d_1$ until it intersects the edge view of plane 2 at b_1, and we find projection b_2 of this point into plane 2 where the projector meets line $A_2 d_2$ extended. Now both points a and b lie in plane 2, and any line drawn in plane 2 is perpendicular to $\mathbf{R}_{BA}$. Therefore, line ab is $\mathbf{V}_{BA}$, and the second auxiliary view of that line shows its true length. Line $A_2 b_2$ is the projection of $\mathbf{V}_B$ on the second auxiliary plane, but not in its true length, since point A is not in plane 2.
7. To simplify the reading of the drawing, step 7 is not shown; those who follow the first six steps carefully should have no difficulty with the seventh. We can project the three vectors into the top, front, and profile views. Velocity V_B can be measured from the profile view where it appears in true length. The result is

$$\mathbf{V}_B = 200\hat{\mathbf{j}} - 158\hat{\mathbf{k}} \text{ in/s.} \qquad \textit{Ans.}$$

When all vectors have been projected into these three views, we can measure their x, y, and z projections directly.

8. If we assume that the idle freedom of spin is not active and, therefore, that the angular velocity vector ω_3 is perpendicular to $\mathbf{R}_{BA}$ and appears in true length in the second auxiliary view,* then the magnitudes of the angular velocities can be determined from the equations

$$\omega_3 = \frac{V_{BA}}{R_{BA}} = \frac{242 \text{ in/s}}{15 \text{ in}} = 16.13 \text{ rad/s},$$

$$\omega_4 = \frac{V_{BO_4}}{R_{BO_4}} = \frac{255 \text{ in/s}}{10 \text{ in}} = 25.50 \text{ rad/s}.$$

* This is the same assumption we used in the analytic solution when we wrote the equation $\boldsymbol{\omega}_3 \times \mathbf{R}_{BA} = \mathbf{0}$.

Therefore, we can draw the angular velocity vectors in the views where they appear in their true lengths and project them into the views where their vector components can be measured. The results are

$$\boldsymbol{\omega}_3 = -7.69\hat{\mathbf{i}} + 13.71\hat{\mathbf{j}} + 3.63\hat{\mathbf{k}} \text{ rad/s}, \qquad \textit{Ans.}$$

$$\boldsymbol{\omega}_4 = 25.50\hat{\mathbf{i}} \text{ rad/s}. \qquad \textit{Ans.}$$

The steps to obtain the solution to the acceleration problem, also not shown, are followed in an identical manner, using the same two auxiliary planes. The equation to be solved is

$$\mathbf{A}^n_{BO_4} + \mathbf{A}^t_{BO_4} = \mathbf{A}^n_{AO_2} + \mathbf{A}^t_{AO_2} + \mathbf{A}^n_{BA} + \mathbf{A}^t_{BA}, \tag{2}$$

where vectors $\mathbf{A}^n_{BO_4}$, $\mathbf{A}^n_{AO_2}$, $\mathrm{A}^t_{AO_2}$, and $\mathbf{A}^n_{BA}$ can be determined now that the velocity analysis has been completed. Also, we see that $\mathbf{A}^t_{BO_4}$ and $\mathbf{A}^t_{BA}$ have known directions. Therefore, the solution can proceed exactly as for the velocity polygon; the only difference in approach is that there are more known vectors. The final results are

$$\boldsymbol{\alpha}_3 = -527\hat{\mathbf{i}} + 619\hat{\mathbf{j}} + 329\hat{\mathbf{k}} \text{ rad/s}^2, \qquad \textit{Ans.}$$

$$\boldsymbol{\alpha}_4 = -865\hat{\mathbf{i}} \text{ rad/s}^2, \qquad \textit{Ans.}$$

$$\mathbf{A}_B = 2\,750\hat{\mathbf{j}} - 10\,450\hat{\mathbf{k}} \text{ in/s}^2. \qquad \textit{Ans.}$$

ANALYTIC VELOCITY AND ACCELERATION SOLUTIONS

From the given information and the constraints, the angular velocities and accelerations can be written as

$$\begin{aligned} \boldsymbol{\omega}_2 &= 40\hat{\mathbf{k}} \text{ rad/s}, & \boldsymbol{\alpha}_2 &= \mathbf{0}, \\ \boldsymbol{\omega}_3 &= \omega_3^x\hat{\mathbf{i}} + \omega_3^y\hat{\mathbf{j}} + \omega_3^z\hat{\mathbf{k}}, & \boldsymbol{\alpha}_3 &= \alpha_3^x\hat{\mathbf{i}} + \alpha_3^y\hat{\mathbf{j}} + \alpha_3^z\hat{\mathbf{k}}, \\ \boldsymbol{\omega}_4 &= \omega_4\hat{\mathbf{i}}, & \boldsymbol{\alpha}_4 &= \alpha_4\hat{\mathbf{i}}. \end{aligned}$$

First, we find the velocity of point A as the velocity difference from point O_2. Thus,

$$\begin{aligned} \mathbf{V}_A = \mathbf{V}_{AO_2} = \boldsymbol{\omega}_2 \times \mathbf{R}_{AO_2} &= \left(40\hat{\mathbf{k}} \text{ rad/s}\right) \times \left(2.000\hat{\mathbf{i}} + 3.464\hat{\mathbf{j}} \text{ in}\right) \\ &= -138.560\hat{\mathbf{i}} + 80.000\hat{\mathbf{j}} \text{ in/s}. \end{aligned} \tag{3}$$

Similarly, for link 3, the velocity difference is

$$\begin{aligned} \mathbf{V}_{BA} &= \left(\omega_3^x\hat{\mathbf{i}} + \omega_3^y\hat{\mathbf{j}} + \omega_3^z\hat{\mathbf{k}}\right) \times \left(10.000\hat{\mathbf{i}} + 2.746\hat{\mathbf{j}} + 10.838\hat{\mathbf{k}} \text{ in}\right) \\ &= \left(10.838\omega_3^y - 2.746\omega_3^z\right)\hat{\mathbf{i}} + \left(10.000\omega_3^z - 10.838\omega_3^x\right)\hat{\mathbf{j}}, \\ &\quad + \left(2.746\omega_3^x - 10.000\omega_3^y\right)\hat{\mathbf{k}} \text{ in} \end{aligned} \tag{4}$$

and for link 4 the velocity of point B is

$$\begin{aligned} \mathbf{V}_B = \mathbf{V}_{BO_4} = \boldsymbol{\omega}_4 \times \mathbf{R}_{BO_4} &= \left(\omega_4\hat{\mathbf{i}}\right) \times \left(6.210\hat{\mathbf{j}} + 7.838\hat{\mathbf{k}} \text{ in}\right) \\ &= -(7.838 \text{ in})\,\omega_4\hat{\mathbf{j}} + (6.210 \text{ in})\,\omega_4\hat{\mathbf{k}}. \end{aligned} \tag{5}$$

The next step is to substitute these into the velocity difference equation,

$$\mathbf{V}_B = \mathbf{V}_A + \mathbf{V}_{BA}. \tag{6}$$

After this, we separate the $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ components to obtain the three algebraic equations:

$$10.838 \text{ in } \omega_3^y - 2.746 \text{ in } \omega_3^z = 138.560 \text{ in/s}, \tag{7}$$

$$-10.838 \text{ in } \omega_3^x + 10.000 \text{ in } \omega_3^z + 7.838 \text{ in } \omega_4 = -80.000 \text{ in/s}, \tag{8}$$

$$2.746 \text{ in } \omega_3^x - 10.000 \text{ in } \omega_3^y - 6.210 \text{ in } \omega_4 = 0.000. \tag{9}$$

We now have three equations; however, we note that there are four unknowns, ω_3^x, ω_3^y, ω_3^z, and ω_4. This would not occur in most problems but does here because of the idle freedom of the coupler to spin about its own axis. Since this spin does not affect the input-output relationship, we will find the same result for ω_4 regardless of the rate of spin. One way to proceed, therefore, would be to choose one component of ω_3 and give it a value, thus setting the rate of spin, and then solve for the other unknowns. Another approach is to set the rate of spin about the coupler link axis to zero by requiring that

$$\boldsymbol{\omega}_3 \times \mathbf{R}_{BA} = 0,$$

$$R_{BA}^x \omega_3^x + R_{BA}^y \omega_3^y + R_{BA}^z \omega_3^z = 0,$$

$$(10.000 \text{ in})\, \omega_3^x + (2.746 \text{ in})\, \omega_3^y + (10.838 \text{ in})\, \omega_3^z = 0. \tag{10}$$

Equations (7) through (10) can now be solved simultaneously for the four unknowns. The result is

$$\boldsymbol{\omega}_3 = -7.692\hat{\mathbf{i}} + 13.704\hat{\mathbf{j}} + 3.625\hat{\mathbf{k}} \text{ rad/s}, \qquad \textit{Ans.}$$

$$\boldsymbol{\omega}_4 = -25.468\hat{\mathbf{i}} \text{ rad/s}. \qquad \textit{Ans.}$$

Substituting these angular velocities into Eq. (5), the velocity of point B is

$$\mathbf{V}_B = 199.618\hat{\mathbf{j}} - 158.156\hat{\mathbf{k}} \text{ in/s}. \qquad \textit{Ans.}$$

Turning next to acceleration analysis, we compute the following components:

$$\mathbf{A}_{AO_2}^n = \boldsymbol{\omega}_2 \times \left(\boldsymbol{\omega}_2 \times \mathbf{R}_{AO_2}\right) = -3\,200.00\hat{\mathbf{i}} - 5\,542.56\hat{\mathbf{j}} \text{ in/s}^2, \tag{11}$$

$$\mathbf{A}_{AO_2}^t = \boldsymbol{\alpha}_2 \times \mathbf{R}_{AO_2} = \mathbf{0}, \tag{12}$$

$$\mathbf{A}_{BA}^n = \boldsymbol{\omega}_3 \times (\boldsymbol{\omega}_3 \times \mathbf{R}_{BA}) = -2\,601.01\hat{\mathbf{i}} - 714.26\hat{\mathbf{j}} - 2\,818.99\hat{\mathbf{k}} \text{ in/s}^2, \tag{13}$$

$$\begin{aligned} \mathbf{A}_{BA}^t &= \boldsymbol{\alpha}_3 \times \mathbf{R}_{BA} = \left(\alpha_3^x\hat{\mathbf{i}} + \alpha_3^y\hat{\mathbf{j}} + \alpha_3^z\hat{\mathbf{k}}\right) \times \left(10.000\hat{\mathbf{i}} + 2.746\hat{\mathbf{j}} + 10.838\hat{\mathbf{k}} \text{ in}\right) \\ &= \left(10.838\alpha_3^y - 2.746\alpha_3^z\right)\hat{\mathbf{i}} + \left(10.000\alpha_3^z - 10.838\alpha_3^x\right)\hat{\mathbf{j}} \\ &\quad + \left(2.746\alpha_3^x - 10.000\alpha_3^y\right)\hat{\mathbf{k}} \text{ in}, \end{aligned} \tag{14}$$

$$\mathbf{A}_{BO_4}^n = \boldsymbol{\omega}_4 \times \left(\boldsymbol{\omega}_4 \times \mathbf{R}_{BO_4}\right) = -4\,028.05\hat{\mathbf{j}} - 5\,084.03\hat{\mathbf{k}} \text{ in/s}^2, \tag{15}$$

$$\mathbf{A}_{BO_4}^t = \boldsymbol{\alpha}_4 \times \mathbf{R}_{BO_4} = -7.838\alpha_4\hat{\mathbf{j}} + 6.210\alpha_4\hat{\mathbf{k}} \text{ in.} \tag{16}$$

These quantities are now substituted into the acceleration difference equation,

$$\mathbf{A}_{BO_4}^n + \mathbf{A}_{BO_4}^t = \mathbf{A}_{AO_2}^n + \mathbf{A}_{AO_2}^t + \mathbf{A}_{BA}^n + \mathbf{A}_{BA}^t, \tag{17}$$

and separated into $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ components. Along with the condition $\boldsymbol{\alpha}_3 \times \mathbf{R}_{BA} = 0$ for the spin of the idle freedom, this results in four equations in four unknowns as follows:

$$\begin{aligned}
10.838 \text{ in } \alpha_3^y - 2.746 \text{ in } \alpha_3^z &= 5\,801.01 \text{ in/s}^2,\\
-10.838 \text{ in } \alpha_3^x + 10.000 \text{ in } \alpha_3^z + 7.838 \text{ in } \alpha_4 &= 2\,228.61 \text{ in/s}^2,\\
2.746 \text{ in } \alpha_3^x - 10.000 \text{ in } \alpha_3^y - 6.210 \text{ in } \alpha_4 &= -2\,265.04 \text{ in/s}^2,\\
10.000 \text{ in } \alpha_3^x + 2.746 \text{ in } \alpha_3^y + 10.838 \text{ in } \alpha_3^z &= 0.00.
\end{aligned}$$

The coefficients (on the left-hand side) of these acceleration equations are identical to those of the velocity equations [Eqs. (7)–(10)]. This is always the case. These equations are now solved to give the desired accelerations.

$$\boldsymbol{\alpha}_3 = -526.94\hat{\mathbf{i}} + 618.71\hat{\mathbf{j}} + 329.43\hat{\mathbf{k}} \text{ rad/s}^2, \qquad \textit{Ans.}$$

$$\boldsymbol{\alpha}_4 = -864.59\hat{\mathbf{i}} \text{ rad/s}^2, \qquad \textit{Ans.}$$

$$\mathbf{A}_B = \mathbf{A}_{BO_4}^n + \mathbf{A}_{BO_4}^t = 2\,748.61\hat{\mathbf{j}} - 10\,453.10\hat{\mathbf{k}} \text{ in/s}^2. \qquad \textit{Ans.}$$

10.5 EULER ANGLES

We saw in Sec. 3.2 that angular velocity is a vector quantity; hence, like all vector quantities, it has components along any set of rectilinear axes,

$$\boldsymbol{\omega} = \omega^x\hat{\mathbf{i}} + \omega^y\hat{\mathbf{j}} + \omega^z\hat{\mathbf{k}},$$

and it obeys the laws of vector algebra. Unfortunately, we also saw in Fig. 3.2 that finite three-dimensional angular displacements do not behave as vectors. Their order of summation is not arbitrary; that is, they are not commutative in addition. Therefore, they do not follow the rules of vector algebra. The inescapable conclusion is that we cannot find a set of three angles, or angular displacements, that specify the three-dimensional orientation of a rigid body and that also have ω^x, ω^y, and ω^z as their time derivatives.

To clarify the problem further, we visualize a rigid body rotating in space about a fixed point, O, that we take as the origin of a grounded or absolute reference frame, xyz. We also visualize a moving reference system, $x'y'z'$, sharing the same origin but attached to and moving with the rotating body. The $x'y'z'$ system is defined as a set of *body-fixed axes*. Our problem here is to determine some way to specify the orientation of the body-fixed axes with respect to the absolute reference axes, a method that is convenient for use with finite three-dimensional rotations.

Let us assume that the stationary axes are aligned along unit vector directions $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ and that the rotated body-fixed axes have directions denoted by $\hat{\mathbf{i}}'$, $\hat{\mathbf{j}}'$, and $\hat{\mathbf{k}}'$. Then, any point in the absolute coordinate system is located by the position vector, $\mathbf{R}$, and in the rotated system by $\mathbf{R}'$. Since both vectors are descriptions of the position of the same point, then we conclude that $\mathbf{R} = \mathbf{R}'$ and

$$R^x\hat{\mathbf{i}} + R^y\hat{\mathbf{j}} + R^z\hat{\mathbf{k}} = R^{x'}\hat{\mathbf{i}}' + R^{y'}\hat{\mathbf{j}}' + R^{z'}\hat{\mathbf{k}}'.$$

Now, taking the vector dot products of this equation with $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$, respectively, gives the transformation equations between the two sets of axes, namely

$$\begin{aligned}
R^x &= \left(\hat{\mathbf{i}}\cdot\hat{\mathbf{i}}'\right)R^{x'} + \left(\hat{\mathbf{i}}\cdot\hat{\mathbf{j}}'\right)R^{y'} + \left(\hat{\mathbf{i}}\cdot\hat{\mathbf{k}}'\right)R^{z'},\\
R^y &= \left(\hat{\mathbf{j}}\cdot\hat{\mathbf{i}}'\right)R^{x'} + \left(\hat{\mathbf{j}}\cdot\hat{\mathbf{j}}'\right)R^{y'} + \left(\hat{\mathbf{j}}\cdot\hat{\mathbf{k}}'\right)R^{z'},\\
R^z &= \left(\hat{\mathbf{k}}\cdot\hat{\mathbf{i}}'\right)R^{x'} + \left(\hat{\mathbf{k}}\cdot\hat{\mathbf{j}}'\right)R^{y'} + \left(\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}'\right)R^{z'}.
\end{aligned} \tag{a}$$

But we remember that the dot product of two unit vectors is equal to the cosine of the angle between them. For example, if the angle from $\hat{\mathbf{i}}$ to $\hat{\mathbf{j}}'$ is denoted by $\theta_{ij'}$, then

$$\left(\hat{\mathbf{i}}\cdot\hat{\mathbf{j}}'\right) = \cos\theta_{ij'}. \tag{b}$$

Thus, the above set of equations becomes

$$\begin{aligned}
R^x &= \cos\theta_{ii'}R^{x'} + \cos\theta_{ij'}R^{y'} + \cos\theta_{ik'}R^{z'},\\
R^y &= \cos\theta_{ji'}R^{x'} + \cos\theta_{jj'}R^{y'} + \cos\theta_{jk'}R^{z'},\\
R^z &= \cos\theta_{ki'}R^{x'} + \cos\theta_{kj'}R^{y'} + \cos\theta_{kk'}R^{z'}.
\end{aligned} \tag{10.3}$$

These equations can be written, in matrix notation, as

$$\begin{bmatrix} R^x \\ R^y \\ R^z \end{bmatrix} = \begin{bmatrix} \cos\theta_{ii'} & \cos\theta_{ij'} & \cos\theta_{ik'} \\ \cos\theta_{ji'} & \cos\theta_{jj'} & \cos\theta_{jk'} \\ \cos\theta_{ki'} & \cos\theta_{kj'} & \cos\theta_{kk'} \end{bmatrix} \begin{bmatrix} R^{x'} \\ R^{y'} \\ R^{z'} \end{bmatrix}. \tag{10.4}$$

This means of defining the orientation of the $x'y'z'$ axes with respect to the xyz axes uses what are called *direction cosines* as coefficients in a set of *transformation equations* between the two coordinate systems. Although we will have use for direction cosines, we note the disadvantage that there are nine of them, whereas only three variables can be independent for spatial rotation. The nine direction cosines are not all independent but are related by six *orthogonality conditions*. We would prefer a technique that had only three independent variables, preferably all angles.

Three angles, called *Euler angles*, can be used to specify the orientation of the body-fixed axes with respect to the reference axes, as shown in Fig. 10.10. To explain the Euler angles, we begin with the body-fixed axes coincident with the reference axes.

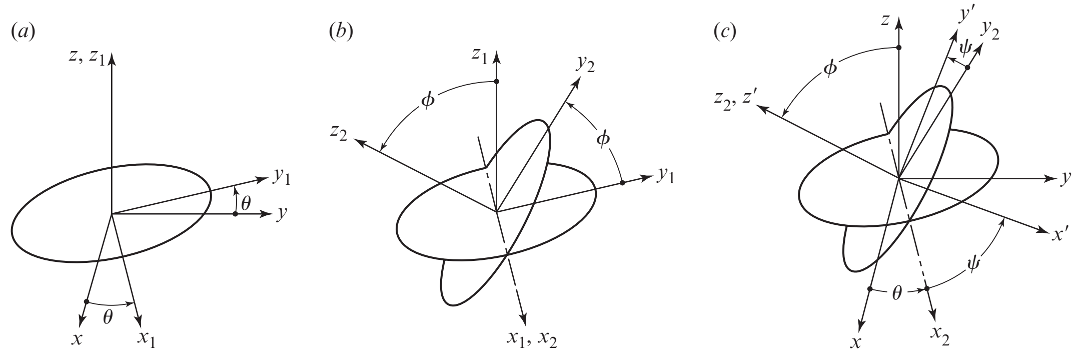

Figure 10.10 Euler angles.

We then specify three successive rotations, θ, ϕ, and ψ, which must occur in the specified order and about the specified axes, to move the body-fixed axes into their final orientation.

The first Euler angle describes a rotation through the angle θ and is taken positive counterclockwise about the positive z axis, as shown in Fig. 10.10*a*. This rotation turns from the xy axes by the angle θ to their new x_1y_1 locations shown, whereas z and z_1 remain fixed and collinear. The transformation equations for this first rotation are

$$\begin{bmatrix} R^x \\ R^y \\ R^z \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R^{x_1} \\ R^{y_1} \\ R^{z_1} \end{bmatrix}. \tag{10.5}$$

The second Euler angle describes a rotation through the angle ϕ and is taken positive counterclockwise about the positive x_1 axis, as shown in Fig. 10.10*b*. This rotation turns from the displaced y_1z_1 axes by the angle ϕ to their new y_2z_2 locations shown, whereas x_1 and x_2 remain fixed. The transformation equations for this second rotation are

$$\begin{bmatrix} R^{x_1} \\ R^{y_1} \\ R^{z_1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} R^{x_2} \\ R^{y_2} \\ R^{z_2} \end{bmatrix}. \tag{10.6}$$

The third Euler angle describes a rotation through the angle ψ and is taken positive counterclockwise about the positive z_2 axis, as shown in Fig. 10.10*c*. This rotation turns from the x_2y_2 axes by the angle ψ to their final $x'y'$ locations shown, whereas z_2 and z' remain fixed. The transformation equations for this third rotation are

$$\begin{bmatrix} R^{x_2} \\ R^{y_2} \\ R^{z_2} \end{bmatrix} = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R^{x'} \\ R^{y'} \\ R^{z'} \end{bmatrix}. \tag{10.7}$$

Now, substituting Eq. (10.6) into Eq. (10.5) gives the transformation from the xyz axes to the $x_2y_2z_2$ axes:

$$\begin{bmatrix} R^x \\ R^y \\ R^z \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta\cos\phi & \sin\theta\sin\phi \\ \sin\theta & \cos\theta\cos\phi & -\cos\theta\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} R^{x_2} \\ R^{y_2} \\ R^{z_2} \end{bmatrix}. \tag{10.8}$$

Finally, substituting Eq. (10.7) into Eq. (10.8) gives the total transformation from the xyz axes to the $x'y'z'$ axes:

$$\begin{bmatrix} R^x \\ R^y \\ R^z \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi - \sin\theta\cos\phi\sin\psi & -\cos\theta\sin\psi - \sin\theta\cos\phi\cos\psi & \sin\theta\sin\phi \\ \sin\theta\cos\psi + \cos\theta\cos\phi\sin\psi & -\sin\theta\sin\psi + \cos\theta\cos\phi\cos\psi & -\cos\theta\sin\phi \\ \sin\phi\sin\psi & \sin\phi\cos\psi & \cos\phi \end{bmatrix} \begin{bmatrix} R^{x'} \\ R^{y'} \\ R^{z'} \end{bmatrix}. \tag{10.9}$$

Since the transformation is expressed as a function of only three Euler angles, θ, ϕ, and ψ, it represents the same information as Eqs. (10.3) and (10.4), but without the difficulty of having nine variables (direction cosines) related by six (orthogonality) conditions. Thus, Euler angles have become a useful tool in treating problems involving three-dimensional rotation.

Note, however, that the form of the transformation equations given here depends on using precisely the conventions for the particular set of Euler angles defined in Fig. 10.10. Unfortunately, there appears to be little agreement among different authors regarding how these angles should be defined. A wide variety of other definitions, differing in the axes about which the rotations are to be measured, or in their order, or in the sign conventions for positive values of the angles, are to be found in other references. The differences are not great, but are sufficient to frustrate easy comparison of the formulae derived.

We must also remember that the time derivatives of the Euler angles, that is, $\dot{\theta}$, $\dot{\phi}$, and $\dot{\psi}$, are *not* the components of the angular velocity, $\boldsymbol{\omega}$, of the body-fixed axes. We see that these angular velocities act about different axes and are expressed in different coordinate systems. From Fig. 10.10, we can see that the angular velocity of the body-fixed coordinate system is

$$\boldsymbol{\omega} = \dot{\theta}\hat{\mathbf{k}} + \dot{\phi}\hat{\mathbf{i}}_1 + \dot{\psi}\hat{\mathbf{k}}'.$$

When these different unit vectors are all transformed into the fixed reference frames and then added, the three components of the angular velocity vector can be written as

$$\begin{aligned} \omega^x &= \dot{\phi}\cos\theta + \dot{\psi}\sin\theta\sin\phi, \\ \omega^y &= \dot{\phi}\sin\theta - \dot{\psi}\cos\theta\sin\phi, \\ \omega^z &= \dot{\theta} + \dot{\psi}\cos\phi. \end{aligned} \tag{10.10}$$

On the other hand, we can also transform the unit vectors into the body-fixed axes; then

$$
\begin{aligned}
\omega^{x'} &= \dot{\theta}\sin\phi\sin\psi + \dot{\phi}\cos\psi, \\
\omega^{y'} &= \dot{\theta}\sin\phi\cos\psi - \dot{\phi}\sin\psi, \\
\omega^{z'} &= \dot{\theta}\cos\phi + \dot{\psi}.
\end{aligned}
\tag{10.11}
$$

10.6 DENAVIT-HARTENBERG PARAMETERS

The transformation equations of the previous section dealt only with three-dimensional rotations about a fixed point. Yet the same approach can be generalized to include both translations and rotations and thus to treat general spatial motion. This has been done, and much literature dealing with these techniques exists, usually under titles such as "matrix methods." Most modern publications on this approach stem from the work of Denavit and Hartenberg [5]. They developed a notation for labeling all single-loop lower pair linkages and also devised a transformation matrix technique for their analysis.

In the Denavit-Hartenberg approach, we start by numbering the joints of the linkage, usually starting with the input joint and numbering consecutively around a kinematic loop to the output joint. If we assume that the linkage has only a single loop with only j_1 joints and a mobility of $m = 1$, then the Kutzbach criterion indicates that there will be $n = 7$ binary links and $j_1 = 7$ joints numbered from $i = 1$ through $i = 7$.* Figure 10.11 shows a typical joint of this loop, revolute joint number i in this case, and the two links that it connects.

Next, we identify the motion axis of each of the joints, choose a positive orientation on each, and label each as a z_i axis. Although the z_i axes may be skew in space, it is possible to find a common perpendicular between each consecutive pair and label these as x_i axes such

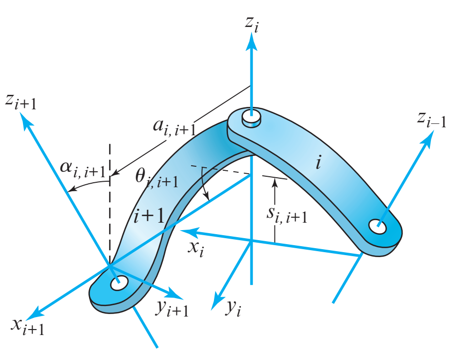

Figure 10.11 Definitions of Denavit-Hartenberg parameters.

* The treatment presented here deals only with j_1 joints; multi-degree-of-freedom lower pairs may be represented as serial combinations of revolute joints and prismatic joints, however, with fictitious links between them.

that each x_i axis is the common perpendicular between z_{i-1} and z_i, choosing an arbitrary positive orientation.*

Having done this, we can now identify and label y_i axes such that there is a right-handed Cartesian coordinate system $x_i y_i z_i$ associated with each joint of the loop. Careful study of Fig. 10.11 and visualization of the motion allowed by the joint indicates that coordinate system $x_i y_i z_i$ remains fixed to the link carrying joint $i-1$ and joint i, whereas $x_{i+1} y_{i+1} z_{i+1}$ moves with and remains fixed to the link carrying joint i and joint $i+1$. Thus, we have a basis for assigning numbers to the links that correspond to the numbers of the coordinate system attached to each link.

The key to the Denavit-Hartenberg approach comes in the standard method they defined for determining the dimensions of the links. The relative location and orientation (posture) of any two consecutive coordinate systems placed as described above can be defined by four parameters, labeled a, α, θ, and s, shown in Fig. 10.11, and defined as follows:

$a_{i,i+1}$ = the distance along x_{i+1} from z_i to z_{i+1} with sign taken from the sense of x_{i+1};
$\alpha_{i,i+1}$ = the angle from positive z_i to positive z_{i+1} taken positive counterclockwise as seen from positive x_{i+1};
$\theta_{i,i+1}$ = the angle from positive x_i to positive x_{i+1} taken positive counterclockwise as seen from positive z_i; and
$s_{i,i+1}$ = the distance along z_i from x_i to x_{i+1} with sign taken from the sense of z_i.

When joint i is a revolute, as depicted in Fig. 10.11, then the $a_{i,i+1}$, $\alpha_{i,i+1}$, and $s_{i,i+1}$ parameters are constants defining the shape of link $i+1$, but $\theta_{i,i+1}$ is a variable; in fact, it serves to measure the joint variable of joint i. When joint i is a prismatic joint, then the $a_{i,i+1}$, $\alpha_{i,i+1}$, and $\theta_{i,i+1}$ parameters are constants, and $s_{i,i+1}$ is the joint variable.

EXAMPLE 10.3

Find the Denavit-Hartenberg parameters for the Hooke, or Cardan, universal joint shown in Fig. 10.12.

SOLUTION

The linkage has four links and four revolute joints (it is identical to the spheric four-bar linkage shown in Fig. 10.2). First, we label the axes of the four revolute joints as z_1, z_2, z_3, and z_4. Next, we identify and label the common perpendiculars between the joint axes, choosing positive orientations for these, and label them x_1, x_2, x_3, and x_4, as shown. Note that, at the posture shown, x_1 and x_3 appear to lie along z_3 and z_1, respectively; this is only a temporary coincidence and changes as the linkage moves.

* When i=1, then $i-1$ must be taken as n, because, in the loop, the last joint is also the joint before the first. Similarly, when i=n, then $i+1$ must be taken as 1.

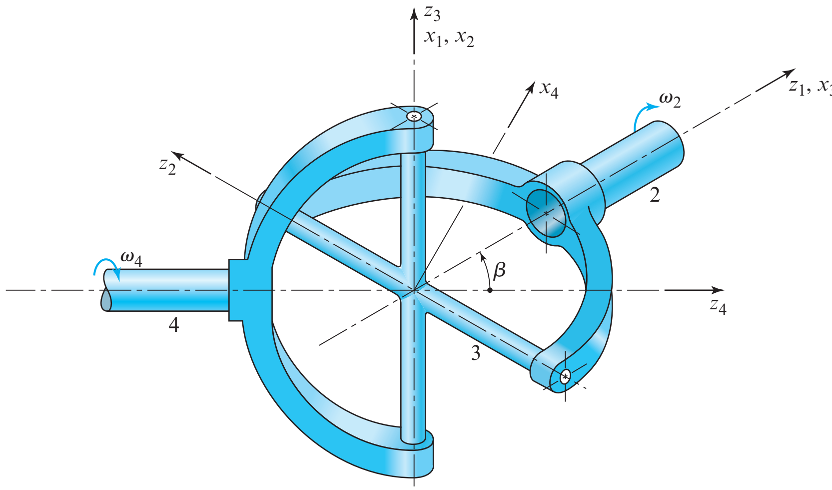

Figure 10.12 Hooke, or Cardan, universal joint.

Now, following the definitions given above, we find the values for the parameters.

$$\begin{array}{llllr} a_{12} = 0 & a_{23} = 0 & a_{34} = 0 & a_{41} = 0 & \qquad \textit{Ans.} \\ \alpha_{12} = 90^\circ & \alpha_{23} = 90^\circ & \alpha_{34} = 90^\circ & \alpha_{41} = \beta & \qquad \textit{Ans.} \\ \theta_{12} = \phi_1 & \theta_{23} = \phi_2 & \theta_{34} = \phi_3 & \theta_{41} = \phi_4 & \qquad \textit{Ans.} \\ s_{12} = 0 & s_{23} = 0 & s_{34} = 0 & s_{41} = 0 & \qquad \textit{Ans.} \end{array}$$

where β is the angle between the two shafts.

We note that all a and s distance parameters are zero; this signifies that the linkage is, indeed, spheric, with all motion axes intersecting at a common center. Also, all θ parameters are joint variables, since this is an *RRRR* linkage. They have been given the symbols ϕ_i rather than numeric values, since they are variables; solutions for their values are found in the next section.

10.7 TRANSFORMATION-MATRIX POSTURE ANALYSIS

The Denavit-Hartenberg parameters provide a standard method for measuring the important geometric characteristics of a linkage, but the value of this approach does not stop there. Having standardized the placement of the coordinate systems on each link, Denavit and Hartenberg also demonstrated that the transformation equations for the postures of successive coordinate systems can be written in a standard matrix format that uses these parameters.

If we know the position coordinates of a point measured in one of the coordinate systems, say R_{i+1}, then we can find the position coordinates of the same point in the previous coordinate system, R_i, as follows:

$$R_i = T_{i,i+1} R_{i+1}, \tag{10.12}$$

where the transformation $T_{i,i+1}$ has the standard form*

$$T_{i,i+1} = \begin{bmatrix} \cos\theta_{i,i+1} & -\cos\alpha_{i,i+1}\sin\theta_{i,i+1} & \sin\alpha_{i,i+1}\sin\theta_{i,i+1} & a_{i,i+1}\cos\theta_{i,i+1} \\ \sin\theta_{i,i+1} & \cos\alpha_{i,i+1}\cos\theta_{i,i+1} & -\sin\alpha_{i,i+1}\cos\theta_{i,i+1} & a_{i,i+1}\sin\theta_{i,i+1} \\ 0 & \sin\alpha_{i,i+1} & \cos\alpha_{i,i+1} & s_{i,i+1} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{10.13}$$

and each point position vector is given by

$$R = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}. \tag{10.14}$$

Now, by applying Eq. (10.12) recursively from one link to the next, for an n-link single-loop mechanism,

$$\begin{aligned} R_1 &= T_{12}R_2 \\ R_1 &= T_{12}T_{23}R_3 \\ R_1 &= T_{12}T_{23}T_{34}R_4 \end{aligned}$$

and, in general,

$$R_1 = T_{12}T_{23}\cdots T_{i-1,i}R_i. \tag{10.15}$$

If we agree on a notation in which a product of these T matrices is still denoted as a T matrix,

$$T_{i,j} = T_{i,i+1}T_{i+1,i+2}\ldots T_{j-1,j}, \tag{10.16}$$

then Eq. (10.15) becomes

$$R_1 = T_{1,i}R_i. \tag{10.17}$$

Finally, since link 1 follows link n at the end of the loop,

$$R_1 = T_{12}T_{23}\cdots T_{n-1,n}T_{n,1}R_1,$$

and, since this equation must be true no matter what point we choose for R_1,

$$T_{1,2}T_{2,3}\ldots T_{n-1,n}T_{n,1} = I, \tag{10.18}$$

where I is the 4×4 identity transformation matrix.

* Note that the first three rows and columns of the T matrix are the direction cosines needed for the rotation between the coordinate systems [for purposes of comparison see Eq. (10.8)]. The fourth column adds the translation terms for the separation of the origins. The fourth row represents a dummy equation, $1 = 1$, but keeps the T matrix square and nonsingular.

This important equation is the transformation matrix form of the *loop-closure equation*. Just as the vector tetrahedron equation, Eq. (10.2), states that the *sum of vectors* around a kinematic loop must equal zero for the loop to close, Eq. (10.18) states that the *product of transformation matrices* around a kinematic loop must equal the identity transformation. Whereas the vector sum ensures that the loop returns to its starting location, the transformation matrix product also ensures that the loop returns to its starting orientation. This becomes critical, for example, in spheric motion problems, and cannot be shown by the vector-tetrahedron equation. The following example illustrates this point.

EXAMPLE 10.4

Develop equations for the positions of all the joint variables of the Hooke, or Cardan, universal joint shown in Fig. 10.12 when the input shaft angle ϕ_1 is specified.

SOLUTION

The Denavit-Hartenberg parameters for this linkage were determined in Example 10.3. Substituting these parameters into Eq. (10.13) gives the individual transformation matrices for each link, namely

$$T_{1,2} = \begin{bmatrix} \cos\phi_1 & 0 & \sin\phi_1 & 0 \\ \sin\phi_1 & 0 & -\cos\phi_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{1}$$

$$T_{2,3} = \begin{bmatrix} \cos\phi_2 & 0 & \sin\phi_2 & 0 \\ \sin\phi_2 & 0 & -\cos\phi_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{2}$$

$$T_{3,4} = \begin{bmatrix} \cos\phi_3 & 0 & \sin\phi_3 & 0 \\ \sin\phi_3 & 0 & -\cos\phi_3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{3}$$

$$T_{4,1} = \begin{bmatrix} \cos\phi_4 & -\cos\beta\sin\phi_4 & \sin\beta\sin\phi_4 & 0 \\ \sin\phi_4 & \cos\beta\cos\phi_4 & -\sin\beta\cos\phi_4 & 0 \\ 0 & \sin\beta & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

Although we could now use Eq. (10.18) directly, that is,

$$T_{1,2}T_{2,3}T_{3,4}T_{4,1} = I,$$

the number of computations can be reduced if we first rearrange the equation to the form

$$T_{2,3}T_{3,4} = T_{1,2}^{-1}T_{4,1}^{-1} = \left(T_{4,1}T_{12}\right)^{-1}. \tag{5}$$

The inverse matrix on the right-hand side can easily be found here by simply transposing the matrix, that is, by switching rows and columns.* Therefore, substituting and carrying out the matrix computations, Eq. (5) becomes

$$\begin{bmatrix} \cos\phi_2\cos\phi_3 & \sin\phi_2 & \cos\phi_2\sin\phi_3 & 0 \\ \sin\phi_2\cos\phi_3 & -\cos\phi_2 & \sin\phi_2\sin\phi_3 & 0 \\ \sin\phi_3 & 0 & -\cos\phi_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} \begin{matrix}\cos\phi_1\cos\phi_4 \\ -\cos\beta\sin\phi_1\sin\phi_4\end{matrix} & \begin{matrix}\cos\phi_1\sin\phi_4 \\ +\cos\beta\sin\phi_1\cos\phi_4\end{matrix} & \sin\beta\sin\phi_1 & 0 \\ \sin\beta\sin\phi_4 & -\sin\beta\cos\phi_4 & \cos\beta & 0 \\ \begin{matrix}\sin\phi_1\cos\phi_4 \\ +\cos\beta\cos\phi_1\sin\phi_4\end{matrix} & \begin{matrix}\sin\phi_1\sin\phi_4 \\ -\cos\beta\cos\phi_1\cos\phi_4\end{matrix} & -\sin\beta\cos\phi_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

Since corresponding row and column elements on both sides of this equation must be equal, then equating the third-row elements of the third-column we have

$$\cos\phi_3 = \sin\beta\cos\phi_1 \quad \text{and} \quad \sin\phi_3 = \sqrt{1-\sin^2\beta\cos^2\phi_1} = \sigma, \qquad \textit{Ans.}$$

where we define the new symbol σ. Next, equating the first- and second-row elements of column three gives

$$\cos\phi_2 = \frac{\sin\beta\sin\phi_1}{\sigma} \quad \text{and} \quad \sin\phi_2 = \frac{\cos\beta}{\sigma}. \qquad \textit{Ans.}$$

Once we have solved for the sine and cosine of ϕ_2 and ϕ_3, we can equate the elements of the second column, second row and the first column, second row. Using the known solutions, this gives

$$\cos\phi_4 = \frac{\sin\phi_1}{\sigma} \quad \text{and} \quad \sin\phi_4 = \frac{\cos\beta\cos\phi_1}{\sigma}. \qquad \textit{Ans.}$$

Since both the sine and cosine of each variable is now known, the arctangent function gives both the magnitude and the proper quadrant of each angle.

10.8 MATRIX VELOCITY AND ACCELERATION ANALYSES

The power of the matrix method of posture analysis was demonstrated in the previous section. However, the method is not limited to posture analysis. The same standardized approach has been extended to velocity and acceleration analyses. To see this, let us start by noting that, of the four Denavit-Hartenberg parameters, three are constants describing

* Although this is true in this special case, where all translation terms are zero, it is not true for all (4×4) transformation matrices.

the link shape and one is the joint variable. Therefore, in the basic transformation of Eq. (10.13), that is,

$$T = \begin{bmatrix} \cos\theta & -\cos\alpha\sin\theta & \sin\alpha\sin\theta & a\cos\theta \\ \sin\theta & \cos\alpha\cos\theta & -\sin\alpha\cos\theta & a\sin\theta \\ 0 & \sin\alpha & \cos\alpha & s \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{10.19}$$

there is only one variable, and it is either the θ or the s parameter, depending on the type of joint.

If we consider the case where the joint variable is the angle θ, then the derivative of T with respect to this variable is

$$\frac{dT}{d\theta} = \begin{bmatrix} -\sin\theta & -\cos\alpha\cos\theta & \sin\alpha\cos\theta & -a\sin\theta \\ \cos\theta & -\cos\alpha\sin\theta & \sin\alpha\sin\theta & a\cos\theta \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{10.20}$$

On the other hand, if the joint variable is the distance s, as in a prismatic joint, then the derivative of T with respect to this variable is

$$\frac{dT}{ds} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{10.21}$$

It is interesting that both of these derivatives can be taken by the same formula,

$$\frac{dT_{i,i+1}}{d\phi_i} = Q_i T_{i,i+1}, \tag{10.22}$$

where we understand that when joint i is a revolute, then $\phi_i = \theta_{i,i+1}$, and we use

$$Q_i = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \tag{10.23}$$

and when joint i is prismatic, then $\phi_i = s_{i,i+1}$, and we use

$$Q_i = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{10.24}$$

For velocity analysis, we need derivatives with respect to time rather than with respect to the joint variables. Therefore, using Eq. (10.22),

$$\frac{dT_{i,i+1}}{dt} = Q_i T_{i,i+1}\dot{\phi}_i, \tag{10.25}$$

where $\dot{\phi}_i = d\phi_i/dt$. In general, these are unknown variables. However, we will present a general procedure by which these variables can be determined. We start by differentiating the loop-closure conditions, Eq. (10.18), with respect to time. Using the chain rule with Eq. (10.25) to differentiate each factor, we get

$$\sum_{i=1}^{n} T_{12}T_{23}\cdots T_{i-1,i}Q_iT_{i,i+1}\cdots T_{n-1,n}T_{n1}\dot{\phi}_i = 0,$$

and using the more condensed notation of Eq. (10.16), this becomes

$$\sum_{i=1}^{n} T_{1i}Q_iT_{i1}\dot{\phi}_i = 0. \tag{10.26}$$

If we now define the symbol

$$D_i = T_{1i}Q_iT_{i1}, \tag{10.27a}$$

and take note that the loop-closure condition indicates that this is the same as

$$D_i = T_{1i}Q_iT_{1i}^{-1}, \tag{10.27b}$$

then Eq. (10.26) can be written as

$$\sum_{i=1}^{n} D_i\dot{\phi}_i = 0. \tag{10.28}$$

This equation contains all the conditions that the joint variable velocities must obey to fit the mechanism in question and be compatible with each other. Note that once the posture analysis is completed, the D_i matrices can be evaluated from known information. Some of the joint variable velocities will be given as input information, namely, the m input variables; all other $\dot{\phi}_i$ values may then be determined from Eq. (10.28).

To make the above procedure clear, we continue Example 10.4.

EXAMPLE 10.5

Given the angular velocity of the input shaft, $\dot{\phi}_1$, determine the angular velocity of the output shaft, $\dot{\phi}_4$, for the Hooke, or Cardan, universal joint of Example 10.4.

SOLUTION

From the results of the previous example, we find the following relationships between the input shaft angle, ϕ_1, and the other pair variables,

$$\begin{aligned}
\sin\phi_2 &= \frac{\cos\beta}{\sigma} & \cos\phi_2 &= \frac{\sin\beta\sin\phi_1}{\sigma}\\
\sin\phi_3 &= \sigma & \cos\phi_3 &= \sin\beta\cos\phi_1\\
\sin\phi_4 &= \frac{\cos\beta\cos\phi_1}{\sigma} & \cos\phi_4 &= \frac{\sin\phi_1}{\sigma}
\end{aligned}$$

where

$$\sigma = \sqrt{1 - \sin^2\beta\cos^2\phi_1}.$$

Substituting these relationships into Eq. (10.13) gives the transformation matrices as functions of ϕ_1 alone. These, in turn, can be substituted into Eq. (10.27) to give the derivative operator matrices D_i, which become

$$D_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_2 = \begin{bmatrix} 0 & 0 & -\cos\phi_1 & 0 \\ 0 & 0 & -\sin\phi_1 & 0 \\ \cos\phi_1 & \sin\phi_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_3 = \begin{bmatrix} 0 & \dfrac{\sin\beta\sin\phi_1}{\sigma} & \dfrac{\cos\beta\sin\phi_1}{\sigma} & 0 \\ -\dfrac{\sin\beta\sin\phi_1}{\sigma} & 0 & \dfrac{-\cos\beta\cos\phi_1}{\sigma} & 0 \\ -\dfrac{\cos\beta\sin\phi_1}{\sigma} & \dfrac{\cos\beta\cos\phi_1}{\sigma} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_4 = \begin{bmatrix} 0 & -\cos\beta & \sin\beta & 0 \\ \cos\beta & 0 & 0 & 0 \\ -\sin\beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

These can now be used in Eq. (10.28). Taking the elements from row 3, column 2, then from row 1, column 3, and finally from row 2, column 1, gives the three equations:

$$(\sin\phi_1)\dot{\phi}_2 + \left(\frac{\cos\beta\cos\phi_1}{\sigma}\right)\dot{\phi}_3 = 0,$$

$$(-\cos\phi_1)\dot{\phi}_2 + \left(\frac{\cos\beta\sin\phi_1}{\sigma}\right)\dot{\phi}_3 + (\sin\beta)\dot{\phi}_4 = 0,$$

$$\left(-\frac{\sin\beta\sin\phi_1}{\sigma}\right)\dot{\phi}_3 + (\cos\beta)\dot{\phi}_4 = -\dot{\phi}_1.$$

Solving these equations simultaneously gives

$$\dot{\phi}_2 = \frac{-\sin\beta\cos\beta\cos\phi_1}{1 - \sin^2\beta\cos^2\phi_1}\dot{\phi}_1,$$

$$\dot{\phi}_3 = \frac{\sin\beta\sin\phi_1}{\sqrt{1-\sin^2\beta\cos^2\phi_1}}\dot{\phi}_1,$$

$$\dot{\phi}_4 = \frac{-\cos\beta}{1-\sin^2\beta\cos^2\phi_1}\dot{\phi}_1. \qquad \textit{Ans.}$$

Note that when the input shaft has constant angular velocity, the angular velocity of the output shaft is not constant unless the shafts are in line (that is, unless the shaft angle $\beta = 0$). If β is not equal to zero, the output angular velocity fluctuates as the shaft rotates. Since the shaft angle β is constant but usually not zero, the maximum value of the output/input velocity ratio, $\dot{\phi}_4/\dot{\phi}_1$, occurs when $\cos\phi_1 = 1$, that is, when $\phi_1 = 0°, 180°, 360°, 540°$, and so on; the minimum value of this velocity ratio occurs when $\cos\phi_1 = 0$. If the speed fluctuation, that is, the difference between the maximum and minimum velocity ratio is expressed as a percentage and plotted against the shaft angle, β, the result is the graph shown in Fig. 10.13. This curve, showing the percentage speed fluctuation, is useful in evaluating Hooke, or Cardan, universal shaft coupling applications.

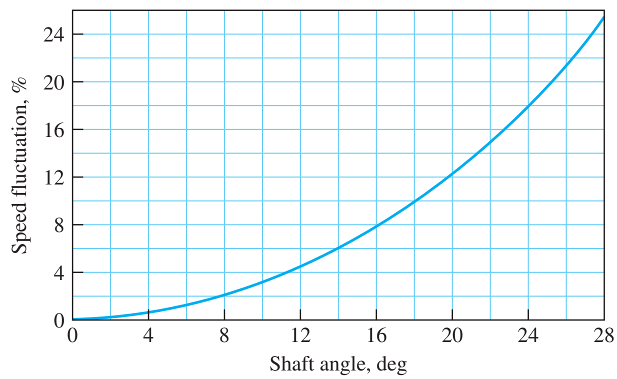

Figure 10.13 Relationship between shaft angle and speed fluctuation.

If we want to find the velocity of a point attached to link i, we can differentiate the equation for the position of the point, Eq. (10.15), with respect to time. Using the operator matrices to do this, we are led to defining a set of velocity operator matrices,

$$\boldsymbol{\omega}_1 = 0, \qquad (10.29a)$$

$$\boldsymbol{\omega}_{i+1} = \boldsymbol{\omega}_i + D_i\dot{\phi}_i, \quad i = 1, 2, \ldots, n, \qquad (10.29b)$$

and then the velocity of a point on link i is given by

$$\dot{R}_i = \boldsymbol{\omega}_i T_{1,i} R_i. \qquad (10.30)$$

Acceleration analysis follows the same approach as for the velocity analysis. Without showing all of the details, the important equations are presented here; for further details, see [22]. First, we must find a formula for taking the derivative of a matrix.

$$\frac{dD_i}{dt} = (\boldsymbol{\omega}_i D_i - D_i \boldsymbol{\omega}_i), \quad i = 1, 2, \ldots, n. \tag{10.31}$$

Differentiating Eq. (10.28) again with respect to time, by means of Eq. (10.31), and rearranging, we get a set of equations relating the pair-variable accelerations:

$$\sum_{i=1}^{n} D_i \ddot{\phi}_i = -\sum_{i=1}^{n} (\boldsymbol{\omega}_i D_i - D_i \boldsymbol{\omega}_i)\, \dot{\phi}_i. \tag{10.32}$$

These equations can be solved for the dependent $\ddot{\phi}_i$ joint variable acceleration values once the input position, velocity, and acceleration values are known. The solution process is identical to that for Eq. (10.28); in fact, the matrix of coefficients of the unknowns is identical for both sets of equations.

This matrix of coefficients, called the *Jacobian*, is essential for the solution of any set of derivatives of the joint variables. As indicated in Sec. 3.20, if this matrix becomes singular, there is no unique solution for the velocities (or accelerations) of the joint variables. If this occurs, such a posture of the mechanism is called a *singular* posture; one example would be a dead-center posture.

The derivatives of the velocity operator matrices of Eq. (10.29) give rise to the definition of a set of acceleration operator matrices, that is

$$\alpha_1 = 0, \tag{10.33a}$$

$$\alpha_{i+1} = \alpha_i + D_i \ddot{\phi}_i + (\boldsymbol{\omega}_i D_i - D_i \boldsymbol{\omega}_i)\, \dot{\phi}_i. \tag{10.33b}$$

From these, the acceleration of a point on link i is given by

$$\ddot{R}_i = (\alpha_i + \boldsymbol{\omega}_i \boldsymbol{\omega}_i)\, T_{1,i} R_i. \tag{10.34}$$

Much greater detail and more power have been developed using this transformation-matrix approach for the kinematic and dynamic analysis of rigid-body systems. These methods [22] go far beyond the scope of this book. Further examples dealing with robotics, however, are presented later in this chapter.

10.9 GENERALIZED MECHANISM ANALYSIS COMPUTER PROGRAMS

It will probably be observed that the methods taken for the solution of each new problem are quite similar from one problem to the next. However, particularly in three-dimensional analysis, the number and complexity of the calculations make solution by hand a very tedious task. These characteristics indicate that a general computer program might have a broad range of applications and that the development costs for such a program might be justified through repeated use, increased accuracy, relief of drudgery, and elimination of error. General computer programs for the simulation of rigid-body kinematic and dynamic systems have been under development for several decades, and many are available and currently used in industrial settings, particularly in the automotive and aerospace industries.

The first widely available computer program for mechanism analysis was named Kinematic Analysis Method (KAM) and was written and distributed by IBM. It included capabilities for position, velocity, acceleration, and force analysis of both planar and spatial mechanisms and was developed using the Chace vector-tetrahedron equation solutions discussed in Sec. 10.3. Released in 1964, this program was the first to recognize the need for a general program for mechanical systems exhibiting large geometric movements. Being first, however, it had limitations and has been superseded by more powerful programs, including those described here.

Powerful generalized programs have also been developed using finite element and finite difference methods; NASTRAN and ANSYS are two examples. In the realm of mechanical systems, these programs have been developed primarily for stress analysis and have excellent capabilities for static- and dynamic-force analysis. These allow the links of the simulated system to deflect under load and are capable of solving statically indeterminate force problems. They are very powerful programs with wide application in industry. Although they are sometimes used for mechanism analysis, they are limited in their ability to simulate the geometric changes typical of kinematic systems.

The most widely used generalized programs for kinematic and dynamic simulation of three-dimensional rigid-body mechanical systems are ADAMS, DADS, and IMP. The MSC ADAMS® program, standing for Automatic Dynamic Analysis of Mechanical Systems, grew from the research efforts of Chace, Orlandea, and others at the University of Michigan [16] and is now available from MSC Software.* DADS, an acronym for Dynamic Analysis and Design System, was originally developed by Haug and others at the University of Iowa [13] and is now available through LMS International.† The Integrated Mechanisms Program (IMP) was developed by Uicker, Sheth, and others at the University of Wisconsin, Madison [19]. These and other similar programs are all applicable to single- or multiple-degree-of-freedom systems in both open- and closed-loop configurations. All operate on workstations, and some operate on microprocessors; all can display results with graphic animation. All are capable of solving position, velocity, acceleration, and static and dynamic force analyses. All can formulate the dynamic equations of motion and can predict the system response to a given set of initial conditions with prescribed motions or forces that may be functions of time. Some of these programs include multibody collision detection, the ability to simulate impact, elasticity, or control system effects. Other commercial software of this type include Mechanism Dynamics†† and MSC Working Model® § systems.

A typical application for any of these programs is the simulation of the half-front automotive suspension shown in Fig. 10.14. The graphs show the comparison of experimental test data and numeric results as a suspension encounters a hole as simulated

* MSC Software Corp., 2 MacArthur Place, Santa Ana, CA 92707, USA.

† LMS International, Researchpark Z1, Interleuvenlaan 68, 3001 Leuven, Belgium; LMS CAE Division, 2651 Crosspark Road, Coralville, IA 52241, USA.

†† Parametric Technologies, Corp., 140 Kendrick Street, Needham, MA 02494, USA.

§ MSC. Software Corp., *ibid.*

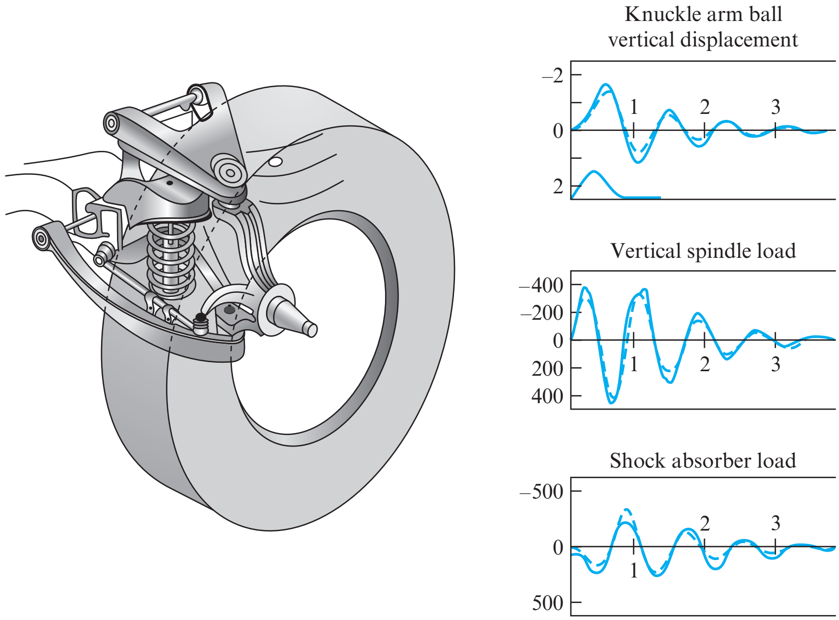

Figure 10.14 Half-front automotive suspension. (Mechanical Dynamics, Inc., Ann Arbor, MI, and JML Research, Inc., Madison, WI.)

by both the ADAMS and IMP programs.* Simulations of this type have been performed with several of these programs, and they compare well with experimental data.

Another type of generalized program available today is intended for kinematic synthesis. The earliest such program was KINSYN (KINematic SYNthesis) [15],† which was followed by LINCAGES (Linkage INteractive Computer Assisted Geometrically Enhanced Synthesis) [7],†† RECSYN (RECtified SYNthesis) [4], and others. These systems are directed toward the kinematic synthesis of planar linkages using methods analogous to those presented in Chap. 9. Users may input their motion requirements through a graphic user interface (GUI); the computer accepts a sketch and provides the required design information on the display screen. Another system of this type is the WATT Mechanism Design Tool from Heron Technologies,§ a spin-off company from Twente

* Simulations of the system shown in Fig. 10.14 were performed using the ADAMS and IMP programs in 1974 for the Strain History Prediction Committee of the Society of Automotive Engineers. Vehicle data and experimental test results were provided by Chevrolet Division, General Motors Corp.

† This paper was accompanied by the presentation of a 16-mm motion picture of 28 minutes' duration. The final version of the program, KINSYN7, was developed and marketed by KINTECH, Inc., until it closed in 1989.

†† LINCAGES is available through Dr. A. G. Erdman by sending e-mail to *agerdman@me.umn.edu*.

§ Heron Technologies b.v., Reekalf 10, 7908 XG Hoogeveen, The Netherlands.

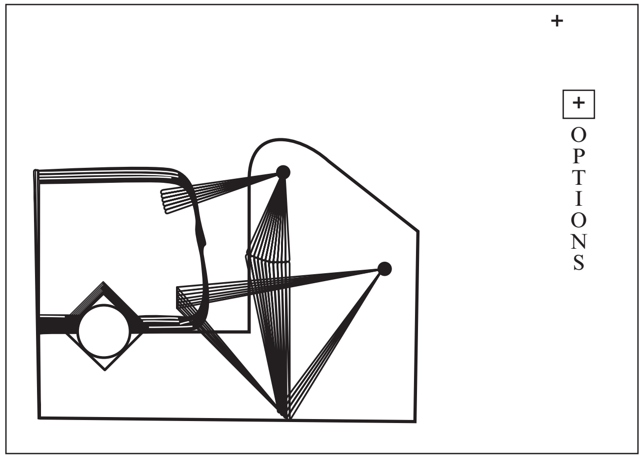

Figure 10.15 This pipe-clamp mechanism. (Courtesy of Dr. Kaufman.)

University in the Netherlands. An example, shown in Fig. 10.15, shows a pipe-clamp mechanism designed in about 15 minutes using KINSYN III, developed at MIT under the direction of Dr. R. E. Kaufman, Professor Emeritus, The George Washington University.

10.10 INTRODUCTION TO ROBOTICS

In preceding chapters, we have studied methods for analyzing the kinematics of machines. First, we studied planar kinematics at great length, justifying this emphasis by observing that over 90% of all mechanisms in use today have planar motion. Then, in the earlier sections of this chapter, we demonstrated how these methods extend to problems with spatial motion. Although the algebra became lengthier with spatial problems, we determined that there was no significant new block of theory required. We also concluded that, as the mathematics becomes more tedious, there is a role for the computer to relieve the drudgery. Still, we reported only a few problems with both practical application and spatial motion.

However, within the past few decades, advancing technology has given considerable attention to the development of robotic devices. Most are spatial mechanisms and require the attendant, more extensive, calculations. Fortunately, however, they also carry on-board computing capability and can deal with this added complexity. The one remaining requirement is that the engineers and designers of the robot itself have a clear understanding of their characteristics and have appropriate tools for their analysis. That is the purpose of the remaining sections of this chapter.

The term *robot* is of Slavic origin; it comes from the Czech word *robota*, meaning work or labor, and was introduced into the English vocabulary by Czechoslovakian dramatist Karl Capek in the early part of the 20th century. This word has been applied to a wide variety of computer-controlled electromechanical systems, from autonomous landrovers to underwater vehicles to teleoperated arms in industrial manipulators. The

Robot Institute of America (RIA) defines a robot as *a reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.* Many industrial manipulators bear a strong resemblance in their conceptual design to a human arm. However, this is not always true and is not essential; the key to the above definition is that a robot has flexibility through its programming, and its motion can be adapted to fit a variety of tasks.

10.11 TOPOLOGICAL ARRANGEMENTS OF ROBOTIC ARMS

Up to this point, we have studied machines with only a few degrees of freedom. With traditional types of machines, it is usually desirable to drive the entire machine from a single source of power; thus they are designed to have mobility of $m = 1$. In keeping with the idea of flexibility of application, however, robots must have more. We know that, if a robot is to reach an arbitrary location in three-dimensional space, it must have mobility of at least $m = 3$ to adjust to proper values of x, y, and z. In addition, if the robot is to be able to manipulate a tool or an object it carries into an arbitrary orientation, once reaching the desired location, an additional three degrees of freedom are required, giving a desired mobility of at least $m = 6$.

Along with the desire for six or more degrees of freedom, we also strive for simplicity, not only for reasons of good design and reliability, but also to minimize the computing burden in the control of the robot. Therefore, we often find that only three of the freedoms are designed into the robot arm and that another two or three may be included in the wrist. Since the tool, or *end effector*, usually varies with the task to be performed, this portion of the total robot may be made interchangeable; the same basic robot arm might carry any of several different end effectors specially designed for particular tasks.

Based on the first three freedoms of the arm, one common arrangement of joints for a manipulator is an *RRR* linkage, also called an *articulated* configuration. Two examples are the Model T26 Cincinnati Milacron T^3 robot shown in Fig. 10.16 and the Intelledex articulated robot shown in Fig. 10.17.

The Selectively Compliant Assembly Robot for Assembly, also called a SCARA robot, is a more recent but popular configuration based on the *RRP* linkage. An example is shown in Fig. 10.18. Although there are other *RRP* robot configurations, note that the SCARA robot has all three joint variable axes parallel (usually parallel to the direction of gravity), restricting its freedom of movement but particularly suiting it to assembly operations.

A manipulator based on the *PPP* chain is shown in Fig. 10.19. It has three mutually perpendicular prismatic joints and, therefore, is called a *Cartesian configuration* or a *gantry* robot. The kinematic analysis of, and the programming for, this style of robot is particularly simple because of the perpendicularity of the three joint axes. It has applications in table-top assembly and in transfer of cargo or material.

The arms depicted up to this point are simple, series-connected open kinematic chains. This is desirable since it reduces their kinematic complexity and eases their design and programming. However, this is not always true; there are robotic arms that include closed kinematic loops in their basic topology. One example is the robot shown in Fig. 10.20.

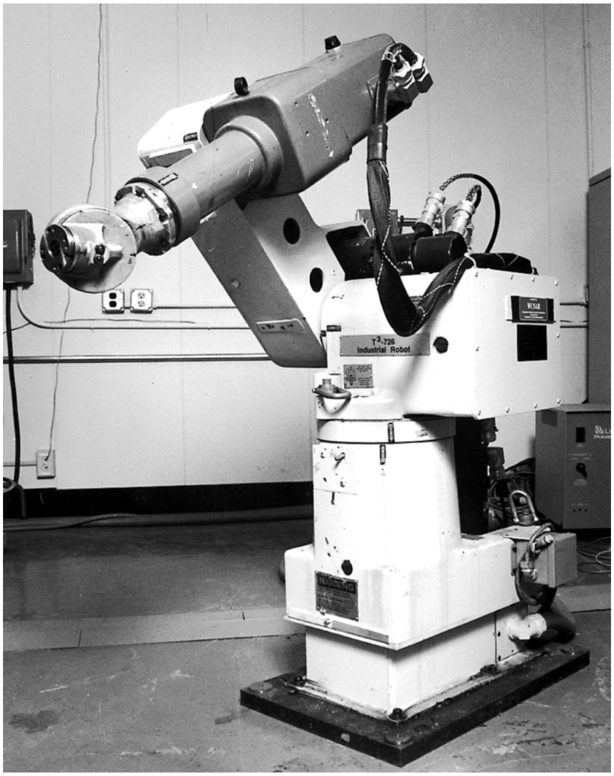

Figure 10.16 Model T26 Cincinnati Milacron T^3 articulated six-axis robot.

In fact, much attention has been given to the design of parallel robot topologies. One example is the Gough/Stewart platform [9][20] shown in Fig. 10.21. Although parallel chains are more complex for analysis and control, they do provide better stiffness, larger payloads, and increased positional accuracy.

10.12 FORWARD KINEMATICS PROBLEM

The kinematic analysis problem addressed here is finding the posture of the end effector (tool) of a robot once we are given the geometry of each component and the positions of the several actuators controlling the degrees of freedom. This problem is commonly referred to as the *forward* or *direct* kinematics problem. For serial robots, the solution can be obtained from the methods on spatial mechanisms presented earlier in this chapter. First, we should recognize three important characteristics of most robotics problems that are different from the spatial mechanisms of earlier sections in this chapter.

1. Knowledge of the position of a single point at the tip of the tool (often called the *tool point*) is often not sufficient. To have a complete knowledge of the end

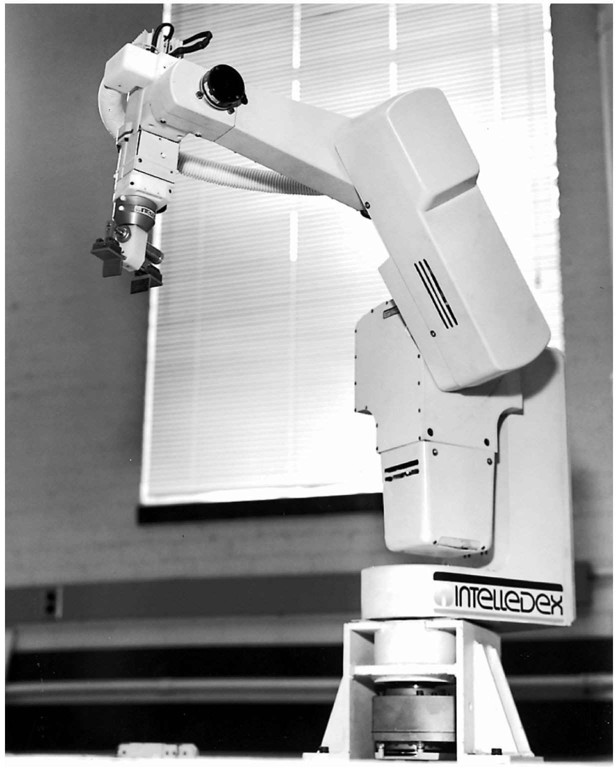

Figure 10.17 Intelledex articulated robot.

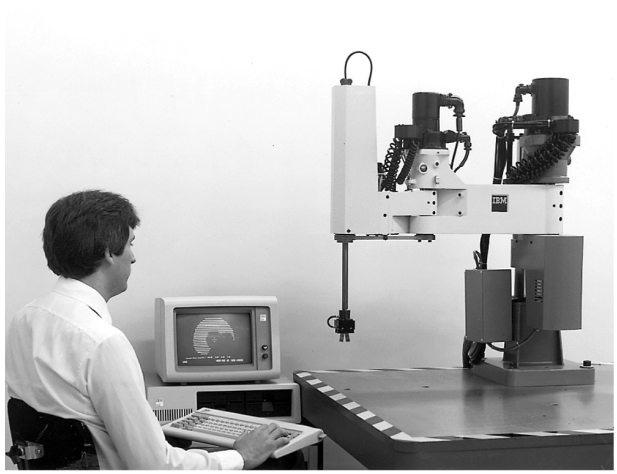

Figure 10.18 IBM model 7525 Selectively Compliant Articulated Robot for Assembly (SCARA). (Courtesy of IBM Corp., Rochester, MN.)

Figure 10.19 IBM model 7650 gantry robot.

effector, we must know the posture of a coordinate system attached to the end effector. This implies that vector methods are not sufficient and that the matrix methods of Sec. 10.7 are better suited.

2. Perhaps because of the previous observation, many robot manufacturers supply the Denavit-Hartenberg parameters of Sec. 10.6 for their particular robot designs. Then, use of the transformation-matrix approach is straightforward.
3. All joint variables of a serially connected chain are independent and are degrees of freedom. Thus, in the forward kinematics problem being discussed now, all joint variables are actuator variables and have given values; there are no loop-closure conditions and no dependent joint variable values to be determined.

The conclusion implied by these three observations combined is that finding the posture of the end effector for given positions of the actuators is a straightforward application of Eq. (10.17). The absolute position of any chosen point R_{n+1} in the end effector coordinate system attached to link $n+1$ is given by

$$R_1 = T_{1,n+1}R_{n+1}, \tag{10.35}$$

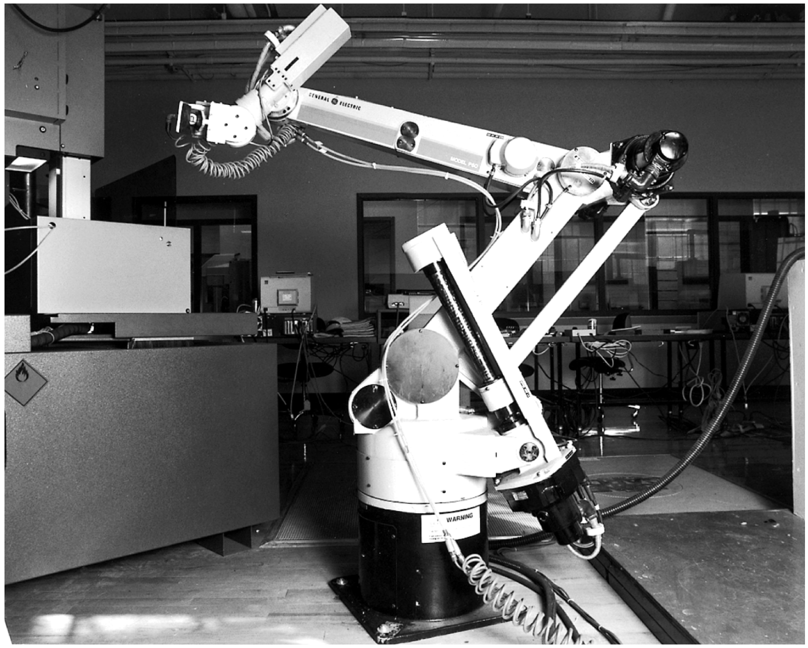

Figure 10.20 General Electric model P80 robotic manipulator.

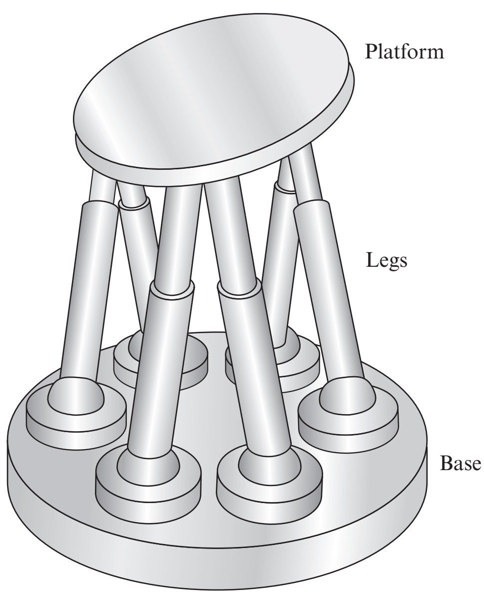

Figure 10.21 Gough/Stewart platform manipulator.

where

$$T_{1,n+1} = T_{1,2}T_{2,3}\dots T_{n,n+1}, \tag{10.36}$$

and each T matrix is given by Eq. (10.13) once the Denavit-Hartenberg parameters (including the actuator positions) are known.

Of course, if the robot has $n+1=7$ links ($n=6$ joints), then symbolic multiplication of these matrices may become lengthy, unless some of the (constant) shape parameters are conveniently set to "nice" values. However, remembering that the robot has computing capability on-board, numeric evaluation of Eq. (10.36) is no great challenge for a particular set of actuator values. Still, because of speed requirements for real-time computer control, it is desirable to simplify these expressions as much as possible before programming. Toward this goal, most robot manufacturers have chosen simplified designs (having "nice" shape parameters) to simplify these expressions. Many have also worked out the final expressions for their particular robots and make these available in the technical documentation.

EXAMPLE 10.6

For the Microbot model TCM five-axis robot shown in Fig. 10.22, find the transformation matrix T_{16} relating the posture of the end effector coordinate system to the ground coordinate system when the actuators are set to position values $\phi_1 = 30°$, $\phi_2 = 60°$, $\phi_3 = -30°$, $\phi_4 = \phi_5 = 0°$. Also, find the absolute position of the tool point that has local coordinates of $x_6 = y_6 = 0$, $z_6 = 2.5$ in.

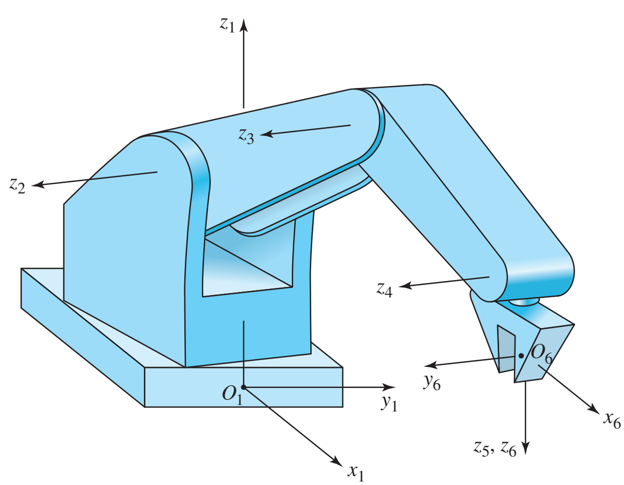

Figure 10.22 Microbot model TCM five-axis robot.

SOLUTION

The coordinate system axes are shown Fig. 10.22. Note that the $x_6y_6z_6$ coordinate system is not based on a joint axis, but was chosen arbitrarily to form a convenient tool coordinate system. Based on these axes and the data in the manufacturer's documentation, we find the Denavit-Hartenberg parameters to be as follows:

$$
\begin{array}{llll}
a_{12} = 0, & \alpha_{12} = 90^\circ, & \theta_{12} = \phi_1 = 30^\circ, & s_{12} = 7.68 \text{ in}, \\
a_{23} = 7.00 \text{ in}, & \alpha_{23} = 0^\circ, & \theta_{23} = \phi_2 = 60^\circ, & s_{23} = 0, \\
a_{34} = 7.00 \text{ in}, & \alpha_{34} = 0^\circ, & \theta_{34} = \phi_3 = -30^\circ, & s_{34} = 0, \\
a_{45} = 0, & \alpha_{45} = 90^\circ, & \theta_{45} = \phi_4 = 0^\circ, & s_{45} = 0, \\
a_{56} = 0, & \alpha_{56} = 0^\circ, & \theta_{56} = \phi_5 = 0^\circ, & s_{56} = 3.80 \text{ in}.
\end{array}
$$

Using Eqs. (10.13) and (10.16) we can now form the following matrices and matrix products:

$$
T_{12} = \begin{bmatrix} 0.866 & 0 & 0.500 & 0 \\ 0.500 & 0 & -0.866 & 0 \\ 0 & 1 & 0 & 7.68 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
T_{23} = \begin{bmatrix} 0.500 & -0.866 & 0 & 3.50 \text{ in} \\ 0.866 & 0.500 & 0 & 6.06 \text{ in} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
T_{13} = \begin{bmatrix} 0.433 & -0.750 & 0.500 & 3.03 \text{ in} \\ 0.250 & -0.433 & -0.866 & 1.75 \text{ in} \\ 0.866 & 0.500 & 0 & 13.74 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
T_{34} = \begin{bmatrix} 0.866 & 0.500 & 0 & 6.06 \text{ in} \\ -0.500 & 0.866 & 0 & -3.50 \text{ in} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
T_{14} = \begin{bmatrix} 0.750 & -0.433 & 0.500 & 8.28 \text{ in} \\ 0.433 & -0.250 & -0.866 & 4.78 \text{ in} \\ 0.500 & 0.866 & 0 & 17.24 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
T_{45} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
T_{15} = \begin{bmatrix} 0.750 & 0.500 & 0.433 & 8.28 \text{ in} \\ 0.433 & -0.866 & 0.250 & 4.78 \text{ in} \\ 0.500 & 0 & -0.866 & 17.24 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
T_{56} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3.80 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
T_{16} = \begin{bmatrix} 0.750 & 0.500 & 0.433 & 9.93 \text{ in} \\ 0.433 & -0.866 & 0.250 & 5.73 \text{ in} \\ 0.500 & 0 & -0.866 & 13.95 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

Ans.

The absolute position of the specified tool point is now found from Eq. (10.35):

$$
R_1 = \begin{bmatrix} 0.750 & 0.500 & 0.433 & 9.93 \text{ in} \\ 0.433 & -0.866 & 0.250 & 5.73 \text{ in} \\ 0.500 & 0 & -0.866 & 13.95 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2.5 \text{ in} \\ 1 \end{bmatrix} = \begin{bmatrix} 11.00 \text{ in} \\ 6.36 \text{ in} \\ 11.78 \text{ in} \\ 1 \end{bmatrix}. \qquad \textit{Ans.}
$$

Note that the direct computation of the velocity and acceleration of an arbitrary point on a moving link of the robot can also be accomplished in a straightforward manner by the matrix methods of Sec. 10.8. In particular, Eqs. (10.29), (10.30), (10.33), and (10.34) are of direct use once the actuator velocities and accelerations are known. A continuation of the previous example makes this clear.

EXAMPLE 10.7

For the Microbot model TCM robot of Fig. 10.22 in the posture described in Example 10.6, find the instantaneous velocity and acceleration of the same tool point $x_6 = y_6 = 0$, and $z_6 = 2.5$ in if the actuators are given constant velocities of $\dot{\phi}_1 = 0.20$ rad/s, $\dot{\phi}_4 = -0.35$ rad/s, and $\dot{\phi}_2 = \dot{\phi}_3 = \dot{\phi}_5 = 0$.

SOLUTION

Using Eq. (10.27) and the results from Example 10.6, we find

$$D_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad D_4 = \begin{bmatrix} 0 & 0 & -0.866 & 14.93\text{ in} \\ 0 & 0 & -0.500 & 8.62\text{ in} \\ 0.866 & 0.500 & 0 & -9.56\text{ in} \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and then, from Eq. (10.29), we obtain

$$\omega_1 = 0,$$

$$\omega_2 = \omega_3 = \omega_4 = \begin{bmatrix} 0 & -0.200\text{ rad/s} & 0 & 0 \\ 0.200\text{ rad/s} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\omega_5 = \omega_6 = \begin{bmatrix} 0 & -0.200\text{ rad/s} & 0.303\text{ rad/s} & -5.225\text{ in/s} \\ 0.200\text{ rad/s} & 0 & 0.175\text{ rad/s} & -3.017\text{ in/s} \\ -0.303\text{ rad/s} & -0.175\text{ rad/s} & 0 & 3.346\text{ in/s} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Now, using Eq. (10.30), we find that the velocity of the tool point is

$$\dot{R}_6 = \omega_6 T_{16} R_6 = \begin{bmatrix} -2.93\text{ in/s} \\ 1.24\text{ in/s} \\ -1.10\text{ in/s} \\ 0 \end{bmatrix}, \qquad \textit{Ans.}$$

which in vector notation is

$$\mathbf{V} = -2.93\hat{\mathbf{i}}_1 + 1.24\hat{\mathbf{j}}_1 - 1.10\hat{\mathbf{k}}_1\text{ in/s}. \qquad \textit{Ans.}$$

The accelerations can be obtained in similar fashion. From Eq. (10.33), we obtain

$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0,$$

$$\alpha_5 = \alpha_6 = \begin{bmatrix} 0 & 0 & -0.035 \text{ rad/s}^2 & 0.603 \text{ in/s}^2 \\ 0 & 0 & 0.152 \text{ rad/s}^2 & -1.045 \text{ in/s}^2 \\ 0.035 \text{ rad/s}^2 & -0.152 \text{ rad/s}^2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Finally, using Eq. (10.34), we find that the acceleration of the tool point is

$$\ddot{R}_6 = (\alpha_6 + \omega_6\omega_6)T_{16}R_6 = \begin{bmatrix} -0.390 \text{ in/s}^2 \\ -0.033 \text{ in/s}^2 \\ 0.088 \text{ in/s}^2 \\ 0 \end{bmatrix}, \qquad \textit{Ans.}$$

which in vector notation is

$$\mathbf{A} = -0.390\hat{\mathbf{i}}_1 - 0.033\hat{\mathbf{j}}_1 + 0.088\hat{\mathbf{k}}_1 \text{ in/s}^2. \qquad \textit{Ans.}$$

10.13 INVERSE KINEMATICS PROBLEM

The previous section demonstrates the basic procedure for finding the positions, velocities, and accelerations of arbitrary points on the moving links of a serial robot once the positions, velocities, and accelerations of the joint actuators are known. The procedures are straightforward and simply applied if, as is usual, the robot is a simply connected chain and has no closed loops. In this simple case, every joint variable is an independent degree of freedom, and no loop-closure equations need be solved. This avoids a major complication.

However, even though the robot itself might have no closed loops, the manner in which a problem or question is presented can sometimes lead to the same complications. Consider, for example, an open-loop robot that we wish to control so that the tool point follows a specified path; we are given the desired path, but we do not know the actuator values (or, more precisely, the time functions) required to achieve this. This problem cannot be solved by the methods of the previous section. When the joint variables (or their derivatives) are the unknowns of the problem rather than given information, the problem is called the *inverse kinematics problem*. In general, this is a more complex problem to solve, and is commonly encountered in robot applications.

When the inverse kinematics problem arises, we must find a set of equations that describe the given situation and that can be solved. In robotics, this usually means that there are one or more constraint equations, not necessarily defined by closed loops within the robot topology, but perhaps defined by the manner in which the problem is posed. For example, if we are told that the tool point is to travel along a certain path with certain timing, then we are being told the values required for the transformation T_{1n} as known functions of time. Then Eq. (10.36) becomes a set of required constraints that must be satisfied and that must be solved to find the joint actuator values or functions of time. An example will make the procedure clear.

EXAMPLE 10.8

The Microbot robot of Fig. 10.22 described in Example 10.6 is to be guided such that: (1) tool point O_6 follows the straight line path given by

$$\mathbf{R}_{O_6}(t) = (4.0 + 1.6t)\hat{\mathbf{i}}_1 + (3.0 + 1.2t)\hat{\mathbf{j}}_1 + 2.0\hat{\mathbf{k}}_1 \text{ in},$$

with t varying from 0.0 to 5.0 s; and (2) the orientation of the end effector is to remain constant with $\hat{\mathbf{i}}_6 = \hat{\mathbf{k}}_1$ (vertical) and $\hat{\mathbf{k}}_6$ radially outward from the base of the robot. Find expressions for how each of the joint actuators must be driven, as functions of time, to achieve this motion.

SOLUTION

From the problem requirements stated, we can express the required time history of the tool coordinate system by the transformation matrix

$$T_{1,6}(t) = \begin{bmatrix} 0 & 0.600 & 0.800 & 4.0 + 1.6t \text{ in} \\ 0 & -0.800 & 0.600 & 3.0 + 1.2t \text{ in} \\ 1 & 0 & 0 & 2.0 \text{ in} \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

We can also multiply out the matrix description for T_{16} from Eq. (10.36); this gives

$$T_{1,6} = \begin{bmatrix} \begin{matrix}\cos\phi_1\cos\beta\cos\phi_5 \\ -\sin\phi_1\sin\phi_5\end{matrix} & \begin{matrix}-\cos\phi_1\cos\beta\sin\phi_5 \\ +\sin\phi_1\cos\phi_5\end{matrix} & \cos\phi_1\sin\beta & \begin{matrix}7\cos\phi_1\cos\phi_2 \\ +7\cos\phi_1\cos(\phi_2+\phi_3) \\ +3.8\cos\phi_1\sin\beta\end{matrix} \\ \begin{matrix}\sin\phi_1\cos\beta\cos\phi_5 \\ -\cos\phi_1\sin\phi_5\end{matrix} & \begin{matrix}-\sin\phi_1\cos\beta\sin\phi_5 \\ -\cos\phi_1\cos\phi_5\end{matrix} & \sin\phi_1\sin\beta & \begin{matrix}7\sin\phi_1\cos\phi_2 \\ +7\sin\phi_1\sin(\phi_2+\phi_3) \\ +3.8\sin\phi_1\sin\beta\end{matrix} \\ \sin\beta\cos\phi_5 & -\sin\beta\sin\phi_5 & -\cos\beta & \begin{matrix}7\sin\phi_2 \\ +7\sin(\phi_2+\phi_3) \\ -3.8\cos\beta\end{matrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{2}$$

where, to save space, we have adopted the definition

$$\beta = \phi_2 + \phi_3 + \phi_4. \tag{3}$$

Now, to achieve the problem requirements, we must find expressions for the actuator variables ϕ_i that ensure that the elements of Eq. (2) are equal to those of Eq. (1) for all values of time; this is our required constraint condition. Equating the ratios of the elements of the second row, third column with the elements of the first row, third column of Eqs. (2) and (1) gives

$$\tan\phi_1 = \frac{0.600}{0.800} = 0.750.$$

Therefore, the first joint variable is

$$\phi_1 = \tan^{-1} 0.750 = 36.87°. \qquad \textit{Ans.}$$

Similarly, from the ratios of the elements of the third row, second column with the elements of the third row, first column we find

$$\tan\phi_5 = 0,$$
$$\phi_5 = 0° \text{ or } \pm 180°.$$

Since the problem gives no preference for the orientation of the end effector in the horizontal plane, we are free to choose

$$\phi_5 = 0°. \qquad \textit{Ans.}$$

Next, from the third row, third column elements, we obtain

$$\cos\beta = 0$$
$$\beta = \phi_2 + \phi_3 + \phi_4 = \pm 90°.$$

Since the orientation of the end effector in the horizontal plane is not specified, we arbitrarily choose

$$\beta = \phi_2 + \phi_3 + \phi_4 = 90°. \tag{4}$$

Turning now to the elements of the fourth column, and simplifying according to already known results, we obtain two more independent equations:

$$7\cos\phi_2 + 7\cos(\phi_2 + \phi_3) = 1.2 + 2t, \tag{5}$$
$$7\sin\phi_2 + 7\sin(\phi_2 + \phi_3) = -5.68. \tag{6}$$

Squaring and adding these two equations gives

$$98 + 98\cos\phi_3 = 33.7024 + 4.8t + 4t^2. \tag{7}$$

Therefore, a solution for ϕ_3 is

$$\phi_3 = \cos^{-1}\left(-0.656\,10 + 0.048\,98t + 0.040\,816t^2\right). \qquad \textit{Ans.}\ (8)$$

Since this form admits to multiple values, we take the smallest solution with $\phi_3 \leq 0$ so that the elbow is kept above the path. At time $t = 0$, for example, the value of ϕ_3 is $\phi_3 = \cos^{-1}(-0.656\,10) = -131.00°$.

Knowing the solution for ϕ_3, we now rewrite Eqs. (5) and (6) as follows:

$$\sin\phi_2 = \frac{-5.68}{14} - \frac{(1.2 + 2t)\sin\phi_3}{14\,(1 + \cos\phi_3)}, \tag{9}$$

$$\cos\phi_2 = \frac{(1.2 + 2t)}{14} - \frac{5.68\sin\phi_3}{14\,(1 + \cos\phi_3)}. \tag{10}$$

Since the third actuator variable ϕ_3 now has a known solution, either of these equations can be used to find ϕ_2. Actually, both should be evaluated to ensure that the proper quadrant is found. These two equations can be numerically evaluated at any value of time, and they are considered solutions even though they are not explicit, closed-form functions of time.

Similarly, from Eq. (4), we now find the fourth joint variable:

$$\phi_4 = 90° - \phi_2 - \phi_3. \qquad \textit{Ans.}\ (11)$$

Graphs of the five actuator variables versus time are shown in Fig. 10.23. Note that for the path being followed, the actuator variables follow very smooth but nonlinear curves. Thus, it appears that the actuator velocities and accelerations will not be zero but should fall within very reasonable limits.

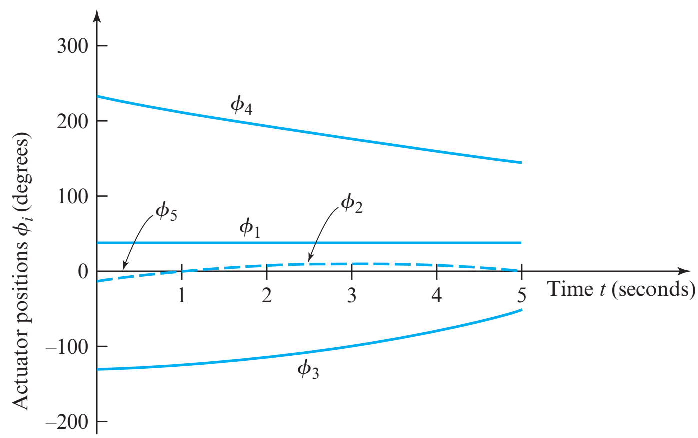

Figure 10.23 Graphs of the actuator variables versus time.

10.14 INVERSE VELOCITY AND ACCELERATION ANALYSES

The previous section presents the general approach to the inverse posture analysis problem. However, at least for control purposes, it may also be important to find the required actuator variable velocities and accelerations. One way to do this is to directly differentiate the actuator variable solution equations with respect to time. However, another, more general approach is to differentiate our constraint conditions.

The constraint equations are given by Eq. (10.36), and, for convenience, can be written as

$$T_{1,2}T_{2,3}\ldots T_{n,n+1} = T_{1,n+1}(t),$$

where, for the inverse kinematics problem, the right-hand side of this equation consists of known functions of time given by the problem specifications, namely, the trajectory (or path) to be followed. Now, using Eq. (10.25) and the D_i differentiation operator matrices of Eq. (10.27b), we can differentiate the left-hand side of the equation, and by direct differentiation with respect to time, we can also differentiate the right-hand side. Thus, we obtain

$$\sum_{i=1}^{n} D_i \dot{\phi}_i T_{1,n+1} = \frac{dT_{1,n+1}(t)}{dt}. \tag{10.37}$$

If we post-multiply both sides of Eq. (10.37) by $T_{1,n+1}^{-1}$, we obtain

$$\sum_{i=1}^{n} D_i \dot{\phi}_i = \frac{dT_{1,n+1}(t)}{dt} T_{1,n+1}^{-1}. \tag{10.38}$$

For convenience, we define the right-hand side by

$$\Omega = \frac{dT_{1,n+1}(t)}{dt} T_{1,n+1}^{-1}, \tag{10.39}$$

where Ω is a new (velocity) operator matrix. Then, substituting Eq. (10.39) into Eq. (10.38) gives

$$\sum_{i=1}^{n} D_i \dot{\phi}_i = \Omega, \tag{10.40}$$

which, as we will see, can be solved for the actuator velocities, $\dot{\phi}_i$.

Let us now look more carefully at the nature of Eq. (10.40). Each of the matrices has four rows and four columns. Yet, since they represent velocities in a single chain in three dimensions, there should only be six independent equations, three for translations and three for rotations. Looking at the definitions of the Q_i differentiation operator matrices in Eq. (10.23) and reviewing the form of the D_i matrices in Example 10.5 and Example 10.7, we discover which of the elements of these matrices contain the six independent equations. Although the proof is left as an exercise, it can be demonstrated that the D_i matrices always have a bottom row of zeroes and the upper-left three rows and three columns are always skew-symmetric; that is, they are always of the form [22]

$$D_i = \begin{bmatrix} 0 & -f & +e & a \\ +f & 0 & -d & b \\ -e & +d & 0 & c \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{10.41}$$

As expected, there are only six independent values (a, b, c, d, e, and f) in each of these matrices. It is now convenient to rearrange these into a single column and to give this column a new symbol:

$$\{D_i\} = \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix}, \tag{10.42}$$

where the braces indicate that, without loss of generality, the elements of the (4×4) matrix have been rearranged into a (6×1) column matrix. There is actually much more mathematical and physical significance to this (6×1) column vector than is explained here; they form the *Plücker coordinates* or *line coordinates* of the motion axis of the joint variable's freedom for motion [14]. The bottom three elements form the three-dimensional unit vector for the orientation of the motion axis, whereas the top three uniquely locate that axis in the absolute coordinate system.

If we now reorganize all matrices in Eq. (10.40) into this column vector form, it becomes

$$\sum_{i=1}^{n} \{D_i\}\dot{\phi}_i = \{\Omega\}, \tag{10.43}$$

which is a set of six simultaneous algebraic equations relating the joint variable velocities. If we write the equations out explicitly, they appear as

$$\begin{aligned}
a_1\dot{\phi}_1 + a_2\dot{\phi}_2 + \cdots + a_n\dot{\phi}_n &= a, \\
b_1\dot{\phi}_1 + b_2\dot{\phi}_2 + \cdots + b_n\dot{\phi}_n &= b, \\
c_1\dot{\phi}_1 + c_2\dot{\phi}_2 + \cdots + c_n\dot{\phi}_n &= c, \\
d_1\dot{\phi}_1 + d_2\dot{\phi}_2 + \cdots + d_n\dot{\phi}_n &= d, \\
e_1\dot{\phi}_1 + e_2\dot{\phi}_2 + \cdots + e_n\dot{\phi}_n &= e, \\
f_1\dot{\phi}_1 + f_2\dot{\phi}_2 + \cdots + f_n\dot{\phi}_n &= f,
\end{aligned}$$

where a, b, and so on are the elements of $\{\Omega\}$ taken in the order consistent with Eqs. (10.41) and (10.42). Regrouping the coefficients of this set of equations into matrix form, we get

$$J = \begin{bmatrix} a_1 & a_2 & \dots & a_n \\ b_1 & b_2 & \dots & b_n \\ c_1 & c_2 & \dots & c_n \\ d_1 & d_2 & \dots & d_n \\ e_1 & e_2 & \dots & e_n \\ f_1 & f_2 & \dots & f_n \end{bmatrix}, \tag{10.44}$$

which is the *Jacobian* (Sec. 10.8). Using this matrix, Eq. (10.43) now becomes

$$J\left\{\dot{\phi}\right\} = \{\Omega\}, \tag{10.45}$$

where $\{\dot{\phi}\}$ is a column of the unknown actuator velocities, $\dot{\phi}_i$, taken in order. This set of equations specifies the constraint conditions that must exist between the actuator velocities to trace the specified path according to the given functions of time.

Assuming that the posture analysis has already been completed, the Ω and D_i matrices are all known functions of time, and this set of equations can now be solved for the actuator velocities as functions of time. If the robot has six independent joint variables, then the solution can be determined by inverting the Jacobian. In this case, the solution is

$$\{\dot{\phi}\} = J^{-1}\{\Omega\}, \tag{10.46}$$

where J^{-1} is the inverse of the Jacobian of Eq. (10.44).

If the Jacobian is not a square matrix, then two cases are possible: (1) When there are fewer than six joint variables, other solution methods, such as Gaussian elimination, must be used. (2) When there are more than six joint variables, the problem has no unique solution; further equations, perhaps even arbitrary conditions, must be supplied. If the Jacobian is square, but singular or if the equations are inconsistent, this shows that there is no unique finite solution for velocities; the problem specifications are not realistic for this robot to perform the task specified. If none of these problems arises, the actuator velocities become known functions of time.

Joint actuator accelerations can be determined by completely analogous methods. Taking the next time derivative of Eq. (10.40), we see that

$$\sum_{i=1}^{n} D_i\dot{\phi}_i = \frac{d\Omega}{dt} - \sum_{i=1}^{n} (\boldsymbol{\omega}_i D_i - D_i\boldsymbol{\omega}_i)\dot{\phi}_i = A, \tag{10.47}$$

where this equation defines a new matrix A, and the $\boldsymbol{\omega}_i$ matrices are defined as in Eqs. (10.29). Extracting the same six independent equations, in the same order as before, we get

$$\sum_{i=1}^{n} \{D_i\}\ddot{\phi}_i = \{A\}. \tag{10.48}$$

We note that the coefficients of the unknown $\ddot{\phi}_i$ joint variable accelerations are identical to those of Eq. (10.43); thus, we can use the same Jacobian defined in Eq. (10.44). The solution to these equations becomes

$$\{\dot{\phi}\} = J^{-1}\{A\}, \tag{10.49}$$

where $\{\dot{\phi}\}$ is a column of the unknown joint actuator accelerations, $\ddot{\phi}_i$, taken in order. This set of equations specifies the relations that must exist between the joint actuator accelerations required to trace the specified path according to the given functions of time.

As with the direct kinematics problem, once the actuator velocities and accelerations are known, the velocity and acceleration of a point on one of the moving links of the robot can be determined from Eqs. (10.29) through (10.34).

EXAMPLE 10.9

Continue Example 10.8 and find the velocities required at the actuators as functions of time to achieve the motion described.

SOLUTION

Using Eq. (10.27*b*) and the solutions for the posture analysis from the above example, we find the D_i matrices. These are

$$D_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 0 & 0 & -0.8 & 6.144\text{ in} \\ 0 & 0 & -0.6 & 4.608\text{ in} \\ 0.8 & 0.6 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_3 = \begin{bmatrix} 0 & 0 & -0.8 & (5.6\sin\phi_2 + 6.144)\text{ in} \\ 0 & 0 & -0.6 & (4.2\sin\phi_2 + 4.608)\text{ in} \\ 0.8 & 0.6 & 0 & -7\cos\phi_2\text{ in} \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_4 = \begin{bmatrix} 0 & 0 & -0.8 & 1.6\text{ in} \\ 0 & 0 & -0.6 & 1.2\text{ in} \\ 0.8 & 0.6 & 0 & -(2t+1.2)\text{ in} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad D_5 = \begin{bmatrix} 0 & 0 & 0.6 & -1.2\text{ in} \\ 0 & 0 & -0.8 & 1.6\text{ in} \\ -0.6 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{12}$$

Using Eq. (10.39) and the trajectory specified in Example 10.8, we also find the Ω matrix.

$$\Omega = \begin{bmatrix} 0 & 0 & 0 & 1.6\text{ in/s} \\ 0 & 0 & 0 & 1.2\text{ in/s} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{13}$$

From these and Eq. (10.43), we construct the set of equations relating the actuator velocities:

$$\begin{bmatrix} 0 & 6.144\text{ in} & (5.6\sin\phi_2 + 6.144)\text{ in} & 1.6\text{ in} & -1.2\text{ in} \\ 0 & 4.608\text{ in} & (4.2\sin\phi_2 + 4.608)\text{ in} & 1.2\text{ in} & 1.6\text{ in} \\ 0 & 0 & -7\cos\phi_2\text{ in} & -(2t+1.2)\text{ in} & 0 \\ 0 & 0.6 & 0.6 & 0.6 & 0.8 \\ 0 & -0.8 & -0.8 & -0.8 & 0.6 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot\phi_1 \\ \dot\phi_2 \\ \dot\phi_3 \\ \dot\phi_4 \\ \dot\phi_5 \end{bmatrix} = \begin{bmatrix} 1.6\text{ in/s} \\ 1.2\text{ in/s} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \tag{14}$$

Here we see the Jacobian for the end effector, such that the tool point follows the specified trajectory. However, in this example, the Jacobian is not square and cannot be inverted. This occurred since this robot has only five degrees of freedom, and, therefore, the end effector is not capable of arbitrary spatial motion. Note that there are six equations and only five unknowns, namely, the five actuator velocities.

Fortunately, however, the specified trajectory is within the capability of this robot. There is a set of solutions for the five unknown actuator velocities that fits all six equations; these solutions are:

$$\dot{\phi}_1 = 0, \qquad \textit{Ans.}$$

$$\dot{\phi}_2 = \frac{(2t+1.2) - 7\cos\phi_2}{\Delta} \text{ rad/s}, \qquad \textit{Ans.}$$

$$\dot{\phi}_3 = \frac{-(2t+1.2)}{\Delta} \text{ rad/s}, \qquad \textit{Ans.}$$

$$\dot{\phi}_4 = \frac{7\cos\phi_2}{\Delta} \text{ rad/s}, \qquad \textit{Ans.}$$

$$\dot{\phi}_5 = 0, \qquad \textit{Ans.}$$

where

$$\Delta = -3.5\,(2t+1.2)\sin\phi_2 - 19.9\cos\phi_2. \tag{15}$$

Since the parameter ϕ_2 is known from the posture analysis, the above equations are solutions for the velocity analysis.

10.15 ROBOT ACTUATOR FORCE ANALYSIS

Of course, the purpose of moving the tool point along the planned trajectory is to perform some useful function, and this almost certainly requires the expenditure of work or power. This work or power must come from the joint actuators. Therefore, it is very helpful to find the sizes of forces and/or torques that must be exerted by the actuators to perform the given task. Conversely, it is useful to know how much force is produced at the end effector for a given set of forces or torques applied by the actuators.

If the force or torque capacity of the actuators is less than those demanded by the task attempted, the robot is not capable of performing the task. It will probably not fail catastrophically, but it will deviate from the desired trajectory and perform a different motion than that specified. The purpose of this section is to find a means of evaluating the forces required of the actuators to perform a given trajectory with a given task loading, so that overloading of the actuators can be anticipated and avoided.

The entire subject of force analysis in mechanical systems is covered in much more depth in Chaps. 11 and 12. Therefore, the treatment here is limited by simplifying assumptions. It is suited specifically to the study of robots performing specified tasks with specified loads at low speeds with no friction or other losses. If these assumptions do not apply, the more extensive treatments of the later chapters should be employed.

The interaction of the robot with the task being performed produces a set of forces and torques at the end effector. Let us assume that these required forces and torques are known functions of time and are grouped into a given six-element column, $\{F\}$, in the following order:

$$\{F\} = \begin{bmatrix} F^{x_1} \\ F^{y_1} \\ F^{z_1} \\ M^{x_1} \\ M^{y_1} \\ M^{z_1} \end{bmatrix}, \tag{10.50}$$

where the first three elements are the components of the required end effector force along the x_1, y_1, z_1 axes, and the next three are the components of the required torque about the same axes.

The source of energy for these needs is a set of unknown forces or torques, τ_i, each one applied by the actuator at the ith joint. We arrange these into another column matrix, τ, in the order of numbering of the joint variables:

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \vdots \\ \tau_n \end{bmatrix}. \tag{10.51}$$

Next, we define a small (absolute) displacement of the end effector against the loads by another six-element column matrix, $\{\delta R\}$, arranged in the same order as the load vector, $\{F\}$:

$$\{\delta R\} = \begin{bmatrix} \delta R^{x_1} \\ \delta R^{y_1} \\ \delta R^{z_1} \\ \delta \theta^{x_1} \\ \delta \theta^{y_2} \\ \delta \theta^{z_1} \end{bmatrix}, \tag{10.52}$$

and yet another column matrix, $\delta\phi$, of displacements of the joint variables, $\delta\phi_i$ during this small displacement.

Now, if the end effector is to undergo the small displacement $\{\delta R\}$ against loads $\{F\}$, the work must come from the torques, τ, of the actuators acting through their small displacements, $\delta\phi$. Therefore, since work input must equal work output in the absence of other losses, we have

$$\{F\}^T \{\delta R\} = \tau^T \delta\phi, \tag{a}$$

where the superscript T signifies the transpose of the matrix. But we also know by integrating Eq. (10.45) over a short time interval δt that the task displacement $\{\delta R\}$ is related to the joint actuator displacements $\{\delta\phi\}$ by

$$\{\delta R\} = J\delta\phi. \tag{10.53}$$

Substituting this relationship into Eq. (*a*) and rearranging, we get

$$\left[\tau^T - \{F\}^T J\right] \delta\phi = 0. \tag{b}$$

Since we assume that the small displacement $\delta\phi$ is not zero, we can solve for the actuator torques, τ, by setting the leading factor to zero. This shows that

$$\tau = J^T \{F\}, \tag{10.54}$$

which is the relationship we are seeking. Under the simplifying conditions assumed above, we have determined that: (*a*) the Jacobian relates the motions of the actuators to the absolute motion of the end effector [Eqs. (10.46), (10.49), and (10.53)]; and (*b*) the transpose of the Jacobian relates the actuator torques to the loads on the end effector [Eq. (10.54)].

EXAMPLE 10.10

Continue with Example 10.9 and assume that the end effector is working against a time-varying load of $10\hat{\mathbf{i}}_1 + 5t\hat{\mathbf{k}}_1$ lb in addition to a constant torque of $25\hat{\mathbf{k}}_1$ in · lb. Find the torques required at the actuators as functions of time to achieve the motion described.

SOLUTION

From the data given, the tool force vector of Eq. (10.50) is

$$\{F\} = \begin{bmatrix} 10 \text{ lb} \\ 0 \\ 5t \text{ lb} \\ 0 \\ 0 \\ 25 \text{ in} \cdot \text{lb} \end{bmatrix}. \tag{16}$$

Since the Jacobian was already found in Example 10.9, Eq. (14), we can now use Eq. (10.54) to find the actuator torques directly. We find them to be

$$\tau_1 = 25.0 \text{ in} \cdot \text{lb}, \qquad \textit{Ans.}$$

$$\tau_2 = 61.4 \text{ in} \cdot \text{lb}, \qquad \textit{Ans.}$$

$$\tau_3 = (-35t\cos\phi_2 + 56.0\sin\phi_2 + 61.4) \text{ in} \cdot \text{lb}, \qquad \textit{Ans.}$$

$$\tau_4 = \left(-10t^2 - 6t + 16.0\right) \text{ in} \cdot \text{lb}, \qquad \textit{Ans.}$$

$$\tau_5 = -12.0 \text{ in} \cdot \text{lb}. \qquad \textit{Ans.}$$

Recall that, since ϕ_2 is known from the posture analysis, then the above equations are solutions for the actuator torques.

10.16 REFERENCES

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PROBLEMS

10.1 Use the Kutzbach criterion to determine the mobility of the *SSC* linkage. Identify any idle freedoms and state how they can be removed. What is the nature of the path described by point *B*?

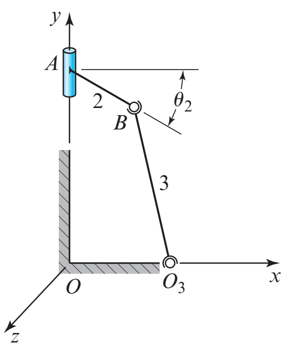


Figure P10.1 $R_{BA} = R_{O_3O} = 75$ mm, $R_{BO_3} = 150$ mm, and $\theta_2 = 30°$.

10.2 For the *SSC* linkage shown in Fig. P10.1, express the posture of each link in vector form.

10.3 For the linkage of Fig. P10.1 with $V_A = -50\hat{\mathbf{j}}$ mm/s, use vector analysis to find the angular velocities of links 2 and 3 and the velocity of point *B* at the posture specified.

10.4 Solve Prob. 10.3 using graphic techniques.

10.5 For the spheric *RRRR* linkage in the posture shown, use vector algebra to make complete velocity and acceleration analyses at the posture indicated.

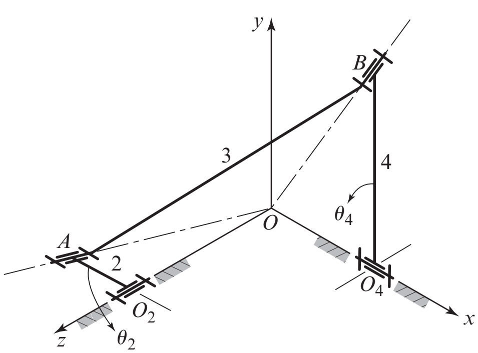


Figure P10.5 $\mathbf{R}_{O_2O} = 7\hat{\mathbf{k}}$ in, $\mathbf{R}_{O_4O} = 2\hat{\mathbf{i}}$ in, $\mathbf{R}_{AO_2} = -3\hat{\mathbf{i}}$ in, $\mathbf{R}_{BO_4} = 9\hat{\mathbf{j}}$ in, $\mathbf{R}_{BA} = 5\hat{\mathbf{i}} + 9\hat{\mathbf{j}} - 7\hat{\mathbf{k}}$ in, and $\boldsymbol{\omega}_2 = -60\hat{\mathbf{k}}$ rad/s.

10.6 Solve Prob. 10.5 using graphic techniques.

10.7 Solve Prob. 10.5 using transformation matrix techniques.

10.8 Solve Prob. 10.5 except with $\theta_2 = 90°$.

10.9 Determine the advance-to-return ratio for Prob. 10.5. What is the total angle of oscillation of link 4?

10.10 For the spheric *RRRR* linkage determine whether the crank is free to turn through a complete revolution. If so, find the angle of oscillation of link 4 and the advance-to-return ratio.

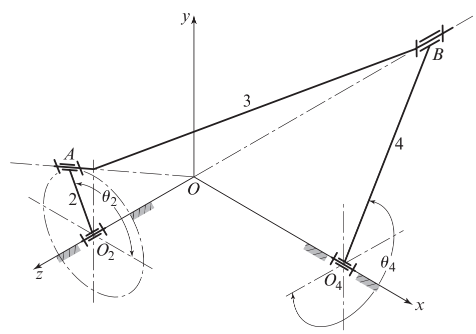


Figure P10.10 $R_{O_2O} = 150$ mm, $R_{O_4O} = 225$ mm, $R_{AO_2} = 37.5$ mm, $R_{BO_4} = 262$ mm, $R_{BA} = 412$ mm, $\theta_2 = 120°$, and $\boldsymbol{\omega}_2 = 30\hat{\mathbf{k}}$ rad/s.

10.11 Use vector algebra to make complete velocity and acceleration analyses of the linkage of Fig. P10.10 at the posture specified.

10.12 Solve Prob. 10.11 using graphic techniques.

10.13 Solve Prob. 10.11 using transformation matrix techniques.

10.14 The figure shows the top, front, and auxiliary views of a spatial slider-crank *RSSP* linkage. In the construction of many such linkages, a provision is made to vary the angle β; thus, the stroke of slider 4 becomes adjustable from zero when $\beta = 0$, to twice the crank length when $\beta = 90°$. With $\beta = 30°$, use vector algebra to make a complete velocity analysis of the linkage at the given posture.

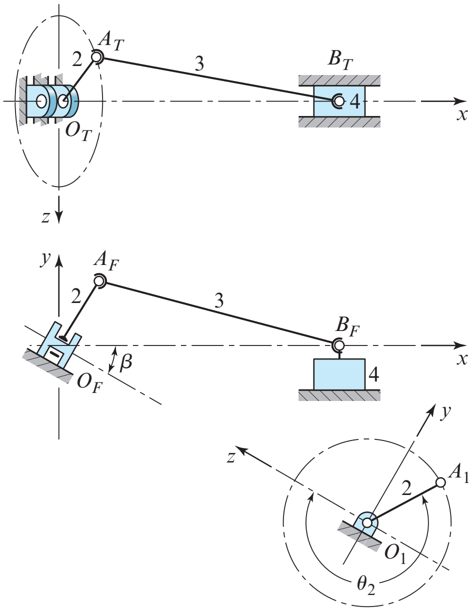


Figure P10.14 $R_{AO} = 2$ in, $R_{BA} = 6$ in, $\theta_2 = 240°$, and $\boldsymbol{\omega}_2 = 24\hat{\mathbf{i}}$ rad/s.

10.15 Solve Prob. 10.14 using graphic techniques.

10.16 Solve Prob. 10.14 using transformation matrix techniques.

10.17 Solve Prob. 10.14 with $\beta = 60°$ using vector algebra.

10.18 Solve Prob. 10.14 with $\beta = 60°$ using graphic techniques.

10.19 Solve Prob. 10.14 with $\beta = 60°$ using transformation matrix techniques.

10.20 The figure shows the top, front, and profile views of an *RSRC* crank and oscillating-slider linkage. Link 4, the oscillating slider, is rigidly attached to a round rod that rotates and slides in the two bearings. (*a*) Use the Kutzbach criterion to find the mobility of this linkage. (*b*) With crank 2 as the driver, find the total angular and linear travel of link 4. (*c*) Write the loop-closure equation for this linkage and use vector algebra to solve it for all unknown position data.

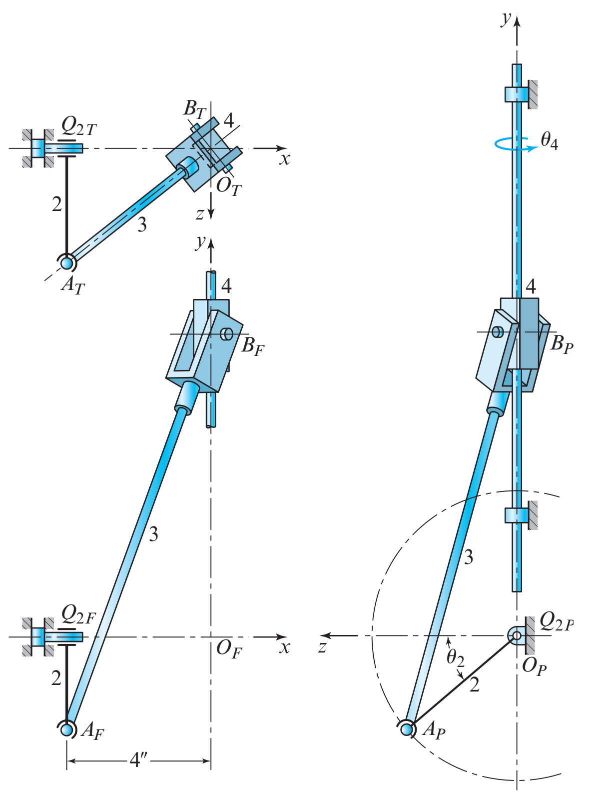


Figure P10.20 $R_{AO} = 4$ in, $R_{BA} = 12$ in, $\theta_2 = 40°$, and $\boldsymbol{\omega}_2 = -48\hat{\mathbf{i}}$ rad/s.

10.21 Use vector algebra to find V_B, ω_3, and ω_4 for Prob. 10.20.

10.22 Solve Prob. 10.21 using graphic techniques.

10.23 Solve Prob. 10.21 using transformation matrix techniques.

†10.24 For the SCARA robot, find the transformation matrix, T_{15}, relating the posture of the tool coordinate system to the ground coordinate system when the joint actuators are set to the values $\phi_1 = 30°$, $\phi_2 = -60°$, $\phi_3 = 2$ in, and $\phi_4 = 0$. Also find the absolute position of the tool point that has coordinates $x_5 = y_5 = 0$, $z_5 = 1.5$ in.

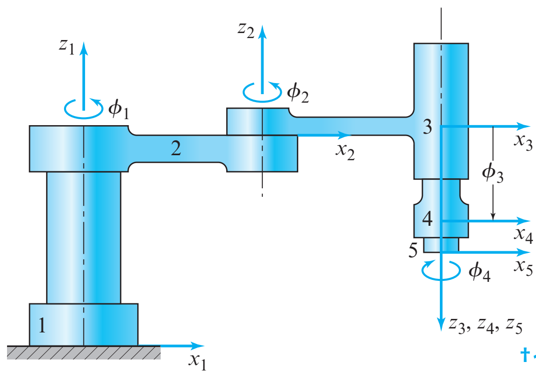


Figure P10.24 $a_{12} = a_{23} = 10$ in, $a_{34} = a_{45} = 0$, $\alpha_{12} = \alpha_{34} = \alpha_{45} = 0$, $\alpha_{23} = 180°$, $\theta_{12} = \phi_1$, $\theta_{23} = \phi_2$, $\theta_{34} = 0$, $\theta_{45} = \phi_4$, $s_{12} = 12$ in, $s_{23} = 0$, $s_{34} = \phi_3$, and $s_{45} = 2$ in.

†10.25 Solve Prob. 10.24 using arbitrary (symbolic) values for the joint variables.

†10.26 For the gantry robot shown, find the transformation matrix T_{15} relating the posture of the tool coordinate system to the ground coordinate system when the joint actuators are set to the values $\phi_1 = 450$ mm, $\phi_2 = 180$ mm, $\phi_3 = 50$ mm, and $\phi_4 = 0$. Also, find the absolute position of the tool point that has coordinates $x_5 = y_5 = 0$, $z_5 = 45$ mm.

†10.27 Repeat Prob. 10.26 using arbitrary (symbolic) values for the joint variables.

†10.28 For the SCARA robot of Prob. 10.24 in the posture described, find the instantaneous velocity and acceleration of the same tool point, $x_5 = y_5 = 0$, $z_5 = 1.5$ in, if the actuators have (constant) velocities of $\dot{\phi}_1 = 0.20$ rad/s, $\dot{\phi}_2 = -0.35$ rad/s, and $\dot{\phi}_3 = \dot{\phi}_4 = 0$.

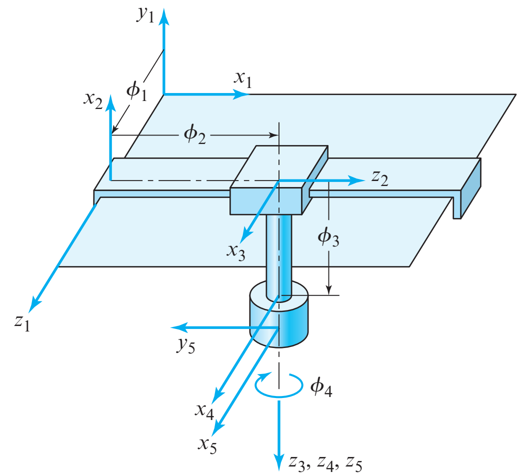


Figure P10.26 $a_{12} = a_{23} = a_{34} = a_{45} = 0$, $\alpha_{12} = 90°$, $\alpha_{23} = -90°$, $\alpha_{34} = \alpha_{45} = 0$, $\theta_{12} = \theta_{23} = 90°$, $\theta_{34} = 0$, $\theta_{45} = \phi_4$, $s_{12} = \phi_1$, $s_{23} = \phi_2$, $s_{34} = \phi_3$, and $s_{45} = 50$ mm.

†10.29 For the gantry robot of Prob. 10.26 in the posture described, find the instantaneous velocity and acceleration of the same tool point, $x_5 = y_5 = 0$, $z_5 = 45$ mm, if the actuators have (constant) velocities of $\dot{\phi}_1 = \dot{\phi}_2 = 0$, $\dot{\phi}_3 = 40$ mm/s, and $\dot{\phi}_4 = 20$ rad/s.

†10.30 The SCARA robot of Prob. 10.24 is to be guided along a path for which the origin of the end effector O_5 follows the straight line given by

$$\mathbf{R}_{O_5}(t) = (1.6t + 4.0)\hat{\mathbf{i}}_1 + (1.2t + 3.0)\hat{\mathbf{j}}_1 + 2.0\hat{\mathbf{k}}_1 \text{ in},$$

with t varying from 0.0 to 5.0 s; the orientation of the end effector is to remain constant with $\hat{\mathbf{k}}_5 = -\hat{\mathbf{k}}_1$ (vertically downward) and $\hat{\mathbf{i}}_5$ radially outward from the base of the robot. Find expressions for how each of the actuators must be driven, as functions of time, to achieve this motion.

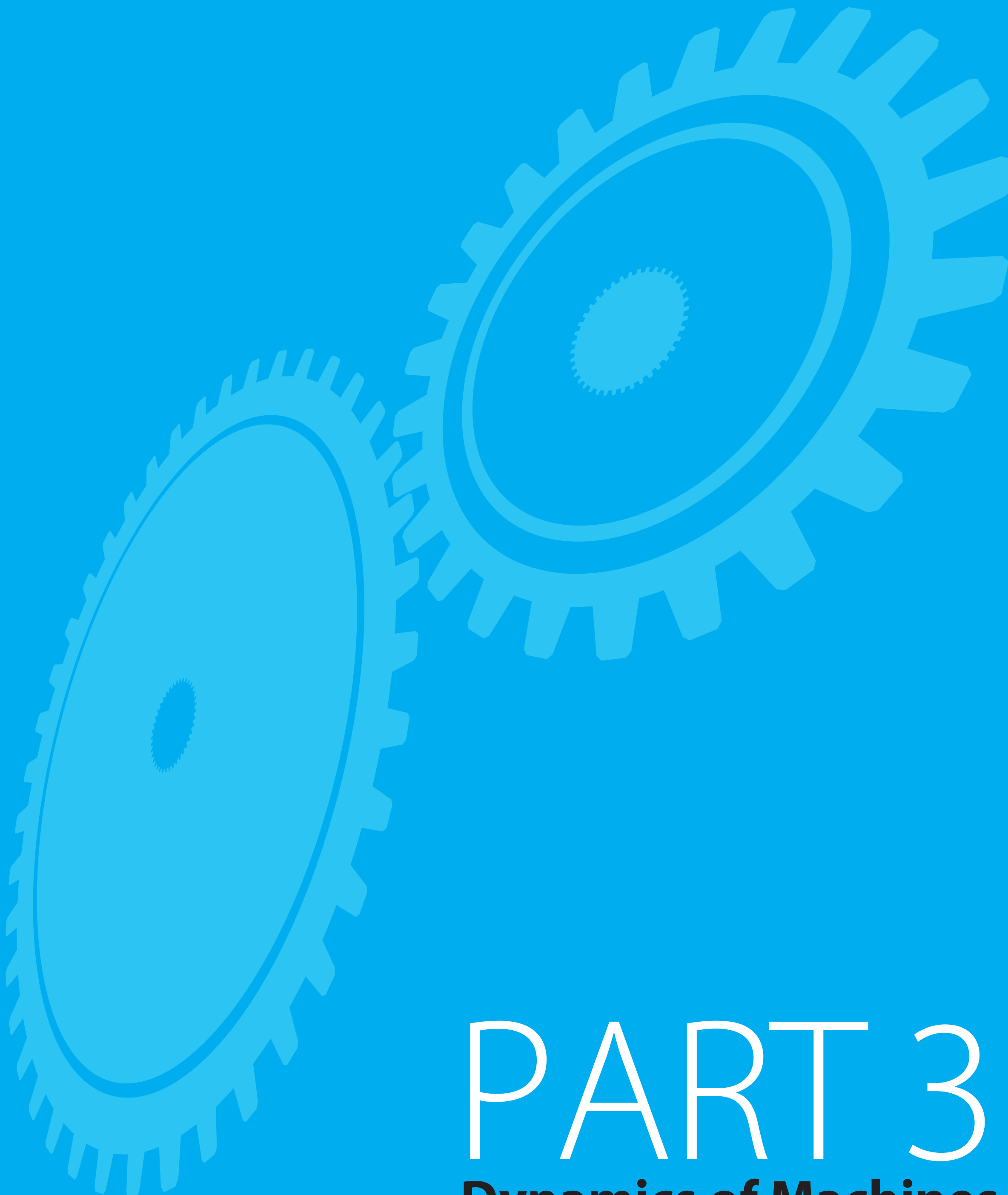
†**10.31** The gantry robot of Prob. 10.26 is to travel a path for which the origin of the end effector O_5 follows the straight line given by

$$\mathbf{R}_{O_5}(t) = (120t + 300)\,\hat{\mathbf{i}}_1 - 150\hat{\mathbf{j}}_1 + (90t + 225)\,\hat{\mathbf{k}}_1 \text{ mm},$$

with t varying from 0.0 to 4.0 s; the orientation of the end effector is to remain constant with $\hat{\mathbf{k}}_5 = -\hat{\mathbf{j}}_1$ (vertically downward) and $\hat{\mathbf{i}}_5 = \hat{\mathbf{i}}_1$. Find expressions for the positions of each of the actuators, as functions of time, for this motion.

†**10.32** The end effector of the SCARA robot of Prob. 10.24 is working against a force loading of $10\hat{\mathbf{i}}_1 + 5t\hat{\mathbf{k}}_1$ lb and a constant torque loading of $25\hat{\mathbf{k}}_1$ in · lb as it follows the trajectory described in Prob. 10.30. Find the torques required at the actuators, as functions of time, to achieve the motion described.

†**10.33** The end effector of the gantry robot of Prob. 10.26 is working against a force loading of $20\hat{\mathbf{i}}_1 + 10t\hat{\mathbf{j}}_1$ N and a constant torque loading of $5\hat{\mathbf{j}}_1$ N · m as it follows the trajectory described in Prob. 10.31. Find the torques required at the actuators, as functions of time, to achieve the motion described.



PART 3
Dynamics of Machines

11 Static Force Analysis

11.1 INTRODUCTION

In our studies of kinematic analysis in the previous chapters, we limited ourselves to consideration of the geometry of the motions and of the relationships between displacement and time. The forces required to produce those motions or the motion that would result from the application of a given set of forces were not considered. We are now ready for a study of the dynamics of machines and systems. Such a study is usually simplified by starting with the statics of such systems.

In the design of a machine, consideration of only those effects that are described by units of *length* and *time* is a tremendous simplification. It frees the mind of the complicating influence of many other factors that ultimately enter into the problem, and it permits our attention to be focused on the primary goal, that of designing a mechanism to obtain a desired motion. That was the problem of *kinematics*, which was covered in the previous chapters of this book.

The fundamental units in kinematic analysis are length and time; in dynamic analysis, they are *length*, *time*, and *force*.

Forces are transmitted between machine members through mating surfaces, that is, from a gear to a shaft or from one gear through meshing teeth to another gear, from a connecting rod through a bearing to a lever, from a V-belt to a pulley, from a cam to a follower, or from a brake drum to a brake shoe. It is necessary to know the magnitudes of these forces for a variety of reasons. The distribution of these forces at the boundaries of mating surfaces must be reasonable, and their intensities must remain within the working limits of the materials composing the surfaces. For example, if the force operating on a sleeve bearing becomes too high, it will squeeze out the oil film and cause metal-to-metal contact, overheating, and rapid failure of the bearing. If the forces between gear teeth are too large, the oil film may be squeezed out from between them. This could result in rough motion, noise, vibration, flaking and spalling of the metal, and eventual failure. In our study of dynamics, we are interested principally in determining the magnitudes, directions, and locations of the forces, but not in sizing the members on which they act. The determination of the sizes of machine members is the subject of books usually titled "machine design" or "mechanical design," such as [5].

Some of the key terms used in this phase of our study are defined as follows.

Force Our earliest perception of force arose from our desire to push, lift, or pull various objects. Thus, force is the action of one body acting on another. Our intuitive concept of force includes such ideas as *magnitude*,*direction*, and *point of application*, and these are commonly referred to as the *characteristics* of the force.

Matter Matter is a material substance. If it is closed and retains its shape, it is called a *body*.

Mass Newton defined the mass of a body as the *quantity of matter*. He recognized the fact that all bodies possess some inherent property that signifies the amount of matter and that is different than weight. Thus, a moon rock has a certain fixed amount of matter, although its moon weight is different from its earth weight. The amount of matter, or quantity of substance, is called the *mass* of the rock.

Inertia Inertia is the property of mass that causes it to resist any effort to change its motion.

Weight Weight is the force that results from gravity acting upon a mass.

Particle A particle is a body whose dimensions can be neglected (Sec. 2.1). The dimensions may or may not be small; if the body is in translation, then the motion of all points of the particle are equal, and it may be considered to be located at a single point. A particle is not a point, however, in the sense that a particle can consist of matter and can have mass, whereas a point cannot.

Rigid Body All real bodies are either elastic or plastic and deform, although perhaps only slightly, when acted upon by forces. When the deformation of a body is small enough to be neglected, such a body is frequently assumed to be *rigid* (that is, incapable of deformation) to simplify the analysis. This assumption of rigidity is the key step that allows the treatment of kinematics without consideration of forces (Sec. 1.3). Without this simplifying assumption of rigidity, forces and motions are interdependent, and kinematic and dynamic analyses require simultaneous solution.

Deformable Body The rigid-body assumption cannot be maintained when internal stresses and strains caused by applied forces and/or moments are to be analyzed. If stress is to be found, we must admit to the existence of strain; thus, we must consider the body to be capable of deformation, even though small. If deformations are small in comparison to the gross dimensions and motion of the body, we can still treat the body as rigid while performing the kinematic (motion) analysis, but must consider it deformable while stresses are determined. (The only place in this text where bodies are considered deformable is in the study of buckling in Secs. 11.14 through 11.18). Using the additional assumption that forces and stresses remain within the elastic range, such analysis is frequently called *elastic-body analysis*.

11.2 NEWTON'S LAWS

Newton's three laws [4] can be stated as follows:

Law 1 If all the forces acting on a body are balanced, that is, they sum to zero, then the body either remains at rest or continues to move in a straight line at a uniform velocity.

Law 2 If the forces acting on a body are not balanced, the body experiences an acceleration proportional to, and in the direction of, the resultant force.

Law 3 When two bodies interact, a pair of reaction forces come into existence between them; these forces have equal magnitudes and opposite senses.

Newton's first two laws can be summarized by the equation

$$\sum \mathbf{F}_{ij} = m_j \mathbf{A}_{G_j}, \tag{11.1}$$

which is called the *equation of motion* for body j. In this equation, $\mathbf{A}_{G_j}$ is the absolute acceleration of the center of mass G_j of body j that has mass m_j, and this acceleration is produced by the sum of all forces acting on the body, that is, all values of the index i. Both $\mathbf{F}_{ij}$ and $\mathbf{A}_{G_j}$ are vector quantities.

In general, this chapter focuses on the first and third of Newton's laws; Chap. 12 concentrates on the second law. For all three laws, the issue of units is critical to successful solution of physical problems; therefore, this topic is addressed in the following section.

11.3 SYSTEMS OF UNITS

An important use of Eq. (11.1) occurs in the standardization of systems of units. Let us employ the following symbols to designate the units associated with physical quantities:

Length	L,
Time	T,
Mass	M,
Force	F.

These symbols are each to stand for any unit we may choose to use for that respective quantity. Thus, possible choices for L are inches, feet, miles, millimeters, meters, or kilometers. The symbols L, T, M, and F are not numeric values; however, they can be entered into an equation as if they were. The equality sign then states that the units on one side must be equivalent to those on the other. Making the indicated entries into Eq. (11.1) gives

$$F = MLT^{-2}, \tag{11.2}$$

since acceleration, **A**, has units of length divided by time squared. Equation (11.2) expresses an equivalence relationship among the four units: force, mass, length, and time. We are free to choose units for three of these, and then the units used for the fourth depend on those used for the chosen three. For this reason, the three units chosen are called *basic units*, whereas the fourth is called the *derived unit*.

When force, length, and time are chosen as basic units, then mass is the derived unit, and the system is called a *gravitational system of units*. When mass, length, and time are chosen as basic units, then force is the derived unit, and the system is called an *absolute system of units*.

In some English-speaking countries, the US customary foot-pound-second (fps) system and the inch-pound-second (ips) system are two gravitational systems of units that are still in use by some practicing engineers.* In the fps system, the (derived) unit of mass is

$$M = FT^2/L = \text{ lbf} \cdot \text{s}^2/\text{ft} = \text{slug}. \tag{11.3}$$

Thus, force, length, and time are the three basic units in the US customary system. The unit of force is the pound, more properly the pound force. We sometimes, but not always, abbreviate this unit *lbf*, but the abbreviation *lb* is also permissible,† since we shall be dealing with US customary gravitational systems. Finally, we note in Eq. (11.3) that the derived unit of mass in the fps gravitational system is the $\text{lbf} \cdot \text{s}^2/\text{ft}$, sometimes referred to as a *slug*; there is no abbreviation for slug.

The derived unit of mass in the ips gravitational system is

$$M = FT^2/L = \text{lbf} \cdot \text{s}^2/\text{in}. \tag{11.4}$$

Note that this unit of mass has not been given an official name.

The International System (SI)†† of units is an absolute system. The three basic units are the meter (m), the kilogram mass (kg), and the second (s). The unit of force is derived and is called a *Newton* (N) to distinguish it from the kilogram, which, as indicated, is a unit of mass. The dimensions of the Newton are

$$F = ML/T^2 = \text{kg} \cdot \text{m}/\text{s}^2 = \text{N}. \tag{11.5}$$

The weight of a body is the force exerted upon it by gravity. Designating the weight as **W** and the acceleration due to gravity as **g**, Eq. (11.1) becomes

$$\mathbf{W} = m\mathbf{g}.$$

In the fps system, standard gravity is $g = 32.174\,0\text{ft/s}^2$, which for most cases is approximated as 32.2ft/s^2. Therefore, the weight of a mass of one slug, under standard gravity, is

* Some prefer to use gravitational systems; this helps to explain some of the resistance to the use the International System (SI), which is an absolute system. However, most US engineering companies with international markets have already required this change.

† The abbreviation *lb* for pound comes from the Latin word, *Libra*, the balance, the seventh sign of the zodiac, which is represented as a pair of scales.

†† SI stands for Système Internationale d'Unités (French).

$$W = mg = (1 \text{ lbf} \cdot \text{s}^2/\text{ft})(32.2 \text{ ft/s}^2) = 32.2 \text{ lb}.$$

In the ips system, standard gravity is 386.088 in/s^2, or approximately 386 in/s^2. Thus, in standard gravity, a unit mass weighs

$$W = mg = (1 \text{ lbf} \cdot \text{s}^2/\text{in})(386 \text{ in/s}^2) = 386 \text{ lb}.$$

With SI units, standard gravity is 9.806 65 m/s^2, or approximately 9.81 m/s^2. Therefore, the weight of a mass of one kilogram, under standard gravity, is

$$W = mg = (1 \text{ kg})(9.81 \text{ m/s}^2) = 9.81 \text{ kg} \cdot \text{m/s}^2 = 9.81 \text{ N}.$$

It is convenient to remember that the weight of a large apple is approximately 1 N.

11.4 APPLIED AND CONSTRAINT FORCES

A great many, if not most, of the forces with which we are concerned in mechanical equipment occur through direct physical or mechanical contact. However, examples of forces that may be applied without physical contact are electrical, magnetic, and gravitational forces. When two or more bodies are connected to form a group or system, the forces acting on this system from outside are called *applied* (or *external*) forces. The action and reaction forces between any two of the connected bodies are called *constraint* (or *internal*) forces. These constraint forces constrain the connected bodies to behave in a specific manner defined by the nature of the connection.

As shown in the next section, the constraint forces of action and reaction at a mechanical contact occur in equal but opposite pairs and thus have no *net* force effect on the system of bodies being considered. Such pairs of constraint forces, although they clearly exist and may be large, are usually not considered further when both the action and the reaction forces act on bodies of the system being considered. However, when we consider a body, or a system of bodies, to be isolated from its surroundings, only one of each pair of constraint forces acts on the system being considered at a point of contact on the separation boundary. The other constraint force, the reaction, is left acting on the surroundings. When we isolate the system being considered, these constraint forces at points of separation must be clearly identified, since they are essential to further study of the isolated system.

As indicated earlier, the *characteristics* of a force are its *magnitude*, its *direction*, and its *point of application*. The direction of a force includes the concept of a line along which the force acts, and a *sense*. Therefore, a force may be directed either positively or negatively along its line of action.

The notation used for force vectors is shown in Fig. 11.1. Boldface letters are used for force vectors and lightface letters for their magnitudes. Thus, the components of a force vector are

$$\mathbf{F} = F^x\hat{\mathbf{i}} + F^y\hat{\mathbf{j}} + F^z\hat{\mathbf{k}}. \qquad (a)$$

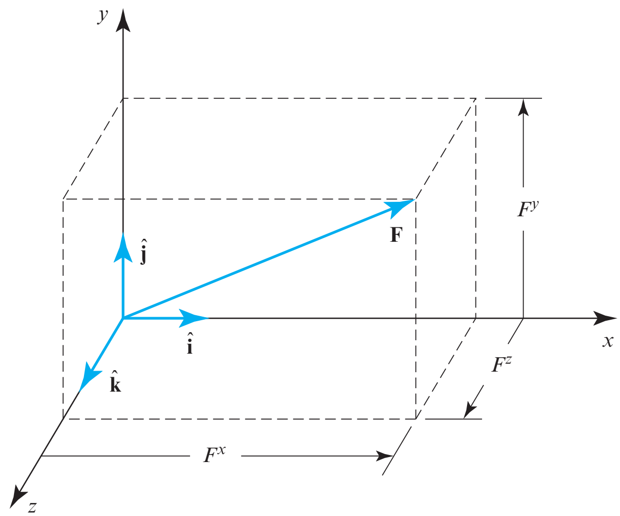


Figure 11.1 Rectangular components of a force vector.

Note that, as in earlier chapters of this text, the directions of components are indicated by superscripts, not subscripts.

Two equal and opposite forces along two parallel but *noncollinear* straight lines in a body cannot be combined to produce a single (null) resultant force on the body. Any two such forces acting on the body constitute a*couple*. For example, consider the two forces, $\mathbf{F}$ and $\mathbf{F}'$, shown in Fig. 11.2*a*. The *arm of the couple* is the perpendicular distance between their lines of action, denoted as h, and *the plane of the couple* is the plane containing the two lines of action. The free vector $\mathbf{M}$ is the *moment* of the couple formed by $\mathbf{F}$ and $\mathbf{F}'$. This vector is directed normal to the plane of the couple, and the sense is in accordance with the right-hand rule. The magnitude of the moment is the product of the arm, h, and the magnitude of one of the forces, say F. Thus,

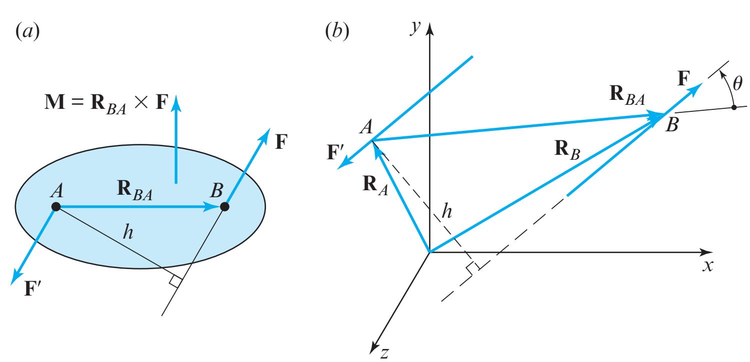


Figure 11.2 (*a*) $\mathbf{R}_{BA}$ is a position-difference vector, whereas $\mathbf{F}$ and $\mathbf{F}'$ are force vectors; (*b*) force couple formed by $\mathbf{F}$ and $\mathbf{F}'$.

$$M = hF. \tag{11.6}$$

In vector form, the moment, $\mathbf{M}$ is the cross-product of the position-difference vector $\mathbf{R}_{BA}$ and the force vector $\mathbf{F}$, and so it is defined by the equation

$$\mathbf{M} = \mathbf{R}_{BA} \times \mathbf{F}. \tag{11.7}$$

Some interesting properties of couples can be determined by examination of Fig. 11.2*b*. Here, $\mathbf{F}$ and $\mathbf{F}'$ are two equal-sized, opposite, and parallel forces. We can choose any point on each line of action and define the positions of these points by vectors $\mathbf{R}_A$ and $\mathbf{R}_B$. Then, the "relative" position, or position-difference vector (Sec. 2.3), is

$$\mathbf{R}_{BA} = \mathbf{R}_B - \mathbf{R}_A. \tag{b}$$

The moment of the couple is the sum of the moments of the two forces and is

$$\mathbf{M} = \mathbf{R}_A \times \mathbf{F}' + \mathbf{R}_B \times \mathbf{F}. \tag{c}$$

But $\mathbf{F}' = -\mathbf{F}$, and, therefore, using Eq. (*b*), Eq. (*c*) can be written as

$$\mathbf{M} = (\mathbf{R}_B - \mathbf{R}_A) \times \mathbf{F} = \mathbf{R}_{BA} \times \mathbf{F}. \tag{d}$$

Equation (*d*) indicates the following:

1. The value of the moment of the couple is independent of the choice of the reference point about which the moments are taken, since vector $\mathbf{R}_{BA}$ is the same for all positions of the origin.
2. Since $\mathbf{R}_A$ and $\mathbf{R}_B$ define any choices of points on the lines of action of the two forces, vector $\mathbf{R}_{BA}$ is not restricted to perpendicularity with $\mathbf{F}$ and $\mathbf{F}'$. This is a very important characteristic of the vector product, since it demonstrates that the value of the moment is independent of how $\mathbf{R}_{BA}$ is chosen. The magnitude of the moment can be obtained as follows: Resolve $\mathbf{R}_{BA}$ into two components $\mathbf{R}^n_{BA}$ and $\mathbf{R}^t_{BA}$, perpendicular and parallel to $\mathbf{F}$, respectively. Then,

   $$\mathbf{M} = (\mathbf{R}^n_{BA} + \mathbf{R}^t_{BA}) \times \mathbf{F}.$$

   But $\mathbf{R}^n_{BA}$ is the perpendicular between the two lines of action, and $\mathbf{R}^t_{BA}$ is parallel to $\mathbf{F}$. Therefore, $\mathbf{R}^t_{BA} \times \mathbf{F} = \mathbf{0}$, and

   $$\mathbf{M} = \mathbf{R}^n_{BA} \times \mathbf{F} \tag{e}$$

   is the moment of the couple. Since $R^n_{BA} = R_{BA} \sin\theta$, where θ is the angle between $\mathbf{R}_{BA}$ and $\mathbf{F}$, the magnitude of the moment is

   $$M = (R_{BA} \sin\theta)F = hF. \tag{f}$$

3. The moment vector $\mathbf{M}$ is independent of any particular origin or line of application and is thus a *free vector*.

4. The forces of a couple can be rotated together within their plane, keeping their magnitudes and the distance between their lines of action constant, or they can be translated to any parallel plane, without changing the magnitude, direction, or sense of the moment vector.
5. Two couples are equal if they have the same moment vectors, regardless of the forces or moment arms. This means that it is the vector product of the two that is significant and not the individual values.

11.5 FREE-BODY DIAGRAMS

The term "body" as used here may consist of an entire machine, several connected parts of a machine, a group of parts, a single part, or a portion of a machine part. A *free-body diagram* is a sketch or drawing of a body or bodies, isolated from the rest of the machine and its surroundings, upon which the forces and moments are shown in action. It is often desirable to include on the diagram the known magnitudes and directions as well as other pertinent information.

The diagram so obtained is called "free," since the machine part, or parts, or portion of the system has been freed (or isolated) from the remaining machine elements, and their effects have been replaced by forces and moments. If the free-body diagram is of a machine part or system of parts, the forces shown on it are the applied (external) forces and moments exerted by adjacent or connecting parts that are not part of this free-body diagram. If the diagram is of a portion of a part, the forces and moments shown acting on the cut (separation) surface are constraint (internal) forces and moments.

The construction and presentation of clear, detailed, and neatly drawn free-body diagrams represent the heart of engineering communication. This is true, since they report a part of the thinking process, whether they are actually placed on paper or not, and since the construction of these diagrams is the *only* way the results of thinking can be communicated to others. It is important to acquire the habit of drawing free-body diagrams no matter how simple the problem may appear. They are a means of storing ideas while concentrating on the next step in the solution of a problem. Construction of, or, at the very minimum, sketching of free-body diagrams speeds up the problem-solving process and dramatically decreases the chances of mistakes.

The advantages of using free-body diagrams can be summarized as follows:

1. They make it easier to translate words, thoughts, and ideas into physical models.
2. They assist in seeing and understanding all facets of a problem.
3. They show the constraint forces and moments acting on the system.
4. They help in planning an approach to a problem.
5. They make mathematical relations easier to see or formulate.
6. Their use makes it easy to keep track of progress and helps in making simplifying assumptions.
7. They are useful for storing methods of solution for future reference.
8. They assist our memory and make it easier to present and explain our work to others.

In analyzing the forces in a machine, we almost always need to separate the machine into its individual components or subsystems and construct free-body diagrams indicating the forces that act on each. Many of these components are connected to each other by kinematic pairs (joints). Accordingly, Fig. 11.3 shows the constraint forces acting between the elements of the six lower pairs (Sec. 1.4) when all friction forces are assumed to be zero. Note that these are not complete free-body diagrams in that they show only the forces on the mating pair surfaces and do not show the forces where the partial links have been separated from the remaining portions of the links.

Upon careful examination of Fig. 11.3, we see that there is no component of a constraint force or moment transmitted along an axis where motion is possible, that is, along with the direction of a joint variable. This is consistent with the assumption of no friction. If one of these pair elements were disposed to transmit a force, or a moment, to its mating element in the direction of the joint variable, in the absence of friction, the tendency would result in motion of the joint variable rather than in the transmission of a force or moment. Similarly, in the case of higher pairs, the constraint forces are always normal to the contacting surfaces in the absence of friction.

The notation shown in Fig. 11.3 is used consistently throughout the text. The force that link i exerts onto link j is denoted $\mathbf{F}_{ij}$, whereas the reaction to this force is denoted $\mathbf{F}_{ji}$ and is the force from link j acting back onto link i.

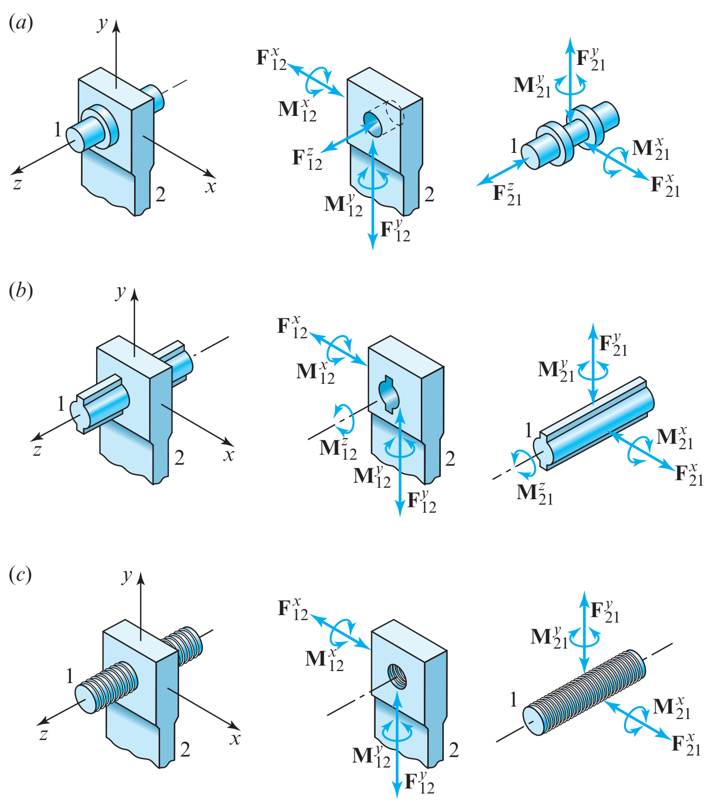


Figure 11.3 (*a*) Revolute or turning pair; (*b*) prism or prismatic pair; (*c*) screw or helical pair; (*d*) cylinder or cylindric pair; (*e*) sphere or globular pair; (*f*) flat or planar pair. Continues on page 576.

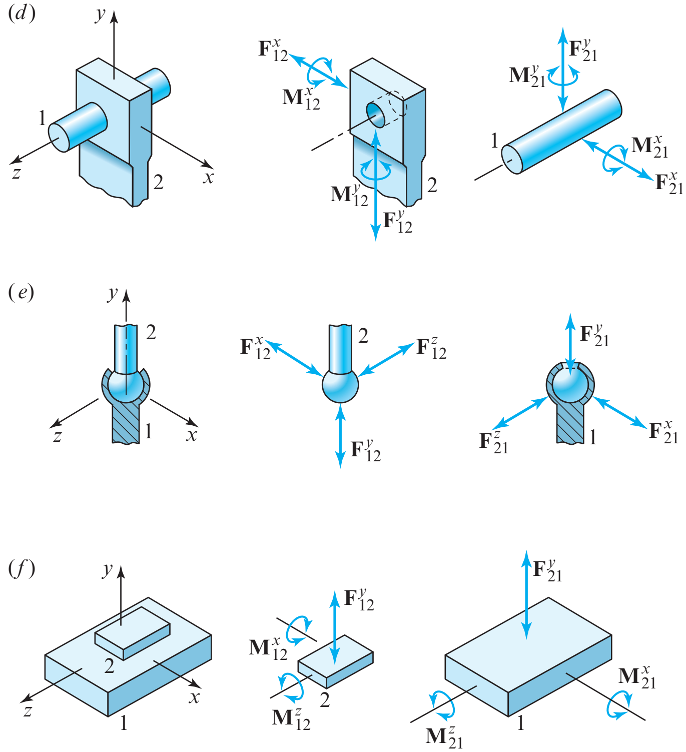


Figure 11.3 (*Continued*).

Note that, when the sense of a force or moment is unknown, the diagram showing this force or moment is drawn with arrowheads at both ends (as in Fig. 11.3). When the orientation is also unknown, the diagram either shows a broken (zig-zag) line for the shaft of the arrow (as in Fig. 11.4*a*), or it shows multiple components for the force or moment (as in Fig. 11.1). These symbolic conventions are used throughout the remainder of this text.

11.6 CONDITIONS FOR EQUILIBRIUM

A body or group of bodies is said to be in equilibrium if all the forces exerted on the system are in balance. In such a situation, Newton's laws, as expressed in Eq. (11.1), indicate that no acceleration results. This may imply that no motion takes place, meaning that all velocities are zero. If so, the system is said to be in *static equilibrium* [2]. On the other hand, no acceleration may imply that velocities do exist but remain constant; the system is then said to be in *dynamic equilibrium*. In either case, Eq. (11.1) demonstrates that a system of bodies is in equilibrium if and only if:

1. *The vector sum of all forces acting upon it is zero.*
2. *The vector sum of the moments of all forces acting about an arbitrary axis is zero.*

Mathematically, these two statements are expressed as

$$\sum \mathbf{F} = \mathbf{0} \quad \text{and} \quad \sum \mathbf{M} = \mathbf{0}. \tag{11.8}$$

Note how these statements are a result of Newton's first and third laws, it being understood that a body is composed of a collection of particles.

Many problems have forces acting in a single plane. When this is true, it is convenient to choose this as the xy plane. Then, Eqs. (11.8) can be simplified by taking the components in that plane as follows:

$$\sum F^x = 0, \quad \sum F^y = 0, \quad \text{and} \quad \sum M = 0,$$

where the z direction for the moment M component is implied by the fact that all forces act only in the xy plane.

11.7 TWO- AND THREE-FORCE MEMBERS

A *member* is the name given to a rigid body subjected only to forces, that is, with no applied moment(s). The free-body diagram of Fig. 11.4*a* shows a two-force member, that is an arbitrarily shaped body acted upon by two forces, $\mathbf{F}_A$ and $\mathbf{F}_B$, acting at points A and B, respectively. The forces are shown with zig-zag lines as a signal that the magnitudes and orientations of these forces are not yet known, and with arrowheads at both ends to remind us that the sense of each is also not yet known.

Assuming that the free-body diagram is complete, that is, no other forces are active on the body, then the first of Eqs. (11.8) gives

$$\sum \mathbf{F} = \mathbf{F}_A + \mathbf{F}_B = \mathbf{0}.$$

This requires that $\mathbf{F}_A$ and $\mathbf{F}_B$ have *equal magnitudes*, *parallel lines of action*, and *opposite senses*. Using the second of Eqs. (11.8) about either point A or point B demonstrates that $\mathbf{F}_A$ and $\mathbf{F}_B$ must also share the *same line of action*; otherwise, their moments cannot sum to zero. Thus, for any two-force member, we have learned that the forces must be oppositely directed along the unique line of action defined by the points A and B, as shown in Fig. 11.4*b*. When the magnitude of one force becomes known, the magnitude of the second force must be equal to the first force but of the opposite sense (simply stated, the two forces must be equal, opposite, and collinear).

Note that in order for a rigid body with two active forces and an applied moment to be in static equilibrium, the two forces cannot be coaxial; that is, they must be equal, opposite, and parallel. Also, because of the applied moment, the body is not referred to as a member.

Next, consider a three-force member, that is, a body with three active forces and no applied moments, as shown in Fig. 11.4*c*. Here, we also assume that the three forces, $\mathbf{F}_A$, $\mathbf{F}_B$, and $\mathbf{F}_C$, are coplanar, that is, they are all known to act in the plane defined by the three points A, B, and C. Suppose further that we also know two of the lines of action,

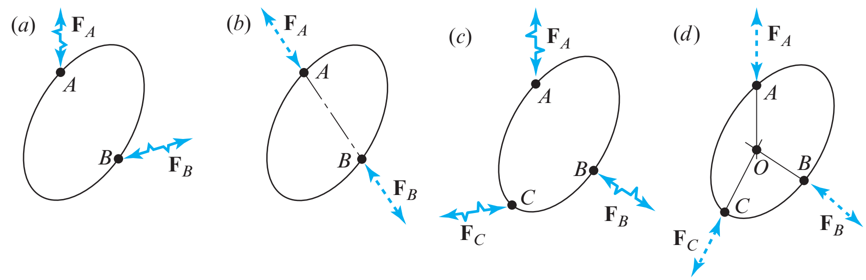


Figure 11.4 (*a*) Two-force member is not in equilibrium; (*b*) two-force member is in equilibrium; (*c*) three-force member is not in equilibrium; (*d*) three-force member is in equilibrium.

say, those of $\mathbf{F}_A$ and $\mathbf{F}_B$, and that they intersect at some point, O. Since neither $\mathbf{F}_A$ nor $\mathbf{F}_B$ exerts any moment about point O, applying $\sum \mathbf{M} = \mathbf{0}$ demonstrates that the moments of each of the three forces (including $\mathbf{F}_C$) about point O must be zero. Thus, the lines of action of all three forces must intersect at the common point O; that is, the three forces acting on the body must be *concurrent*, as shown in Fig. 11.4*d*. This explains why, in two dimensions, a three-force member can be solved for only two force magnitudes, even though there are three scalar equations implied in Eqs. (11.8); the moment equation has already been satisfied once the lines of action become known. If one of the three force magnitudes is known, then the other two magnitudes can be found by using $\sum \mathbf{F} = \mathbf{0}$. This is demonstrated in the following example.

EXAMPLE 11.1

The four-bar linkage shown in Fig. 11.5*a* has an external load $\mathbf{P} = 120\ \text{lb}\angle 220^\circ$ acting on link 4 at point Q. For the posture shown, find the constraint forces and the crank torque $\mathbf{T}_{12}$ to hold the linkage in static equilibrium.

ASSUMPTIONS

1. Gravitational forces are small in comparison with applied force $\mathbf{P}$ and can be ignored.
2. Frictional forces can be neglected. (Frictional forces are addressed in Sec. 11.9.)

GRAPHIC SOLUTION

1. Draw the linkage and the given force or forces to scale, as shown in Fig. 11.5*a*.
2. Draw the free-body diagrams of links 2, 3, and 4, as shown in Figs. 11.5*b*, 11.5*c*, and 11.5*d*. At this stage, unknown information, such as applied forces and torques on each link, can be sketched lightly on the free-body diagrams but will be redrawn to scale once they become known.
3. From the free-body diagram, observe that link 3 is a two-force member. Thus, according to the above discussion, the lines of action of $\mathbf{F}_{23}$ and $\mathbf{F}_{43}$ are shown

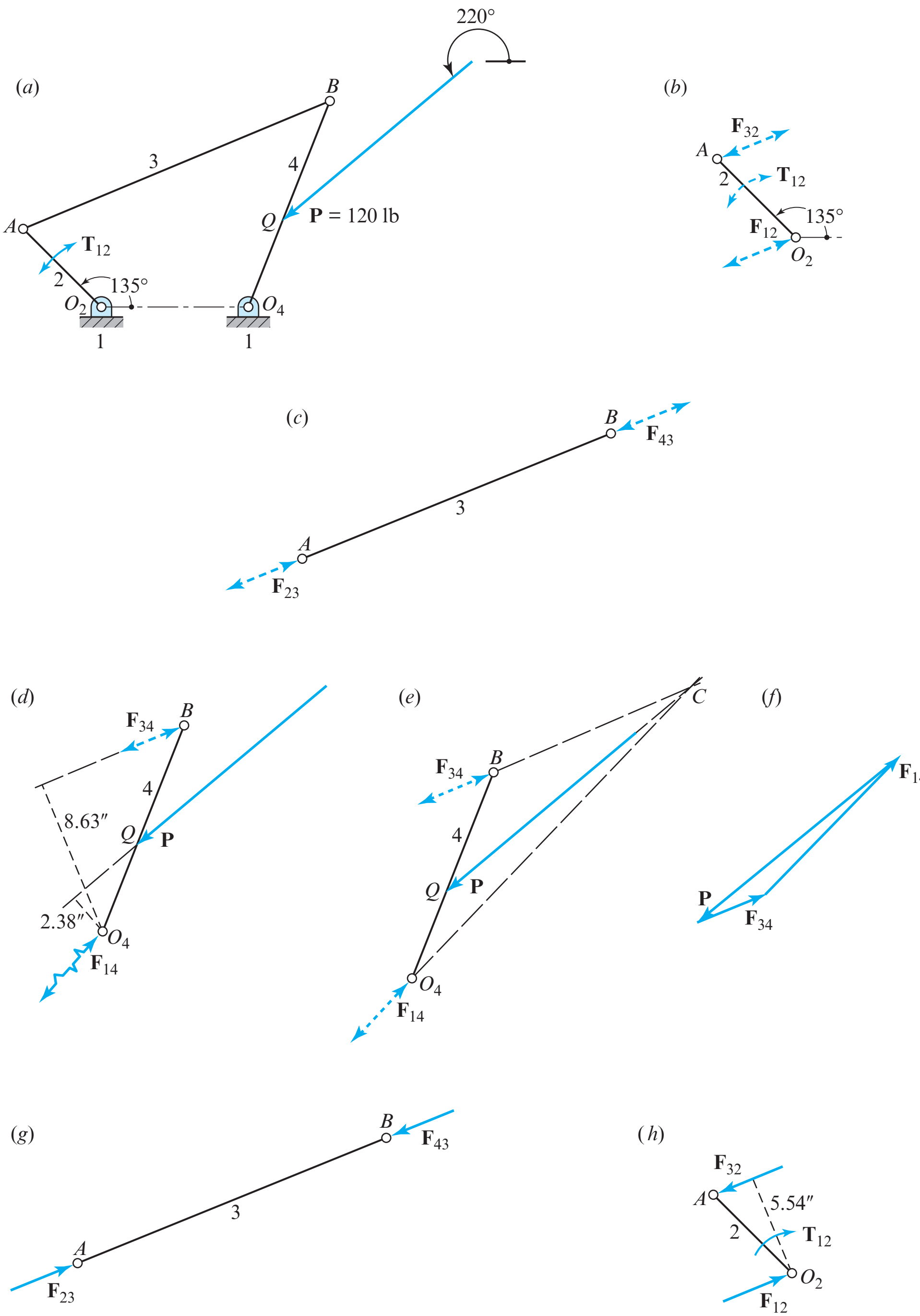


Figure 11.5 $R_{AO_2} = 6$ in, $R_{BA} = 18$ in, $R_{O_4O_2} = 8$ in, $R_{BO_4} = 12$ in, and $R_{QO_4} = 5$ in.

along the axis defined by points A and B in Fig. 11.5c. Since action and reaction forces must be equal and opposite, these also define the lines of action of $\mathbf{F}_{32}$ on link 2 in Fig. 11.5b and $\mathbf{F}_{34}$ on link 4 in Figs. 11.5d and 11.5e.

4a. Measure the moment arms of forces $\mathbf{P}$ and $\mathbf{F}_{34}$ about O_4, which are found to be 2.38 in and 8.63 in, respectively. Then, assuming that F_{34} is positive to the right, and taking counterclockwise moments as positive, Eq. (11.8) gives

$$\sum M_{O_4} = (2.38 \text{ in})(120 \text{ lb}) - (8.63 \text{ in})F_{34} = 0.$$

The solution is $F_{34} = 33.09$ lb, where the fact that the result is positive confirms the sense assumed for $\mathbf{F}_{34}$. If F_{34} were assumed positive to the left, then the solution would give $F_{34} = -33.09$ lb.

4b. As an alternative approach to Step 4a, note that link 4 is a three-force member; therefore, when the lines of action of $\mathbf{P}$ and $\mathbf{F}_{34}$ are extended, they intersect at the concurrency point C and define the line of action of $\mathbf{F}_{14}$, as shown in Fig. 11.5e.

5. No matter whether Step 4a or Step 4b is used, the force polygon for link 4 shown in Fig. 11.5f can be used to graphically solve the equation

$$\sum \mathbf{F} = \mathbf{P} + \mathbf{F}_{34} + \mathbf{F}_{14} = \mathbf{0}.$$

Note that Step 4b avoids the need for assuming a positive sense for F_{34} or for use of the moment equation of Step 4a; these are satisfied by use of the concurrency point C. The force polygon gives $F_{34} = 33.09$ lb and $F_{14} = 89.0$ lb with orientations and senses as indicated.

6. From the free-body diagram of link 3, Fig. 11.5g, and the nature of action and reaction forces, note that $\mathbf{F}_{23} = -\mathbf{F}_{43} = \mathbf{F}_{34}$. For a two-force member, it is important to note whether the member is in tension or in compression. In the case of a deformable body, a member in tension, if overloaded, would fail due to yielding; however, if the member is in compression, it could fail either by yielding or by buckling. Hence, buckling becomes an important topic for deformable bodies and is addressed in Secs. 11.14 through 11.18. In this example, link 3 is in compression.

7. Show the known force $\mathbf{F}_{32} = -\mathbf{F}_{23}$ on the free-body diagram of link 2 in Fig. 11.5h. Since link 2 is subjected to two forces and a torque, $\mathbf{F}_{12}$ must be equal, opposite, and parallel to, (but not collinear with) $\mathbf{F}_{32}$. Alternatively, from the summation of forces on the free-body diagram of link 2 in Fig. 11.5h, $\mathbf{F}_{12} = -\mathbf{F}_{32}$. Measure the moment arm of $\mathbf{F}_{32}$ about point O_2; it is found to be 5.54 in. Taking counterclockwise moments as positive gives

$$\sum M_{O_2} = (5.54 \text{ in})F_{32} - T_{12} = 0.$$

Therefore, the crank torque to maintain static equilibrium is

$$T_{12} = (5.54 \text{ in})(33.09 \text{ lb}) = 183.3 \text{ in} \cdot \text{lb}, \qquad \textit{Ans.}$$

where the positive sign confirms the clockwise sense shown in Fig. 11.5h.

ANALYTIC SOLUTION

1. First, use Eqs. (2.25) through (2.30) to make a position analysis of the linkage at the posture of interest to determine the angular orientation of each link. The angles are shown in Fig. 11.6*a*. In addition, note that link 3 is a two-force member, assume that F_{34} is positive to the right, and find

$$\mathbf{R}_{AO_2} = 6.0 \text{ in}\angle 135^\circ = -4.242\,64\hat{\mathbf{i}} + 4.242\,64\hat{\mathbf{j}} \text{ in},$$

$$\mathbf{R}_{BO_4} = 12.0 \text{ in}\angle 68.65^\circ = 4.368\,18\hat{\mathbf{i}} + 11.176\,72\hat{\mathbf{j}} \text{ in},$$

$$\mathbf{R}_{QO_4} = 5.0 \text{ in}\angle 68.65^\circ = 1.820\,08\hat{\mathbf{i}} + 4.656\,96\hat{\mathbf{j}} \text{ in},$$

$$\mathbf{F}_{34} = F_{34}\angle 22.66^\circ = 0.922\,81F_{34}\hat{\mathbf{i}} + 0.385\,26F_{34}\hat{\mathbf{j}},$$

$$\mathbf{P} = 120 \text{ lb}\angle 220^\circ = -91.925\,33\hat{\mathbf{i}} - 77.134\,51\hat{\mathbf{j}} \text{ lb}.$$

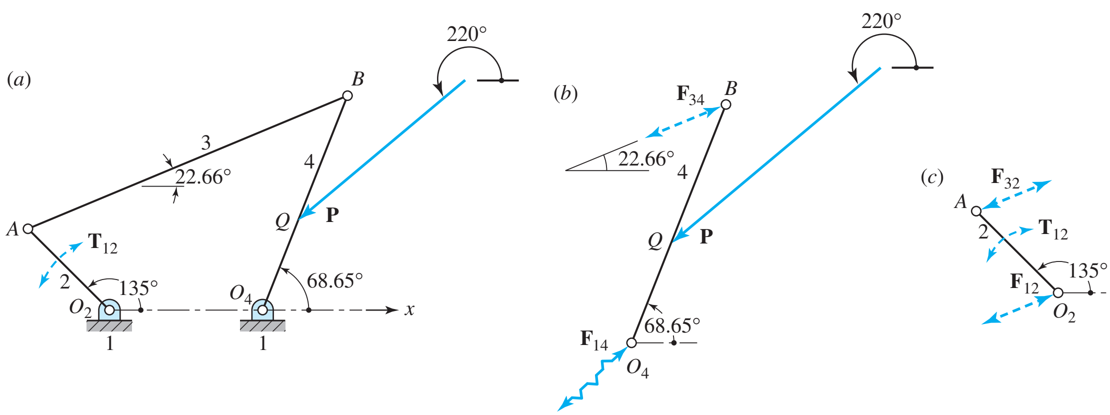


Figure 11.6 Analytic solution.

2. Referring to Fig. 11.6*b*, sum moments about point O_4. Thus,

$$\sum \mathbf{M}_{O_4} = \mathbf{R}_{QO_4} \times \mathbf{P} + \mathbf{R}_{BO_4} \times \mathbf{F}_{34} = \mathbf{0}.$$

Perform the cross-product operations, showing the first term to be $\mathbf{R}_{QO_4} \times \mathbf{P} = 287.701\,61\hat{\mathbf{k}}$ in · lb, whereas the second term is $\mathbf{R}_{BO_4} \times \mathbf{F}_{34} = -8.631\,10F_{34}\hat{\mathbf{k}}$ in · lb. Substitute these values into the above equation and solve giving

$$\mathbf{F}_{34} = 33.333\,13 \text{ lb}\angle 22.66^\circ = 30.760\,05\hat{\mathbf{i}} + 12.841\,98\hat{\mathbf{j}} \text{ lb}. \qquad \textit{Ans.}$$

3. Next, determine the force $\mathbf{F}_{14}$ from the equation

$$\sum \mathbf{F} = \mathbf{P} + \mathbf{F}_{34} + \mathbf{F}_{14} = \mathbf{0}.$$

Solve, giving

$$\mathbf{F}_{14} = 61.165\ 28\hat{\mathbf{i}} + 64.292\ 53\hat{\mathbf{j}}\ \text{lb} = 88.739\ 62\ \text{lb}\angle 46.43^\circ. \qquad \textit{Ans.}$$

4. Next, from the free-body diagram of link 2 (Fig. 11.6*c*), write

$$\sum \mathbf{M}_{O_2} = \mathbf{T}_{12} + \mathbf{R}_{AO_2} \times \mathbf{F}_{32} = \mathbf{0}.$$

Using $\mathbf{F}_{32} = -\mathbf{F}_{34} = -30.760\ 05\hat{\mathbf{i}} - 12.841\ 98\hat{\mathbf{j}}$ lb, find

$$\mathbf{T}_{12} = -184.987\ 72\hat{\mathbf{k}}\ \text{in} \cdot \text{lb}. \qquad \textit{Ans.}$$

Note that the graphic solution is in good agreement with the analytic result. Also note that in both the graphic and analytic solutions, the free-body diagram of the frame, link 1, is not drawn, since it is not helpful in the solution. If drawn, force $\mathbf{F}_{21} = -\mathbf{F}_{12}$ would be shown at O_2, force $\mathbf{F}_{41} = -\mathbf{F}_{14}$ at O_4, and reaction moment $\mathbf{M}_{21} = -\mathbf{T}_{12}$ would be shown. In addition, a minimum of two more force components and a moment would be shown where link 1 is anchored to keep it stationary. Since this would introduce at least three additional unknowns and since Eqs. (11.8) would not allow the solution for more than three, drawing the free-body diagram of the frame is of no benefit in the solution process. The only case for which this is necessary is when these anchoring forces, called "shaking forces," and moments are sought as part of the solution. This is discussed in more detail in Sec. 12.7.

In the preceding example, it was assumed that all forces act in the same plane. For the connecting link 3, for example, it was assumed that the line of action of the forces and the centerline of the link are coincident. A careful designer sometimes takes extreme measures to approach these conditions as closely as possible. Note that if the pin connections are arranged as shown in Fig. 11.7*a*, such conditions are obtained theoretically. If, on the other hand, the connections are made as shown in Fig. 11.7*b*, the pin itself as well as the link

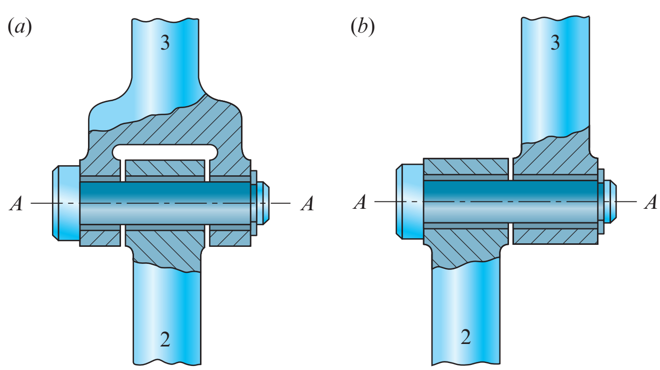


Figure 11.7 (*a*) Balanced revolute joint connection; (*b*) unbalanced revolute joint connection.

have moments acting upon them. If the forces are not in the same plane, then moments exist proportional to the distance between the force planes.

EXAMPLE 11.2

For the slider-crank linkage in the posture shown in Fig. 11.8*a*, an external load $P = 100$ lb acts horizontally on link 4 at point Q, which is 1 in above the centerline. Block 4 is 8 in wide by 3 in high with pin B centrally located. Determine the torque $\mathbf{T}_{12}$ that must be applied to link 2 to hold the linkage in static equilibrium.

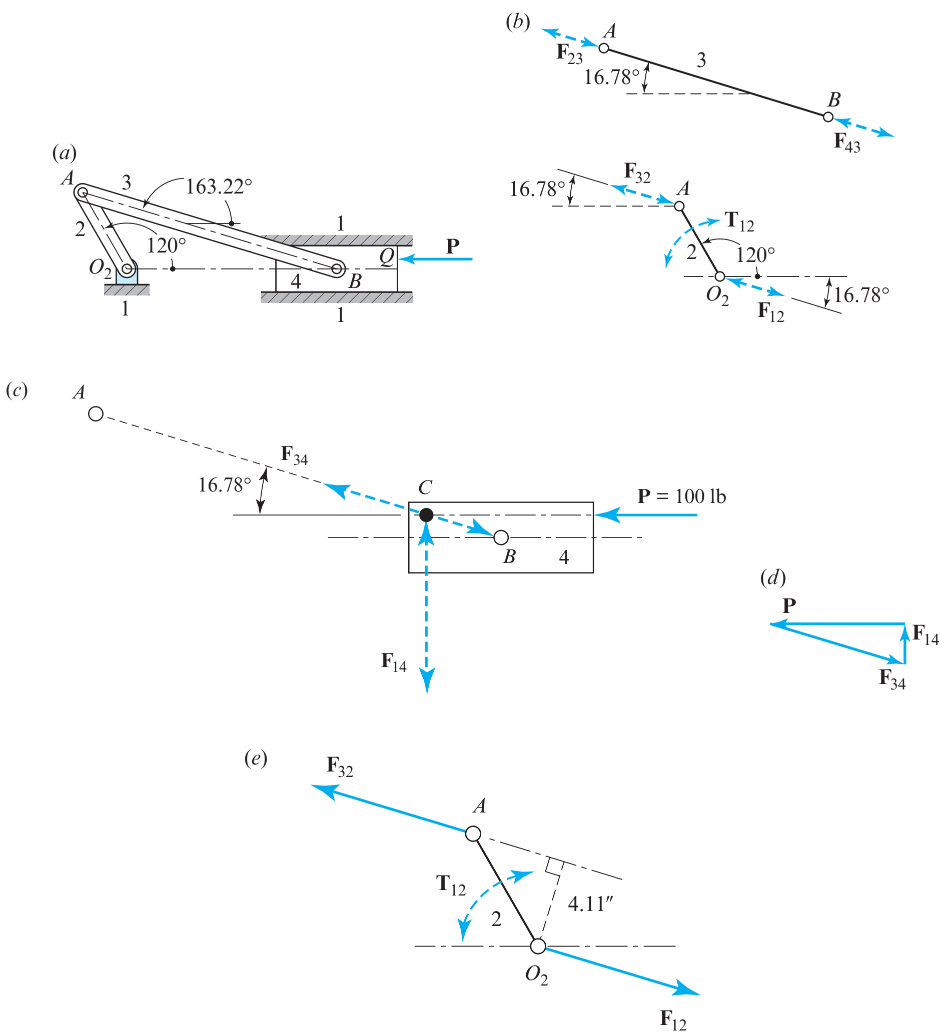


Figure 11.8 $R_{AO_2} = 6$ in and $R_{BA} = 18$ in.

ASSUMPTIONS

1. Gravitational forces are small in comparison with the applied force **P** and can be ignored.
2. Frictional forces can be neglected.

GRAPHIC SOLUTION

1. Draw the linkage and the given force to scale, as shown in Fig. 11.8*a*. (The size scale shown here is about 13 in/in and the force scale shown is about 140 lb/in. The numeric values given below, however, were obtained from larger figures with a size scale of 2 in/in and a force scale of 10 lb/in.)
2. Draw the free-body diagram of each moving link, as shown in Figs. 11.8*b* and 11.8 *c*.
3. Observe that link 3 is a two-force member. Show the lines of action of $\mathbf{F}_{23}$ and $\mathbf{F}_{43}$ along the axis defined by points A and B in Fig. 11.8*b*. Also show the lines of action of $\mathbf{F}_{34}$ on link 4 in Fig. 11.8*c* and $\mathbf{F}_{32}$ on link 2 in Fig. 11.8*e*.
4. For link 4, neither the magnitude nor the location of the frame reaction force $\mathbf{F}_{14}$ is known as yet. However, the orientation of this force must be perpendicular to the sliding surface, since friction is neglected. Also, link 4 is a three-force member. Therefore, when the lines of action of **P** and $\mathbf{F}_{34}$ are extended, they intersect at concurrency point C and define the line of action of $\mathbf{F}_{14}$, as shown in Fig. 11.8*c*.
5. Draw the force polygon for link 4, as shown in Fig. 11.8*d*, to graphically solve the equation

$$\sum \mathbf{F} = \mathbf{P} + \mathbf{F}_{34} + \mathbf{F}_{14} = \mathbf{0}.$$

   From this polygon, measure

$$F_{34} = 105 \text{ lb} \quad \text{and} \quad F_{14} = 31 \text{ lb},$$

   and also recognize the sense of each.
6. From the free-body diagram of link 3, Fig. 11.8*b*, and the nature of action and reaction forces, note that $\mathbf{F}_{23} = -\mathbf{F}_{43} = \mathbf{F}_{34}$. Also, note that link 3 is in compression; therefore, if link 3 is a deformable body, the problem of buckling should be investigated (Secs. 11.14–11.18).
7. From the free-body diagram of link 2, Fig. 11.8*e*, and the nature of action and reaction forces, note that $\mathbf{F}_{32} = -\mathbf{F}_{23}$. Also, from the summation of forces, $\mathbf{F}_{12} = -\mathbf{F}_{32}$. When the moment arm of $\mathbf{F}_{32}$ about point O_2 is measured, it is found to be 4.11 in. Taking counterclockwise moments as positive gives

$$\sum M_{O_2} = (4.11 \text{ in})F_{32} + T_{12} = 0.$$

Therefore, the torque acting on link 2 to maintain static equilibrium is

$$T_{12} = -(4.11\text{ in})(105\text{ lb}) = -432\text{ in}\cdot\text{lb (cw)}, \qquad \textit{Ans.}$$

where the negative sign indicates a clockwise sense for the torque T_{12}.

ANALYTIC SOLUTION

1. From a position analysis of the linkage at the posture of interest, find the angles indicated in Fig. 11.9*a*. Thus,

$$\mathbf{R}_{AO_2} = 6.0\text{ in}\angle 120^\circ = -3.000\,00\hat{\mathbf{i}} + 5.196\,15\hat{\mathbf{j}}\text{ in.}$$

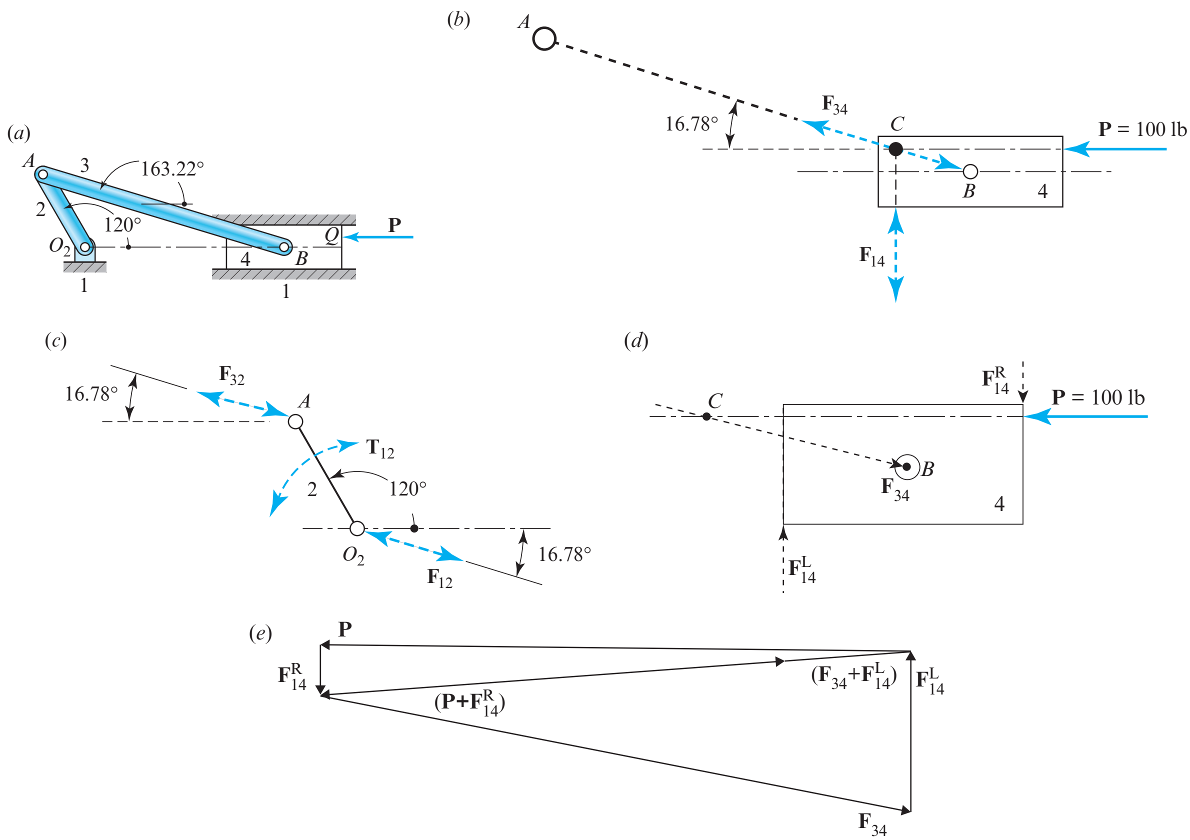


Figure 11.9 Analytic solution.

2. Write $\mathbf{F}_{14} = F_{14}\hat{\mathbf{j}}$ and

$$\mathbf{F}_{34} = F_{34}\angle -16.78^\circ = 0.957\,42F_{34}\hat{\mathbf{i}} - 0.288\,70F_{34}\hat{\mathbf{j}}.$$

Then, considering Fig. 11.9*b*, the forces $\mathbf{F}_{34}$ and $\mathbf{F}_{14}$ are determined from the equation

$$\sum \mathbf{F} = \mathbf{P} + \mathbf{F}_{34} + \mathbf{F}_{14} = \mathbf{0}.$$

This equation can be written as

$$-100.00\hat{\mathbf{i}}\text{ lb} + 0.957\,42F_{34}\hat{\mathbf{i}} - 0.288\,70F_{34}\hat{\mathbf{j}} + F_{14}\hat{\mathbf{j}} = \mathbf{0}.$$

Then, equating the $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ components, respectively, gives

$$F_{34} = \frac{100.00\text{ lb}}{0.957\,42} = 104.45\text{ lb},$$
$$F_{14} = 0.288\,70F_{34} = 30.15\text{ lb}.$$

These forces can be written as

$$\mathbf{F}_{34} = 100.00\hat{\mathbf{i}} - 30.15\hat{\mathbf{j}}\text{ lb} = 104.45\text{ lb}\angle -16.78^\circ,$$
$$\mathbf{F}_{14} = 30.15\hat{\mathbf{j}}\text{ lb}.$$

3. Next, from the free-body diagram of link 2 (Fig. 11.9*c*), write

$$\sum \mathbf{M}_{O_2} = \mathbf{T}_{12} + \mathbf{R}_{AO_2} \times \mathbf{F}_{32} = \mathbf{0}.$$

Using $\mathbf{F}_{32} = -\mathbf{F}_{34} = -100.00\hat{\mathbf{i}} + 30.15\hat{\mathbf{j}}$ lb and solving, find

$$\mathbf{T}_{12} = -429.17\hat{\mathbf{k}}\text{ in}\cdot\text{lb (cw)}. \qquad \textit{Ans.}$$

Note that the graphic solution is in good agreement with the analytic result.

The graphic and the analytic solutions both encounter difficulties in any of the following three situations: (*a*) if block 4 were of lesser width; or (*b*) if force P were applied at a point of application even higher on link 4; or (*c*) if the connecting rod were at a shallower angle, as it would be for input crank angles between 135° and 225°. If any of these situations occur, then concurrency point C lies to the left of link 4, and the point of application of force F_{14} cannot be as shown in Fig. 11.9*b*. In such a case, block 4 would tip counterclockwise unless prevented by its top-right corner being constrained by the upper sliding surface; the single force, F_{14}, would then be replaced by a couple (F_{14}^{L}, and F_{14}^{R}) at the lower-left and top-right corners of block 4, as shown in Fig. 11.9*d*

Block 4 would then become a four-force member, that is, P, F_{34}, F_{14}^{L}, and F_{14}^{R}. The method of analysis will be studied in the following section, and this example will be concluded, as shown in the force polygon of Fig. 11.9*e*.

11.8 FOUR- AND MORE-FORCE MEMBERS

When all free-body diagrams of a system have been constructed, at least in planar problems, we usually find that one or more represent either two- or three-force members, and the techniques of the last section can be used for their solution. We start by seeking out such members and using them to establish known lines of action for other unknown forces. In doing this, we are implicitly enforcing the summation-of-moments equation for such members. Then, by noting that action and reaction forces share the same line of action, we proceed to establish more lines of action. Finally, upon reaching a body where one or more forces and all lines of action are known, we apply the summation-of-forces equations to find the magnitudes and senses of the unknown forces. Proceeding from body to body in this way, we find solutions for the unknown forces.

It is always good practice to replace multiple known forces on the same free-body diagram by a single force representing the sum of known forces. This sometimes simplifies the diagrams to reveal two- and three-force members that are otherwise not noticed.

Sometimes it is helpful to combine two links, or even three links, and to draw a free-body diagram of the combination. This approach can sometimes be used to eliminate unknown forces on the individual links, which combine with their reactions as pairs of constraint forces within the combined assembly.

In some problems, we find that the solution cannot proceed in this fashion, for example, when two- or three-force members cannot be found, and lines of action remain unknown. It is then wise to count the number of unknown quantities (magnitudes and directions) to be determined for each free-body diagram. In planar problems, it is clear that Eqs. (11.8), the equilibrium conditions, cannot be solved for more than three unknowns on any single free-body diagram.

In planar problems, the most general case of a system of forces that is solvable is one in which the three unknowns are the magnitudes of three forces. If all other forces are combined into a single force, we then have a four-force member. In order for such a member to be in static equilibrium, the resultant of any two of these forces must be equal, opposite, and collinear with the resultant of the other two forces.

The following example demonstrates the solution of a four-force member.

EXAMPLE 11.3

A cam with a reciprocating roller follower is shown in Fig. 11.10*a*. The follower is held in contact with the cam by a spring pushing downward at *C* with a spring force of $F_C = 12$ N at this particular posture. Also, an external load $F_E = 35$ N acts on the follower at *E* in the direction indicated. For the posture shown, determine the follower pin force at *A* and the bearing reactions at *B* and *D*.

ASSUMPTIONS

1. Gravitational forces are small in comparison with the applied force **P** and can be ignored.
2. Frictional forces can be neglected.

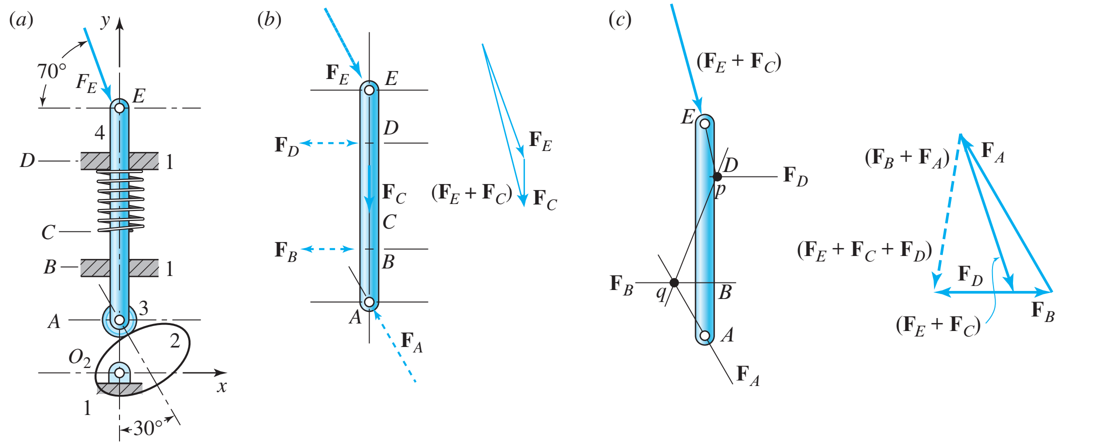


Figure 11.10 (*a*) $R_{AO_2} = 15$ mm, $R_{BO_2} = 30$ mm, $R_{CA} = 25$ mm, $R_{DB} = 30$ mm, and $R_{EA} = 60$ mm; (*b*) free-body diagram of link 4, a five-force member, and the force polygon; (*c*) free-body diagram of link 4, reduced to a four-force member, and the force polygon.

SOLUTION

The free-body diagram of the follower (link 4) is shown in Fig. 11.10*b*. Note that the follower is a five-force member ($\mathbf{F}_A$, $\mathbf{F}_B$, $\mathbf{F}_C$, $\mathbf{F}_D$, and $\mathbf{F}_E$). However, forces $\mathbf{F}_C$ and $\mathbf{F}_E$ are known, and their sum is obtained from the force polygon in Fig. 11.10*b*. The free-body diagram of link 4 is redrawn to indicate this in Fig. 11.10*c*. The resultant force $(\mathbf{F}_E + \mathbf{F}_C)$ is shown with its point of application at E, where the original lines of action of $\mathbf{F}_E$ and $\mathbf{F}_C$ intersect, and with its line of action dictated by the direction of their vector sum. This placement of the point of application at E is not arbitrary, but is required. Since the original forces, $\mathbf{F}_E$ and $\mathbf{F}_C$, each have no moment about point E, their resultant must have no moment about this point. The result of combining these two forces is that the free-body diagram of link 4 in Fig. 11.10*c* is now reduced to a four-force member with one known force, the resultant $(\mathbf{F}_E + \mathbf{F}_C)$, and three unknown force magnitudes (F_A, F_B, and F_D).

In a similar manner, if the magnitude of $\mathbf{F}_D$ were known, it could be added to $(\mathbf{F}_E + \mathbf{F}_C)$ to produce a new resultant, $(\mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_D)$, which would act through the intersection point, p.

Now, consider the moment equation about point q, the intersection of $\mathbf{F}_A$ and $\mathbf{F}_B$. If we now write $\sum \mathbf{M}_q = \mathbf{0}$, we see that this equation can only be satisfied if the resultant, $(\mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_D)$, has no moment about point q and thus has pq as its line of action. This is the basis for the solution shown in Fig. 11.10*c*. The direction of the line of action pq of the resultant, $(\mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_D)$, is used in the force polygon to determine the magnitude of force $\mathbf{F}_D$; the solution gives $F_D = 29.2$ N. The force polygon is then completed by finding $\mathbf{F}_A$ and $\mathbf{F}_B$, whose lines of action are known. The solutions are $F_A = 51.8$ N and $F_B = 43.1$ N when rounded to three significant figures.

Note that this approach defines a general concept, useful in either the graphic or the analytic approach. When there are three unknown force magnitudes on a member, we choose a point such as q where the lines of action of two of the unknown forces intersect and write the moment equation about that point, that is, $\sum \mathbf{M}_q = \mathbf{0}$. This equation will have only one unknown remaining and can be solved directly. Only then should $\sum \mathbf{F} = \mathbf{0}$ be written, since the member has then been reduced to two unknowns.

11.9 FRICTION-FORCE MODELS

Over the years, there has been much interest in the subjects of friction and wear, and many papers and books have been devoted to these subjects. It is not our purpose to explore the mechanics of friction here, but to present classical simplifications that have been used to analyze the performance of mechanical devices with friction. The results of any such analysis may not be theoretically ideal, but they do correspond closely to experimental measurements, so that reliable decisions can be made regarding a design and its operating performance.

Consider two bodies constrained to remain in contact with each other, such as the surfaces of block 3 and link 2 shown in Fig. 11.11*a*. A force, $\mathbf{F}_{43}$, may be exerted on block 3 by link 4, tending to cause block 3 to slide relative to link 2. Without the presence of friction between the interface surfaces of links 2 and 3, block 3 cannot transmit the component of force $\mathbf{F}^t_{32}$ tangent to the surfaces. Instead of transmitting this component of force onto link 2, block 3 slides in the direction of this unbalanced force component;

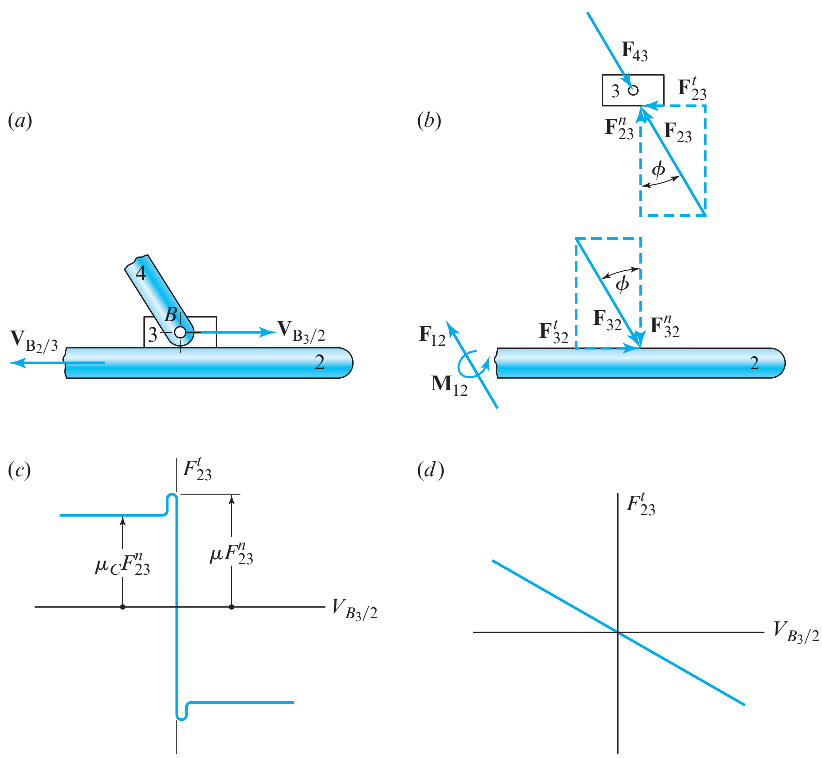


Figure 11.11 (*a*) Physical system; (*b*) free-body diagrams; (*c*) static and Coulomb friction models; (*d*) viscous friction model.

equilibrium is not possible without friction unless the line of action of $\mathbf{F}_{32}$ and its reaction $\mathbf{F}_{23}$ are normal to the surface.

With friction, however, a resisting force $\mathbf{F}_{23}^t$ is developed at the contact surface, as shown in the free-body diagrams of Fig. 11.11*b*. This friction force component $\mathbf{F}_{23}^t$ acts in addition to the usual constraint force component $\mathbf{F}_{23}^n$ across the surface of the sliding joint; together, these two force components form the total constraint force $\mathbf{F}_{23}$ that balances the constraint force $\mathbf{F}_{43}$ when block 3 is in static equilibrium. Of course, the reaction force components, $\mathbf{F}_{32}^n$ and $\mathbf{F}_{32}^t$ (or their sum), also act simultaneously onto link 2, as shown in the other free-body diagram of Fig. 11.11*b*. The force component $\mathbf{F}_{23}^t$ and its reaction $\mathbf{F}_{32}^t$ are called *friction forces*.

Depending on the materials of links 2 and 3, there is a limit to the size of force component $\mathbf{F}_{23}^t$ that can be transmitted by friction while maintaining static equilibrium. This limit is expressed by the relationship

$$F_{23}^t \leq \mu F_{23}^n, \tag{11.9}$$

where μ, referred to as the *coefficient of static friction*, is a characteristic property of the contacting materials. Values of the coefficient μ have been determined experimentally for many material combinations and are found in many engineering handbooks, for example [3].

If constraint force $\mathbf{F}_{43}$ is tipped too much, so that its tangential component and therefore the friction force component F_{23}^t becomes too large to satisfy the inequality of Eq. (11.9), static equilibrium is not possible, and block 3 then slides relative to link 2 with an apparent velocity $\mathbf{V}_{B_3/2}$. When sliding takes place, the friction force takes on the value

$$F_{23}^t = \mu_c F_{23}^n, \tag{11.10}$$

where μ_c is the *coefficient of sliding friction.* This model for approximating sliding friction is called *Coulomb friction*, and we shall use this term to refer to the relationship of Eq. (11.10). The coefficient μ_c can also be found experimentally and for most materials is slightly less than μ, the coefficient of static friction.

To summarize what we have discussed so far, Fig. 11.11*c* shows a graph of the magnitude of the friction force, F_{23}^t, versus the apparent sliding velocity, $V_{B_3/2}$. Here it can be seen that when the apparent sliding velocity is zero, the friction force, F_{23}^t, can have any magnitude between μF_{23}^n and $-\mu F_{23}^n$. When the velocity is not zero, the magnitude of the friction force, F_{23}^t, drops slightly to the value $\mu_c F_{23}^n$ and has a sense opposing that of the apparent sliding motion, $V_{B_3/2}$.

Looking again at the total force $\mathbf{F}_{23}$ in Fig. 11.11*b*, we see that it is inclined at an angle to the surface normal and is equal and opposite to $\mathbf{F}_{43}$ whenever the system is in equilibrium. Thus, the angle ϕ is given by

$$\tan\phi = \frac{F_{23}^t}{F_{23}^n},$$

$$\tan\phi \leq \frac{\mu F_{23}^n}{F_{23}^n} = \mu,$$

$$\phi \leq \tan^{-1}\mu.$$

When $\mathbf{F}_{43}$ is tipped so that block 3 is on the verge of sliding, then

$$\phi = \tan^{-1} \mu. \tag{11.11}$$

This limiting value of the angle ϕ, called the *friction angle*, defines the maximum angle through which force $\mathbf{F}_{23}$ can tip from the surface normal before equilibrium is not possible and sliding begins (commonly referred to as *impending motion*). Note that the limiting value of the friction angle does not depend on the magnitude of force $\mathbf{F}_{23}$ but only on the coefficient of friction of the materials involved.

Note also that, in the above discussion, we have treated only the effects of friction between surfaces that permit sliding motion. We might ask how this can be extended to treat rotational motion, such as in a revolute joint. Since good low-friction rotational bearings are not expensive, and since they can be easily lubricated to reduce friction, this is usually not a problem. Extensions of the above ideas are known for rotational bearings [1]. However, since the results are usually not affected by more than a very small percentage, the additional difficulty does not often warrant their use.

Although the static friction or Coulomb friction models are often used and often do represent good approximations for friction forces in mechanical equipment, they are not the only models. Sometimes—for example, when representing a machine or its dynamic effects by its differential equation of motion—it is more convenient to analyze the machine's performance using a different approximation for friction forces, called *viscous friction* or *viscous damping*. As shown in the graph of Fig. 11.11*d*, this model assumes a linear relationship between the magnitude of the friction force and the sliding velocity. This viscous friction model is especially useful when the dynamic analysis of a machine leads to the use of one or more differential equations—for example, in vibration analysis (Secs. 13.8–13.10). The nonlinear relationship of static and/or Coulomb friction, shown in Fig. 11.11*c*, leads to a nonlinear differential equation that is more difficult to solve.

Whether the friction effect is represented by the static, Coulomb, or viscous friction models, it is important to recognize the sense of the friction force. As a mnemonic device, the rule is often stated that "friction opposes motion," as shown by the free-body diagram of link 3 in Fig. 11.11*b*, where the sense of $\mathbf{F}^t_{23}$ is opposite to that of $\mathbf{V}_{B_3/2}$. This rule of thumb is not wrong if it is applied carefully, but it can be dangerous and even misleading. It will be noted in Fig. 11.11*a* that there are two motions that might be thought of, namely, $\mathbf{V}_{B_3/2}$ and $\mathbf{V}_{B_2/3}$; there are also two friction forces, $\mathbf{F}^t_{23}$ and $\mathbf{F}^t_{32}$. Careful examination of Fig. 11.11*b* indicates that $\mathbf{F}^t_{23}$ opposes the sense of $\mathbf{V}_{B_3/2}$, whereas $\mathbf{F}^t_{32}$ opposes the sense of $\mathbf{V}_{B_2/3}$. In machine systems, particularly where both sides of a sliding joint are in motion, it is very important to understand *which* friction force "opposes" *which* motion.

Of course, in static force analysis, the assumption throughout is that forces are found under conditions of static equilibrium; therefore, there should be no motion. When we say that "friction opposes motion," we are speaking of *impending motion*; that is, we are assuming that the system is on the verge of moving and we are speaking of the movement that would begin if a small change in force were to perturb the equilibrium.

11.10 FORCE ANALYSIS WITH FRICTION

When we begin a force analysis with friction, it is recommended to solve the problem first without friction. The purpose is to find the sense of each of the normal force components We will demonstrate this procedure in the following two examples.

EXAMPLE 11.4

Repeat Example 11.2 (Fig. 11.8*a*) with a coefficient of static friction between the slider and the ground of $\mu = 0.25$.

ASSUMPTIONS

1. The impending motion of the sliding block 4 is to the left.
2. Friction can be neglected in the revolute joints.

SOLUTION

The normal force component $\mathbf{F}^n_{14}$ without friction was obtained in Example 11.2 and was determined (Fig. 11.8*d*) to act vertically upward.

Since the impending motion of the slider is to the left, that is, $\mathbf{V}_{B4/1}$ is to the left, then the friction force, $\mathbf{F}^t_{14}$, must act to the right. We can redraw the free-body diagram of link 4 (Fig. 11.8*c*) and include the friction force as shown in Fig. 11.12*a*. Because of static friction, the line of action of $\mathbf{F}_{14}$ is shown tipped through the friction angle ϕ, which is calculated from Eq. (11.11):

$$\phi = \tan^{-1} 0.25 = 14°.$$

In deciding the direction of tip of the angle ϕ, it is necessary to know both the sense of the normal force (upward) and the sense of the friction force (horizontal to the right) at the point of contact. This explains why the recommended procedure is to first solve the problem without friction.

Once the new line of action of the force $\mathbf{F}_{14}$ is known, the solution proceeds exactly as in Example 11.2. The graphic solution, with friction effects, is shown in Fig. 11.12*b*, and it is found that

$$F_{34} = 97 \text{ lb} \quad \text{and} \quad F_{14} = 29 \text{ lb}. \qquad \textit{Ans.}$$

Note that the line of action of $\mathbf{F}_{34}$ has not changed, but that the magnitude is less than the frictionless case. Therefore, less torque is required on link 2 to maintain the linkage in static equilibrium. We find that

$$T_{12} = (4.11 \text{ in})(97 \text{ lb}) = 399 \text{ in} \cdot \text{lb cw}. \qquad \textit{Ans.}$$

Note that, if the impending motion of sliding block 4 were to the right, then a greater torque would be required on link 2 to maintain the linkage in static equilibrium.

Note also that we have $F^n_{14} = (29 \text{ lb}) \cos 14° = 28.14$ lb, whereas in Example 11.2, the force before friction was included was $F_{14} = 30.15$ lb. This indicates why it would

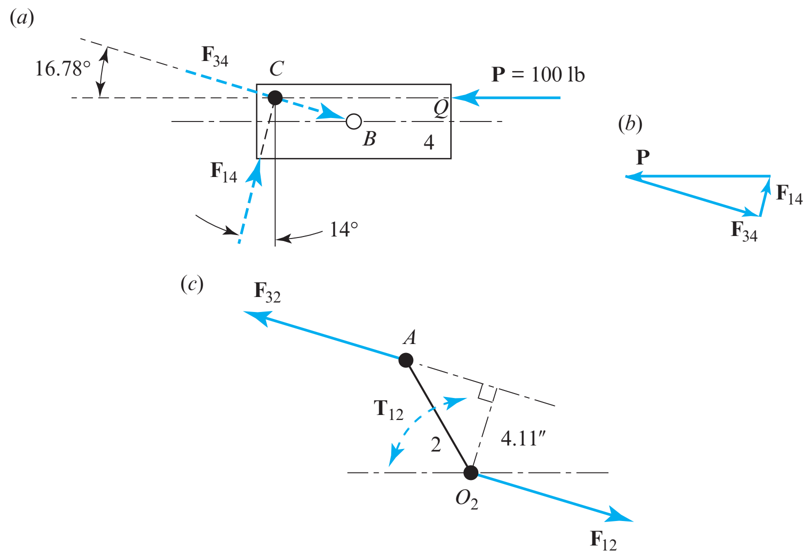


Figure 11.12 Graphic solution.

be totally wrong to multiply the value of F_{14} without friction by μ to obtain F_{14}^t and then to include it by superposition; friction changes both the normal and the tangential components of all forces.

Finally, note that the point of application of force F_{14} is located 3.76 in to the left of point B. Fortunately, this is still within the width of block 4; therefore the solution is valid. However, if the width of block 4 were not 8 in, but only 6 in, then the point of application of force $\mathbf{F}_{14}$ would fall outside of the width of block 4. In this case, block 4 would not be in equilibrium, but would tip counterclockwise until prevented by contact with the ground at both the top-right and bottom-left corners. Block 4 would then become a four-force member with two unknown forces, $\mathbf{F}_{14}^{\mathrm{R}}$ at the top-right corner and $\mathbf{F}_{14}^{\mathrm{L}}$ at the bottom-left corner, each tipped by the friction angle, ϕ, but tipped in opposite directions, since one normal component is downward and one is upward, and a new force polygon would be required for the solution for block 4 under these conditions.

EXAMPLE 11.5

Repeat Example 11.3 (Fig. 11.10*a*) with a coefficient of static friction between links 1 and 4 at both sliding bearings B and D of $\mu = 0.15$. Determine the minimum force necessary at follower pin A to hold the system in static equilibrium.

ASSUMPTIONS

Friction in all other joints is negligible.

SOLUTION

First, we solve the problem without friction to find the sense of each of the normal force components, $\mathbf{F}_B^n$ and $\mathbf{F}_D^n$. This was performed in Example 11.3, where $\mathbf{F}_B$ and $\mathbf{F}_D$ were both found in Fig. 11.10*c*.

The next step is to determine the direction of the impending motion of the follower, link 4. The problem asks for the *minimum* force at *A* necessary for static equilibrium; this implies that if $\mathbf{F}_A$ were any smaller, the system would move downward. Thus, the impending motion of link 4 is downward for both velocities, $\mathbf{V}_{D_4/1}$ and $\mathbf{V}_{B_4/1}$. Therefore, the two friction forces from link 1 onto link 4 at *B* and *D* must both act upward.

Next, we redraw the free-body diagram of link 4 (Fig. 11.10*c*) to include the friction forces as shown in Fig. 11.13*a*. Here, the points of application of forces $\mathbf{F}_B$ and $\mathbf{F}_D$ are on the left and right side of link 4, respectively, at the points of contact with the bearings, not at the centerline of link 4. However, because of static friction, the lines of action of $\mathbf{F}_B$ and $\mathbf{F}_D$ are each shown tipped through the angle ϕ, which is calculated from Eq. (11.11):

$$\phi = \tan^{-1} 0.15 = 8.5°.$$

In deciding the direction of tip of the angle ϕ, it is necessary to know the sense of each friction force (upward) and the sense of each normal force (toward the right at *B* and toward the left at *D*). This demonstrates why the solution without friction must be done first.

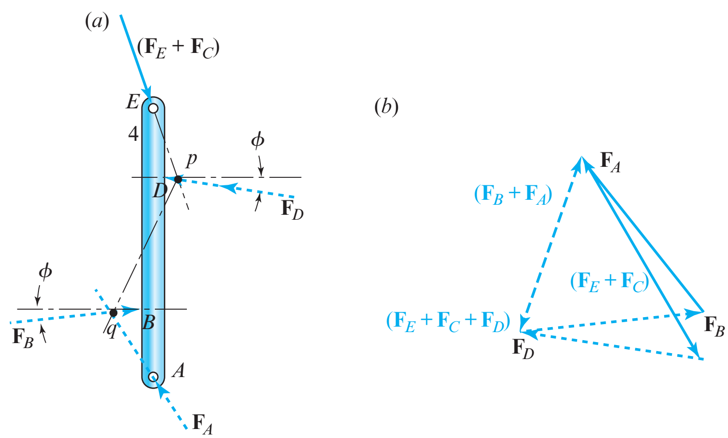


Figure 11.13 (*a*) Free-body diagram of link 4 with static friction; (*b*) force polygon.

Once the new lines of action of forces $\mathbf{F}_B$ and $\mathbf{F}_D$ are known, the solution proceeds exactly as in Example 11.3. The graphic solution with friction effects is shown in Fig. 11.13*b*, where it is found that

$$F_B = 39.1\text{ N}, F_D = 34.7\text{ N}, \quad \text{and} \quad F_A = 39.1\text{ N}. \qquad \textit{Ans.}$$

The normal components of the two forces at B and D are $F_B^n = 37.7$ N and $F_D^n = 33.6$ N. Note that these values are quite different from $F_B = 43.1$ N and $F_D = 29.2$ N, determined without friction in Example 11.3. This should be a warning, whether working graphically or analytically, that it is incorrect to simply multiply the frictionless normal forces by the coefficient of friction to find the friction forces and then to add these to the frictionless solution. All forces may (and usually do) change magnitude when friction is included, and the problem must be completely reworked from the beginning with friction included. The effects of static or Coulomb friction *cannot* be added by superposition.

We note, also, that if the problem statement had asked for the maximum force at A, the impending motion of link 4 would have been upward and the friction forces downward on this link. This would have reversed the tilt of the two lines of action for both $\mathbf{F}_B$ and $\mathbf{F}_D$ and would have totally changed the final results. In practice, if the actual value of force $\mathbf{F}_A$ is between these minimum and maximum values, equilibrium is maintained, and the values of other constraint forces are between the two extreme values found by this type of analysis. Also, if the value of $\mathbf{F}_A$ is slowly increased from the minimum to the maximum value, the follower remains in equilibrium until the maximum value is reached and then begins to move; this discontinuous action is sometimes referred to as "stiction."

11.11 SPUR- AND HELICAL-GEAR FORCE ANALYSIS

Gear problems are typically designed to operate at constant speed, and accelerations do not usually appear, except at startup. Therefore, it is appropriate to analyze their forces in this chapter under these conditions. Recall that, when there are no accelerations, this implies that velocities may exist, but that they remain constant; this is then referred to as *dynamic equilibrium* (Sec. 11.6). Figure 11.14*a* shows a pinion with center O_2 rotating clockwise at a constant speed of ω_2 and driving a gear with center O_3 at a speed of ω_3. As discussed in Chap. 7, the reactions between the teeth occur along the pressure line AB, tipped by the pressure angle ϕ from the common tangent to the pitch circles. Free-body diagrams of the pinion and the gear are shown in Fig. 11.14*b*. The action of the pinion on the gear is indicated by force $\mathbf{F}_{23}$ acting at the pitch point along the pressure line.* Since the gear is supported by its shaft, then from $\sum \mathbf{F} = \mathbf{0}$, an equal and opposite force $\mathbf{F}_{13}$ must act at the centerline of the shaft. A similar analysis of the pinion indicates that the same observations are true. In each case, the forces are equal in magnitude, opposite in direction, parallel, and in the same plane. On either gear, therefore, they form a couple.

* It is true that treating force $\mathbf{F}_{23}$ in this manner ignores the possible effects of friction forces between the meshing gear teeth. Justification for this comes in four forms: (1) The actual point of contact is continually varying during the meshing cycle but always remains near the pitch point, which is the instantaneous center of velocity; thus, the relative motion between the teeth is close to true rolling motion, with only a very small amount of slip. (2) The fact that the friction forces are continually changing in both magnitude and direction throughout the meshing cycle, and that the total force is often shared by more than one tooth in contact, makes a more exact analysis impractical. (3) The machined surfaces of the teeth and the fact that they are usually well-lubricated produces a very small coefficient of friction. (4) Experimental data indicate that gear efficiencies are usually very high, often approaching 99%, demonstrating that any errors produced by ignoring friction are quite small.

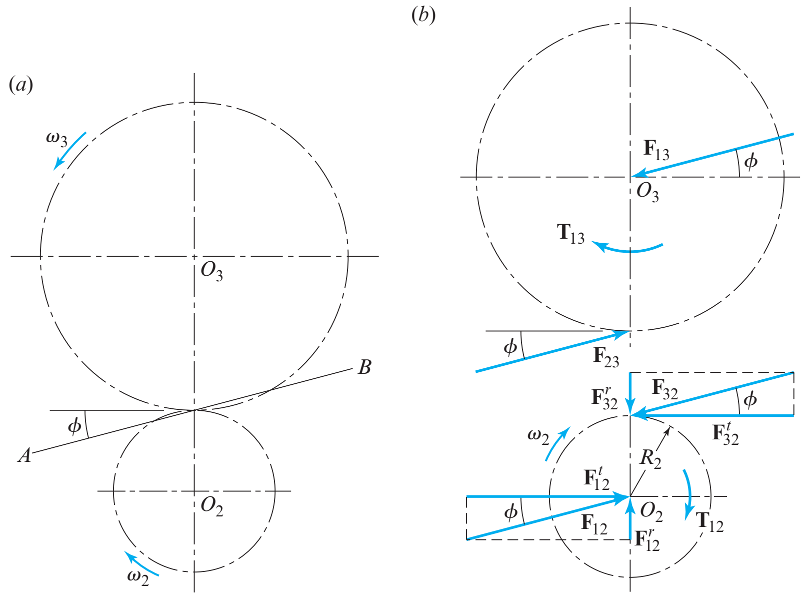


Figure 11.14 Forces on spur gears.

Note that the free-body diagram of the pinion indicates the forces resolved into components. Here, we employ superscripts r and t to indicate the radial and tangential directions with respect to the pitch circle. It is expedient to use the same superscripts for the components of the force, $\mathbf{F}_{12}$, that the shaft exerts on the pinion. The moment of the couple formed by $\mathbf{F}_{32}^t$ and $\mathbf{F}_{12}^t$ is in equilibrium with the torque, $\mathbf{T}_{12}$, that must be applied by the shaft to drive the gearset. When the pitch radius of the pinion is designated R_2, then $\sum \mathbf{M}_{O_2} = \mathbf{0}$ indicates that the magnitude of the torque is

$$T_{12} = R_2 F_{32}^t. \tag{11.12}$$

Note that the radial force component $\mathbf{F}_{32}^r$ serves no purpose as far as the transmission of power is concerned. For this reason, the tangential component $\mathbf{F}_{32}^t$ is frequently called the *transmitted* force.

In applications involving gears, the power transmitted and the shaft speeds are often specified. Remembering that power is the product of force times velocity, or torque times angular velocity, we can find the relation between power and the transmitted force. Using the symbol P to denote power, we obtain

$$P = T_{12}\omega_2 = R_2 F_{32}^t \omega_2. \tag{11.13a}$$

Therefore, the transmitted force can be written as

$$F_{32}^t = \frac{P}{R_2\omega_2}. \tag{11.13b}$$

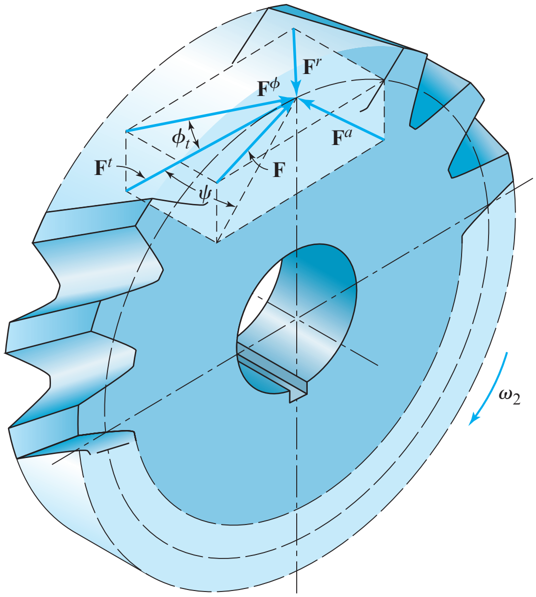


Figure 11.15 Force components on a helical gear at the tooth contact point.

In application of these formulae, it is often necessary to remember that the US customary unit used for power is horsepower, abbreviated hp, where 1 hp = 33 000 ft · lb/min, and in SI units, power is measured in watts (or kilowatts), abbreviated W (or kW), where 1 W = 1 N · m/s.

Once the transmitted force is known, the following relations for spur gears are evident from Fig. 11.14*b*:

$$F_{32}^r = F_{32}^t \tan\phi \quad \text{and} \quad F_{32} = \frac{F_{32}^t}{\cos\phi}.$$

In the treatment of forces on helical gears, it is convenient to determine the axial force, work with it independently, and treat the remaining force components the same as for spur gears. Figure 11.15 is a drawing of a helical gear with a portion of the face removed to show the tooth contact force and its components acting at the pitch point. The gear is imagined to be driven clockwise under load. The driving gear has been removed from the figure and its effect replaced by the forces shown acting on the teeth.

The resultant force, **F,** is shown divided into three components, $\mathbf{F}^a$, $\mathbf{F}^r$, and $\mathbf{F}^t$, which are the axial, radial, and tangential components, respectively. The tangential force component is the transmitted force and is the force that is effective in transmitting torque. When the transverse pressure angle is designated ϕ^t and the helix angle ψ, the following relations are evident from Fig. 11.15:

$$\mathbf{F} = \mathbf{F}^a + \mathbf{F}^r + \mathbf{F}^t, \tag{11.14}$$

$$F^a = F^t \tan\psi, \tag{11.15}$$

$$F^r = F^t \tan\phi^t. \tag{11.16}$$

It is also expedient to make use of the resultant of $\mathbf{F}^r$ and $\mathbf{F}^t$. We shall designate this force $\mathbf{F}^\phi$, defined by the equation

$$\mathbf{F}^\phi = \mathbf{F}^r + \mathbf{F}^t. \tag{11.17}$$

EXAMPLE 11.6

A gear train is composed of three helical gears with shaft centers in line, as shown in Fig. 11.16*a*. The driver is a right-hand helical gear having a pitch radius of 2 in, a transverse pressure angle of 20°, and a helix angle of 30°. An idler gear in the train has the teeth cut left handed and has a pitch radius of 3.25 in. The idler transmits no power to its shaft. The driven gear in the train has the teeth cut right handed and has a pitch radius of 2.50 in. If the transmitted force is 600 lb, find the shaft forces acting on each gear.

ASSUMPTIONS

Gravitational forces can be neglected.

SOLUTION

First, we consider only the axial forces, as previously suggested. For each mesh, the axial component of the reaction, from Eq. (11.15), is

$$F^a = F^t \tan\psi = (600 \text{ lb}) \tan 30° = 346 \text{ lb}.$$

Figure 11.16*a* is a top view of the three gears, looking down on the plane formed by the three axes of rotation. For each gear, rotation is considered to be about the z axis for this problem. In Fig. 11.16*b*, free-body diagrams of each of the three gears are drawn in projection, and the three coordinate axes are shown. As indicated, the idler exerts an axial force $\mathbf{F}^a_{32}$ on the driver. This is resisted by the axial shaft force $\mathbf{F}^a_{12}$. Forces $\mathbf{F}^a_{12}$ and $\mathbf{F}^a_{32}$ form a couple that is resisted by the moment $\mathbf{M}^y_{12}$. Note that this moment is negative (clockwise) about the y axis, and the magnitude of this moment is

$$M^y_{12} = -R_2 F^a_{12} = -(2.00 \text{ in})(346 \text{ lb}) = -692 \text{ in} \cdot \text{lb}.$$

Turning our attention next to the idler, we see from Figs. 11.16*a* and 11.16*b* that the net axial force on the shaft of the idler is zero. The axial component of the force from the driver onto the idler is $\mathbf{F}^a_{23}$ and that from the driven gear onto the idler is $\mathbf{F}^a_{43}$. These two force components are equal and opposite and form a couple that is resisted by the moment on the shaft of the idler of magnitude

$$M^y_{13} = -2R_3 F^a_{23} = -2\,(3.25 \text{ in})\,(346 \text{ lb}) = -2\,249 \text{ in} \cdot \text{lb}.$$

The driven gear has the axial force component $\mathbf{F}^a_{34}$ acting along its pitch line caused by the helix angle of the idler. This is resisted by the axial shaft reaction, $\mathbf{F}^a_{14}$. As indicated, these forces form a couple that is resisted by the moment $\mathbf{M}^y_{14}$. The magnitude of this moment is

$$M^y_{14} = -R_4 F^a_{34} = -(2.50 \text{ in})(346 \text{ lb}) = -865 \text{ in} \cdot \text{lb}.$$

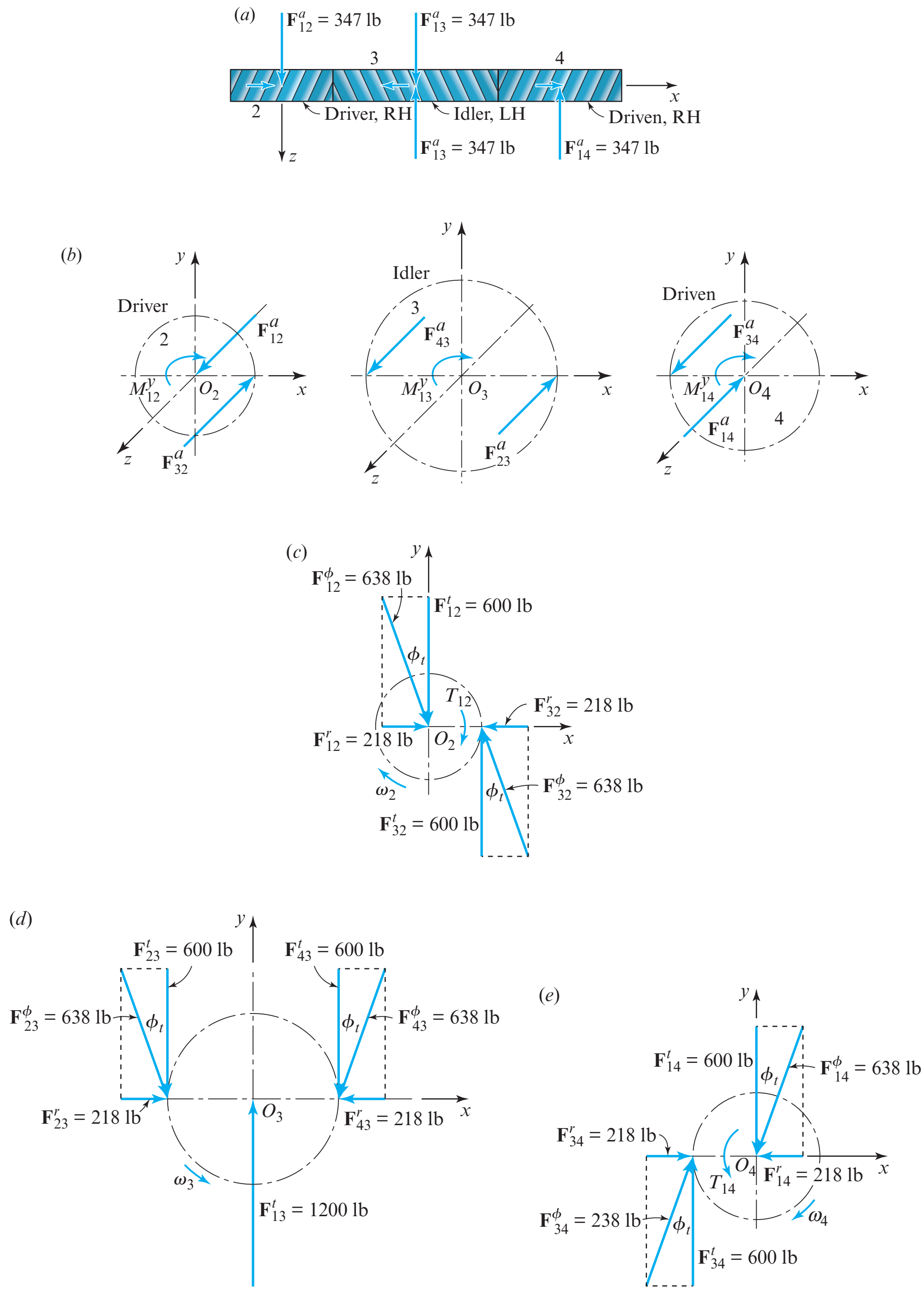


Figure 11.16 (*a*, *b*) Axial forces; (*c*) free-body diagram of driver gear 2; (*d*) free-body diagram of idler gear 3; (*e*) free-body diagram of driven gear 4.

It is emphasized that the three resisting moments, $\mathbf{M}_{12}^y$, $\mathbf{M}_{13}^y$, and $\mathbf{M}_{14}^y$, are caused solely by the axial components of the reaction forces between the gear teeth. These produce static bearing reactions and have no effect on the amount of power transmitted.

Now that all of the reactions due to the axial components have been determined, we turn our attention to the remaining force components and examine their effects as if they were operating independently of the axial forces. Free-body diagrams showing the force components in the plane of rotation for the driver, idler, and driven gears are shown in Figs. 11.16*c*, 11.16*d*, and 11.16*e*, respectively. These force components can be obtained graphically as shown or by applying Eqs. (11.12) through (11.16). Equation (11.12) gives the input torque,

$$T_{12} = R_2 F_{32}^t = (2.00\ \text{in})\,(600\ \text{lb}) = 1\,200\ \text{in} \cdot \text{lb} = 100\ \text{ft} \cdot \text{lb},$$

and the output torque,

$$T_{14} = R_4 F_{34}^t = (2.50\ \text{in})\,(600\ \text{lb}) = 1\,500\ \text{in} \cdot \text{lb} = 125\ \text{ft} \cdot \text{lb}.$$

It is not necessary to combine the components to determine the resultant forces, since the components are exactly those that are desired to proceed with machine design.

EXAMPLE 11.7

For the reverted planetary gear train shown in Fig. 11.17*a*, input shaft 2 is driven by a torque $\mathbf{T}_{12} = -100\hat{\mathbf{k}}$ in · lb. Shafts 2 and 3 are not connected but rotate about the same axis. Gear 6 is fixed to frame 1 (not shown). All gears have a diametral pitch of 10 teeth per in and a pressure angle of 20°. Perform a force analysis on the parts of the train and compute the magnitude and direction of the output torque delivered by shaft 3.

ASSUMPTIONS

The forces act in a single plane, and gravitational forces and centrifugal forces on the planet gears can be neglected.

SOLUTION

The pitch radii of the gears are $R_2 = (20\ \text{teeth})\,/\,(2 \cdot 10\ \text{teeth/in}) = 1.00$ in, $R_4 = 1.50$ in, $R_5 = 0.80$ in, and $R_6 = 1.70$ in. The distance between centers of meshing gear pairs is $(R_2 + R_4) = (1.00\ \text{in} + 1.50\ \text{in}) = 2.50\ \text{in} = (R_5 + R_6)$. The transmitted force is $F_{42}^t = T_{12}/R_2 = (100\ \text{in} \cdot \text{lb})\,/\,(1\ \text{in}) = 100$ lb. Therefore, $F_{42} = F_{42}^t/\cos\phi = 100/\cos 20^\circ = 106$ lb. The free-body diagram of gear 2 is shown in Fig. 11.17*b*. In vector form, the results are

$$\mathbf{F}_{12} = -\mathbf{F}_{42} = 106\ \text{lb}\angle 20^\circ.$$

Figure 11.17*b* also shows the free-body diagram of gear 4. The forces are

$$\mathbf{F}_{24} = -\mathbf{F}_{34} = 106\ \text{lb}\angle 20^\circ,$$

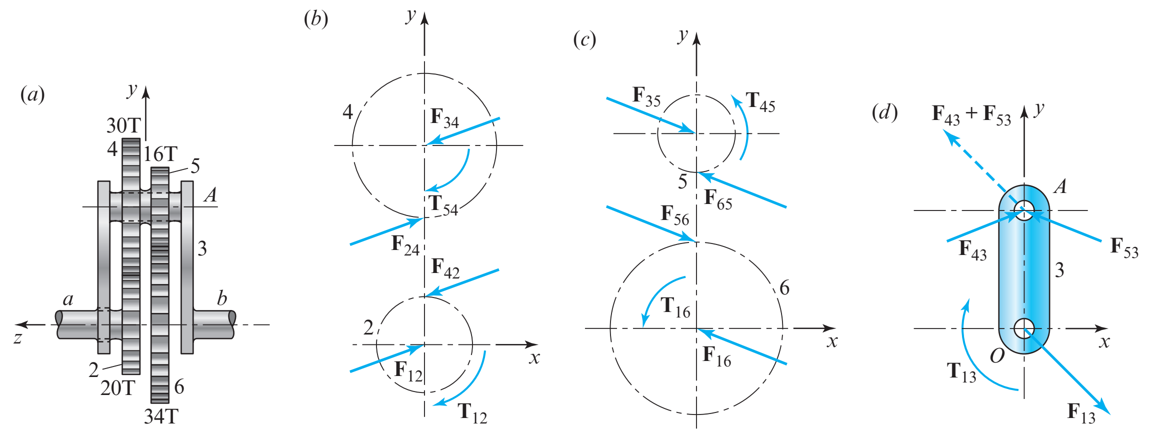


Figure 11.17 (*a*) Planetary gear train with tooth numbers; (*b*) free-body diagrams of gears 2 and 4; (*c*) free-body diagrams of gears 5 and 6; (*d*) free-body diagram of planet carrier arm 3.

where $\mathbf{F}_{34}$ is the force of planet arm 3 onto gear 4. Gears 4 and 5 are connected to each other but turn freely on the planet arm shaft. Thus, $\mathbf{T}_{54}$ is the torque exerted by gear 5 onto gear 4. This torque is $T_{54} = (R_4)\,F_{24}^t = (1.50 \text{ in})\,(100 \text{ lb}) = 150 \text{ in}\cdot\text{lb cw}$.

Turning next to the free-body diagram of gear 5 in Fig. 11.17*c*, we first find $F_{65}^t = T_{45}/R_5 = (150 \text{ in}\cdot\text{lb})\,/\,(0.800 \text{ in}) = 188 \text{ lb}$. Therefore, $F_{65} = (188 \text{ lb})\,/\cos 20° = 200 \text{ lb}$. In vector form, the results for gear 5 are summarized as

$$\mathbf{F}_{65} = -\mathbf{F}_{35} = 200 \text{ lb}\angle 160°, \mathbf{T}_{45} = 150\hat{\mathbf{k}} \text{ in}\cdot\text{lb}.$$

For gear 6, shown in Fig. 11.17*c*, we have

$$\mathbf{F}_{16} = -\mathbf{F}_{56} = 200 \text{ lb}\angle 160°,$$

$$\mathbf{T}_{16} = R_6 F_{56}^t \hat{\mathbf{k}} = (1.70 \text{ in})\left(200\cos 20° \text{ lb}\right)\hat{\mathbf{k}} = 319\hat{\mathbf{k}} \text{ in}\cdot\text{lb}.$$

Note that $\mathbf{F}_{16}$ and $\mathbf{T}_{16}$ are the force and torque, respectively, exerted by the frame on gear 6.

The free-body diagram of arm 3 is shown in Fig. 11.17*d*. As noted earlier, the forces are assumed to act in a single plane. The two forces $\mathbf{F}_{43}$ and $\mathbf{F}_{53}$ are

$$\mathbf{F}_{43} = -\mathbf{F}_{34} = 106 \text{ lb}\angle 20°, \quad \mathbf{F}_{53} = -\mathbf{F}_{35} = 200 \text{ lb}\angle 160°,$$

and can be summed to

$$\mathbf{F}_{43} + \mathbf{F}_{53} = 137 \text{ lb}\angle 130.2°.$$

We now find the output shaft reaction to be

$$\mathbf{F}_{13} = -(\mathbf{F}_{43} + \mathbf{F}_{53}) = 137 \text{ lb}\angle{-49.8°}.$$

Using $\mathbf{R}_{AO} = 2.50\hat{\mathbf{j}}$ in and the equation

$$\sum \mathbf{M}_O = \mathbf{T}_{13} + \mathbf{R}_{AO} \times (\mathbf{F}_{43} + \mathbf{F}_{53}) = \mathbf{0},$$

we find $\mathbf{T}_{13} = -221\hat{\mathbf{k}}$ in · lb. Therefore, the output shaft torque is

$$\mathbf{T}_{13} = 221\hat{\mathbf{k}} \text{ in} \cdot \text{lb.} \qquad \textit{Ans.}$$

11.12 STRAIGHT-TOOTH BEVEL-GEAR FORCE ANALYSIS

In determining the forces on bevel gear teeth, it is customary to use the forces that would occur at the mid-thickness of the tooth on the pitch cone. The resultant tangential force probably occurs somewhere between the midpoint and the large end of the tooth, but there will be only a small error in making this approximation. The tangential or transmitted force is then given by

$$F^t = \frac{T}{R}, \tag{11.18}$$

where R is the mid-radius of the pitch cone, as shown in Fig. 11.18, and T is the shaft torque.

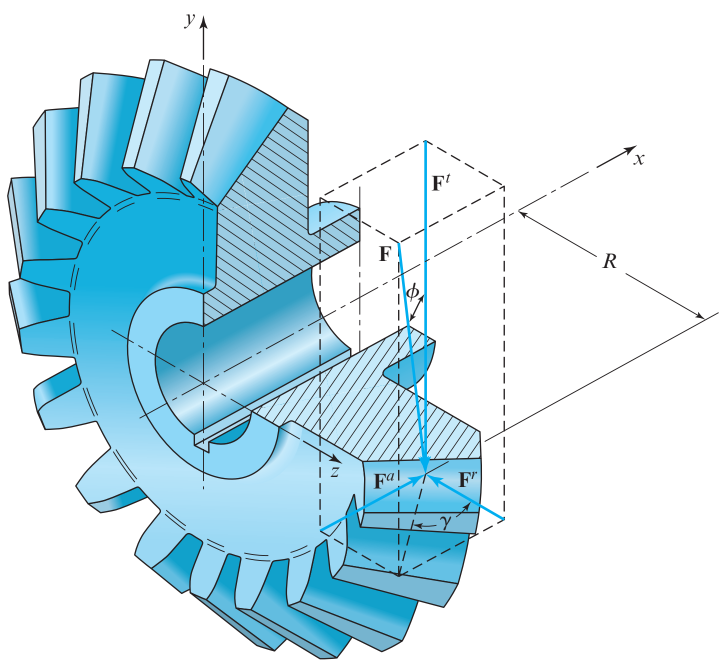


Figure 11.18 Force components on a straight-tooth bevel gear at the tooth contact point.

Figure 11.18 also shows all the components of the resultant force acting at the midpoint of the tooth. The following relationships can be derived by inspection of Fig. 11.18:

$$\mathbf{F} = \mathbf{F}^a + \mathbf{F}^r + \mathbf{F}^t, \tag{11.19}$$

$$F^r = F^t \tan\phi \cos\gamma, \tag{11.20}$$

$$F^a = F^t \tan\phi \sin\gamma. \tag{11.21}$$

Note that the axial force $\mathbf{F}^a$ results in a couple on the shaft.

EXAMPLE 11.8

The bevel pinion shown in Fig. 11.19 rotates at 600 rev/min in the direction indicated and transmits 5 hp to the gear. The mounting distances are shown, together with the locations of the bearings on each shaft. Bearings *A* and *C* are capable of taking both radial and axial loads, whereas bearings *B* and *D* are designed to receive only radial loads. The teeth of the gears have a 20° pressure angle. Find the components of the forces that the bearings exert on the shafts in the *x*, *y*, and *z* directions.

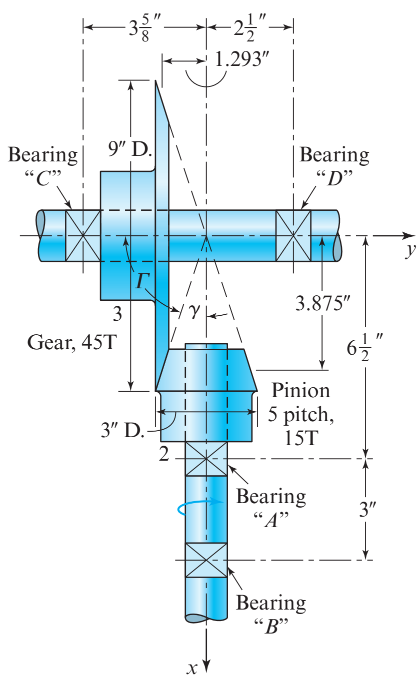


Figure 11.19 Bevel gearset and bearing locations.

SOLUTION

The pitch angles for the pinion and gear are

$$\gamma = \tan^{-1}\left(\frac{1.5 \text{ in}}{4.5 \text{ in}}\right) = 18.4^\circ,$$

$$\Gamma = \tan^{-1}\left(\frac{4.5 \text{ in}}{1.5 \text{ in}}\right) = 71.6^\circ.$$

The radii to the midpoints of the teeth are shown on the drawing and are $R_2 = 1.293$ in and $R_3 = 3.875$ in for the pinion and gear, respectively.

Let us first determine the forces acting on the pinion. Using Eq. (13.13*b*), we find the transmitted force to be

$$F_{32}^t = \frac{P}{R_2\omega_2} = \frac{(5 \text{ hp})\,(33\,000 \text{ ft}\cdot\text{lb/min/hp})\,(12 \text{ in/ft})}{(1.293 \text{ in})\,(600 \text{ rev/min})\,(2\pi \text{ rad/rev})} = 406 \text{ lb}.$$

This force acts in the negative z direction, that is, into the plane of Fig. 11.19. The radial and axial components of $\mathbf{F}_{32}$ are obtained from Eqs. (11.20) and (11.21),

$$F_{32}^r = 406 \text{ lb} \tan 20^\circ \cos 18.4^\circ = 140 \text{ lb},$$

$$F_{32}^a = 406 \text{ lb} \tan 20^\circ \sin 18.4^\circ = 46.6 \text{ lb},$$

where $\mathbf{F}_{32}^r$ acts in the positive y direction and $\mathbf{F}_{32}^a$ in the positive x direction.

These three forces are components of the total force, $\mathbf{F}_{32}$. Thus,

$$\mathbf{F}_{32} = 46.6\hat{\mathbf{i}} + 140\hat{\mathbf{j}} - 406\hat{\mathbf{k}} \text{ lb}.$$

The torque applied to the pinion shaft must be

$$\mathbf{T}_{12} = -\mathbf{R}_2 \times \mathbf{F}_{32}^t = -\left(-1.293\hat{\mathbf{j}} \text{ in}\right) \times \left(-406\hat{\mathbf{k}} \text{ lb}\right) = -525\hat{\mathbf{i}} \text{ in}\cdot\text{lb}.$$

A free-body diagram of the pinion and shaft is shown schematically in Fig. 11.20*a*. The dimensions, torque $\mathbf{T}_{12}$, and force $\mathbf{F}_{32}$ are known, and the problem is to determine the bearing reactions $\mathbf{F}_A$ and $\mathbf{F}_B$. To determine $\mathbf{F}_B$, we sum moments about A:

$$\sum \mathbf{M}_A = \mathbf{T}_{12} + \mathbf{R}_{BA} \times \mathbf{F}_B + \mathbf{R}_{PA} \times \mathbf{F}_{32} = \mathbf{0}, \tag{1}$$

where the two position-difference vectors are

$$\mathbf{R}_{BA} = 3.0\hat{\mathbf{i}} \text{ in} \quad \text{and} \quad \mathbf{R}_{PA} = -2.625\hat{\mathbf{i}} - 1.293\hat{\mathbf{j}} \text{ in}.$$

Therefore, the second and third terms of Eq. (1), respectively, are

$$\begin{aligned} \mathbf{R}_{BA} \times F_B &= (3.0 \text{ in})\,\hat{\mathbf{i}} \times (F_B^y\hat{\mathbf{j}} + F_B^z\hat{\mathbf{k}}) \\ &= -(3.0 \text{ in})F_B^z\hat{\mathbf{j}} + (3.0 \text{ in})\,F_B^y\hat{\mathbf{k}} \end{aligned} \tag{2}$$

$$\mathbf{R}_{PA} \times \mathbf{F}_{32} = (-2.625\hat{\mathbf{i}} - 1.293\hat{\mathbf{j}} \text{ in}) \times (46.6\hat{\mathbf{i}} + 140\hat{\mathbf{j}} - 406\hat{\mathbf{k}} \text{ lb})$$
$$= 525\hat{\mathbf{i}} - 1\,066\hat{\mathbf{j}} - 307\hat{\mathbf{k}} \text{ in} \cdot \text{lb}. \tag{3}$$

Substituting the value of $\mathbf{T}_{12}$ and Eqs. (2) and (3) into Eq. (1) and solving, the bearing reaction at B is

$$\mathbf{F}_B = 102\hat{\mathbf{j}} - 355\hat{\mathbf{k}} \text{ lb}. \qquad \textit{Ans.}$$

The magnitude of the bearing reaction at B is 370 lb.

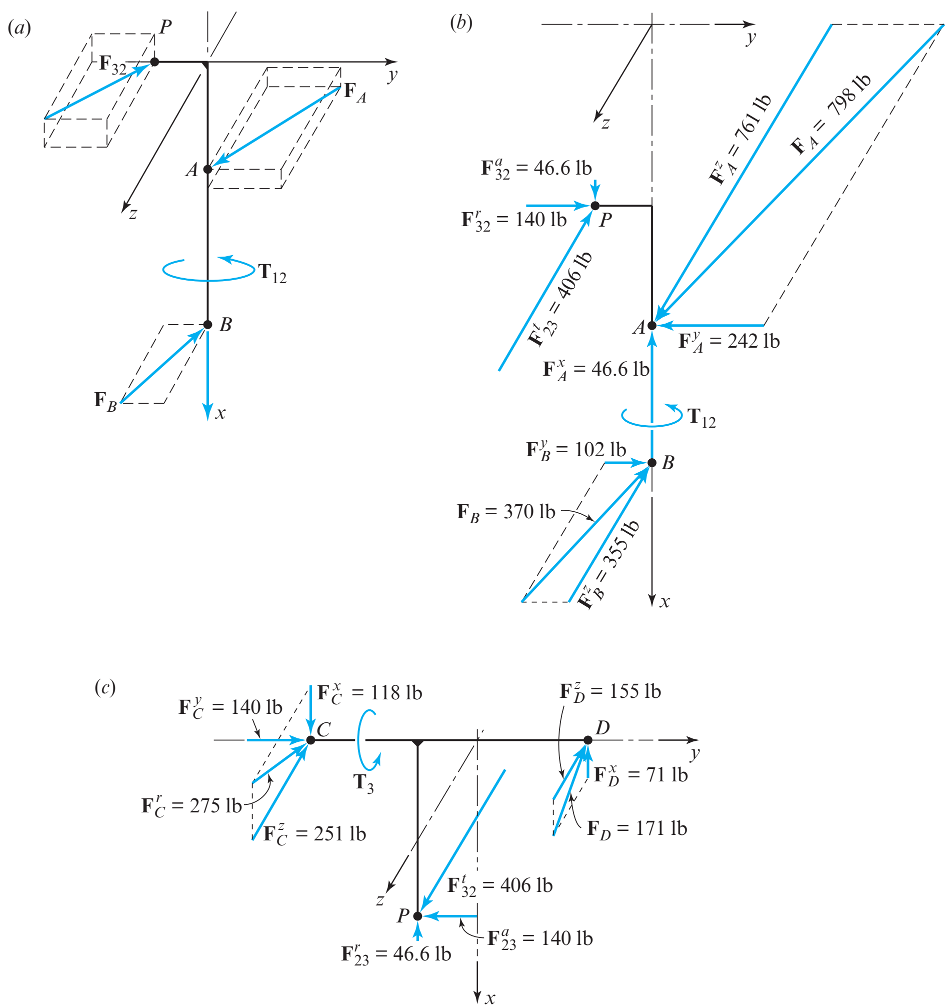


Figure 11.20 (*a*) Free-body diagrams of pinion and input shaft; (*b*) results and component values on pinion and input shaft; (*c*) free-body diagrams of gear and output shaft with results and component values.

Next, to determine the bearing reaction at A, we write

$$\sum \mathbf{F} = \mathbf{F}_{32} + \mathbf{F}_A + \mathbf{F}_B = \mathbf{0}.$$

Then, solving this equation, the bearing reaction at A is

$$\mathbf{F}_A = -46.6\hat{\mathbf{i}} - 242\hat{\mathbf{j}} + 761\hat{\mathbf{k}} \text{ lb.} \qquad \textit{Ans.}$$

The magnitude of the bearing reaction at A is 798 lb. The results are shown in Fig. 11.20*b*.

A similar procedure is used for the gear shaft. The results are displayed in Fig. 11.20*c*.

11.13 METHOD OF VIRTUAL WORK

So far in this chapter, we have learned to analyze problems involving the static equilibrium of mechanical systems by the application of Newton's laws. Another fundamentally different approach to force-analysis problems is based on the principle of virtual work, first proposed by the Swiss mathematician J. Bernoulli in the 18th century.

The method is based on an energy balance of the system that requires that the net change in internal energy during a small displacement must be equal to the difference between the work input to the system and the work output, including the work done against friction, if any. Thus, for a system of rigid bodies in equilibrium under a system of applied forces, if given an arbitrary small displacement from equilibrium, the net change in the internal energy, denoted here by dU, is equal to the work dW done on the system:

$$dU = dW. \tag{11.22}$$

Work and change in internal energy are positive when work is done on the system, giving it increased internal energy, and negative when energy is lost from the system, such as when it is dissipated through friction. If the system has no friction or other dissipation losses, energy is conserved, and the net change in internal energy during a small displacement from equilibrium is zero.

Of course, such a method requires that we know how to calculate the work done by each force during the small *virtual displacement* chosen. If some force $\mathbf{F}$ acts at a point of application Q that undergoes a small displacement $d\mathbf{R}_Q$, then the work done by this force on the system is given by

$$dU = \mathbf{F} \cdot d\mathbf{R}_Q, \tag{11.23}$$

where dU is a scalar value, having units of work or energy, and is positive for work done onto the system and negative for work output from the system.

The displacement considered is called a *virtual* displacement, since it need not be one that truly happens on the physical machine. It need only be a small displacement that is hypothetically possible and consistent with the constraints imposed on the system. The small displacement relationships of the system can be determined through the principles of kinematics covered in Part I of this text, and the input-to-output force relationships of the machine can therefore be determined. This will become clear through the following example.

EXAMPLE 11.9

Using the method of virtual work, determine the input crank torque, $\mathbf{T}_{12}$, required for equilibrium for the four-bar linkage subjected to external load $\mathbf{P}$ as shown in Fig. 11.5*a*.

ASSUMPTIONS

Friction effects and weights of the links can be neglected.

SOLUTION

From Example 11.1, we recall that the position-difference vector from O_4 to Q and the applied force at Q, respectively, are

$$\mathbf{R}_{QO_4} = 1.84\hat{\mathbf{i}} + 4.65\hat{\mathbf{j}} \text{ in}, \tag{1}$$

$$\mathbf{P} = 120 \text{ lb}\angle 220^\circ = -91.93\hat{\mathbf{i}} - 77.13\hat{\mathbf{j}} \text{ lb}. \tag{2}$$

Figure 11.21 presents a scale diagram of the system at the posture in question. If a small virtual displacement $d\theta_2$ is given to the input crank 2, this results in a small displacement $d\theta_4$ of the output crank 4, along with a small displacement $d\mathbf{R}_{QO_4}$ of point Q, as shown.

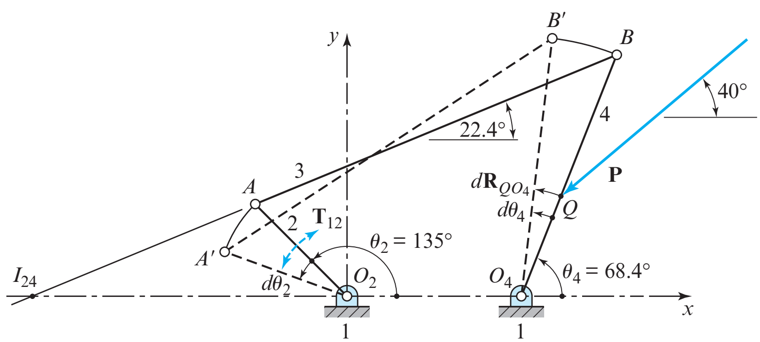


Figure 11.21 Virtual-work solution.

Assuming that these virtual displacements are small and happen in a short time increment, dt, the velocity difference equation, Eq. (3.3), can be multiplied by dt to yield the relationship between $d\mathbf{R}_{QO_4}$ and $d\theta_4$, as follows:

$$d\mathbf{R}_{QO_4} = d\theta_4\hat{\mathbf{k}} \times \mathbf{R}_{QO_4} = \left(d\theta_4\hat{\mathbf{k}} \text{ rad}\right) \times \left(1.84\hat{\mathbf{i}} + 4.65\hat{\mathbf{j}} \text{ in}\right)$$

$$= \left(-4.65\hat{\mathbf{i}} + 1.84\hat{\mathbf{j}} \text{ in}\right) d\theta_4. \tag{3}$$

Since the input torque, $\mathbf{T}_{12}$, and the applied force, $\mathbf{P}$, are the only forces doing work during the chosen displacement, Eq. (11.22) can be written for this problem as

$$dU = T_{12}\hat{\mathbf{k}} \cdot d\theta_2\hat{\mathbf{k}} + \mathbf{P} \cdot d\mathbf{R}_{QO_4} = 0.$$

Substituting Eqs. (2) and (3) into this equation gives

$$T_{12}d\theta_2 + \left(-91.93\hat{\mathbf{i}} - 77.13\hat{\mathbf{j}}\text{ lb}\right) \cdot \left(-4.65\hat{\mathbf{i}}+1.84\hat{\mathbf{j}}\text{ in}\right)d\theta_4 = 0.$$

Then, dividing by $d\theta_2$, rearranging, and recognizing the first-order kinematic coefficient, we find

$$T_{12} = (-285.6\text{ in}\cdot\text{lb})\,\frac{d\theta_4}{d\theta_2} = -\,285.6\theta_4'\text{ in}\cdot\text{lb}. \tag{4}$$

By dividing the numerator and denominator of this angular displacement ratio by dt, we recognize that this is equal to the angular-velocity ratio ω_4/ω_2, which can be determined from the angular velocity ratio theorem of Eq. (3.28).* Using the location of the instant center I_{24} and measurements from Fig. 11.21, we find

$$\theta_4' = \frac{d\theta_4}{d\theta_2} = \frac{\omega_4}{\omega_2} = \frac{R_{I_{24}I_{12}}}{R_{I_{24}I_{14}}} = \frac{14.65\text{ in}}{22.65\text{ in}} = 0.646\,8\text{ rad/rad}.$$

Finally, substituting this result into Eq. (4) gives the input torque,

$$\mathbf{T}_{12} = \left(-285.6\hat{\mathbf{k}}\text{ in}\cdot\text{lb}\right)\left(0.646\,8\text{ rad/rad}\right) = -184.7\hat{\mathbf{k}}\text{ in}\cdot\text{lb}. \qquad \textit{Ans.}$$

One primary advantage of the method of virtual work for force analysis over the other methods demonstrated above comes in problems where input-to-output force relationships are sought. Note that the constraint forces are not required in this solution technique, since both their action and reaction forces are internal to the system; both move through identical displacements, and thus their virtual work contributions cancel each other. This would not be true for friction forces where the displacements would be different for the action and reaction force components, with this work difference representing the energy dissipated. Otherwise, constraint forces need not be considered, since they cause no net virtual work.

SUMMARY

The following is a summary of the procedures for the methods of static force analysis that have been presented up to this point in Chap. 11.

---

* This technique of treating small displacement ratios as velocity ratios or first-order kinematic coefficients can often be extremely helpful, because, as we remember from Sec. 3.8, velocity relationships lead to linear equations rather than the nonlinear relations implied by position or displacement. Velocity polygons and instant-center methods are also helpful.

1. We draw complete free-body diagrams of each member in the mechanism. It is good practice to combine all known forces on a free-body diagram into a single vector. We classify each member; that is, we identify two-force members, three-force members, four-force members, and so on. We then commence with the member with the lowest number of forces. That is, we draw the two-force members first, then we draw the three-force members, and, finally, we draw any four-force members. If a member is acted upon by more than four forces, then either (*a*) it can be reduced to one of the above cases, or (*b*) it has more than three unknowns and is not yet solvable.
2. Using the definitions of two-force, three-force, and four-force members, we apply the following rules:
   (*a*) For a two-force member, the two forces must be equal, opposite, and collinear. Note that for a link with two forces and a moment, the forces are equal, opposite, and parallel.
   (*b*) For a three-force member, the three forces must intersect at a single point.
   (*c*) For a four-force member, the resultant of any two forces must be equal, opposite, and collinear with the resultant of the other two forces.
3. We draw force polygons for three- and four-force members, clearly stating the scale of each force polygon. Note that, in general, it is less confusing if force polygons are not superimposed on the mechanism diagram. To draw a force polygon for a three-force member, we must have only two unknown pieces of information, both of which are force magnitudes. For a four-force member, we may have three unknown pieces of information, all of which are force magnitudes.
4. If desirable, multiple links can be grouped together and treated as a single free-body diagram.

11.14 INTRODUCTION TO BUCKLING

Throughout previous chapters and sections of this text, we have assumed that all bodies are completely rigid and cannot deform. In fact, we noted in Sec. 1.4 that this assumption of rigidity is mandatory to allow the separation of the study of kinematics from the treatment of dynamics. But we also noted that this assumption of rigidity prevents the study of strain or deformation and therefore of stress and strength.

Sections 11.14 through 11.18 deviate from the assumption of rigidity. In these sections we consider a nonrigid, large deformation effect called *buckling*, which is the sudden failure of a long, slender member subjected to compressive loading. When a member buckles, it experiences severe bending and loses its strength. This loss in strength can be very dramatic, and for this reason, a buckling failure in a structure can occur without warning and is often catastrophic. Figure 11.22 shows the buckling collapse of steel beams in the Cortland St. subway station in New York. Here we can see that buckling is a very important problem. It is also a fascinating topic in mechanical design and is the subject of the following four sections.

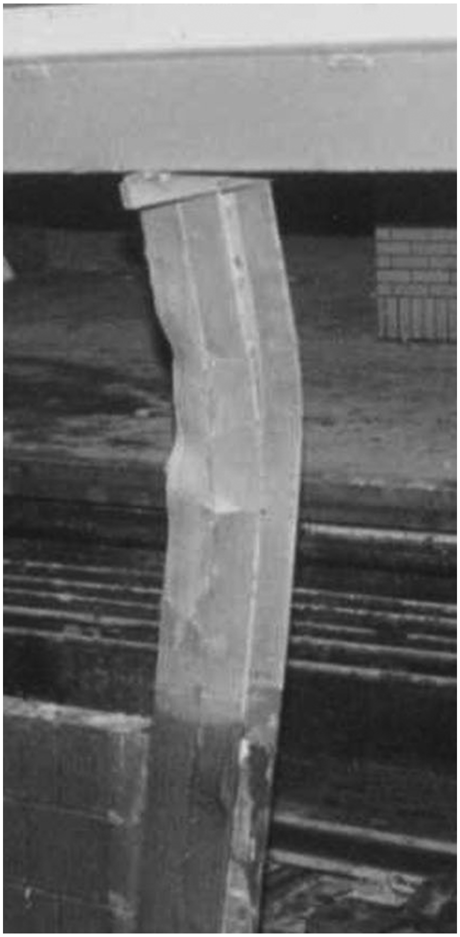


Figure 11.22 A catastrophic failure due to buckling in the Cortland St. subway station in New York, NY (circa September 2001).

11.15 EULER COLUMN FORMULA

Consider a long, slender member of length L with fixed-free end conditions, as shown in Fig. 11.23. Assume that the member is subject to the central compressive load P shown in the figure. The first problem is to derive an equation for the critical value of the load, P, that is, the load that will cause the member to become unstable. This load is referred to as the *critical load*.

For small deflections and for homogeneous and isotropic materials, the curvature of the member caused by compressive load P can be written as

$$\kappa = \frac{d^2y}{dx^2} = \frac{M}{EI}, \tag{11.24}$$

where d^2y/dx^2 is the second derivative of the transverse deflection, M is the bending moment at an arbitrary location x along the member, E is Young's modulus, and I is the

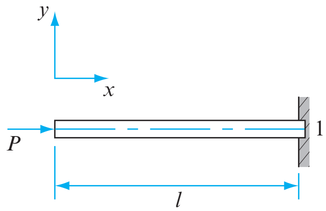


Figure 11.23 A long, slender member subjected to a central compressive load, P.

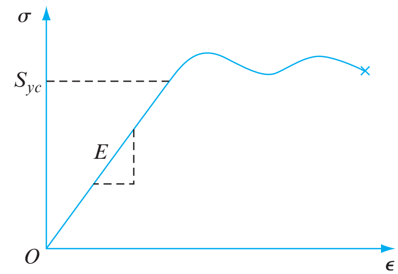


Figure 11.24 Stress versus strain for a mild steel bar in tension.

second moment of area (sometimes called the *area moment of inertia*) of the member cross section at location x. The bending moment can be written as

$$M = -Py, \tag{11.25}$$

where y is the transverse deflection of the member at location x.

Young's modulus E is the modulus of elasticity of the material of the member; it has units of stress (that is, lb/in^2 or Pa) and is a measure of the stiffness of the material. Consider a typical plot of stress, σ, versus strain, ε, for a mild steel bar in tension; Fig. 11.24. Hooke's law states that, in the elastic region, the stress in a mild steel bar is proportional to the strain, with a proportionality constant of the modulus of elasticity. This can be written as

$$E = \sigma/\varepsilon.$$

As a rule of thumb, the modulus of elasticity for a steel alloy is

$$E = 30 \text{ Mpsi} = 30 \times 10^6 \text{ lb/in}^2 \quad \text{or} \quad E = 207 \text{ GPa} = 207 \times 10^9 \text{ N/m}^2,$$

and for stainless steel, it is

$$E = 27.5 \text{ Mpsi} \quad \text{or} \quad E = 190 \text{ GPa}.$$

Substituting Eq. (11.25) into Eq. (11.24), the second derivative of the transverse deflection of the member can be written as

$$\frac{d^2y}{dx^2} = -\frac{Py}{EI}.$$

Rearranging this equation gives the second-order differential equation for the member:

$$\frac{d^2y}{dx^2} + \left(\frac{P}{EI}\right) y = 0. \tag{11.26}$$

Figure 11.25 A member with pinned-pinned ends.

The solution to this differential equation, that is, the transverse deflection of the member, is

$$y = A\sin\left(\sqrt{\frac{P}{EI}}x\right) + B\cos\left(\sqrt{\frac{P}{EI}}x\right), \tag{11.27}$$

where A and B are the constants of integration and are obtained from the support conditions at the ends of the member.

Pinned-pinned end conditions Consider the case where both ends of the member are pinned, as shown in Fig. 11.25 (also referred to in civil engineering as *rounded-rounded ends*).

The end conditions are: (a) at $x = 0$, the deflection is $y = 0$, and (b) at $x = L$, the deflection is $y = 0$. Substituting $y = 0$ at $x = 0$ into Eq. (11.27) gives

$$0 = A\sin 0 + B\cos 0.$$

Therefore, one constant of integration is $B = 0$. Substituting this into Eq. (11.27) gives

$$y = A\sin\left(\sqrt{\frac{P}{EI}}x\right). \tag{11.28}$$

Then, substituting $y = 0$ at $x = L$ into this equation gives

$$0 = A\sin\left(\sqrt{\frac{P}{EI}}L\right).$$

There are two solutions to this equation. Either (a) the constant of integration $A = 0$, which is a trivial solution (it implies that the deflection is $y = 0$ for all values of x) and can be ignored, or (b)

$$\sin\left(\sqrt{\frac{P}{EI}}L\right) = 0.$$

The solution to this equation is

$$\sqrt{\frac{P_{cr}}{EI}}L = n\pi \quad (n = 1, 2, 3, \cdots), \tag{11.29}$$

where P_{cr} is a compressive load that places the member in a condition of unstable equilibrium and is referred to as the *critical load*. The first critical load (that is, the lowest

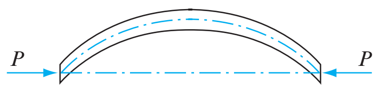


Figure 11.26 Deflection curve is a half sine wave.

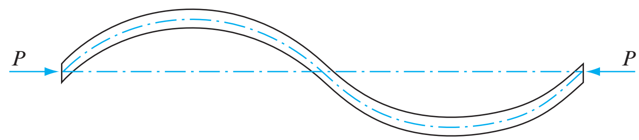


Figure 11.27 Deflection curve is a full sine wave.

or minimum value of P_{cr}) is the most important from physical considerations. For $n = 1$, the critical load is

$$P_{cr} = \left(\frac{\pi^2}{L^2}\right) EI. \tag{11.30}$$

This equation is commonly referred to as *Euler's column formula* for a long, slender member with pinned-pinned ends. Note an interesting result, that the strength of the material is not a factor; that is, the critical load depends only on: (*a*) the length, L; (*b*) the second moment of area, I; and (*c*) Young's modulus, E. Therefore, if the member is steel, then using a stronger steel (with a higher compressive yield strength, S_{yc}) does not help, since all steel alloys have essentially the same modulus of elasticity ($E = 30$ Mpsi or 207 GPa).

Substituting Eq. (11.30) into Eq. (11.28), the transverse deflection of the member at the critical load can be written as

$$y = A \sin\left(\frac{\pi x}{L}\right), \tag{11.31}$$

which indicates that the deflection curve of the member (for pinned-pinned ends) is a half sine wave, as shown in Fig. 11.26.

Values of $n > 1$ in Eq. (11.29) result in deflection curves that cross the X axis at points of inflection and are multiples of half sine waves. For $n = 2$, the deflection curve is a full sine wave, as shown in Fig. 11.27.

11.16 CRITICAL UNIT LOAD

Next, we hope to obtain an expression for the critical load per unit cross-sectional area of the member, referred to as the *critical unit load*, which has the units of stress or strength. The second moment of area (also called the *area moment of inertia**) of the member cross

* Note that the *area* moment of inertia, shown here, is quite different from the *mass* moment of inertia of Sec. 12.3. Note the dimensions of the fourth power of length compared to mass length squared.

section in a plane perpendicular to the axis of the member can be written as

$$I = Ak^2,$$

where k is called the *radius of gyration*.* Rearranging this equation, the radius of gyration of the member cross section can be written as

$$k = \sqrt{\frac{I}{A}}.$$

From this we define a nondimensional parameter called the *slenderness ratio*,

$$S_\mathrm{r} = \frac{L}{k}. \tag{11.32}$$

The slenderness ratio, rather than the actual length, L, is commonly used to classify members in compression into length categories.

Substituting this definition into Eq. (11.30), the first *critical load* can be written as

$$P_\mathrm{cr} = \frac{\pi^2 EA}{S_\mathrm{r}^2}.$$

The *critical unit load* is defined by dividing this critical load by the cross-sectional area,

$$\frac{P_\mathrm{cr}}{A} = \frac{\pi^2 E}{S_\mathrm{r}^2}. \tag{11.33}$$

The critical unit load, rather than the strength of the material from which it is made, that is, the yield strength or the ultimate strength, represents the strength of a particular member under compression.

Including the pinned-pinned end conditions presented above, there are four other common pairs of end conditions for a member. These four are: (*a*) fixed-fixed ends, (*b*) fixed-pinned ends, (*c*) pinned-pinned ends, and (*d*) fixed-free ends. The critical unit load for each of these four cases can be written in the form of Eq. (11.33) using an end-condition constant, C; that is,

$$\frac{P_\mathrm{cr}}{A} = C\frac{\pi^2 E}{S_\mathrm{r}^2}. \tag{11.34}$$

This equation is commonly referred to as the Euler column formula for different end conditions.

To determine the end-condition constant, C, the right-hand side of this equation can be written as

$$C\frac{\pi^2 E}{S_\mathrm{r}^2} = \frac{\pi^2 E}{(L_\mathrm{eff}/k)^2},$$

* Note that this is *not* the same as the term with a similar name defined in Eq. (12.10).

where L_{eff} denotes the effective length of the member. The effective length is the length of the half sine wave for the equivalent member. Therefore, by rearranging this equation, the end-condition constant can be expressed in terms of the effective length of the member as $C = (L/L_{\text{eff}})^2$. Also, using the American Institute of Steel Construction (AISC)-recommended column end-condition effective length factor, $\alpha = L_{\text{eff}}/L$, the end-condition constant can be written as

$$C = 1/\alpha^2.$$

Case (*a*). Fixed-fixed ends are shown in Fig. 11.28.

For the case of a member with fixed-fixed ends, the effective length of the member is $L_{\text{eff}} = 0.5L$. Substituting this value into the above equations, the end-condition constant is $C = 4$.

Case (*b*). Fixed-pinned ends are shown in Fig. 11.29.

For the case of a member with fixed-pinned ends, the effective length of the member is $L_{\text{eff}} = 0.707L$. Using this value, the above equations give the end-condition constant $C = 2$.

Case (*c*). Pinned-pinned ends are shown in Fig. 11.25.

The case of a member with pinned-pinned ends is the case used for the above equations. The effective length of the member is $L_{\text{eff}} = L$, and the end-condition constant is $C = 1$.

Case (*d*). Fixed-free ends are shown in Fig. 11.30.

For the case of a member with fixed-free ends, the effective length of the member is $L_{\text{eff}} = 2L$. With this value in the above equations, the end-condition constant is $C = 1/4$.

When analyzing buckling in links of a planar mechanism, we often find that the end conditions are pinned-pinned for bending in the xy plane. Recall that the end-condition constant for a pinned-pinned link is $C = 1$. Note, however, that if bending of such a link is assumed to be out of the xy plane, then the end conditions of the link are not known. It is common for out of the plane bending to assume that the ends of the link are fixed-fixed. In such a case, the end-condition constant for a fixed-fixed link is $C = 4$.

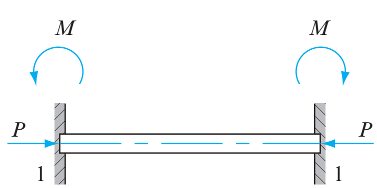


Figure 11.28 Fixed-fixed ends.

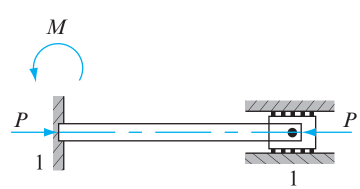


Figure 11.29 Fixed-pinned ends.

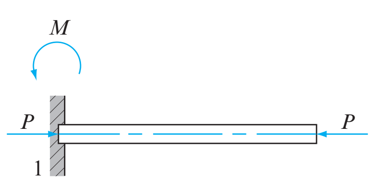


Figure 11.30 Fixed-free ends.

11.17 CRITICAL UNIT LOAD AND SLENDERNESS RATIO

A typical plot of critical unit load versus slenderness ratio of a long, slender member subjected to an axial compressive load P is shown in Fig. 11.31.

Note that there are two distinct curves shown in Fig. 11.31: (a) Euler's column formula, and (b) *Johnson's parabolic equation.** The criterion for using Euler's column formula, Eq. (11.34), is

$$S_r > (S_r)_D, \tag{11.35}$$

where $(S_r)_D$ is the slenderness ratio at the point of tangency, D, between Euler's column formula and Johnson's parabolic equation. The critical unit load at the point of tangency is

$$\frac{P_{cr}}{A} = 0.5 S_{yc}.$$

Substituting this equation into Eq. (11.34) and rearranging, the slenderness ratio at the point of tangency, that is, when the slenderness ratio $S_r = (S_r)_D$, can be written as

$$(S_r)_D = \pi\sqrt{\frac{2CE}{S_{yc}}}. \tag{11.36}$$

If the slenderness ratio of the member is less than this value, $S_r < (S_r)_D$, then the criterion that is most commonly used for buckling is Johnson's parabolic equation.

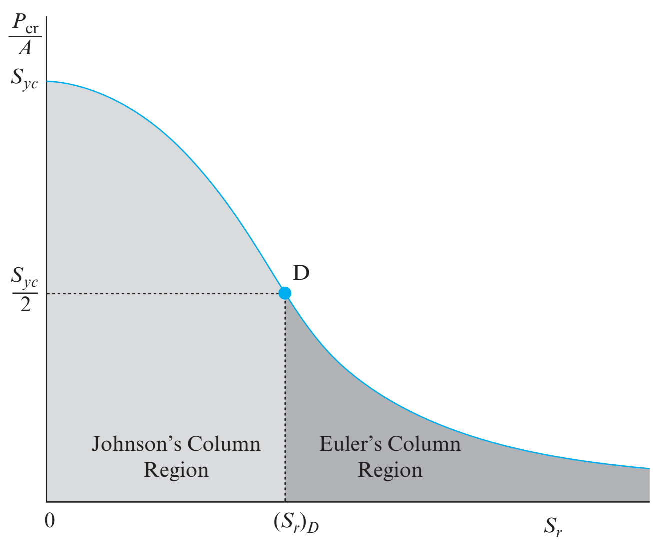


Figure 11.31 Critical unit load versus slenderness ratio.

* Proposed by Prof. J. B. Johnson in 1898 and, independently, by Prof. A. Ostenfeld of Copenhagen.

11.18 JOHNSON'S PARABOLIC EQUATION

The equation of the parabola in the left side of Fig. 11.31 can be written as

$$\frac{P_{\mathrm{cr}}}{A} = a + bS_{\mathrm{r}}^2. \tag{11.37}$$

The focus here is to determine constants a and b. There are two known data points in Fig. 11.31; namely, (a) for the slenderness ratio $S_{\mathrm{r}} = 0$, the critical unit load is $P_{\mathrm{cr}}/A = S_{yc}$, and ($b$) for the slenderness ratio $S_{\mathrm{r}} = (S_{\mathrm{r}})_D$, the critical load is $P_{\mathrm{cr}}/A = 0.5S_{yc}$.

Substituting the first of these data points into Eq. (11.37), we find the coefficient

$$a = S_{yc}.$$

Substituting this result into Eq. (11.37), the critical unit load can be written as

$$\frac{P_{\mathrm{cr}}}{A} = S_{yc} + bS_{\mathrm{r}}^2. \tag{11.38}$$

Then, substituting the second data point and Eq. (11.36) into Eq. (11.38) gives

$$0.5S_{yc} = S_{yc} + b\pi^2 \frac{2CE}{S_{yc}}.$$

Finally, rearranging this equation, coefficient b can be written as

$$b = -\frac{S_{yc}^2}{4CE\pi^2}.$$

Substituting this result into Eq. (11.38), Johnson's parabolic equation for the critical unit load can be written as

$$\frac{P_{\mathrm{cr}}}{A} = S_{yc} - \frac{1}{CE}\left(\frac{S_{yc}S_{\mathrm{r}}}{2\pi}\right)^2. \tag{11.39}$$

Therefore, the critical load for a member using Johnson's parabolic equation is

$$P_{\mathrm{cr}} = A\left[S_{yc} - \frac{1}{CE}\left(\frac{S_{yc}S_{\mathrm{r}}}{2\pi}\right)^2\right]. \tag{11.40}$$

EXAMPLE 11.10

For the four-bar linkage in the posture shown in Fig. 11.32, a vertical load P is acting at point E on horizontal coupler link 3, that is, the link is parallel to the x axis. The coupler link is pinned to vertical links 2 and 4 at points A and B. The links are made of a steel alloy with a compressive yield strength $S_{yc} = 60$ kpsi and a modulus of elasticity $E =$ 30 Mpsi. Links 2 and 4 have hollow circular cross sections with 2-in outside diameters and 0.25-in wall thicknesses. Assume that the value for the end-condition constant for both vertical links is $C = 1$. The known lengths are $R_{AO} = 8$ ft, $R_{BA} = 6$ ft, $R_{BC} = 5$ ft, and $R_{EA} = 4$ ft. Determine: (*a*) the radii of gyration for links 2 and 4; (*b*) the values of the slenderness ratios for links 2 and 4 and at the point of tangency between Euler's column formula and Johnson's parabolic equation; (*c*) the critical loads for links 2 and 4; and (*d*) the maximum value of load P if the factor of safety guarding against buckling for links 2 and 4 are both specified as $N = 2$.

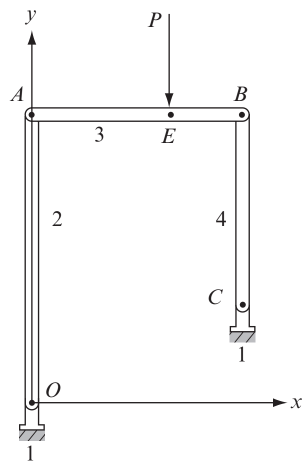


Figure 11.32 Four-bar linkage subjected to a vertical load P.

SOLUTION

(*a*) The cross-sectional area of both links 2 and 4 is

$$A = \frac{\pi}{4}\left(D^2 - d^2\right) = \frac{\pi}{4}\left[(2\text{ in})^2 - (1.5\text{ in})^2\right] = 1.374\text{ in}^2. \tag{1}$$

The area moment of inertia for the hollow circular cross section of links 2 and 4 is (Appendix A, Table 4)

$$I = \frac{\pi}{64}\left(D^4 - d^4\right) = \frac{\pi}{64}\left[(2\text{ in})^4 - (1.5\text{ in})^4\right] = 0.537\text{ in}^4. \tag{2}$$

The radius of gyration for links 2 and 4 is

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{0.537\text{ in}^4}{1.374\text{ in}^2}} = 0.625\text{ in.} \qquad \textit{Ans.}\ (3)$$

(*b*) From the given end conditions for links 2 and 4, that is, pinned-pinned ends, the end condition constant is $C = 1.0 = (1/\alpha)^2$ and $\alpha = 1.0 = L/L_{\text{eff}}$. This indicates that the effective length, L_{eff}, is equal to the actual length, L, for both link 2 and link 4. Therefore, the slenderness ratio of link 2 is

$$(S_r)_2 = \frac{(L_{\text{eff}})_2}{k} = \frac{(8\text{ ft})\ 12\text{ in/ft}}{0.625\text{ in}} = 153.60, \qquad \textit{Ans.}\ (4a)$$

and the slenderness ratio of link 4 is

$$(S_r)_4 = \frac{(L_{eff})_4}{k} = \frac{(5\text{ ft})\ 12\text{ in/ft}}{0.625\text{ in}} = 96.00. \qquad \textit{Ans.}\ (4b)$$

The slenderness ratio for either link at the point of tangency is

$$(S_r)_D = \pi\sqrt{\frac{2CE}{S_{yc}}} = \pi\sqrt{\frac{2\cdot 1.0\left(30\times 10^6\text{ psi}\right)}{60\times 10^3\text{ psi}}} = 99.35. \qquad \textit{Ans.}\ (5)$$

(*c*) The critical load is determined by first determining whether each link is considered an Euler column or a Johnson column. Note that the slenderness ratios given by Eqs. (4*a*), (4*b*), and (5) are

$$(S_r)_2 = 153.60, \quad (S_r)_4 = 96.00, \quad \text{and} \quad (S_r)_D = 99.35.$$

The criterion for using Euler's column formula is given in Eq. (11.35) as $S_r > (S_r)_D$. Therefore, because $153.60 > 99.35$, the conclusion is that link 2 is regarded as an Euler column. The critical unit load for link 2, from Euler's column formula, Eq. (11.34), can be written as

$$\frac{(P_{cr})_2}{A} = \frac{C\pi^2 E}{(S_r)_2^2}. \qquad (6)$$

Substituting the known values into Eq. (6) gives

$$\frac{(P_{cr})_2}{1.374\text{ in}^2} = \frac{1.0\pi^2\left(30\times 10^6\text{ psi}\right)}{153.60^2}. \qquad (11.41)$$

Therefore, the critical load for link 2 is

$$(P_{cr})_2 = 17\,244\text{ lb}. \qquad \textit{Ans.}\ (7)$$

The criterion for using Johnson's parabolic equation is $S_r < (S_r)_D$. Therefore, since $96.00 < 99.35$, Johnson's parabolic equation must be used for link 4. The critical load for link 4 from Johnson's parabolic equation is given by Eq. (11.40), that is,

$$(P_{cr})_4 = A\left[S_{yc} - \frac{1}{CE}\left(\frac{S_{yc}S_r}{2\pi}\right)^2\right].$$

Substituting the known values into this equation gives

$$(P_{cr})_4 = \left(1.374\text{ in}^2\right)\left[\left(60\times 10^3\text{ psi}\right) - \frac{1}{1.0\left(30\times 10^6\text{ psi}\right)}\left(\frac{\left(60\times 10^3\text{ psi}\right)96.00}{2\pi}\right)^2\right].$$

Therefore, the critical load for link 4 is

$$(P_{\text{cr}})_4 = 43\,950 \text{ lb.} \qquad \textit{Ans.} \ (8)$$

(*d*) For the given factor of safety guarding against buckling, an equation can be written for the applied load on either of the two side links as follows:

$$P_{\text{app}} = \frac{P_{\text{cr}}}{N} = \frac{P_{\text{cr}}}{2}.$$

The following two cases must be compared.

Case (*a*): When the load acting at point A from link 3 onto link 2 is a maximum, the maximum applied load at point A can be written from Eq. (7) as

$$(P_{\text{app}})_2 = \frac{(P_{\text{cr}})_2}{2} = \frac{17\,244 \text{ lb}}{2} = 8\,622 \text{ lb.}$$

To determine the load on link 2 from the applied load P, we take moments about point B. Therefore, the applied load P necessary to cause buckling of link 2 is

$$(P)_2 = \frac{R_{AB}(P_{\text{app}})_2}{R_{EB}} = \frac{6 \text{ ft } (8\,622 \text{ lb})}{2 \text{ ft}} = 25\,866 \text{ lb.} \qquad (9a)$$

Case (*b*): When the load acting at point B from link 3 onto link 4 is a maximum, the maximum applied load at point B can be written from Eq. (8) as

$$(P_{\text{app}})_4 = \frac{(P_{\text{cr}})_4}{2} = \frac{43\,950 \text{ lb}}{2} = 21\,975 \text{ lb.}$$

To determine the load on link 4 from the applied load P, we take moments about point A. Therefore, the applied load P necessary to cause buckling of link 4 is

$$(P)_4 = \frac{R_{BA}(P_{\text{app}})_4}{R_{EA}} = \frac{6 \text{ ft } (21\,975 \text{ lb})}{4 \text{ ft}} = 32\,963 \text{ lb.} \qquad (9b)$$

To ensure a factor of safety of at least 2 on both members, we must take the minimum of the applied loads from Cases (*a*) and (*b*). That is, in comparing Eqs. (9*a*) and (9*b*), since $(P)_2 < (P)_4$ (that is, since 25 866 lb < 32 963 lb), the maximum load P to guard against buckling of either link with a factor of safety of 2 is

$$P < 25\,866 \text{ lb.} \qquad \textit{Ans.}$$

EXAMPLE 11.11

A force P is applied vertically upward on the horizontal link 4 that is pinned to the ground at point B and pinned to links 2 and 3 at point A, as shown in Fig. 11.33. Link 2 is fixed to the ground at point D. Assume that links 2 and 4 are in static equilibrium. The distances are $R_{BA} = 1$ m and $R_{CA} = 3$ m, and links 2 and 4 are made from a steel alloy with a compressive yield strength $S_{yc} = 415$ MPa and a modulus of elasticity $E = 207$ GPa. The length of the vertical link 2 is $R_{DA} = 2.75$ m, and this link has a hollow circular cross section with an outside diameter of 50 mm and a wall thickness of 6 mm. Using the theoretic value for the end-condition constant for link 2, determine: (*a*) the value of the slenderness ratio at the point of tangency between Euler's column formula and Johnson's parabolic formula; (*b*) the critical load acting on link 2 and the critical unit load on link 2; and (*c*) the force P for the factor of safety of link 2 to guard against buckling to be $N = 2$. (*d*) If link 2 is replaced with a solid circular cross section of the same material with a diameter of 40 mm and force $P = 70$ kN, then determine the maximum length of link 2 for the factor of safety to guard against buckling of the link to be $N = 2$.

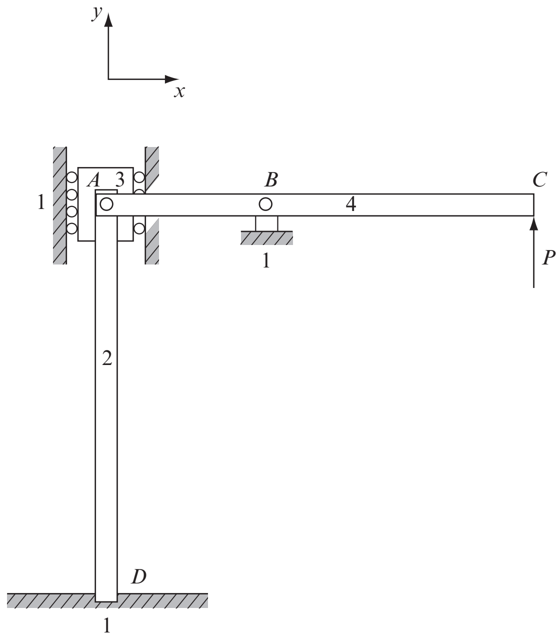


Figure 11.33

SOLUTION

(*a*) From the fixed-pinned end conditions of link 2, the end-condition constant for this link is $C = 2$. Therefore, the slenderness ratio for link 2 at the point of tangency is

$$(S_r)_D = \pi\sqrt{\frac{2CE}{S_{yc}}} = \pi\sqrt{\frac{2 \cdot 2\left(207 \times 10^9 \text{ Pa}\right)}{415 \times 10^6 \text{ Pa}}} = 140.3. \qquad \textit{Ans.}\ (1)$$

(*b*) The cross-sectional area of link 2 can be written as

$$A = \frac{\pi}{4}\left(D^2 - d^2\right) = \frac{\pi}{4}\left[(0.050 \text{ m})^2 - (0.038 \text{ m})^2\right] = 0.000\,829 \text{ m}^2.$$

The second moment of area of link 2 can be written as (Appendix A, Table 4)

$$I = \frac{\pi}{64}\left(D^4 - d^4\right) = \frac{\pi}{64}\left[(0.050\text{ m})^4 - (0.038\text{ m})^4\right] = 0.2044 \times 10^{-6}\text{ m}^4.$$

The radius of gyration of link 2 can be written as

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{0.2044 \times 10^{-6}\text{ m}^4}{0.000\,829\text{ m}^2}} = 0.0157\text{ m}.$$

The slenderness ratio of link 2 is

$$S_\text{r} = \frac{L}{k} = \frac{2.75\text{ m}}{0.0157\text{ m}} = 175.2. \tag{2}$$

To determine the critical load on link 2, we must first determine whether this link is an Euler column or a Johnson column. The criterion for using Johnson's parabolic equation is $S_\text{r} < (S_\text{r})_D$. From Eqs. (1) and (2), we have $S_r > (S_r)_D$, that is, $175.2 > 140.3$. Therefore, link 2 is an Euler column.

The critical load on link 2, from the Euler column equation, is

$$P_{cr} = A\left[\frac{C\pi^2 E}{S_r^2}\right] = \left(0.000\,829\text{ m}^2\right)\left[\frac{2\pi^2\left(207 \times 10^9\text{ Pa}\right)}{175.2^2}\right] = 110.35\text{ kN}.$$

Ans.

The critical unit load on link 2 is defined as

$$\frac{P_{cr}}{A} = \frac{110.35\text{ kN}}{0.000\,829\text{ m}^2} = 133.1\text{ MPa}.$$ *Ans.*

(c) Links 2 and 4 are in static equilibrium. The free-body diagram of link 4 is shown in Fig. 11.34.

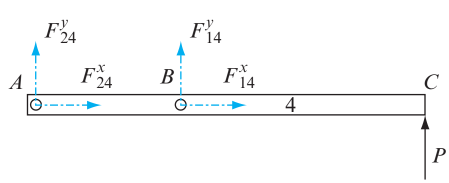


Figure 11.34
Free-body diagram of link 4.

Taking moments about pin B gives

$$\mathbf{R}_{CB} \times \mathbf{P} = \mathbf{R}_{AB} \times \mathbf{F}^y_{24}.$$

Substituting the given data into this equation gives

$$P = \frac{(1\text{ m})\,F_{24}^{y}}{2\text{ m}} = \frac{F_{24}^{y}}{2}. \tag{3}$$

Link 2 is in compression as shown in Fig. 11.35.

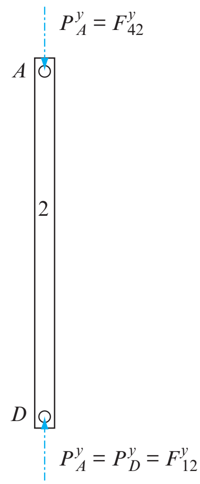


Figure 11.35
Link 2 is in compression.

The reaction force $P_A^y = F_{42}^y$ onto link 2 is opposite to the reaction force F_{24}^y; therefore, the magnitude of P_A^y is equal to the magnitude of F_{24}^y. Equation (3) can be written as

$$P = \frac{P_A^y}{2}. \tag{4}$$

The factor of safety guarding against buckling for link 2 is defined as

$$N = \frac{P_{\text{cr}}}{P_A^y}, \tag{5}$$

where $P_A^y = F_{42}^y$ is the compressive load at point D on link 2, as shown in Fig. 11.35. Therefore, rearranging Eq. (5), the force, P_A^y, is

$$P_A^y = \frac{P_{\text{cr}}}{N} = \frac{110.35\text{ kN}}{2} = 55.175\text{ kN}. \tag{6}$$

Finally, substituting Eq. (6) into Eq. (4) gives

$$P = \frac{P_A^y}{2} = \frac{55.175\text{ kN}}{2} = 27.588\text{ kN}. \qquad \textit{Ans.}$$

(d) Link 2 is a solid circular column with a factor of safety $N = 2$ and $P = 70$ kN. Rearranging Eq. (4), the compressive force is

$$P_A^y = 2P = 2\,(70\text{ kN}) = 140\text{ kN}.$$

Rearranging Eq. (5), the critical load applied on link 2 is

$$P_{\text{cr}} = P_A^y \cdot N = 140\text{ kN} \cdot 2 = 280\text{ kN}.$$

First, assume the link is an Euler column. Then from Euler's column formula, Eq. (11.34), the critical load applied on link 2 is

$$P_{\text{cr}} = A\left[\frac{C\pi^2 E}{S_{\text{r}}^2}\right]. \tag{7}$$

The cross-sectional area of link 2 is

$$A = \frac{\pi}{4}\left(D^2\right) = \frac{\pi}{4}\,(0.040\text{ m})^2 = 0.001\ 26\text{ m}^2.$$

The second moment of area of the cross section of link 2 is

$$I = \frac{\pi}{64}\left(D^4\right) = \frac{\pi}{64}\,(0.040\text{ m})^4 = 0.126 \times 10^{-6}\text{ m}^4.$$

The radius of gyration of the cross section of link 2 is

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{0.126 \times 10^{-6}\text{ m}^4}{0.001\ 26\text{ m}^2}} = 0.010\text{ m}.$$

Rearranging Eq. (7), the slenderness ratio of link 2 is

$$(S_{\text{r}})_{\text{Euler}} = \sqrt{\frac{C\pi^2 EA}{P_{\text{cr}}}} = \sqrt{\frac{2\pi^2\left(207 \times 10^9\text{ Pa}\right)\left(0.001\ 26\text{ m}^2\right)}{280 \times 10^3\text{ N}}} = 135.6.$$

Next, assume the link is a Johnson column. The critical unit load can be written from Johnson's parabolic equation, Eq. (11.39), as

$$\frac{P_{\text{cr}}}{A} = \left[S_{yc} - \frac{1}{CE}\left(\frac{S_{yc}S_{\text{r}}}{2\pi}\right)^2\right]. \tag{8}$$

Rearranging Eq. (8), the slenderness ratio can be written as

$$(S_{\text{r}})_{\text{Johnson}} = \frac{2\pi}{S_{yc}}\sqrt{\left(S_{yc} - \frac{P_{\text{cr}}}{A}\right)CE}.$$

Substituting the known data into this equation gives

$$(S_r)_{\text{Johnson}} = \frac{2\pi}{(415 \times 10^6 \text{ Pa})}\sqrt{\left[(415 \times 10^6 \text{ Pa}) - \frac{280\,000 \text{ N}}{0.001\,26 \text{ m}^2}\right] 2\,(207 \times 10^9 \text{ Pa})}$$
$$= 135.3.$$

The slenderness ratio at the point of tangency can be written as

$$(S_r)_D = \pi\sqrt{\frac{2CE}{S_{yc}}}.$$

Since the material properties are the same, this gives the same result as Eq. (1), that is $(S_r)_D = 140.33$. Therefore, since $(S_r)_{\text{Johnson}} < (S_r)_{\text{Euler}} < (S_r)_D$, the link is a Johnson column. Using the slenderness ratio for the case of a Johnson column; that is, $S_r = (S_r)_{\text{Johnson}} = 135.3$ and rearranging Eq. (2), the maximum possible length of link 2 is

$$L = kS_r = (0.010 \text{ m})\,135.3 = 1.353 \text{ m}. \qquad \textit{Ans.}$$

EXAMPLE 11.12

A load P_C is acting on the horizontal link 2 at point C of a simple truss, as shown in Fig. 11.36. Link 2 is pinned to the ground at point O_2 and to the inclined link 3 at point B, whereas link 3 is pinned to the ground at point O_3. The lengths are $R_{CO_2} = 5$ ft, $R_{BO_2} = 4$ ft, and $R_{BO_3} = 5$ ft. The two links each have a solid circular cross section with a diameter of $D = 2$ in. Also, the two links are made from a steel alloy with a compressive yield strength $S_{yc} = 60$ kpsi, a tensile yield strength $S_{yt} = 50$ kpsi, and a modulus of elasticity $E = 30$ Mpsi.

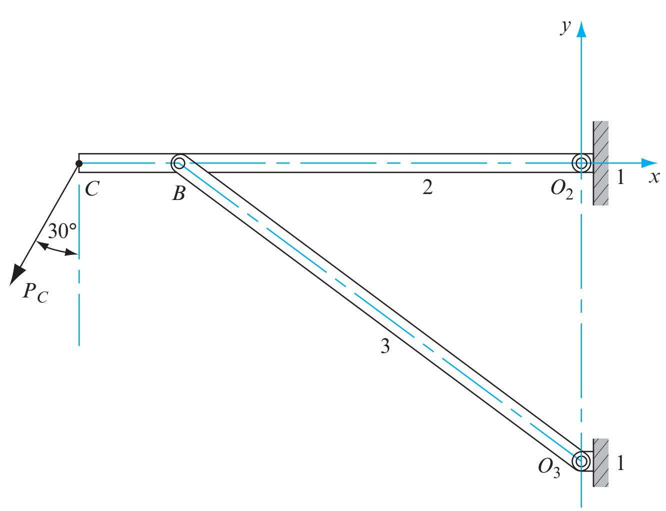


Figure 11.36 A simple pinned truss.

Part 1: Using the theoretic value for the end-condition constants for the two links, determine: (*a*) the value of the slenderness ratio at the point of tangency between Euler's column formula and Johnson's parabolic formula for link 3; (*b*) the critical load and the critical unit load acting on link 3; and (*c*) the maximum load P_C for the factor of safety to guard against buckling of link 3 to be $N = 1$.

Part 2: If the load acting at point C is specified as $P_C = 24\,000$ lb, determine: (*a*) the factor of safety for link 2; (*b*) the factor of safety for link 3; and (*c*) which link is most likely to fail first. Briefly explain the answer.

SOLUTION

Part 1.

(*a*) Links 2 and 3 are in static equilibrium. The free-body diagram of link 2 is shown in Fig. 11.37. Note that the link is in tension due to the external load P_C.

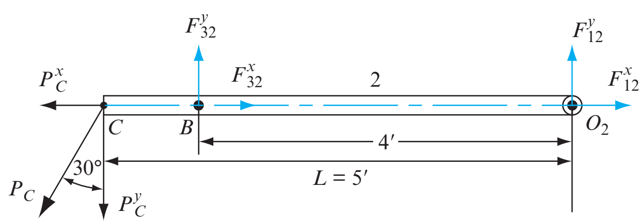


Figure 11.37 Free-body diagram of link 2.

Taking moments about the pin O_2 gives the y component of the load at pin B as

$$F^y_{32} = \frac{5\text{ ft}}{4\text{ ft}} P^y_C \quad \text{or} \quad F^y_{23} = -\frac{5}{4} P^y_C.$$

The free-body diagram of link 3 is as shown in Fig. 11.38.

Taking moments about the pin O_3 gives

$$F^x_{23} = -\frac{4\text{ ft}}{3\text{ ft}} F^y_{23} = -\frac{4}{3}\left(-\frac{5}{4} P^y_C\right) = \frac{5}{3} P^y_C.$$

The magnitude of the force at pin B can be written as

$$F_{23} = \sqrt{\left(\frac{5}{3} P^y_C\right)^2 + \left(-\frac{5}{4} P^y_C\right)^2} = 5P^y_C \sqrt{\frac{1}{9} + \frac{1}{16}} = 5P^y_C \sqrt{\frac{16+9}{144}} = \frac{25}{12} P^y_C. \tag{1}$$

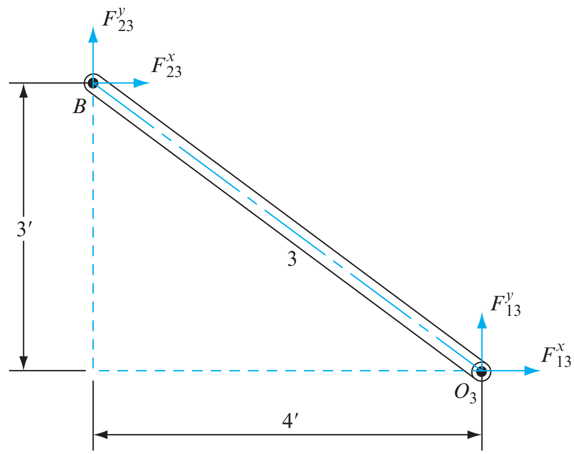


Figure 11.38 Free-body diagram of link 3.

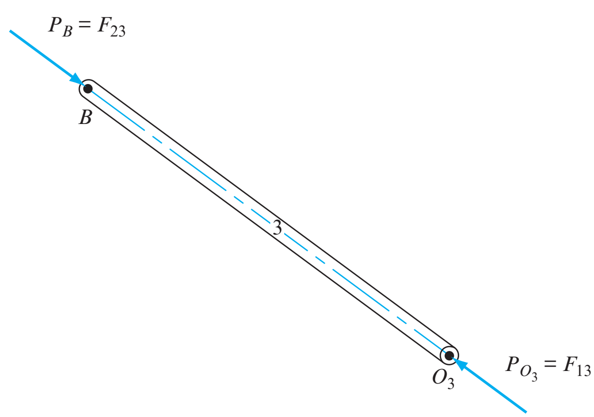


Figure 11.39 Link 3 is in compression.

The direction of this force at pin B, that is, the line of action, is

$$\angle F_{23} = \tan^{-1}\left(\frac{F_{23}^y}{F_{23}^x}\right) = \tan^{-1}\left(\frac{-5/4P_C^y}{5/3P_C^y}\right) = \tan^{-1}\left(\frac{-3}{4}\right) = -36.87^\circ.$$

Since link 3 is a two-force member, forces F_{23} and F_{13} must be equal, opposite, and collinear; that is, $F_{23} = F_{13}$, and link 3 is in compression because of the internal load P_B (Fig. 11.39). Therefore, it is possible that link 3 could fail because of buckling.

We must check link 3 for the factor of safety to guard against buckling to be $N = 1$. The cross-sectional area of link 3 is

$$A = \frac{\pi}{4}D^2 = \frac{\pi}{4}(2\text{ in})^2 = 3.1416\text{ in}^2$$

The second moment of area of link 3 can be written as

$$I = \frac{\pi}{64} D^4 = \frac{\pi}{64} (2 \text{ in})^4 = 0.785\,4 \text{ in}^4.$$

The radius of gyration of link 3 is

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{0.785\,4 \text{ in}^4}{3.141\,6 \text{ in}^2}} = 0.500 \text{ in}.$$

The slenderness ratio of link 3 is

$$S_r = \frac{L}{k} = \frac{5 \text{ ft} \cdot 12 \text{ in/ft}}{0.500 \text{ in}} = 120. \tag{2}$$

From the pinned-pinned end conditions, the end-condition constant for link 3 is $C = 1.0$. Therefore, the slenderness ratio of link 3 at the point of tangency is

$$(S_r)_D = \pi \sqrt{\frac{2CE}{S_{yc}}} = \pi \sqrt{\frac{2 \cdot 1.0\left(30 \times 10^6 \text{ psi}\right)}{60 \times 10^3 \text{ psi}}} = 99.35. \qquad \textit{Ans.} \ (3)$$

(*b*) To determine the critical load on link 3, we must first determine whether this link is an Euler column or a Johnson column. The criterion for using Johnson's parabolic equation is $S_r < (S_r)_D$. From Eqs. (2) and (3), we have $120 > 99.35$; that is, $S_r > (S_r)_D$. Therefore, link 3 is an Euler column. The critical load on link 3 is

$$P_{cr} = A\left[\frac{C\pi^2 E}{S_r^2}\right] = \left(3.141\,6 \text{ in}^2\right)\left[\frac{1.0\pi^2\left(30 \times 10^6 \text{ psi}\right)}{120^2}\right] = 64\,596 \text{ lb}. \qquad \textit{Ans.} \ (4)$$

The critical unit load is

$$\frac{P_{cr}}{A} = \frac{64\,596 \text{ lb}}{3.141\,6 \text{ in}^2} = 20\,562 \text{ lb/in}^2. \qquad \textit{Ans.}$$

(*c*) If we identify $P_B = F_{23}$ as the compressive load at point B on link 3 (Fig. 11.40), then, from the given factor of safety of $N = 1$ guarding against buckling for link 3, the applied load at pin B on link 3 can be written as

$$P_B = \frac{P_{cr}}{N} = \frac{P_{cr}}{1} = 64\,596 \text{ lb}. \tag{5}$$

Therefore, load $P_B (= F_{23})$ is equal to the critical load P_{cr}, as shown in Fig. 11.40. Note that a compressive load $P_B = 64\,596$ lb is the limiting load on link 3 to avoid buckling, with a factor of safety of $N = 1$.

Rearranging Eq. (1), the vertical component of the load at point C is

$$P_C^y = \frac{12}{25} F_{23} = \frac{12}{25} P_B. \tag{6}$$

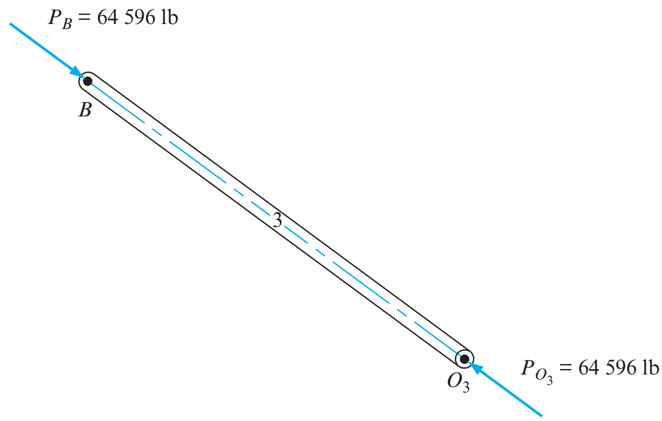


Figure 11.40 The compressive load acting on link 3.

Substituting Eq. (5) into Eq. (6), the vertical component of the load at point C is

$$P_C^y = \frac{12}{25}(64\ 596\ \text{lb}) = 31\ 006\ \text{lb}.$$

Finally, the load acting at point C is

$$P_C = \frac{P_C^y}{\cos 30°} = \frac{31\ 006\ \text{lb}}{0.866} = 35\ 803\ \text{lb}. \qquad \textit{Ans.}$$

Note that the load $P_C = 35\ 803$ lb is the load that ensures that the factor of safety to guard against buckling of link 3 is $N = 1$. A load $P_C > 35\ 803$ lb would cause link 3 to buckle; that is, the factor of safety would be $N < 1$.

Part 2.

(*a*) The x and y components of the load $P_C = 24\ 000$ lb are

$$P_C^x = P_C \sin 30° = (24\ 000\ \text{lb})\,0.500 = 12\ 000\ \text{lb},$$
$$P_C^y = P_C \cos 30° = (24\ 000\ \text{lb})\,0.866 = 20\ 785\ \text{lb}. \qquad (7)$$

Link 2 is subject to the tensile load P_C^x, which creates a tensile stress in the link. The factor of safety guarding against yielding of link 2 is defined as

$$N = \frac{S^{yt}}{\sigma},$$

where the normal stress due to the tensile load can be can be written as

$$\sigma = \frac{P_C^x}{A} = \frac{12\ 000\ \text{lb}}{3.141\ 6\ \text{in}^2} = 3\ 819.7\ \text{lb/in}^2.$$

Therefore, the factor of safety guarding against yielding of link 2 is

$$N = \frac{50\ 000 \text{ psi}}{3\ 819.7 \text{ psi}} = 13.1. \qquad \textit{Ans.}\ (8)$$

Note that this relatively high factor of safety indicates safety against tensile failure.

(*b*) Substituting Eq. (7) into Eq. (6), the compressive force F_{23} exerted on link 3 is

$$F_{23} = \frac{25}{12} P_C^y = \frac{25}{12} (20\ 785 \text{ lb}) = 43\ 302 \text{ lb}, \qquad (9)$$

where force F_{23} is equal to force P_B. For link 3, the factor of safety to guard against buckling can be calculated by substituting Eqs. (4) and (9) into Eq. (5); that is,

$$N = \frac{P_{\text{cr}}}{P_B} = \frac{P_{\text{cr}}}{F_{23}} = \frac{64\ 596 \text{ lb}}{43\ 302 \text{ lb}} = 1.49. \qquad \textit{Ans.}\ (10)$$

(*c*) Note that the factor of safety guarding against yielding of link 2 [the tensile member; Eq. (8)] is much higher than the factor of safety guarding against buckling of link 3 [the compressive member; Eq. (10)]. Therefore, the prediction is that link 3 will fail before link 2. *Ans.*

EXAMPLE 11.13

A vertically downward force $F = 15\ 000$ lb is applied at point B on member 2, which is pinned to the ground at point O_2 and to the vertical member 3 at point A, as shown in Fig. 11.41. The length of member 2, that is R_{BO_2}, is 60 in, and the ground pin O_3 is located at a distance $b = 30$ in along the x axis from the ground pin O_2. Members 2 and 3 are made from a steel alloy with a compressive yield strength $S_{yc} = 60$ kpsi and a modulus of elasticity $E = 30$ Mpsi. For buckling in the plane, member 3 behaves as a pinned-pinned member. Therefore, the theoretic end-condition constant is $C = 1$. For buckling out of the plane, member 3 behaves as a fixed-fixed member with a theoretic end-condition constant $C = 4$.

Part A: Assuming that the slenderness ratio of member 3 is such that Euler's column formula is valid, determine whether the following statements are true or false; give a brief explanation for each.

1. For members with a slenderness ratio greater than 10, it is safe to assume that the members designed for failure against yielding are automatically safe against buckling failure.
2. With all other design parameters unchanged, a member with fixed-free end conditions has a larger factor of safety guarding against buckling than the same member with pinned-pinned end conditions.
3. Two members with the same length, cross section, and modulus of elasticity, but with different compressive yield strengths, have the same critical buckling load.

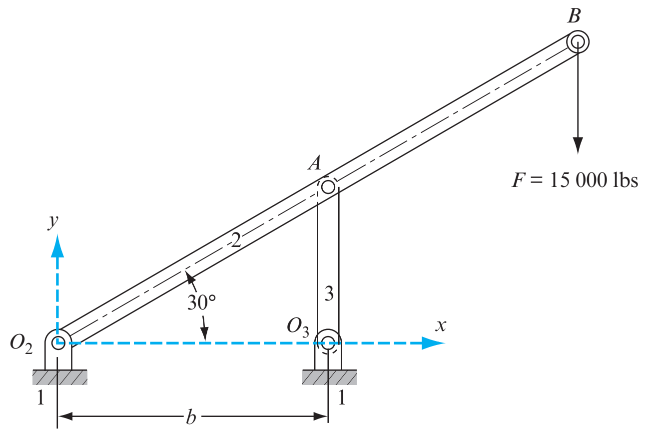


Figure 11.41 The structure for buckling analysis.

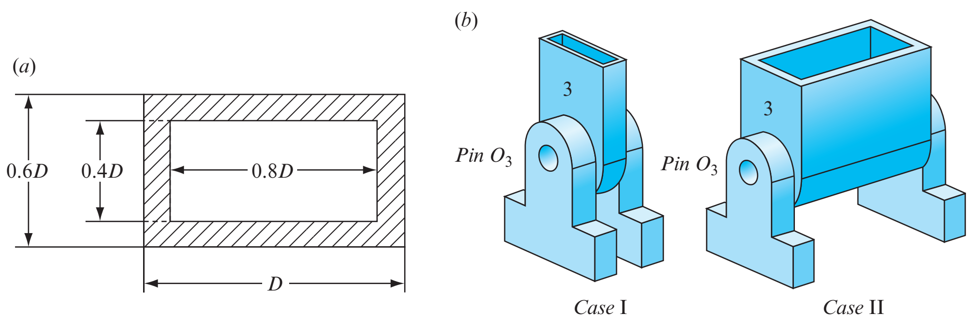


Figure 11.42 (*a*) Cross section of member 3; (*b*) Case I and Case II.

Part B: Assume that member 3 has a hollow rectangular cross section as shown in Fig. 11.42*a*. Initially, it is not known whether member 3 will be attached to the ground at pin O_3 as in either Case I or Case II shown in Fig. 11.42*b*. Using symbolic arguments (no numeric calculations are required), determine whether Case I or Case II leads to a higher factor of safety guarding against buckling. Assume the possibility of buckling failure in the plane of Fig. 11.41.

Part C: Assuming the worst case scenario, that is, the case with the lower factor of safety guarding against buckling, design the cross section of member 3, that is, determine the dimension, D, such that member 3 has a factor of safety guarding against buckling failure of $N = 2$. Note that designing the cross section for the case with the lower factor of safety automatically ensures a safe design for the case with the higher factor of safety.

Part D: Determine the factor of safety of member 3 guarding against buckling failure. Use the dimensions of the rectangular cross section obtained in Part C. Assume that member 3 is attached as in Case I and that there is the possibility of buckling out of the plane.

Part E: Assume that member 3 is attached as shown in Case II and that there is the possibility of buckling in the plane. The problem is to determine a new value for b; that is, to change the location of pin O_3. Distance b cannot be less than 5 in and cannot be greater than 45 in. Using the dimensions of the rectangular cross section obtained in Part C, determine an expression for the factor of safety guarding against buckling failure as a function of distance b. Then, plot a graph of N versus b. Using this graph, determine the minimum value of b above which the member is safe against buckling failure.

SOLUTION

Part A:

1. For members with a slenderness ratio greater than 10, it is safe to assume that members designed for failure against yielding are automatically safe against buckling failure.
   FALSE. *Ans.*

2. With all other design parameters unchanged, a member with fixed-free end conditions has a larger factor of safety than the same member with pinned-pinned end conditions.
   FALSE. *Ans.*

3. Two members with the same length, cross section, and modulus of elasticity but with different compressive yield strengths have the same value of critical buckling load.
   TRUE. *Ans.*

Part B:

For Case I, the second moment of area for member 3 can be written as

$$I_{\text{Case I}} = \frac{0.6D\,(D)^3}{12} - \frac{0.4D\,(0.8D)^3}{12} = \frac{0.395\;2D^4}{12} = 0.032\;93D^4.$$

For Case II, the second moment of area for member 3 can be written as

$$I_{\text{Case II}} = \frac{D\,(0.6D)^3}{12} - \frac{0.8D\,(0.4D)^3}{12} = \frac{0.164\;8D^4}{12} = 0.013\;73D^4. \tag{1}$$

Therefore, the second moment of area of Case I is greater than the second moment of area of Case II; that is,

$$I_{\text{Case I}} > I_{\text{Case II}}. \tag{2}$$

Since the cross-sectional area is the same in both cases, Eq. (2) implies that the radius of gyration of Case I is greater than the radius of gyration of Case II; that is,

$$k_{\text{Case I}} > k_{\text{Case II}}. \tag{3}$$

Since the length of the member is the same in both cases, Eq. (3) implies that the slenderness ratio in Case I is less than the slenderness ratio in Case II; that is,

$$(S_r)_{\text{Case I}} < (S_r)_{\text{Case II}}.$$

Therefore, for either Euler's column formula or Johnson's parabolic equation, the critical load of Case I is greater than the critical load of Case II, that is,

$$(P_{cr})_{\text{Case I}} > (P_{cr})_{\text{Case II}}.$$

Since the applied load is the same in both cases, the factor of safety guarding against buckling for Case I is greater than the factor of safety guarding against buckling for Case II, that is,

$$(N)_{\text{Case I}} > (N)_{\text{Case II}}. \qquad \textit{Ans.}$$

This implies that, if the cross-sectional area of the member is designed for Case II, then the member will automatically be safe against buckling for Case I.

Part C:

The cross-sectional area of member 3, that is, a hollow rectangular cross section, can be written as

$$A = D(0.6D) - 0.8D(0.4D) = 0.28D^2.$$

From Eq. (1), the second moment of area for member 3 can be written as $I = 0.013\,73D^4$. The radius of gyration of member 3 can be written as

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{0.013\,73D^4}{0.28D^2}} = 0.221\,5D. \tag{4}$$

Since $b = 30$ in, the length of member 3 is

$$L = b\tan 30° = 17.321 \text{ in}.$$

Therefore, the slenderness ratio of the member is

$$S_r = \frac{L}{k} = \frac{17.321 \text{ in}}{0.221\,5D} = \frac{78.199 \text{ in}}{D}.$$

The slenderness ratio at the point of tangency can be written as

$$(S_r)_D = \pi\sqrt{\frac{2CE}{S_{yc}}}.$$

From the specified fixed-fixed end conditions, the end-condition constant is $C = 1$. Therefore, the slenderness ratio at the point of tangency is

$$(S_r)_D = \pi\sqrt{\frac{2 \cdot 1\left(30 \times 10^6 \text{ psi}\right)}{60 \times 10^3 \text{ psi}}} = 99.3.$$

To determine the applied force at point A, we take moments about point O_2, which can be written as

$$(60\text{ in})\cos 30^\circ F = bP_{\text{app}}.$$

Therefore, the applied force at point A is

$$P_{\text{app}} = \frac{(60\text{ in})\cos 30^\circ\,(15\,000\text{ lb})}{30\text{ in}} = 25\,981\text{ lb}.$$

The factor of safety guarding against buckling is defined as

$$N = \frac{P_{\text{cr}}}{P_{\text{app}}}.$$

Since $N = 2$, the critical load is

$$P_{\text{cr}} = P_{\text{app}}N = (25\,981\text{ lb})\,2 = 51\,962\text{ lb}.$$

Note: At this point, it is not known whether Euler's column formula or Johnson's parabolic equation is the proper equation to determine the dimensions of the cross section of the member. If we assume that Johnson's parabolic equation is the proper equation, then the critical load is

$$P_{\text{cr}} = A\left[S_{yc} - \frac{1}{CE}\left(\frac{S_{yc}S_{\text{r}}}{2\pi}\right)^2\right].$$

Substituting the known values into this equation gives

$$51\,962\text{ lb} = 0.28D^2\left[\left(60\times 10^3\text{ psi}\right) - \frac{1}{1\left(30\times 10^6\text{ psi}\right)}\left(\frac{\left(60\times 10^3\text{ psi}\right)78.199}{2\pi D}\right)^2\right].$$

Rearranging this equation gives

$$(16\,800\text{ psi})\,D^2 = 51\,962\text{ lb} + 5\,204\text{ lb} = 57\,166\text{ lb}.$$

Solving this equation, dimension D is

$$D = 1.845\text{ in}. \qquad \textit{Ans.}$$

As a check, using Eq. (4) the slenderness ratio is

$$S_{\text{r}} = \frac{L}{k} = \frac{17.321\text{ in}}{0.221\,5\,(1.845\text{ in})} = 42.384 < (S_{\text{r}})_D = 99.3.$$

Therefore, Johnson's parabolic equation is proper for this case.

Part D:

Since the dimensions of the rectangular cross section are now known, the cross-sectional area of member 3 can be written as

$$A = 0.28D^2 = 0.953\ 1\ \text{in}^2.$$

From Eq. (1), the moment of inertia of the hollow rectangular member can be written as

$$I = 0.013\ 73D^4 = 0.159\ 1\ \text{in}^4.$$

From Eq. (4), the radius of gyration for the member can be written as

$$k = 0.221\ 5D = 0.408\ 7\ \text{in}.$$

Since $b = 30$ in, the length of the member is

$$L = b \tan 30° = 17.310\ \text{in}.$$

The slenderness ratio of the member is

$$S_\text{r} = \frac{L}{k} = \frac{17.310\ \text{in}}{0.408\ 7\ \text{in}} = 42.354.$$

Since the member is attached as shown in Case I, and there is the possibility of buckling out of the plane of Fig. 11.42, the end-condition constant is $C = 4$. Since $S_\text{r} < (S_\text{r})_D$, Johnson's parabolic equation is appropriate and the critical load is

$$P_\text{cr} = A\left[S_{yc} - \frac{1}{CE}\left(\frac{S_{yc}S_\text{r}}{2\pi}\right)^2\right],$$

which gives

$$P_\text{cr} = 0.953\ 1\ \text{in}^2\left[60\ 000\ \text{psi} - \frac{1}{4\left(30 \times 10^6\ \text{psi}\right)}\left(\frac{60\ 000\ \text{psi} \cdot 42.353}{2\pi}\right)^2\right]$$

$$= 55\ 886\ \text{lb}.$$

Since the applied load $P_\text{app} = 25\ 981$ lb, the factor of safety is

$$N = \frac{55\ 886\ \text{lb}}{25\ 981\ \text{lb}} = 2.15. \qquad \textit{Ans.}$$

Part E:

Let distance b between pins O_2 and O_3 be varied between 5 in and 45 in. The goal is to determine the minimum value of b such that the member does not fail in buckling. Assume that member 3 is located at an unknown distance b from the pin O_2. Then, force P_app acting at point A can be obtained by taking moments about O_2, that is,

$$P_\text{app} = \frac{(60\ \text{in} \cos 30°)\ 15\ 000\ \text{lb}}{b} = \frac{779\ 420\ \text{in} \cdot \text{lb}}{b}. \qquad (5)$$

As in Part D, the cross-sectional area of member 3 is $A = 0.953\ 1\ \text{in}^2$, the moment of inertia is $I = 0.159\ 1\ \text{in}^4$, and the radius of gyration of the member is $k = 0.408\ 7$ in.

The length of the member is

$$L = b \tan 30^\circ = 0.577\ 4b.$$

The slenderness ratio of the member is

$$S_r = \frac{L}{k} = \frac{0.577\ 4b}{0.408\ 7\ \text{in}} = \left(1.413\ \text{in}^{-1}\right) b.$$

Note: Since b can vary between 5 in and 45 in, the maximum value of the slenderness ratio of the member is $S_r = 63.58$, which is still less than $(S_r)_D = 99.3$. This implies that Johnson's parabolic equation is valid for all possible values of b. Therefore, the critical load can be written as

$$P_{cr} = A\left[S_{yc} - \frac{1}{CE}\left(\frac{S_{yc} S_r}{2\pi}\right)^2\right].$$

Substituting the known data into this equation, the critical load can be written as

$$P_{cr} = 0.953\ 1\ \text{in}^2 \left[\left(60 \times 10^3\ \text{psi}\right) - \frac{1}{1\left(30 \times 10^6\ \text{psi}\right)} \left(\frac{\left(60 \times 10^3\ \text{psi}\right)\left(1.413\ \text{in}^{-1}\right) b}{2\pi}\right)^2\right]$$

or as

$$P_{cr} = 57\ 186\ \text{lb} - (5.784\ \text{psi})\, b^2. \tag{6}$$

The factor of safety guarding against buckling is defined as

$$N = \frac{P_{cr}}{P_{app}}. \tag{7}$$

Substituting Eqs. (5) and (6) into Eq. (7), the factor of safety guarding against buckling can be written as

$$N = \frac{57\ 186\ \text{lb} - (5.784\ \text{psi})\, b^2}{779\ 420\ \text{in} \cdot \text{lb}/b} = \frac{(57\ 186\ \text{lb})\, b - (5.784\ \text{psi})\, b^3}{779\ 420\ \text{in} \cdot \text{lb}}$$

or as

$$N = \left(0.073\ 37\ \text{in}^{-1}\right) b - \left(7.421 \times 10^{-6}\ \text{in}^{-3}\right) b^3.$$

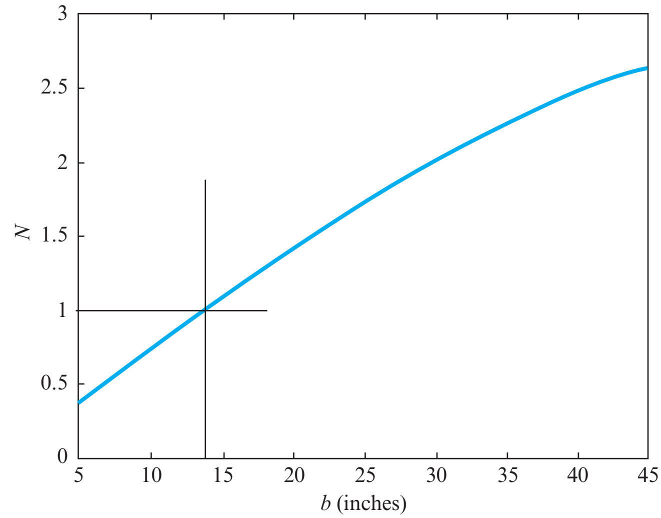


Figure 11.43 Plot of the factor of safety N versus dimension b.

The member will fail in buckling for all values of b for which the value of N is less than 1. A plot of N versus b is shown in Fig. 11.43. We observe from this plot that the factor of safety guarding against buckling is less than 1 for values of b less than 13.9 in. This implies failure in the buckling mode.

EXAMPLE 11.14

A solid aluminum member is subjected to concentric axial load $P = 3$ kN, as shown in Fig. 11.44. The length of the member is $L = 180$ mm, and the cross section is rectangular with dimensions $b = 20$ mm and $h = 4$ mm. The aluminum has a modulus of elasticity $E = 75$ GPa and a compressive yield strength $S_{yc} = 150$ MPa. Use the AISC recommended end-condition constant for fixed-pinned end conditions.

(*a*) Is the member an Euler column or a Johnson column?
(*b*) Determine the maximum load, P_{cr}, that can be applied before buckling occurs.
(*c*) Determine the factor of safety guarding against buckling of the member.

SOLUTION

The cross-sectional area of the member is

$$A = bh = (0.020\ \mathrm{m})\,(0.004\ \mathrm{m}) = 0.000\,080\ \mathrm{m}^2.$$

The second moment of area (Table 4, Appendix A) is

$$I = \frac{bh^3}{12} = \frac{(0.020\ \mathrm{m})\,(0.004\ \mathrm{m})^3}{12} = 1.067 \times 10^{-10}\mathrm{m}^4.$$

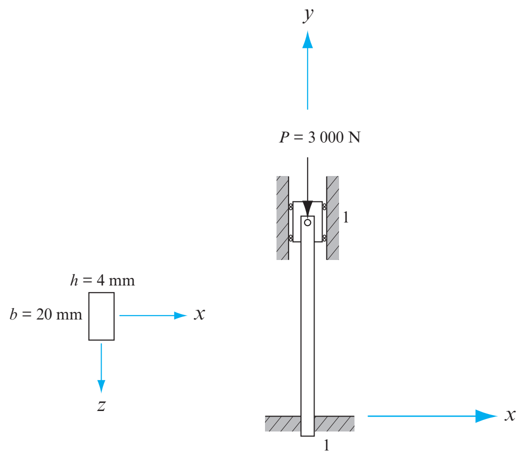


Figure 11.44 A pinned-fixed member.

Note here that the second moment of area has the smallest value allowable, since the member will buckle in the weakest plane. For this example, the member will buckle in the plane of Fig. 11.44.

The radius of gyration of the member is

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{1.067 \times 10^{-10}\ \text{m}^4}{0.000\,080\ \text{m}^2}} = 1.155\ \text{mm}.$$

Using the AISC recommended end-condition effective length factor for fixed-pinned end conditions (that is, $\alpha = L_{\text{eff}}/L = 0.707$, Fig. 11.29), the end-condition constant is

$$C = \frac{1}{\alpha^2} = \frac{1}{0.707^2} = 2.$$

(*a*) The slenderness ratio of the member is

$$S_{\text{r}} = \frac{L}{k} = \frac{0.180\ \text{m}}{0.001\,155\ \text{m}} = 155.84.$$

The slenderness ratio at the point of tangency is

$$(S_{\text{r}})_D = \pi\sqrt{\frac{2EC}{S_{yc}}} = \pi\sqrt{\frac{2\left(75 \times 10^9\ \text{Pa}\right)2}{150 \times 10^6\ \text{Pa}}} = 140.5.$$

Since the slenderness ratio $S_{\text{r}} \geq (S_{\text{r}})_D$, the member is an Euler column. *Ans.*

(*b*) Using Euler's column formula, the critical load is

$$P_{\text{cr}} = A\left[\frac{C\pi^2 E}{S_{\text{r}}^2}\right] = \left(0.000\,080\ \text{m}^2\right)\left[\frac{2\pi^2\left(75 \times 10^9\ \text{Pa}\right)}{155.84^2}\right] = 4\,877\ \text{N}. \quad \textit{Ans.}$$

(*c*) The factor of safety guarding against buckling is

$$N = \frac{P_{\mathrm{cr}}}{P} = \frac{4\,877\text{ N}}{3\,000\text{ N}}1.63. \qquad \textit{Ans.}$$

Therefore, the member should not fail in buckling.

EXAMPLE 11.15

For the simple truss shown in Fig. 11.45, a vertical load $P = 10$ kN is acting at point A in the negative y direction on the horizontal member 2, which is pinned to the vertical wall at point D. Assume that member 3 is pinned to member 2 at point B and also to the vertical wall at point C. Use the theoretic value for the end-condition constant of member 3, and assume that this member is steel with a yield strength $S_{yc} = 370$ MPa and a modulus of elasticity $E = 207$ GPa.

(*a*) Assume that member 3 has a solid circular cross section with a constant diameter d. Determine the diameter d to ensure that the member has a factor of safety guarding against buckling of $N = 2$. Does Euler's column formula or Johnson's parabolic equation give the correct diameter?

(*b*) Assume that member 3 has a square cross section with a constant width t. Determine the width, t, to ensure that the member has a factor of safety guarding against buckling of $N = 2$. Does Euler's column formula or Johnson's parabolic equation give the correct width?

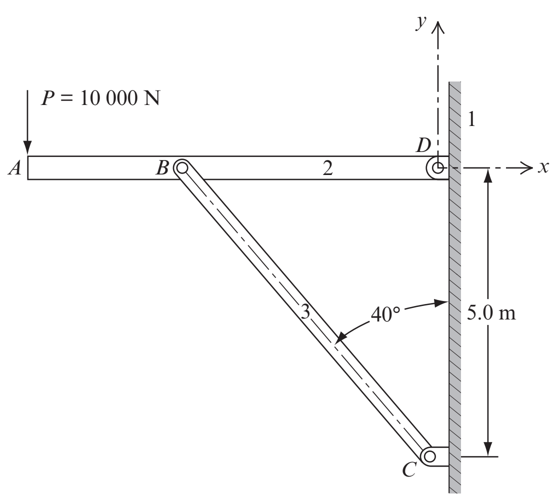


Figure 11.45 A simple truss subjected to a vertical load P.

SOLUTION

First, determine the magnitude of the compressive load that is acting on member 3. The unknown lengths can be determined from the given geometry. The length of member 3 can be obtained from the trigonometric relation

$$L_3 = 5.0 \text{ m}/\cos 40^\circ = 6.527 \text{ m}.$$

The distance from point B to point D is

$$R_{BD} = (5.0 \text{ m}) \tan 40^\circ = 4.196 \text{ m}.$$

The force acting on member 3 can be obtained by summing moments about pin D; that is,

$$(7.0 \text{ m})\,(10\,000 \text{ N}) - (4.196 \text{ m})\, F_B^y = 0.$$

Therefore, the vertical force acting on member 3 is

$$F_B^y = 16\,683 \text{ N}.$$

The total compressive force P acting on member 3 can then be obtained from the geometry; that is,

$$P = \frac{16\,683 \text{ N}}{\cos 40^\circ} = 21\,778 \text{ N}.$$

(a) Member 3 has a circular cross section with constant diameter d. Therefore, from Table 4 of Appendix A, the cross-sectional area and second moment of area of the member are

$$A = \frac{\pi d^2}{4} \quad \text{and} \quad I = \frac{\pi d^4}{64}.$$

The radius of gyration of the member is

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{\pi d^4/64}{\pi d^2/4}} = \frac{d}{4}.$$

The end conditions of member 3 are pinned-pinned. Using the theoretic value for the end-condition constant (that is, $C = 1$ or $\alpha = 1$) gives $L_{\text{eff}} = L$. Therefore, the slenderness ratio of member 3 is

$$S_r = \frac{L_{\text{eff}}}{k} = \frac{6.527 \text{ m}}{d/4} = \frac{26.108 \text{ m}}{d}.$$

The factor of safety guarding against buckling is defined as

$$N = \frac{P_{\text{cr}}}{P}.$$

Rearranging this equation gives the critical load as

$$P_{\mathrm{cr}} = NP = 2\,(21\ 778\ \mathrm{N}) = 43\ 556\ \mathrm{N}.$$

The slenderness ratio at the point of tangency is

$$(S_{\mathrm{r}})_D = \pi\sqrt{\frac{2E}{S_{yc}}} = \pi\sqrt{\frac{2\left(207\times10^9\ \mathrm{Pa}\right)}{370\times10^6\ \mathrm{Pa}}} = 105.1.$$

The diameter d can be obtained from Euler's column formula; that is,

$$\frac{P_{\mathrm{cr}}}{A} = \frac{\pi^2 E}{S_{\mathrm{r}}^2}.$$

Substituting known values gives

$$\frac{43\ 556\ \mathrm{N}}{\pi d^2/4} = \frac{\pi^2\left(207\times10^9\ \mathrm{Pa}\right)}{(26.108\ \mathrm{m}/d)^2}.$$

Solving for diameter d gives

$$d = 0.066\ \mathrm{m}. \tag{1}$$

Check: Diameter d can also be obtained from Johnson's parabolic equation; that is,

$$\frac{P_{\mathrm{cr}}}{A} = S_{yc} - \frac{1}{E}\left(\frac{S_{yc}S_{\mathrm{r}}}{2\pi}\right)^2. \tag{2}$$

Substituting the known values into Eq. (2) gives

$$\frac{43\ 556\ \mathrm{N}}{\left(\pi d^2/4\right)} = \left(370\times10^6\ \mathrm{Pa}\right) - \frac{1}{207\times10^9\ \mathrm{Pa}}\left(\frac{\left(370\times10^6\ \mathrm{Pa}\right)(26.108\ \mathrm{m}/d)}{2\pi}\right)^2.$$

Solving this for diameter d gives

$$d = 0.176\ \mathrm{m}. \tag{3}$$

Now we must check which answer is valid; that is, Eq. (1) or (3). We recall that the slenderness ratio is

$$S_{\mathrm{r}} = \frac{26.108\ \mathrm{m}}{d}.$$

First, we check to determine whether Euler's column result, Eq. (1), is valid. The slenderness ratio is

$$(S_{\mathrm{r}})_{\mathrm{Euler}} = \frac{26.108\ \mathrm{m}}{0.066\ \mathrm{m}} = 395.6.$$

Since the slenderness ratio $(S_{\mathrm{r}})_{\mathrm{Euler}}$ is greater than $(S_{\mathrm{r}})_D$, that is, since 395.6 is greater than 105.1, Euler's column formula gives a valid result. The diameter of the member is

$$d = 0.066 \text{ m}. \qquad \textit{Ans.}$$

Next we check to make sure that Johnson's parabolic equation result is not valid. The slenderness ratio from this result is

$$(S_{\mathrm{r}})_{\mathrm{Johnson}} = \frac{26.108 \text{ m}}{0.176 \text{ m}} = 148.3. \tag{11.42}$$

Since the slenderness ratio, $(S_{\mathrm{r}})_{\mathrm{Johnson}}$, is greater than $(S_{\mathrm{r}})_D$, that is, since 148.3 is greater than 105.1, Johnson's parabolic equation result is *not* valid.

(*b*) Member 3 has a square cross section with constant width t. Therefore, from Table 4, Appendix A, the cross-sectional area and second moment of area of the member are

$$A = t^2 \quad \text{and} \quad I = t^4/12.$$

The radius of gyration of the member is

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{t^4/12}{t^2}} = 0.289t.$$

The slenderness ratio of member 3 is

$$S_{\mathrm{r}} = \frac{L_{\mathrm{eff}}}{k} = \frac{6.527 \text{ m}}{0.289t} = \frac{22.6 \text{ m}}{t}.$$

First, t can be found using the Euler column formula; that is,

$$\frac{P_{\mathrm{cr}}}{A} = \frac{\pi^2 E}{S_{\mathrm{r}}^2}.$$

Substituting the known values into this equation gives

$$\frac{43\,556 \text{ N}}{t^2} = \frac{\pi^2 \left(207 \times 10^9 \text{ Pa}\right)}{(22.6 \text{ m}/t)^2}.$$

Solving for the width, t, gives

$$t = 0.057 \text{ m}.$$

The width, t, can also be obtained from the Johnson parabolic equation; that is,

$$\frac{P_{\mathrm{cr}}}{A} = S_{yc} - \frac{1}{E}\left(\frac{S_{yc} S_{\mathrm{r}}}{2\pi}\right)^2.$$

Substituting the known values into this equation gives

$$\frac{43\ 556\ \text{N}}{t^2} = \left(370 \times 10^6\ \text{Pa}\right) - \frac{1}{\left(207 \times 10^9\ \text{Pa}\right)} \left(\frac{\left(370 \times 10^6\ \text{Pa}\right) 22.6\ \text{m}/t}{2\pi}\right)^2.$$

Therefore, the width, t, is

$$t = 0.152\ \text{m}.$$

To check which answer is valid, we recall that the slenderness ratio is

$$S_r = \frac{22.6\ \text{m}}{t}.$$

First, check to see whether the use of Euler's column formula is valid:

$$(S_r)_{\text{Euler}} = \frac{22.6\ \text{m}}{0.057\ \text{m}} = 396.5.$$

Since the slenderness ratio, $(S_r)_{\text{Euler}}$, is greater than $(S_r)_D$, that is, since 396.5 is greater than 105.1, Euler's column formula gives a valid result. The correct width of the member is

$$t = 0.057\ \text{m}. \qquad \textit{Ans.}$$

Next we check to make sure that Johnson's parabolic equation result is not valid. The slenderness ratio from this result is

$$(S_r)_{\text{Johnson}} = \frac{22.6\ \text{m}}{0.152\ \text{m}} = 148.7.$$

Since the slenderness ratio, $(S_r)_{\text{Johnson}}$, is greater than $(S_r)_D$, that is, since 148.7 is greater than 105.1, use of Johnson's parabolic equation is *not* valid.

11.19 REFERENCES

[1] Beer, F. P., E. R. Johnston, and E. R. Eisenberg, 2007. *Vector Mechanics for Engineers: Statics*, 8th ed., New York: McGraw-Hill, 440–6.
[2] Plesha, M. E., G. L. Gray, and F. Costanzo, 2013. *Engineering Mechanics: Statics*, 2nd ed., New York: McGraw-Hill.
[3] Neale, M. J. (ed.), 1975. *Tribology Handbook*, London: Butterworths, C8.
[4] Newton, I., 1687. *Principia*, translated by A. Motte, 1729.
[5] Shigley, J. E., and C. R. Mischke, 2001. *Mechanical Engineering Design*, 6th ed., New York: McGraw-Hill.

PROBLEMS*

11.1 The figure shows four linkages and the external forces and torques exerted on or by the linkages. Sketch the free-body diagram of each part of each linkage. Do not attempt to show the magnitudes of the forces, except roughly, but do sketch them in their proper locations and orientations.

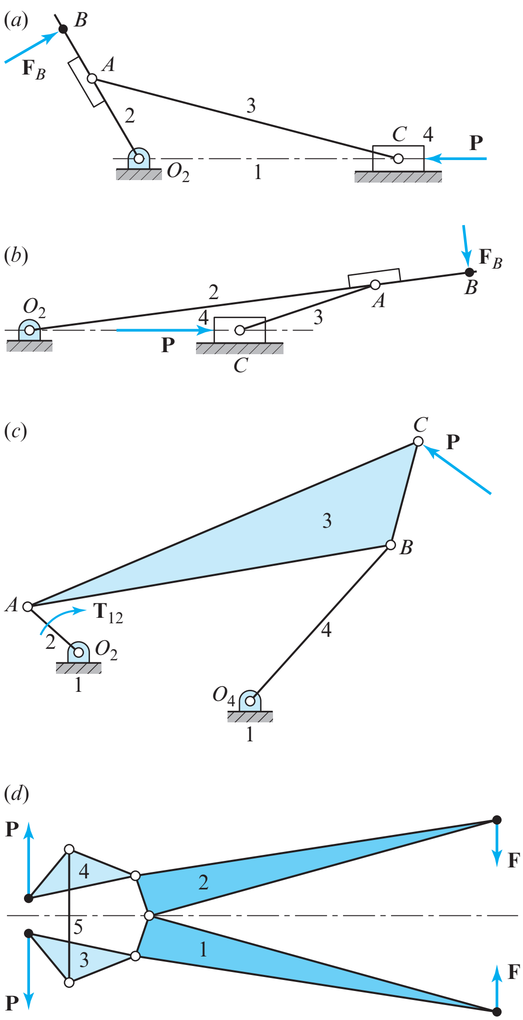


Figure P11.1

†11.2 If force $P = 0.9$ kN, determine the torque, $\mathbf{T}_{12}$, that must be applied to crank 2 to maintain the linkage in static equilibrium.

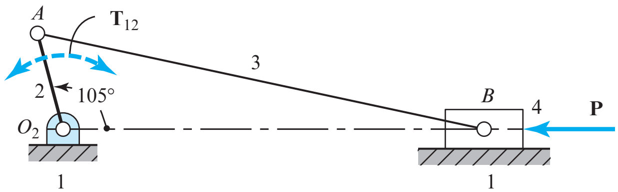


Figure P11.2 $R_{AO_2} = 75$ mm and $R_{BA} = 350$ mm.

†11.3 If $T_{12} = 100$ N · m cw for the linkage shown in Fig. P11.2, determine the force, $\mathbf{P}$, to maintain static equilibrium.

11.4 Determine the forces acting on the ground and the torque, $\mathbf{T}_{12}$, to maintain static equilibrium for the four-bar linkage shown in Fig. P11.4*a*.

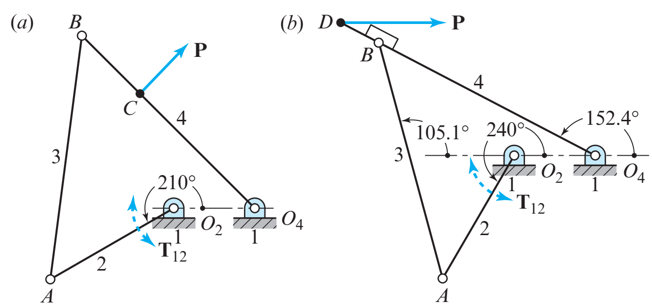


Figure P11.4 $R_{AO_2} = 3.5$ in, $R_{BA} = R_{BO_4} = 6$ in, $R_{CO_4} = 4$ in, $R_{DO_4} = 7$ in, and $R_{O_2O_4} = 2$ in.

11.5 What torque must be applied to link 2 of the linkage shown in Fig. P11.4*b* to maintain static equilibrium?

11.6 Sketch a complete free-body diagram of each link and determine the force, $\mathbf{P}$, to maintain static equilibrium.

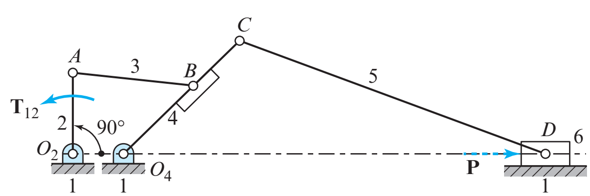


Figure P11.6 $R_{AO_2} = 100$ mm, $R_{BA} = 150$ mm, $R_{BO_4} = 125$ mm, $R_{CO_4} = 200$ mm, $R_{CD} = 400$ mm, and $R_{O_2O_4} = 60$ mm.

* Unless otherwise stated, solve all problems without friction and without gravitational loads.

11.7 Determine the torque, $\mathbf{T}_{12}$, required to drive slider 6 of Fig. P11.7 against a load of $P = 100$ lb at a crank angle of $\theta = 30°$, or as specified by your instructor.

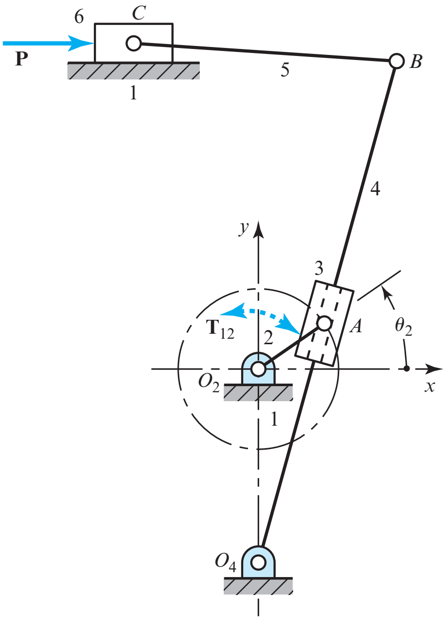


Figure P11.7 $R_{AO_2} = 2.5$ in, $R_{O_2O_4} = 6$ in, $R^y_{CO_2} =$ 10 in, $R_{BO_4} = 16$ in, and $R_{BC} = 8$ in.

†11.8 Sketch complete free-body diagrams of each link and determine the torque, $\mathbf{T}_{12}$, that must be applied to link 2 to maintain static equilibrium at the posture shown.

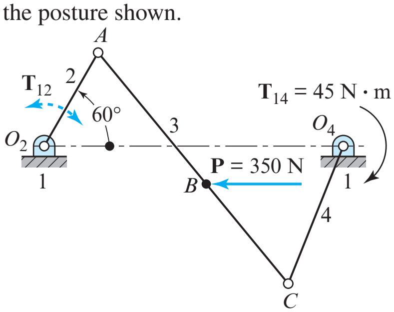


Figure P11.8 $R_{AO_2} = 200$ mm, $R_{BA} = 400$ mm, $R_{CA} = R_{O_4O_2} = 700$ mm, and $R_{CO_4} = 350$ mm.

11.9 Sketch free-body diagrams of each link, and show all the forces acting. Find the magnitude and direction of the torque that must be applied to link 2 at the posture shown to drive the linkage against the forces shown.

11.10 The figure shows a four-bar linkage with external forces applied at points B and C. Draw a free-body

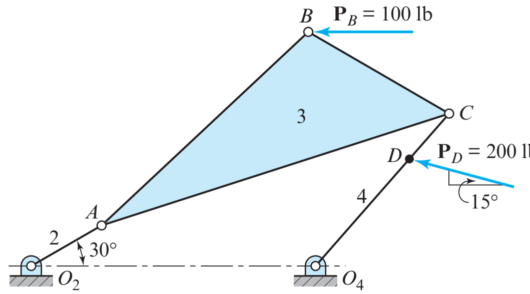


Figure P11.9 $R_{AO_2} = 4$ in, $R_{CA} = 14$ in, $R_{O_4O_2} = 14$ in, $R_{CO_4} = 10$ in, $R_{DO_4} = 7$ in, $R_{BA} = 14$ in, and $R_{BC} = 8$ in.

diagram of each link, and show all the forces acting on each. Find the torque that must be applied to link 2 to maintain static equilibrium at the posture shown.

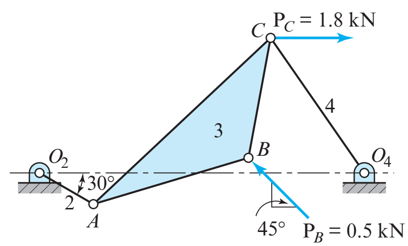


Figure P11.10 $R_{AO_2} = 75$ mm, $R_{CA} = 300$ mm, $R_{O_4O_2} = 400$ mm, $R_{CO_4} = R_{BA} = 200$ mm, and $R_{BC} = 150$ mm.

11.11 Draw a free-body diagram of each member of the linkage, and find the magnitudes and the directions of all forces and moments. Compute the magnitude and direction of the torque that must be applied to link 2 to maintain static equilibrium at the posture indicated.

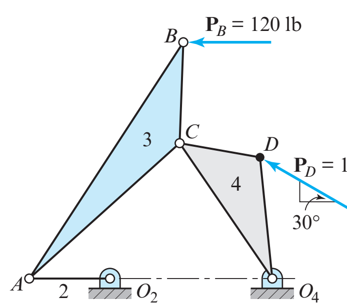


Figure P11.11 $R_{AO_2} = 4$ in, $R_{CA} = 10$ in, $R_{O_4O_2} = R_{CO_4} = 8$ in, $R_{DO_4} = 6$ in, $R_{DC} = 4$ in, $R_{BA} = 14$ in, and $R_{BC} = 5$ in.

11.12 Determine the magnitude and direction of the torque that must be applied to link 2 to maintain static equilibrium at the posture shown.

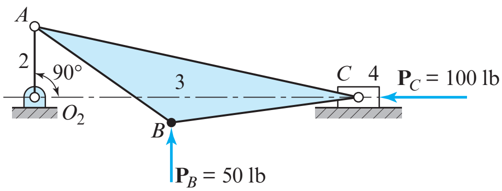


Figure P11.12 $R_{AO_2} = 3$ in, $R_{CA} = 14$ in, $R_{BA} =$ 7 in, and $R_{BC} = 8$ in.

11.13 Figure P11.13*a* shows a Figee floating crane with a lemniscate boom configuration, and Fig. P11.13*b* shows a schematic diagram of the crane with dimensions given in the legend. The lifting capacity is 16 T (where 1 T = 1 metric ton = 1 000 kg) including the grab, which is about 10 T. The maximum outreach is 30 m, which corresponds to the position $\theta_2 = 49°$. The minimum outreach is 10.5 m at $\theta_2 = 132°$. For the maximum outreach posture and a grab load of 10 T (under standard gravity), find the bearing reactions at A, B, O_2, and O_4, as well as the torque, $\mathbf{T}_{12}$, at O_2. Notice that the photograph shows a counterweight on link 2; neglect this weight and also the weights of the members.

11.14 Repeat Prob. 11.13 for the minimum outreach posture.

11.15 Repeat Prob. 11.7 assuming coefficients of Coulomb friction $\mu_c = 0.20$ between links 1 and 6 and $\mu_c = 0.10$ between links 3 and 4. Determine the torque, $\mathbf{T}_{12}$, necessary to drive the system, including friction, against the load, $\mathbf{P}$.

11.16 Repeat Prob. 11.12 assuming a coefficient of static friction $\mu = 0.15$ between links 1 and 4. Determine the torque, $\mathbf{T}_{12}$, necessary to overcome friction.

11.17 In each case shown, pinion 2 is the driver, gear 3 is an idler, and the gears have diametral pitch of 6 and 20° pressure angle. For each case, sketch the free-body diagram of gear 3 and show all forces acting. For (*a*), pinion 2 rotates at 600 rev/min and transmits 18 hp to the gearset. For (*b*) and (*c*), pinion 2 rotates at 900 rev/min and transmits 25 hp to the gearset.

(*a*)

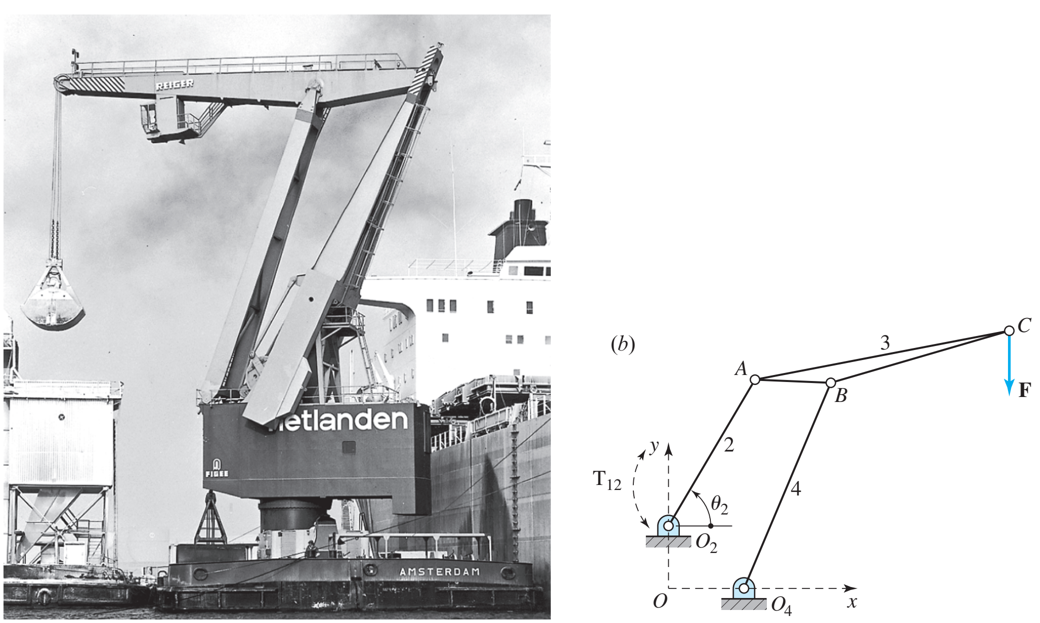


Figure P11.13 (*a*) Photograph; (*b*) $R_{AO_2} = 14.7$ m, $R_{BA} = 6.5$ m, $R_{BO_4} = 19.3$ m, $R_{CA} = 22.3$ m, $R_{CB} = 16$ m, and $\mathbf{R}_{O_2O_4} = -6.4\hat{\mathbf{i}} + 5.3\hat{\mathbf{j}}$ m. (Courtesy of B.V. Machinefabriek Figee, Haarlem, Holland).

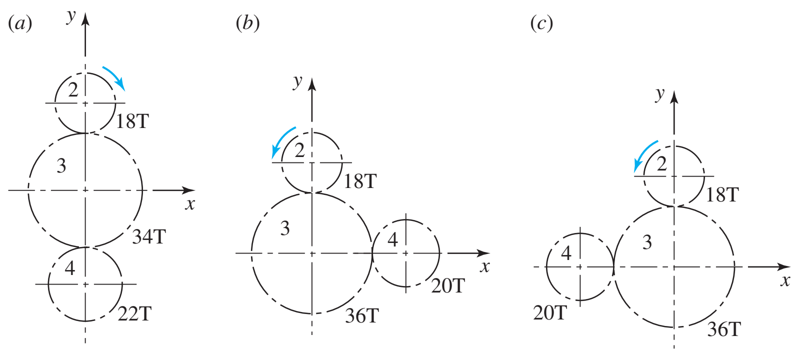


Figure P11.17

11.18 A 15-tooth spur pinion has a diametral pitch of 5 and 20° pressure angle, rotates at 600 rev/min, and drives a 60-tooth gear. The drive transmits 25 hp. Construct a free-body diagram of each gear showing upon it the tangential and radial components of the forces and their proper directions.

11.19 A 16-tooth pinion on shaft 2 rotates at 1 720 rev/min and transmits 5 hp to the double-reduction gear train. All gears have 20° pressure angle. Find the magnitude and direction of the radial force that each bearing exerts against the shaft.

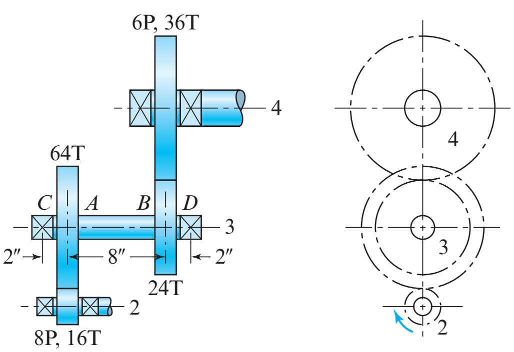


Figure P11.19

11.20 Solve Prob. 11.17 if each pinion has right-hand helical teeth with a 30° helix angle and a 20° pressure angle. All gears in the train are helical, and the normal diametral pitch is 6 teeth/in for each case.

11.21 Analyze the gear shaft of Example 11.8, and find the bearing reactions $\mathbf{F}_C$ and $\mathbf{F}_D$.

11.22 In each of the bevel gear drives shown, bearing A takes both thrust load and radial load, whereas bearing B takes only radial load. The teeth are cut with a 20° pressure angle. For (*a*) $\mathbf{T}_2 = -180\hat{\mathbf{i}}$ in · lb, and for (*b*) $\mathbf{T}_2 = -240\hat{\mathbf{k}}$ in · lb. Compute the bearing loads for each case.

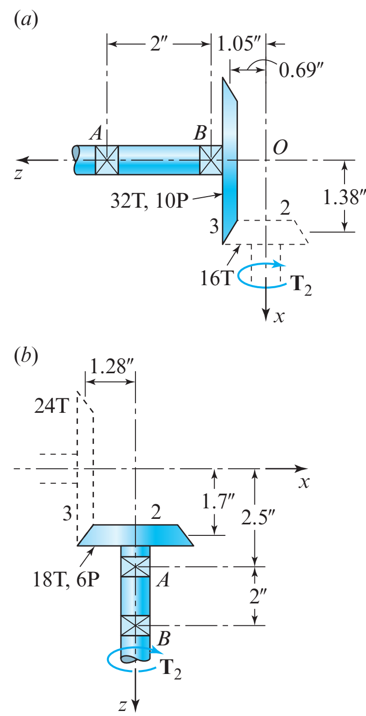


Figure P11.22

11.23 The figure shows a gear train composed of a pair of helical gears and a pair of straight-tooth bevel gears. Shaft 4 is the output of the train and delivers 6 hp to the load at a speed of 370 rev/min. All gears have pressure angles of 20°. If bearing E is to take both thrust load and radial load, whereas

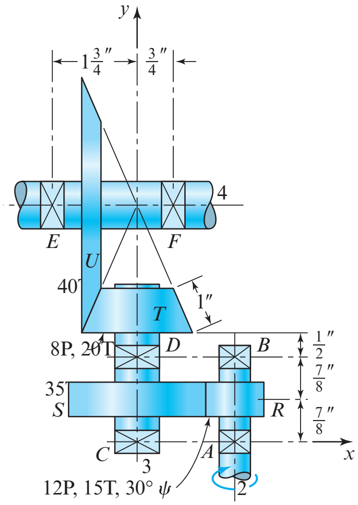


Figure P11.23

bearing F is to take only radial load, determine the force that each bearing exerts against shaft 4.

11.24 Using the data of Prob. 11.23, find the forces exerted by bearings C and D onto shaft 3. Which of these bearings should take the thrust load if the shaft is to be loaded in compression?

11.25 Use the method of virtual work to solve the slider-crank linkage of Prob. 11.2.

11.26 Use the method of virtual work to solve the four-bar linkage of Prob. 11.5.

11.27 Use the method of virtual work to analyze the crank-shaper linkage of Prob. 11.7. Given that the load remains constant at $\mathbf{P} = 100\hat{\mathbf{i}}$ lb, find and plot a graph of the crank torque, T_{12}, for all postures in the cycle using increments of 30° for the input crank.

11.28 Use the method of virtual work to solve the four-bar linkage of Prob. 11.10.

11.29 A car (link 2) that weighs 2 000 lb is slowly backing a 1 000 lb trailer (link 3) up a 30° inclined ramp. The car wheels are of 13-in radius, and the trailer wheels have 10-in radius; the center of the hitch ball is also 13 in above the roadway. The centers of mass of the car and trailer are located at G_2 and G_3, respectively, and gravity acts vertically downward. The weights of the wheels and friction in the bearings are considered negligible. Assume that there are no brakes applied on the car or on the trailer and that the car has front-wheel drive. Determine the loads on each of the wheels and the minimum coefficient of static friction between the driving wheels and the road to avoid slipping.

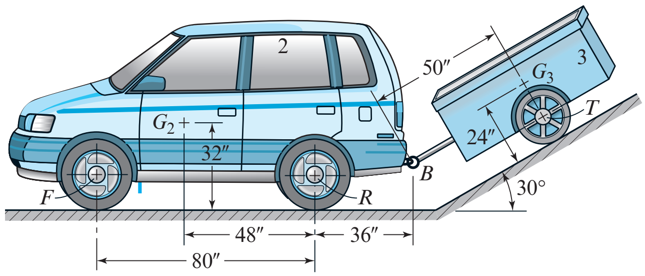


Figure P11.29

11.30 Repeat Prob. 11.29, assuming that the car has rear-wheel drive rather than front-wheel drive.

11.31 The low-speed disk cam with oscillating flat-faced follower is driven at a constant shaft speed. The displacement curve for the cam has a full-rise cycloidal motion, defined by Eq. (6.13) with parameters $L = 30°$, $\beta = 150°$, and a prime circle radius $R_0 = 30$ mm; the instant pictured is at $\theta_2 = 112.5°$. A force of $F_C = 8$ N is applied at point C and continues at 45° from the face of the follower as shown. Use the virtual-work approach to determine the torque, $\mathbf{T}_{12}$, required on the crankshaft at the instant shown to produce this motion.

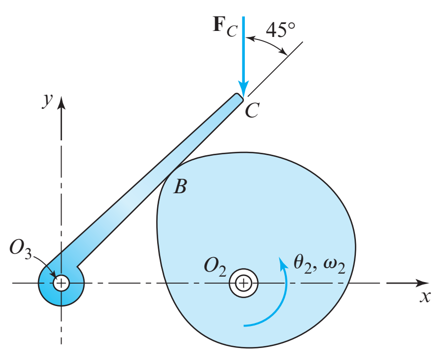


Figure P11.31 $R_{O_2O_3} = 50$ mm, $R_B = 42$ mm, and $R_C = 150$ mm.

11.32 Repeat Prob. 11.31 for the entire lift portion of the cycle, finding $\mathbf{T}_{12}$ as a function of θ_2.

11.33 Disk 3 of radius R is being slowly rolled under pivoted bar 2 driven by an applied torque, T. Assume a coefficient of static friction of μ between the disk and ground and that all other

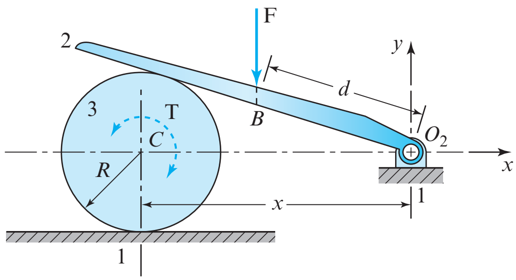


Figure P11.33

joints are frictionless. A force, **F**, is acting vertically downward on the bar at distance d from the pivot, O_2. Assume that the weights of the links are negligible in comparison to F. Find an equation for the torque, **T**, required as a function of distance $x = R_{CO_2}$, and an equation for the final distance, x, that is reached when friction no longer allows further movement.

11.34 For the linkage in the posture shown, an external torque $\mathbf{T}_{14} = -50\hat{\mathbf{k}}$ in · lb is acting on link 4 about O_4, and a horizontal force $\mathbf{F}_C$ is acting at point C on link 3 to hold the linkage in static equilibrium. Assume that gravity is acting into the plane, and the effects of friction can be neglected. (a) Draw free-body diagrams of links 2, 3, and 4. (b) Determine the magnitudes, directions, and locations of all internal reaction forces. (c) Determine the magnitude and direction of the external force acting at point C.

11.35 A horizontal force $F_C = 25$ N is acting at point C on link 4 and an external torque T_2 is acting on link 2. The coefficient of friction between link 4 and the ground link is $\mu = 0.3$, and the coefficient of friction between link 2 and the ground link is not specified, but is large enough that there is no slip at E. Assume that gravity is acting into the plane of the figure. Block 4 is 60 mm wide by 40 mm high, with pin B centrally located. Point C is 5 mm above the centerline. (a) Determine the magnitude and direction of the external torque, T_{12}, necessary to overcome the force, F_C. (b) Determine the magnitudes, the directions, and the locations of the internal reaction forces. (c) Determine whether link 4 is slipping or tipping on the ground link 1.

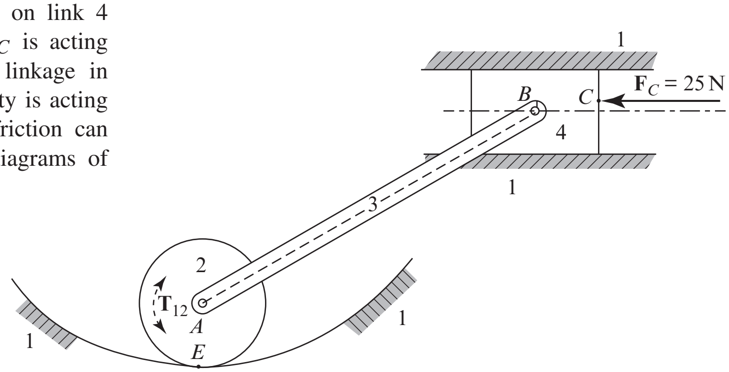


Figure P11.35 $\mathbf{R}_{BA} = 180$ mm $\angle 30°$ and $R_{EA} = 30$ mm.

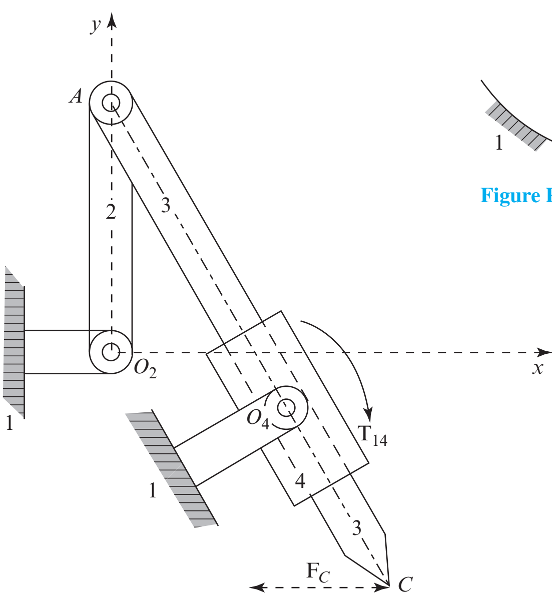


Figure P11.34 $\mathbf{R}_{O_4O_2} = 2\hat{\mathbf{i}} - 0.864\hat{\mathbf{j}}$ in, $R_{AO_2} = 2.6$ in, and $R_{CA} = 6$ in.

11.36 For the linkage in the posture shown, a constant external torque $\mathbf{T}_{12} = 180\hat{\mathbf{k}}$ in · lb is acting on link 2 about shaft O_2, and a horizontal external force $\mathbf{P}$ is acting on link 5 to hold the linkage in static equilibrium. Assume that gravity is acting into the plane and effects of friction in the linkage can be neglected. (*a*) Draw free-body diagrams for links 2, 3, 4, 5, and 6. (*b*) Determine the magnitudes, directions, and locations of the internal reaction forces. (*c*) Determine the magnitude and direction of the external force $\mathbf{P}$.

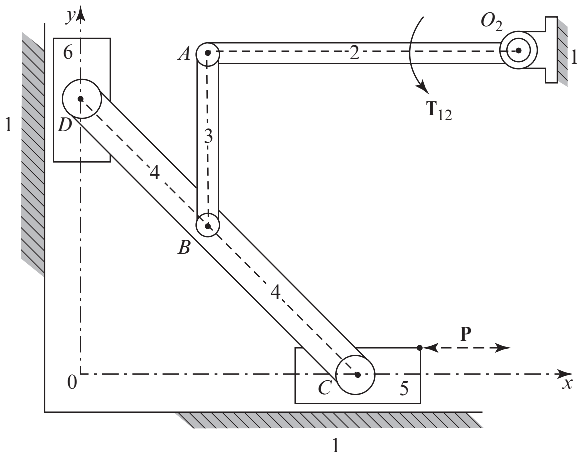


Figure P11.36 $\mathbf{R}_{O_2O} = 7.0\hat{\mathbf{i}} + 5.0\hat{\mathbf{j}}$ in, $R_{AO_2} =$ 5.0 in, $R_{BA} = 2.75$ in, $R_{CD} = 6.0$ in, and $R_{BD} = 2.818$ in. Blocks 5 and 6 are each 2 in by 1 in with the pins centrally located.

11.37 For the linkage in the posture shown, a torque $\mathbf{T}_{12} = 0.24\hat{\mathbf{k}}$ N · m is acting on link 2 at crankshaft O_2. A force $\mathbf{P}$ is applied to point D on link 4 at an angle of 45° to hold the linkage in static equilibrium. Gravity is acting into the plane and friction in the linkage can be neglected. Determine the magnitude and the direction of the force $\mathbf{P}$. Determine the magnitudes, directions, and locations of the internal reaction forces in the linkage. Is link 4 slipping or tipping on the ground link?

11.38 For the linkage in the posture shown, force $\mathbf{P} =$ 10 lb is applied on link 4. A torque $\mathbf{T}_{12}$ is acting on link 2 at the crankshaft O_2 to hold the linkage in static equilibrium. Gravity is acting into the plane and friction in the linkage can be neglected. Is the impending motion of link 3 slipping or tipping in link 4? Is the impending motion of link 4 slipping or tipping on the ground link? Determine the magnitudes, directions, and locations of the internal reaction forces in the linkage. Determine the magnitude and the direction of the torque, $\mathbf{T}_{12}$. $S_{yc} = 200$ MPa. Using the theoretic values for the end-condition constants of each link, determine: (*a*) the slenderness ratio, the critical load, and the factor of safety guarding against buckling of link 2; and (*b*) the minimum diameter D_{min} of link 3 if the static factor of safety guarding against buckling is to be $N = 2$.

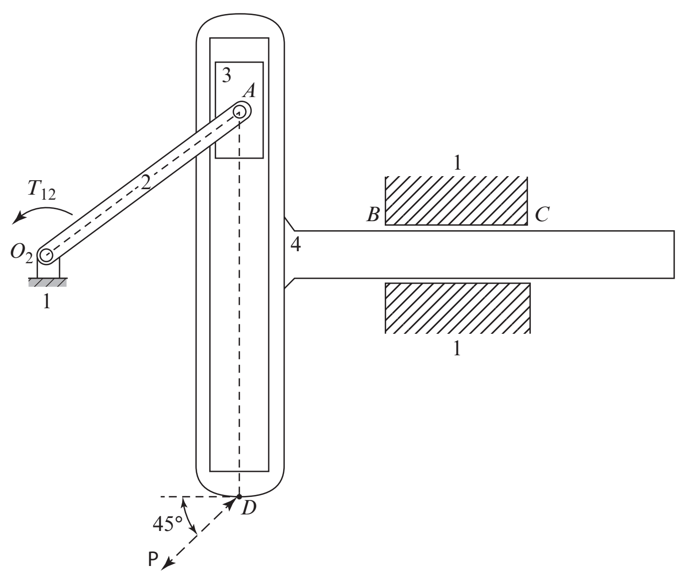


Figure P11.37 $\mathbf{R}_{AO_2} = 40\hat{\mathbf{i}} + 30\hat{\mathbf{j}}$ mm, $\mathbf{R}_{BO_2} = 70\hat{\mathbf{i}}$ mm, $\mathbf{R}_{DO_2} = 40\hat{\mathbf{i}} - 50\hat{\mathbf{j}}$ mm, and $\mathbf{R}_{CB} = 30\hat{\mathbf{i}}$ mm.

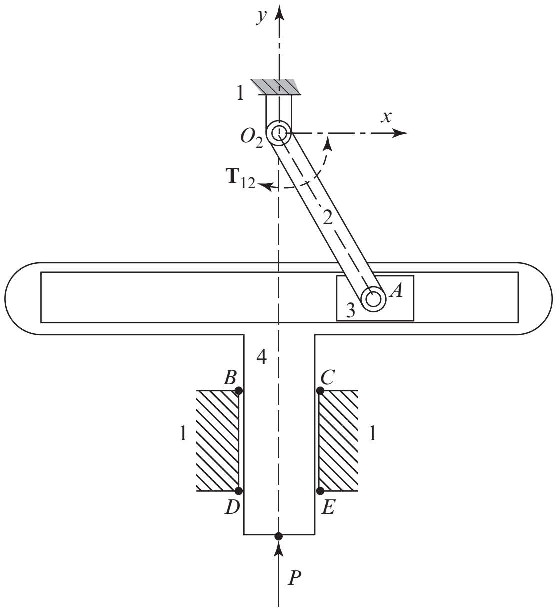


Figure P11.38 $\mathbf{R}_{AO_2} = 5.0 \text{ in}\angle -60°$, $R^y_{CO_2} = -6.6$ in, and $R^y_{EO_2} = -9.2$ in.

11.39 Links 2 and 3 are pinned together at B, and a constant vertical load $P = 800$ kN is applied at B. Link 2 is fixed in the ground at A, and link 3 is pinned to the ground at C. The length of link 2 is 8 m, and it has a 150-mm solid square cross section. The length of link 3 is 0.5 m, and it has a solid circular cross section with diameter D. Both links are made from a steel with a modulus of elasticity

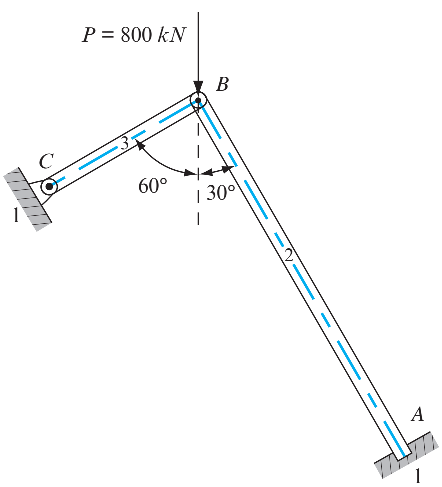


Figure P11.39 Two links, AB and BC, pinned at B.

$E = 207$ GPa and a compressive yield strength Using the theoretic values for the end condition constants of each link, determine: (*a*) the slenderness ratio, the critical load, and the factor of safety guarding against buckling of link 2 and (*b*) the minimum diameter of link 3 if the static factor of safety guarding against buckling is to be $N = 2$.

11.40 Horizontal link 2 is subjected to load $F = 150$ kN at C as shown. The link is supported by the solid circular aluminum link 3. The lengths of the links are $L_2 = R_{CA} = 5$ m, $R_{BA} = 3$ m, and $L_3 = R_{BD} = 3$ m. End D of link 3 is fixed in the ground, and the opposite end, B, is pinned to link 2 (that is, the effective length of the link is $L_{\text{eff}} = 0.5\,L_2$). For aluminum, the yield strength is $S_{yc} = 370$ MPa, and the modulus of elasticity is $E = 207$ GPa. Determine the diameter, d, of the solid circular cross section of link 3 to ensure that the static factor of safety is $N = 2.5$.

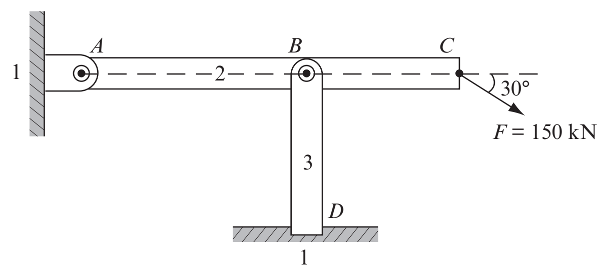


Figure P11.40 Link BD supporting link AC.

11.41 The horizontal link 2 is subjected to the inclined load $F = 8\,000$ N at C as shown. The link is supported by a solid circular cross section link 3 whose length $L_3 = BD = 5$ m. End D of vertical link 3 is fixed in the ground link, and end B supports link 2 (that is, the effective length of the link is $L_{\text{eff}} = 0.5\,L_3$). Link 3 is steel with a compressive yield strength $S_{yc} = 370$ MPa and a modulus of elasticity $E = 207$ GPa. Determine the diameter, d, of link 3 to ensure that the factor of safety guarding against buckling is $N = 2.5$. Also, answer the following statements true or false and briefly give your reasons. (*a*) The slenderness ratio at the point of tangency between Euler's column formula and Johnson's parabolic equation does not depend on the geometry of the column. (*b*) Under the same loading conditions, a link with pinned-pinned ends gives a higher factor of safety against buckling than an identical link with fixed-fixed ends. (*c*) If the slenderness ratio $S_r = (S_r)_D$ at the point of tangency, then the critical unit load does not depend on the yield strength of the column material.

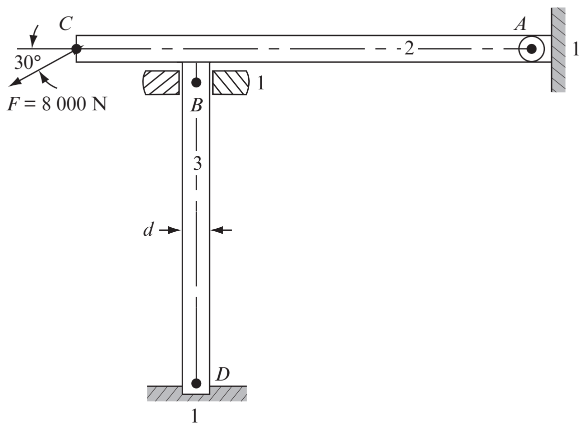


Figure P11.41 Link *AC* supported by link *BD*.

11.42 Load $\mathbf{P}_A$ is acting at *A*, and load $\mathbf{P}_B$ is acting at *B* of horizontal link 3. Link 3 is pinned to vertical link 2 at *O*, and link 2 is fixed in ground link 1 at *D*. The lengths are $AO = 4$ ft, $OB = 2$ ft, and $DO = 6$ ft. Both links have solid circular cross sections with diameter $D = 2$ in and are made from a steel alloy with compressive yield strength $S_{yc} = 85\ 000$ psi, tensile yield strength $S_{yt} = 75\ 000$ psi, and modulus of elasticity $E = 30 \times 10^6$psi. Assuming that links 2 and 3 are in static equilibrium and using the theoretic value for the end-condition constant for link 2, determine: (*a*) the magnitude of force $\mathbf{P}_B$ that is acting as shown at *B* if $P_A = 30\ 000$ lb; (*b*) the critical load, the critical unit load, and the factor of safety to guard against buckling for link 2; and (*c*) the diameter of a solid circular cross section for link 2 that will ensure the factor of safety guarding against buckling of the link is $N = 4$.

11.43 Horizontal link 2 is subjected to load $P = 5\ 000$ N and is supported by vertical link 3, which has a constant circular cross section. The lengths are $AC = 5\ AB = 4$m, and $DB = L_3 = 5$m. For vertical link 3, end *D* is fixed in the ground link, and end *B* supports link 2 (that is, the effective length of link 3 is $L_{\text{eff}} = 0.5L_3$). The yield strength and the modulus of elasticity for the aluminum link 3 are $S_y = 370$ MPa and $E = 207$ GPa, respectively. Determine the diameter, *d*, of link 3 to ensure that the static factor of safety guarding against buckling is $N = 2.5$.

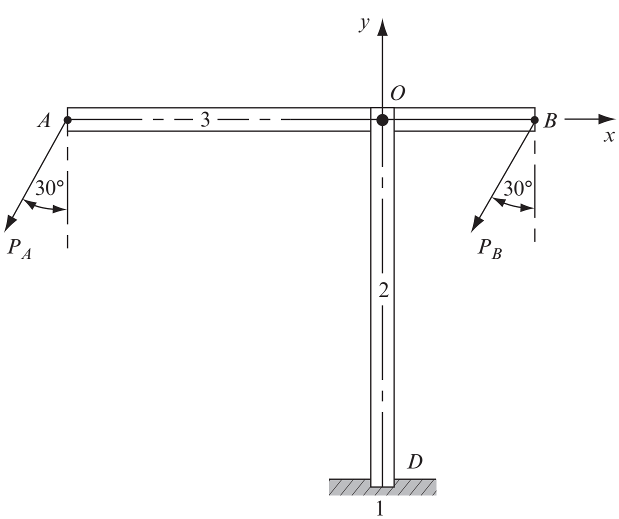


Figure P11.42 Link *AB* supporting two loads.

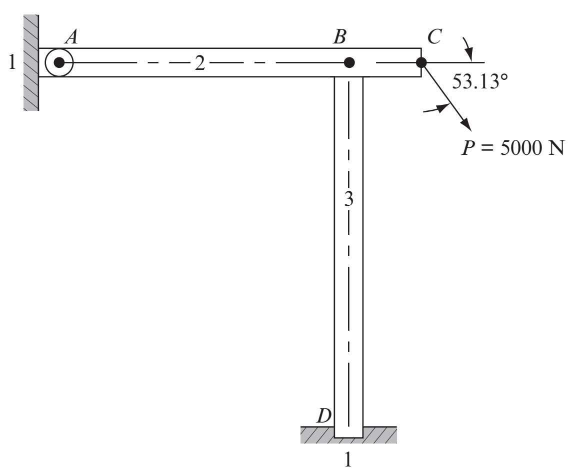


Figure P11.43 Link 3 supporting link *AC*.

11.44 Horizontal link 2 is pinned to the vertical wall at A and pinned to link 3 at B. The opposite end of link 3 is pinned to the wall at C. A vertical force, $P = 25$ kN is acting on link 2 at B. Link 2 has a 20 mm-by-30 mm solid rectangular cross section, and link 3 has a 40 mm-by-40 mm solid square cross section. The length of link 3 is $BC = 1.2$ m, and $\angle ABC = 30°$. The two links are made from a steel alloy with tensile yield strength $S_{yt} = 190$ MPa, compressive yield strength $S_{yc} = 205$ MPa, and modulus of elasticity $E = 207$ GPa. Using the theoretic value for the end-condition constant for link 3, determine: (a) the value of the slenderness ratio at the point of tangency between the Euler column formula and the Johnson parabolic formula; (b) the critical load and the factor of safety guarding against buckling of link 3; and (c) the minimum width of the square cross section of link 3 for the factor of safety to guard against buckling to be $N = 1$.

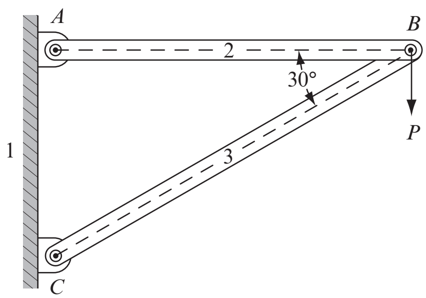


Figure P11.44 Simple truss.

11.45 Link $BC = 1.2$ m and 25 mm square cross section is fixed in the vertical wall at C and pinned at B to a circular steel cable, AB, with diameter $d = 20$ mm. Distance $AC = 0.7$ m. The mass, m, of a container, suspended from pin B, produces a gravitational load at B, which results in the moment at point C in the wall $M_C = 8\,000\text{N}\cdot\text{m}$ ccw. The yield strength and modulus of elasticity of the steel cable AB and the steel link BC are $S_y = 370$ MPa and $E = 207$ GPa, respectively. Given that $m = 2\,000$ kg, determine: (a) the tension in the cable AB and the factor of safety guarding against tensile failure; (b) the compressive load acting in link BC; (c) the factor of safety guarding against buckling of link BC. (Use the theoretic value of the end-condition constant assuming that the link has fixed-pinned ends.) If $M_C = 10\,000\ \text{N}\cdot\text{m}$ ccw, then determine the maximum mass of a container that can be suspended from pin B before buckling of link BC begins (that is, the factor of safety guarding against buckling failure is $N = 1$).

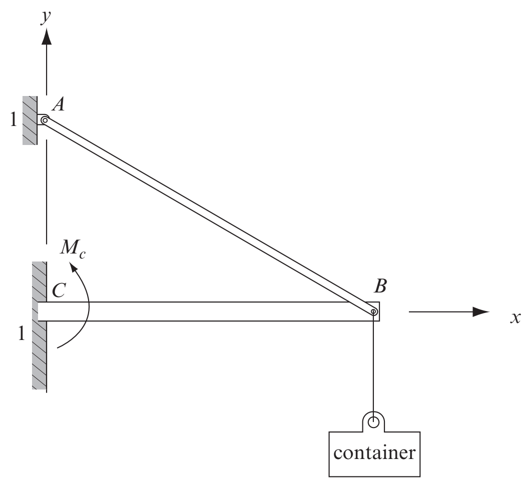


Figure P11.45 Structure supporting a load.

11.46 A vertically upward force, F, is applied at C of horizontal link 4, which is pinned to the ground at B and pinned to vertical links 2 and 3 at A. Lengths $AB = 2$ ft and $AC = 5$ ft and the three links are made from a steel alloy with a compressive yield strength $S_{yc} = 60\,000$ psi and a modulus of elasticity $E = 30 \times 10^6$ psi. Link 2 has a hollow circular cross section with an outside diameter $D = 2$ in, wall thickness $t = 0.25$ in, and length $L = 5$ ft. Using the theoretic value for the end-condition constant for link 2, determine: (a) the value of the slenderness ratio at the point of tangency between the Euler-column formula and the Johnson parabolic formula; (b) the critical load acting on link 2; and (c) the force, F, for the factor of safety of link 2 to guard against buckling to be $N = 1$. (d) If link 2 has a solid circular cross section with diameter $D = 3$ in and $F = 20\,000$ lb, then determine the maximum length of link 2 for the factor of safety to guard against buckling to be $N = 2$.

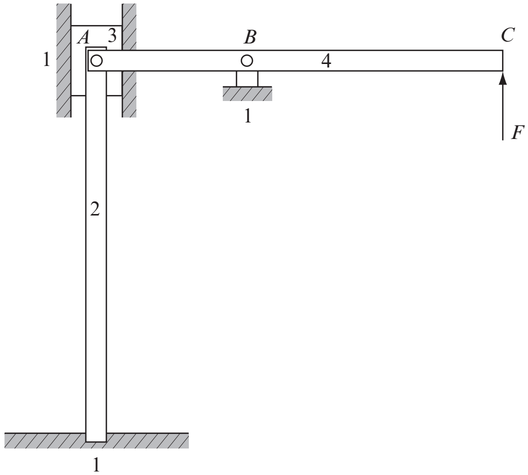


Figure P11.46 Link AC supporting load F.

11.47 A force, F, is acting at C perpendicular to link 2, end A is pinned to the ground, and supporting link 3 is pinned to link 2 at B and pinned to the ground at D. The lengths are $AC = 200$ mm and $AD = 150$ mm. Link 3 has a circular cross section with diameter $D = 5$ mm. Both links are a steel alloy with a compressive yield strength $S_{yc} = 420$ MPa and a modulus of elasticity $E = 206$ GPa. Using the theoretic value for the end-condition constant for link 3, determine: (*a*) the slenderness ratio at the point of tangency between the Euler column formula and the Johnson parabolic formula; (*b*) the critical load and the critical unit load acting on the link; and (*c*) the force, F, for the factor of safety to guard against buckling to be $N = 1$. If force $F = 3\,000$ N, then for link 3 determine: (*d*) the critical load for the factor of safety to guard against buckling $N = 1$; and (*e*) the slenderness ratio.

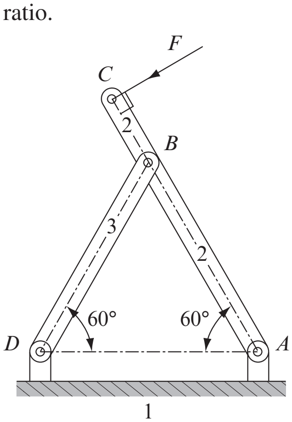


Figure P11.47 Simple truss.

11.48 For the four-bar linkage in the posture shown, the torque acting on link 2 at crankshaft O_2 is $\mathbf{T}_{12} = -6\,700\hat{\mathbf{k}}$ ft · lb. There is also a torque, $\mathbf{T}_{14}$, acting on link 4 at crankshaft O_4 to hold the linkage in static equilibrium. The length of coupler link 3 is $AB = 6$ in, and the cross section is rectangular with width $3t$ and thickness $4t$ (into the plane). The coupler link is a steel alloy with compressive yield strength $S_{yc} = 60$ kpsi and a modulus of elasticity E = 30 Mpsi. If the critical load for the coupler link is $P_{cr} = 150\,000$ lb, and the effects of gravity are ignored, then determine: (*a*) the factor of safety guarding against buckling; (*b*) the slenderness ratio at the point of tangency between the Euler column formula and the Johnson parabolic equation; and (*c*) the numeric value of the parameter t. (*d*) If the rectangular cross section is replaced by a circular cross section of the same material and diameter $d = 0.200$ in, then determine the factor of safety guarding against buckling of the coupler link.

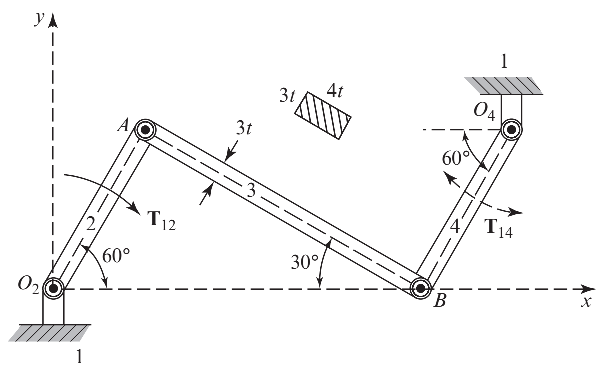


Figure P11.48 Four-bar linkage.

11.49 The single-cylinder engine is in static equilibrium due to external force $\mathbf{F} = -15\hat{\mathbf{i}}$kN acting at point D. The cross section of the connecting rod (link 3) is rectangular with width $4t$ and thickness $2t$, and the material is a steel alloy with compressive yield strength $S_{yc} = 205$ MPa and modulus of elasticity E = 207 GPa. Assume for practical purposes that the thickness $(2t)$ of link 3 must be greater than 5 mm and that buckling will occur in the xy plane. The effects of gravity and friction in the engine can be ignored. Determine: (*a*) the slenderness ratio of link 3 in terms of the unknown thickness parameter, t;

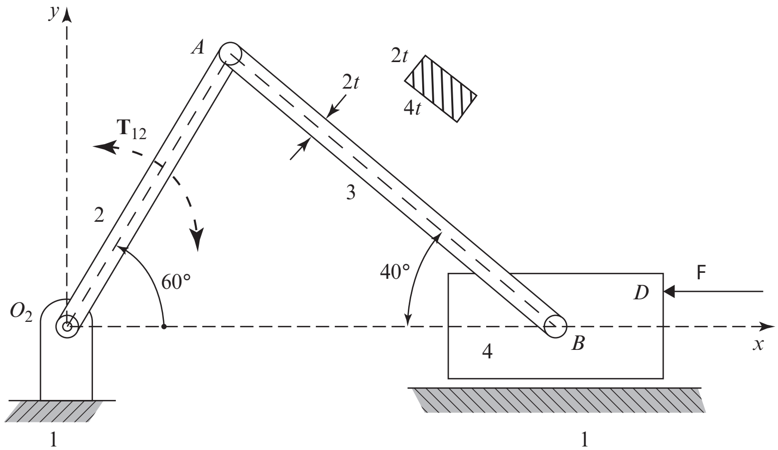


Figure P11.49 Single-cylinder engine.

(*b*) the slenderness ratio of link 3 at the point of tangency between the Euler column formula and the Johnson parabolic equation; and (*c*) the thickness parameter, t, to ensure that the factor of safety guarding against buckling for link 3 is $N = 5$.

11.50 Vertical link 2 is rigidly fixed to the ground (at A) and pinned to horizontal link 3 at B. Link 4 is pinned to link 3 at H, and the mass of link 4 is 200 kg. Vertical link 5 is pinned to link 3 at D and to the ground at O_5. Link 2 has a solid circular cross section with diameter $d_2 =$ 25 mm, compressive yield strength $S_{yc} = 370$ MPa, and modulus of elasticity $E = 205$ GPa. The factor of safety guarding against buckling, for link 2, is $N = 2.5$, and the end-condition constant $C = 2$. Assume that gravity is acting vertically downward, and links 2, 3, and 5 are massless compared to link 4. (*a*) For link 2 determine: (1) the slenderness ratio; and (2) the slenderness ratio at the point of tangency between the Euler column formula and the Johnson parabolic formula. (*b*) Determine the critical unit load acting on link 2. (*c*) If the diameter of link 2 is increased to $d_2 = 90$ mm, then is link 2 an Euler column or a Johnson column?

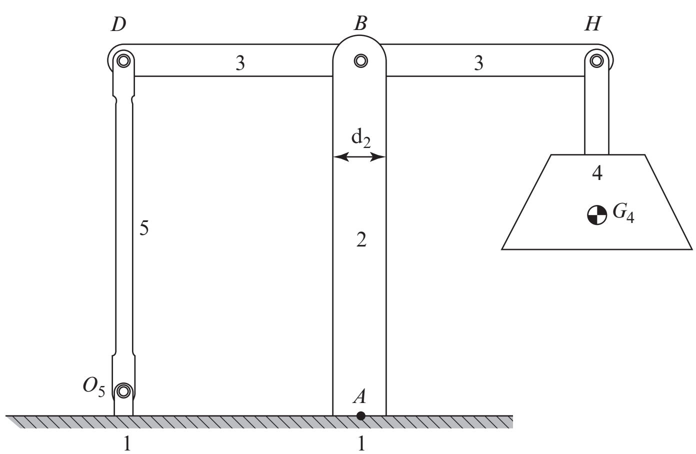


Figure P11.50 $AB = 2.5$ m and $DB = BH = 1.5$ m.

12 Dynamic Force Analysis

12.1 INTRODUCTION

In Chap. 11, we studied the forces in machine systems in which all forces on the bodies were in balance, and therefore the systems were in either static or dynamic equilibrium. However, in real machines, this is seldom, if ever, the case except when the machine is stopped. We learned in Chap. 4 that although the input crank of a machine may be driven at constant speed, this does not mean that all points of the input crank have constant velocity vectors or that other links of the machine operate at constant speeds. In general, there are accelerations, and therefore machines with moving parts having mass are not balanced.

Of course, techniques for static force analysis are important, not only because stationary structures must be designed to withstand their imposed loads, but also because they introduce concepts and approaches that can be built upon and extended to nonequilibrium situations. That introduces the purpose of Chap. 12: to learn how much acceleration will result from a system of unbalanced forces and also to learn how these *dynamic* forces can be assessed for systems that are not in balance.

12.2 CENTROID AND CENTER OF MASS

We recall from Sec. 11.2 that Newton's laws set forth the relationships between the net unbalanced force on a *particle*, its mass, and its acceleration. For Chap. 11, since we were only studying systems in equilibrium, we made use of that relationship for entire rigid bodies, arguing that they are made up of collections of particles and that the action and reaction forces between the particles balance each other. In Chap. 12, we must be more careful. We must remember that each of these particles may have acceleration, and that

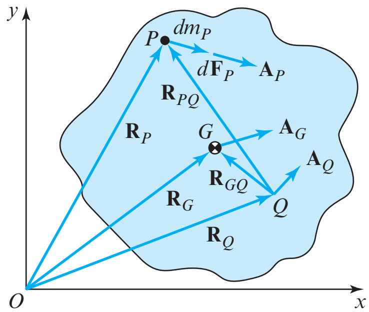


Figure 12.1 Particle P of mass dm_P.

the accelerations of the particles may all be different from each other. Important questions include the following: Which of these many point accelerations are we to use? And why?

Referring to the rigid body shown in Fig. 12.1, we consider an infinitesimal particle with mass dm_P, say at point P. Equation (11.1) tells us that the net unbalanced force, $d\mathbf{F}_P$, on this particle is proportional to its mass and its absolute acceleration, $\mathbf{A}_P$, that is,

$$d\mathbf{F}_P = \mathbf{A}_P dm_P. \tag{a}$$

Our task now is to sum these effects, that is, to integrate over all particles of the body and to put the result in a usable form for rigid bodies other than single particles. As we did in Chap. 11, we can conclude that the action and the reaction forces between particles of the body balance each other and therefore cancel in the process of the summation. The only net forces remaining are the constraint forces, those whose reactions are on some other body than this one. Thus, integrating Eq. (a) over all particles of mass in our rigid body, we obtain

$$\sum \mathbf{F} = \int \mathbf{A}_P dm_P. \tag{b}$$

We find it difficult to perform this integration, since each particle has a different acceleration. However, if we know (or can find) the acceleration of one particular point of the body, say point Q, and also the angular velocity $\boldsymbol{\omega}$ and angular acceleration $\boldsymbol{\alpha}$ of the body, then we can express the various accelerations of all other particles of the body in terms of their acceleration differences from that of point Q. Thus, from Eqs. (4.5), we write

$$\mathbf{A}_P = \mathbf{A}_Q + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PQ}) + \boldsymbol{\alpha} \times \mathbf{R}_{PQ}. \tag{c}$$

Substituting this equation into Eq. (b), the sum of the forces can be written as

$$\sum \mathbf{F} = \mathbf{A}_Q \int dm_P + \boldsymbol{\omega} \times \left(\boldsymbol{\omega} \times \int \mathbf{R}_{PQ} dm_P\right) + \boldsymbol{\alpha} \times \int \mathbf{R}_{PQ} dm_P, \tag{d}$$

where we have factored out of the integrals all quantities that are equal for all particles of the body.

We recognize the first integral of Eq. (*d*); the summation of all particle masses gives the total mass of the body:

$$\int dm_P = m. \tag{12.1}$$

The second and third terms of Eq. (*d*) both contain another integral, which is not as easy to recognize. However, because of its frequent appearance in the study of mechanics, a particular point G having a location given by the position vector $\mathbf{R}_G$ is defined by the integral equation:

$$\int \mathbf{R}_P dm_P = m\mathbf{R}_G. \tag{12.2}$$

This point G is called the *centroid*, or the *center of mass* or, simply, the *mass center* of the body. More will be said about this special point later.

Substituting Eqs. (12.1) and (12.2) into Eq. (*d*) gives

$$\sum \mathbf{F} = m\mathbf{A}_Q + \boldsymbol{\omega} \times [\boldsymbol{\omega} \times (m\mathbf{R}_{GQ})] + \boldsymbol{\alpha} \times (m\mathbf{R}_{GQ}),$$

which can be written as

$$\sum \mathbf{F} = m[\mathbf{A}_Q + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{GQ}) + \boldsymbol{\alpha} \times \mathbf{R}_{GQ}].$$

Finally, recognizing the sum of the three acceleration terms as the acceleration of the mass center, we can write

$$\sum \mathbf{F} = m\mathbf{A}_G. \tag{12.3}$$

This important equation is the integrated form of Newton's law for a particle, now extended to a rigid body. Note that it is the same equation as Eq. (11.1). However, careful derivation of the acceleration term was not presented there, since we were treating static problems where accelerations were zero.

We have now answered the question raised earlier in this section. Recognizing that each particle of a rigid body may have a different acceleration, which one should be used? Equation (12.3) indicates clearly that the absolute acceleration of the center of mass of the body is the proper acceleration to be used in Newton's law. That particular point, and no other, is the proper point for which Newton's law for a rigid body has the same form as for a single particle.

In engineering problems, we frequently find that forces are distributed in some manner along a line, over an area, or over a volume. The resultant of these distributed forces is usually not too difficult to find. To have the equivalent effect, this resultant must act at the centroid of the system. Thus, *the centroid of a system is a point at which a system of distributed forces may be considered concentrated with exactly the same effect* [1].

Instead of a system of forces, we may have a distributed mass, as in the above derivation. Then, by *center of mass* we mean *the point at which the mass may be considered concentrated* and that the effect is the same.

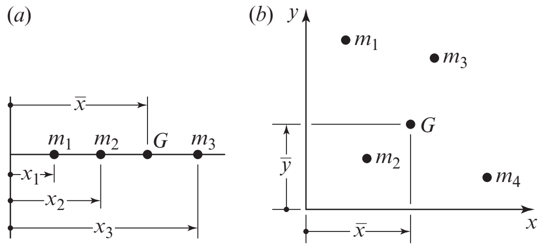


Figure 12.2 Particles of mass distributed: (*a*) along a line; (*b*) over a plane.

In Fig. 12.2*a*, a series of particles with masses are shown located at various positions along a line. The location of the center of mass G is given by

$$\bar{x} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3} = \frac{\sum m_i x_i}{\sum m_i}. \tag{12.4}$$

In Fig. 12.2*b*, the mass particles m_i are located at various positions $\mathbf{R}_i$ over an area. The location of the center of mass G is now given by

$$\mathbf{R}_G = \frac{m_1\mathbf{R}_1 + m_2\mathbf{R}_2 + m_3\mathbf{R}_3 + m_4\mathbf{R}_4}{m_1 + m_2 + m_3 + m_4} = \frac{\sum m_i\mathbf{R}_i}{\sum m_i}. \tag{12.5}$$

This procedure can also be extended to particles distributed over a volume by simply using Eq. (12.5) and treating position vectors $\mathbf{R}_i$ as three dimensional rather than as two dimensional.

When the mass is distributed continuously along a line, over a plane, or over a volume, the concept of summation in Eq. (12.5) is replaced by integration over infinitesimal particles of mass. The result is

$$\mathbf{R}_G = \frac{1}{m}\int \mathbf{R}dm, \tag{12.6}$$

where m is the total mass of the body considered. It is from this definition that Eq. (12.2) was obtained.

When mass is uniformly distributed along a line, over a plane, or volume, the center of mass can often be found by symmetry. Figure 12.3 illustrates the locations of the centers of mass for a circular solid, a rectangular solid, and a triangular solid. Each is assumed to have constant thickness and uniform density distribution.

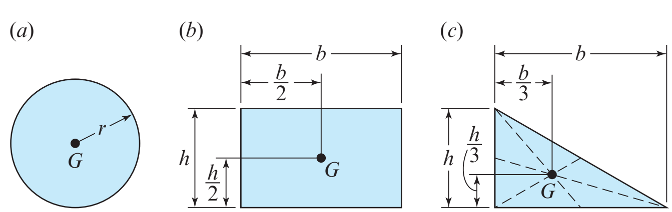


Figure 12.3 Center of mass location for: (*a*) a right circular solid; (*b*) a rectangular solid; (*c*) a triangular prism.

When a body is of a more irregular shape, the center of mass can often still be determined by considering it as a combination of simpler subshapes, as demonstrated in the following example.

EXAMPLE 12.1

The composite shape with dimensions in millimeters shown in Fig. 12.4 consists of a rectangular solid, less a circular through hole, plus a triangular plate of a different thickness. Determine the center of mass of this shape, which is made of a homogeneous material with density of ρ kg/(mm)^3.

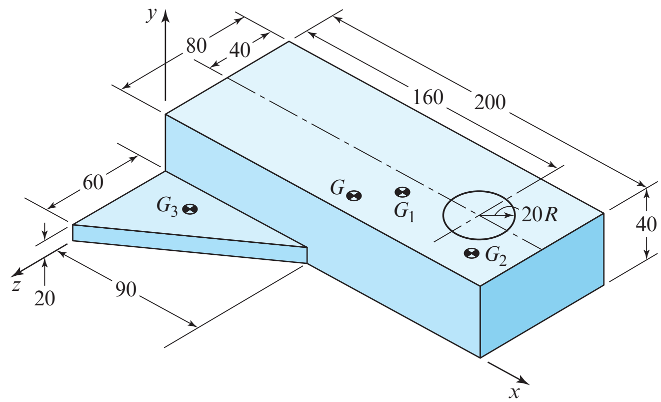


Figure 12.4 Composite shape.

SOLUTION

We can find the masses and the centers of mass of each of the three subshapes with the help of the formulae in Appendix A. For the rectangular solid subshape, we obtain*

$$m_1 = (200 \text{ mm})(40 \text{ mm})(80 \text{ mm})(\rho \text{ kg/(mm)}^3) = 640\,000\rho \text{ kg},$$

$$\mathbf{R}_{G_1} = 100\hat{\mathbf{i}} + 20\hat{\mathbf{j}} - 40\hat{\mathbf{k}} \text{ mm}.$$

For the hole, we treat the mass as negative, giving

$$m_2 = \pi(20 \text{ mm})^2(40 \text{ mm})(-\rho \text{ kg/(mm)}^3) = -50\,265\rho \text{ kg},$$

$$\mathbf{R}_{G_2} = 160\hat{\mathbf{i}} + 20\hat{\mathbf{j}} - 40\hat{\mathbf{k}} \text{ mm}.$$

* Note that in this example, we use SI units. US customary units, pounds, or other units might have been used. However, this is not emphasized here, because the density cancels in the use of Eq. (12.5).

Finally, for the triangular subshape, we find

$$m_3 = 0.5(60\text{ mm})(90\text{ mm})(20\text{ mm})[\rho\text{ kg}/(\text{mm})^3] = 54\,000\rho\text{ kg},$$
$$\mathbf{R}_{G_3} = 30\hat{\mathbf{i}} + 10\hat{\mathbf{j}} + 20\hat{\mathbf{k}}\text{ mm}.$$

Now since we are completing a process of integration, we can combine the values of the subshapes; that is, from Eq. (12.5) we can write

$$\mathbf{R}_G = \frac{m_1\mathbf{R}_1 + m_2\mathbf{R}_2 + m_3\mathbf{R}_3}{m_1 + m_2 + m_3}.$$

This gives the following result:

$$\mathbf{R}_G = 89.4\hat{\mathbf{i}} + 19.2\hat{\mathbf{j}} - 35.0\hat{\mathbf{k}}\text{ mm}. \qquad \textit{Ans.}$$

12.3 MASS MOMENTS AND PRODUCTS OF INERTIA

Another problem that often arises when forces are distributed over an area or volume is that of calculating their moment about a specified point or axis of rotation. Sometimes the force intensity varies according to its distance from the point or axis of rotation. Although we will save a more thorough derivation of these equations until Sec. 12.11 and what follows, we will point out here that such problems always give rise to integrals of the form $\int (\text{distance})^2 dm$.

In three-dimensional problems, three such integrals are defined as follows:*

$$\begin{aligned}
I^{xx} &= \int (\hat{\mathbf{i}} \times \mathbf{R}) \cdot (\hat{\mathbf{i}} \times \mathbf{R}) dm = \int [(R^y)^2 + (R^z)^2] dm,\\
I^{yy} &= \int (\hat{\mathbf{j}} \times \mathbf{R}) \cdot (\hat{\mathbf{j}} \times \mathbf{R}) dm = \int [(R^z)^2 + (R^x)^2] dm,\\
I^{zz} &= \int (\hat{\mathbf{k}} \times \mathbf{R}) \cdot (\hat{\mathbf{k}} \times \mathbf{R}) dm = \int [(R^x)^2 + (R^y)^2] dm.
\end{aligned} \tag{12.7}$$

These three integrals are called the *mass moments of inertia* of the body. Another three similar integrals are

$$\begin{aligned}
I^{xy} = I^{yx} &= \int (\hat{\mathbf{i}} \times \mathbf{R}) \cdot (\hat{\mathbf{j}} \times \mathbf{R}) dm = -\int (R^x R^y) dm,\\
I^{yz} = I^{zy} &= \int (\hat{\mathbf{j}} \times \mathbf{R}) \cdot (\hat{\mathbf{k}} \times \mathbf{R}) dm = -\int (R^y R^z) dm,\\
I^{zx} = I^{xz} &= \int (\hat{\mathbf{k}} \times \mathbf{R}) \cdot (\hat{\mathbf{i}} \times \mathbf{R}) dm = -\int (R^z R^x) dm.
\end{aligned} \tag{12.8}$$

* It should be carefully noted here that these integrals are not the same as those called *area moments of inertia* or *second moments of area,* which are integrals over dA, a differential area, rather than integrals over dm, a differential mass. These other integrals also arise in problems involving forces distributed over an area or volume, but are different. In two-dimensional problems of constant thickness, however, they are easily related, since dA times the thickness times the mass density yields dm for the integral.

and these three integrals are called the *mass products of inertia* of the body. Sometimes it is convenient to arrange these mass moments of inertia and mass products of inertia into a symmetric square array or matrix called the *inertia tensor* of the body [3]:

$$\mathbf{I} = \begin{bmatrix} I^{xx} & I^{xy} & I^{xz} \\ I^{yx} & I^{yy} & I^{yz} \\ I^{zx} & I^{zy} & I^{zz} \end{bmatrix}. \tag{12.9}$$

A careful look at the above integrals show that they represent the mass distribution of the body with respect to the coordinate system about which they are determined, but that they change if evaluated in a different coordinate system. To keep their meaning direct and simple, we assume that the coordinate system chosen for each body is attached to the body in a convenient location and orientation. Therefore, for rigid bodies, the mass moments and products of inertia are constant properties of the body and its mass distribution, and they do not change when the body moves; they do, however, depend on the coordinate system chosen.

An interesting property of these integrals is that it is always possible to choose the coordinate system so that its origin is located at the center of mass of the body and oriented such that all of the mass products of inertia become zero. Such a choice of the coordinate axes of the body is called its *principal axes*, and the corresponding values of Eqs. (12.7) are called the *principal mass moments of inertia*. A variety of simple geometric solids, the orientations of their principal axes, and formulae for their principal mass moments of inertia are included in Appendix A, Table 5.

If we note that mass moments of inertia have units of mass times distance squared, it seems natural to define a radius value for the body as

$$I_G = mk^2 \quad \text{or} \quad k = \sqrt{\frac{I_G}{m}}. \tag{12.10}$$

This distance k is called the *radius of gyration* of the body, and it is always calculated or measured from the center of mass of the body about one of the principal axes. For three-dimensional motions of bodies there are three radii of gyration, k^x, k^y, and k^z, associated with the three principal axes, I^{xx}, I^{yy}, and I^{zz}.

It is often necessary to determine the moments and products of inertia of bodies that are composed of several simpler subshapes for which formulae are known, such as those given in Table 5, Appendix A. The easiest method of finding these is to compute the mass moments about the principal axes of each subshape, then to shift the origin of each to the mass center of the composite body, and then to sum the results. This requires that we develop methods of redefining mass moments and products of inertia when the axes are translated to a new posture. The form of the *transfer*, or *parallel-axis theorem* for a mass moment of inertia, is written

$$I = I_G + md^2, \tag{12.11}$$

where I_G is one of the principal mass moments of inertia about a known principal axis, and I is the mass moment of inertia about a parallel axis at distance d from that principal axis. Equation (12.11) must only be used for *translation* of inertia axes starting from a principal axis. Also, the rotation of these axes results in the introduction of product of inertia terms. More will be said on general transformations of inertia axes in Sec. 12.11.

EXAMPLE 12.2

Figure 12.5 illustrates a connecting rod made of ductile iron with density of 0.260 lb/in^3. Find the mass moment of inertia about the z axis.

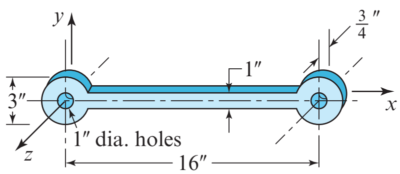


Figure 12.5 Connecting rod.

SOLUTION

First we find the mass moment of inertia of each of the annular cylinders at the ends of the rod and of the central prismatic bar, each taken about its own mass center. Then we use the parallel-axis theorem to transfer each to the z axis.

The mass of each annular cylinder is

$$\begin{aligned} m_{\text{cyl}} &= \rho\pi(r_o^2 - r_i^2)l \\ &= \frac{(0.260\ \text{lb/in}^3)\pi[(1.5\ \text{in})^2 - (0.5\ \text{in})^2](0.75\ \text{in})}{386\ \text{in/s}^2} \\ &= 0.003\ 17\ \text{lb} \cdot \text{s}^2/\text{in} \end{aligned}$$

The mass of the central prismatic bar is

$$\begin{aligned} m_{\text{bar}} = \rho w h l &= \frac{(0.260\ \text{lb/in}^3)(13\ \text{in})(1\ \text{in})(0.75\ \text{in})}{386\ \text{in/s}^2} \\ &= 0.006\ 57\ \text{lb} \cdot \text{s}^2/\text{in} \end{aligned}$$

From Table 5, Appendix A, we find the mass moment of inertia of each cylinder and the bar as

$$\begin{aligned} I_{\text{cyl}} &= \frac{m(r_o^2 + r_i^2)}{2} \\ &= \frac{(0.003\ 17\ \text{lb} \cdot \text{s}^2/\text{in})[(1.5\ \text{in})^2 + (0.5\ \text{in})^2]}{2} \\ &= 0.003\ 96\ \text{in} \cdot \text{lb} \cdot \text{s}^2 \end{aligned}$$

and

$$\begin{aligned} I_{\text{bar}} &= \frac{m(w^2+h^2)}{12} \\ &= \frac{(0.006\,57\ \text{lb}\cdot\text{s}^2/\text{in})[(1\ \text{in})^2+(13\ \text{in})^2]}{12} \\ &= 0.093\,08\ \text{in}\cdot\text{lb}\cdot\text{s}^2. \end{aligned}$$

Then using Eq. (12.11) to transfer the axes, the mass moment of inertia about the z axis is

$$\begin{aligned} I^{zz} &= I_{\text{cyl}} + (I_{\text{bar}} + m_{\text{bar}}d_{\text{bar}}^2) + (I_{\text{cyl}} + m_{\text{cyl}}d_{\text{cyl}}^2) \\ &= (0.003\,96\ \text{in}\cdot\text{lb}\cdot\text{s}^2) + [(0.093\,08\ \text{in}\cdot\text{lb}\cdot\text{s}^2) + (0.006\,57\ \text{lb}\cdot\text{s}^2/\text{in})(8\ \text{in})^2] \\ &\quad + [(0.003\,96\ \text{in}\cdot\text{lb}\cdot\text{s}^2) + (0.003\,17\ \text{lb}\cdot\text{s}^2/\text{in})(16\ \text{in})^2] \\ &= 1.333\ \text{in}\cdot\text{lb}\cdot\text{s}^2. \qquad \textit{Ans.} \end{aligned}$$

It is noted that only one mass moment of inertia, I^{zz}, was requested or found in this example. This does not mean that I^{xx} and I^{yy} are zero, but rather that they are likely not needed for further analysis. In problems with only planar motion, only I^{zz} is needed since I^{xx} and I^{yy} are used only with rotations out of the xy plane. These other mass moments and products of inertia are used in Secs. 12.10 and later, where we treat problems with spatial motion, and they are found in identical fashion.

12.4 INERTIA FORCES AND D'ALEMBERT'S PRINCIPLE

Next let us consider a moving rigid body of mass m acted upon by a system of forces, say, $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$, as shown in Fig. 12.6a. We designate the center of mass of the body as

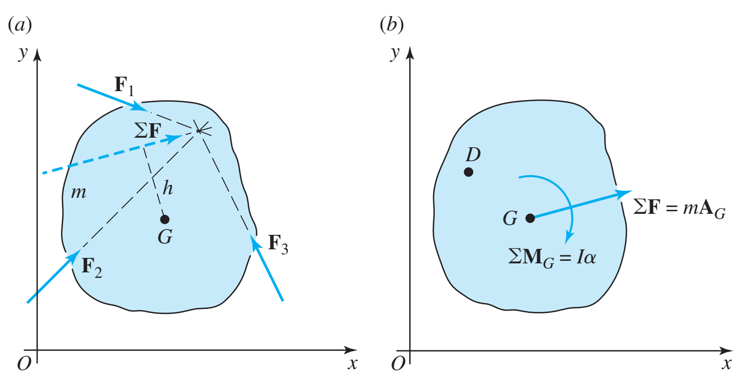


Figure 12.6 (a) Unbalanced system of forces; (b) accelerations that result from the forces.

point G, and we find the resultant of the system of forces from the equation

$$\sum \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3.$$

In the general case, the line of action of this resultant is *not* through the mass center but is displaced by some distance, shown in Fig. 12.6*a* as distance h. We demonstrated in Eq. (12.3) that the effect of this unbalanced force system is to produce an acceleration of the center of mass of the body, that is

$$\sum \mathbf{F} = m\mathbf{A}_G. \tag{12.12}$$

In a similar way, the unbalanced moment effect of this resultant force about the center of mass causes angular acceleration of the body that obeys the equation

$$\sum \mathbf{M}_G = I_G \boldsymbol{\alpha}. \tag{12.13}$$

However, this equation is restricted to use in taking moments about the center of mass, G; that is, it cannot be used for taking moments about an arbitrary point in the body.

The quantity $\Sigma\mathbf{F}$ is the resultant of all external forces acting upon the body, and $\Sigma\mathbf{M}_G$ is the sum of all applied external moments and the moments of all externally applied forces about point G. The mass moment of inertia is designated I_G, signifying that it must be taken with respect to the mass center, G.

Equation (12.12) indicates that when an unbalanced system of forces acts upon a rigid body, the mass center of the body experiences a rectilinear acceleration, $\boldsymbol{A}_G$, in the same direction as the resultant force, $\sum \mathbf{F}$. Also, Eq. (12.13) indicates that the body experiences an angular acceleration, $\boldsymbol{\alpha}$, in the same direction as the resultant moment, $\sum \mathbf{M}_G$, caused by the moments of the forces and the torques about the mass center. This situation is shown in Fig. 12.6*b*. If the forces and moments are known, Eqs. (12.12) and (12.13) may be used to determine the resulting acceleration pattern, that is, the resulting motion of the body.

As pointed out, Eq. (12.13) is restricted to the sum of moments taken about the center of mass. Suppose we wish to take moments about arbitrary point D shown in Fig. 12.6*b*; then, the sum of the moments about point D can be written as

$$\sum \mathbf{M}_D = I_G \boldsymbol{\alpha} + \mathbf{R}_{GD} \times m\mathbf{A}_G. \tag{a}$$

Then, from Sec. 4.3, the acceleration of the center of mass can be expressed in terms of the acceleration of point D as

$$\sum \mathbf{M}_D = I_G \boldsymbol{\alpha} + \mathbf{R}_{GD} \times m\left(\mathbf{A}_D - \omega^2 \mathbf{R}_{GD} + \boldsymbol{\alpha} \times \mathrm{R}_{GD}\right). \tag{b}$$

Simplifying and then rearranging this equation gives

$$\sum \mathbf{M}_D = \left(I_G + mR_{GD}^2\right)\boldsymbol{\alpha} + \mathbf{R}_{GD} \times m\mathbf{A}_D. \tag{c}$$

Then, according to Eq. (12.11), the parallel-axis theorem, Eq. (c) can be written as

$$\sum \mathbf{M}_D = I_D \boldsymbol{\alpha} + \mathbf{R}_{GD} \times m\mathbf{A}_D. \tag{12.14}$$

Comparing the right-hand side of this equation with the right-hand side of Eq. (a), we note a pattern, namely, the point used for for computing the acceleration is the same point that was used for the axis of the second moment of mass. This equation accomplishes our goal that the sum of moments can now be taken about an arbitrary point D; it is no longer restricted to the sum of moments about the center of mass.

During engineering design, however, the motions of the machine members are often specified in advance by other machine requirements. The problem, then, is this: given the motions of the machine parts, what forces are required to produce these motions? The problem requires (1) a kinematic analysis to determine the translational and rotational accelerations of the various links, and (2) definitions of the actual shapes, dimensions, and material specifications to determine the centroids and mass moments of inertia of the links. In the examples presented here, only the results of kinematic analysis are included, since methods of finding these have been presented in Chap. 4. The selection of the materials, shapes, and many of the dimensions of machine parts form the subject of machine design and are also not further discussed here.

In the dynamic analysis of machines, the acceleration vectors are usually known or found by the methods of Chap. 4; therefore, an alternative form of Eqs. (12.12) and (12.13) is often convenient in determining the forces required to produce these known accelerations. Thus, we can write

$$\sum \mathbf{F} + (-m\mathbf{A}_G) = \mathbf{0} \tag{12.15}$$

and

$$\sum \mathbf{M}_G + (-I_G\boldsymbol{\alpha}) = \mathbf{0}. \tag{12.16a}$$

or

$$\sum \mathbf{M}_D + (-I_D\boldsymbol{\alpha}) = \mathbf{0}. \tag{12.16b}$$

These are vector equations applying to the planar motion of a rigid body. Equation (12.15) states that the vector sum of all external forces acting upon the body plus the fictitious force $-m\mathbf{A}_G$ sum to zero. This new fictitious force, $-m\mathbf{A}_G$, is called an *inertia force*. It has the same line of action as the absolute acceleration, $\mathbf{A}_G$, but is opposite in sense. Equation (12.16) states that the sum of all external moments and the moments of all external forces acting upon the body about an axis through G (or D) and perpendicular to the plane of motion plus the fictitious moment $-I_G\boldsymbol{\alpha}$ (or $-I_D\boldsymbol{\alpha}$) sum to zero. This new fictitious moment, $-I_G\boldsymbol{\alpha}$ (or $-I_D\boldsymbol{\alpha}$), is called an *inertia moment*. The inertia moment is opposite in sense to the angular acceleration vector, $\boldsymbol{\alpha}$, of the body. We recall that Newton's first law states that a body perseveres in its state of uniform motion except when compelled to change by impressed forces; in other words, bodies resist any change in their motions. In a sense, we can picture the fictitious inertia force and inertia moment vectors as resistances of a body to the change of motion required by the net unbalanced forces.

Equations (12.15) and (12.16) are known as *d'Alembert's principle*, since d'Alembert* was the first to call attention to the fact that addition of the inertia force and inertia moment

* Jean leRond d'Alembert (1717–1783).

to the real system of forces and moments enables a solution from the equations of static equilibrium. We note that the equations can also be written

$$\sum \mathbf{F} = \mathbf{0} \quad \text{and} \quad \sum \mathbf{M} = \mathbf{0}, \tag{12.17}$$

where it is understood that both the external and the inertia forces and moments are to be included in the summations. Equations (12.17) are useful, since they permit us to take the summation of moments about any axis perpendicular to the plane of motion.

D'Alembert's principle is summarized as follows: The vector sum of all external forces and inertia forces acting upon a system of rigid bodies is zero. The vector sum of all external moments and inertia moments acting upon a system of rigid bodies is also separately zero.

When a graphic solution by a force polygon is desired, the sum of the forces and the sum of the moments given by Eqs. (12.17) can be combined. For example, consider Fig. 12.7*a*, where link 3 is acted upon by the external forces $\mathbf{F}_{23}$ and $\mathbf{F}_{43}$. The resultant $\mathbf{F}_{23} + \mathbf{F}_{43}$ produces an acceleration of the center of mass, $\mathbf{A}_G$, and an angular acceleration of the link, $\boldsymbol{\alpha}_3$, since the line of action of the resultant does not pass through the center of mass. Representing the inertia moment $-I_G\boldsymbol{\alpha}_3$ as a couple, as shown in Fig. 12.7*b*, we intentionally choose the two forces of this couple to be $\pm m\mathbf{A}_G$. For the moment of the couple to be of magnitude $-I_G\alpha_3$, the distance between the forces of the couple must be

$$h = \frac{I_G\alpha_3}{mA_G}. \tag{12.18}$$

Because of this particular choice for the couple, one force of the couple exactly balances the inertia force itself and leaves only a single force, as shown in Fig. 12.7*c*. This force includes the combined effects of the inertia force and the inertia moment, yet appears as only a single inertia force offset by the distance, h, to include the effect of the inertia moment.

To illustrate the graphic approach to dynamic force analysis, we present the following example.

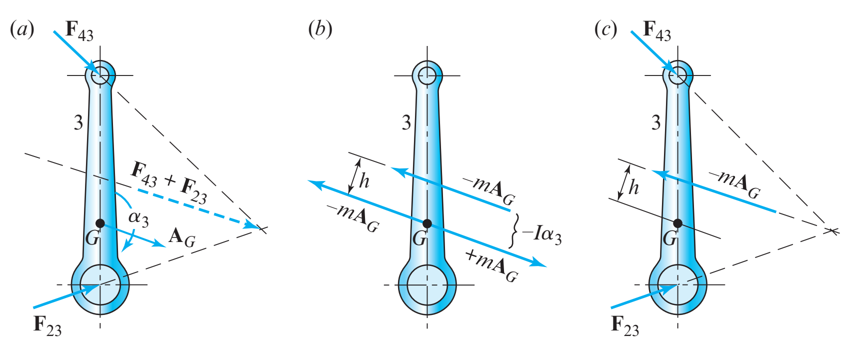


Figure 12.7 (*a*) Unbalanced forces and resulting accelerations; (*b*) inertia force and inertia couple; (*c*) inertia force offset from center of mass.

EXAMPLE 12.3

For the elliptic-trammel linkage shown in Fig. 12.8*a*, determine the force F_A that ensures that point A has the constant velocity $\mathbf{V}_A = 12.6\hat{\mathbf{j}}$ ft/s. The linkage is in the horizontal plane with gravity normal to the plane of motion. The weight and mass moment of inertia of coupler link 3 are $w_3 = 2.20$ lb and $I_{G_3} = 0.047\,9$ in · lb · s^2, respectively.

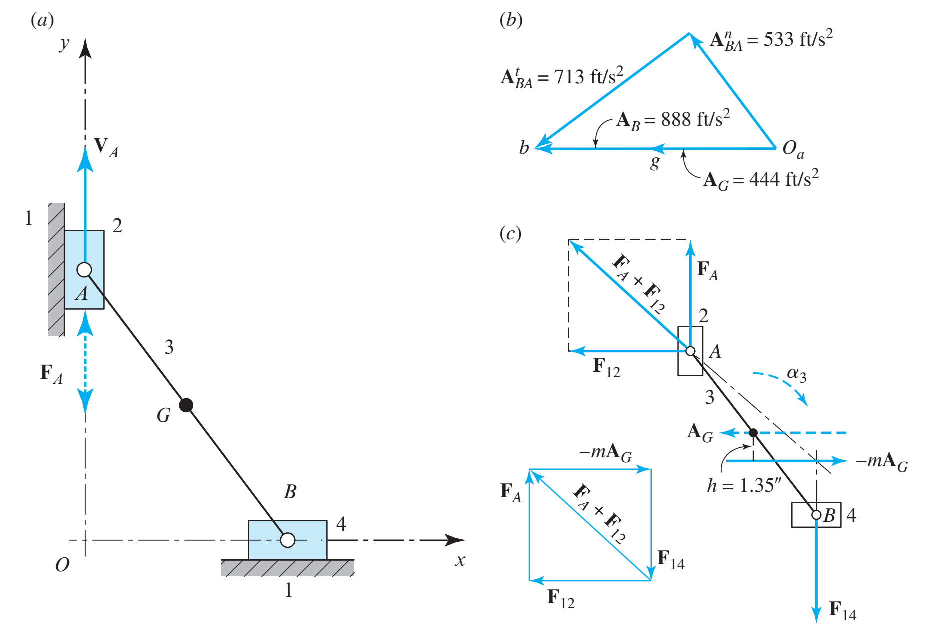


Figure 12.8 (*a*) $R_{BA} = 10$ in, $R_{GA} = 5$ in, $R_{AO} = 8$ in, and $R_{BO} = 6$ in; (*b*) acceleration polygon; (*c*) free-body diagram and force polygon.

ASSUMPTIONS

The weights of links 2 and 4 and the friction in the linkage are negligible.

SOLUTION

A kinematic analysis provides the acceleration information shown in the polygon of Fig. 12.8*b*. The acceleration of the mass center of the coupler link is $A_G = 444$ ft/s^2, and the angular acceleration of this link is

$$\alpha_3 = \frac{A^t_{BA}}{R_{BA}} = \frac{(713\ \text{ft/s}^2)(12\ \text{in/ft})}{10\ \text{in}} = 856\ \text{rad/s}^2\ \text{cw}.$$

The mass of the coupler link is

$$m_3 = \frac{w_3}{g} = \frac{2.20\ \text{lb}}{32.17\ \text{ft/s}^2\,(12\ \text{in/ft})} = 0.005\,70\ \text{lb}\cdot\text{s}^2/\text{in}.$$

Substituting the known information into Eq. (12.18), the offset distance is

$$h = \frac{(0.047\,9\ \text{in} \cdot \text{lb} \cdot \text{s}^2)(856\ \text{rad/s}^2)}{(0.005\,70\ \text{lb} \cdot \text{s}^2/\text{in})(444\ \text{ft/s}^2)(12\ \text{in/ft})} = 1.35\ \text{in.}$$

The free-body diagram of the assembly of links 2, 3, and 4 and the resulting force polygon are shown in Fig. 12.8*c*. Note that the inertia force, $-m\mathbf{A}_G$, is offset from G by the distance, h, on the proper side of (below) G to include a counterclockwise moment $-I_G\alpha_3$ (opposite to α_3) about G, and with the inertia force in the opposite sense to $\mathbf{A}_G$. The constraint reaction at B is $\mathbf{F}_{14}$ and, with no friction, is vertical. The forces at A are the horizontal constraint reaction, $\mathbf{F}_{12}$, and the vertical actuating force, $\mathbf{F}_A$. Recognizing this as a four-force member, as demonstrated in Sec. 11.8, we find the concurrency point at the intersection of the offset inertia force and $\mathbf{F}_{14}$, with the directions of both being known. The line of action of the total force, $\mathbf{F}_{12} + \mathbf{F}_A$, at A must pass through this point of concurrency. This fact permits construction of the force polygon, where the unknown forces, $\mathbf{F}_{12}$ and $\mathbf{F}_A$, having known directions, are found as components of $\mathbf{F}_{12} + \mathbf{F}_A$. The actuating force, $\mathbf{F}_A$, is found by measurement to be

$$\mathbf{F}_A = 27\hat{\mathbf{j}}\ \text{lb.} \qquad \textit{Ans.}$$

To illustrate the analytic approach to dynamic force analysis of a mechanism, we present another example.

EXAMPLE 12.4

Determine the reaction forces and the driving torque for the four-bar linkage in the posture shown in Fig. 12.9. The input crank is operating at a constant angular velocity $\omega_2 = 48$ rad/s ccw, and the applied force acting at point C is $\mathbf{F}_C = -0.8\hat{\mathbf{j}}$ kN. The masses of links 3 and 4 are $m_3 = 1.5$ kg and $m_4 = 5$ kg, respectively, and the second moments of mass of links 2, 3, and 4 are $I_{G_2} = 0.025\ \text{kg} \cdot \text{m}^2$, $I_{G_3} = 0.012\ \text{kg} \cdot \text{m}^2$, and $I_{G_4} = 0.054\ \text{kg} \cdot \text{m}^2$, respectively.

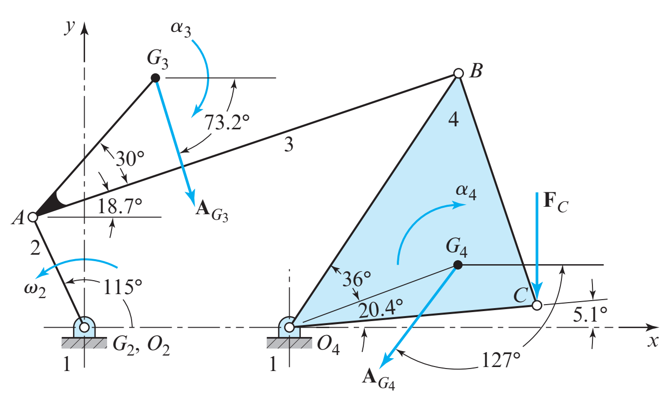


Figure 12.9 $R_{AO_2} = 60$ mm, $R_{O_4O_2} = 100$ mm, $R_{BA} = 220$ mm, $R_{BO_4} = 150$ mm, $R_{CO_4} = R_{CB} = 120$ mm, $R_{G_3A} = 90$ mm, and $R_{G_4O_4} = 90$ mm.

ASSUMPTIONS

Gravity and friction effects are negligible.

SOLUTION

The information given in the figure caption in vector form is:

$$\mathbf{R}_{AO_2} = 60 \text{ mm}\angle 115^\circ = -25.4\hat{\mathbf{i}} + 54.4\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{G_3A} = 90 \text{ mm}\angle 48.7^\circ = 59.4\hat{\mathbf{i}} + 67.6\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{BA} = 220 \text{ mm}\angle 18.7^\circ = 208.4\hat{\mathbf{i}} + 70.5\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{BO_4} = 150 \text{ mm}\angle 61.5^\circ = 71.6\hat{\mathbf{i}} + 131.8\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{G_4O_4} = 90 \text{ mm}\angle 25.5^\circ = 81.2\hat{\mathbf{i}} + 38.7\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{CO_4} = 120 \text{ mm}\angle 5.1^\circ = 119.5\hat{\mathbf{i}} + 10.7\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{O_4O_2} = 100 \text{ mm}\angle 0^\circ = 100.0\hat{\mathbf{i}} \text{ mm}.$$

A kinematic analysis of the linkage gives: $\boldsymbol{\alpha}_3 = -119\hat{\mathbf{k}}$ rad/s^2, $\boldsymbol{\alpha}_4 = -625\hat{\mathbf{k}}$ rad/s^2, $\mathbf{A}_{G_3} = 162$ m/s$^2\angle -73.2^\circ = 46.8\hat{\mathbf{i}} - 155\hat{\mathbf{j}}$ m/s^2, and $\mathbf{A}_{G_4} = 104$ m/s$^2\angle -127^\circ = -62.6\hat{\mathbf{i}} - 83.1\hat{\mathbf{j}}$ m/s^2.

Next, we calculate the inertia forces and inertia moments using Eqs (12.12) and (12.13):

$$-m_2\mathbf{A}_{G_2} = \mathbf{0},$$

$$-m_3\mathbf{A}_{G_3} = -(1.5 \text{ kg})(46.8\hat{\mathbf{i}} - 155\hat{\mathbf{j}} \text{ m/s}^2) = -70.2\hat{\mathbf{i}} + 233\hat{\mathbf{j}} \text{ N},$$

$$-m_4\mathbf{A}_{G_4} = -(5.0 \text{ kg})(-62.6\hat{\mathbf{i}} - 83.1\hat{\mathbf{j}} \text{ m/s}^2) = 313\hat{\mathbf{i}} + 415\hat{\mathbf{j}} \text{ N},$$

$$-I_{G_2}\boldsymbol{\alpha}_2 = \mathbf{0},$$

$$-I_{G_3}\boldsymbol{\alpha}_3 = -(0.012 \text{ kg}\cdot\text{m}^2)(-119\hat{\mathbf{k}} \text{ rad/s}^2) = 1.43\hat{\mathbf{k}} \text{ N}\cdot\text{m},$$

and

$$-I_{G_4}\boldsymbol{\alpha}_4 = -(0.054 \text{ kg}\cdot\text{m}^2)(-625\hat{\mathbf{k}} \text{ rad/s}^2) = 33.8\hat{\mathbf{k}} \text{ N}\cdot\text{m}.$$

Since the solution here is analytic, we do not need to calculate offset distances using Eq. (12.18).

For our next step, we make sketches of the free-body diagrams of link 3 and link 4. From the two free-body diagrams, we note that there are a total of six unknown force components. Since there are a total of six scalar equations available, two components of the forces and one for the sum of the moments about the mass center of each link, then the problem is solvable. However, these six equations must be solved simultaneously to find the six unknowns. In an effort to decouple the equations, an alternative strategy is to take the sum of the moments about different points in the links.

Considering the free-body diagram of link 3, we formulate the summation of moments about point A:

$$\Sigma \mathbf{M}_A = \mathbf{R}_{G_3A} \times (-m_3\mathbf{A}_{\mathbf{G_3}}) + (-I_{G_3}\boldsymbol{\alpha}_3) + \mathbf{R}_{BA} \times \mathbf{F}_{43} = \mathbf{0}. \tag{1}$$

Also, considering the free-body diagram of link 4, we formulate the summation of moments about point O_4:

$$\Sigma \mathbf{M}_{O_4} = \mathbf{R}_{G_4O_4} \times \left(-m_4\mathbf{A}_{G_4}\right) + (-I_{G_4}\boldsymbol{\alpha}_4) + \mathbf{R}_{CO_4} \times \mathbf{F}_C + \mathbf{F}_{BO_4} \times \mathbf{F}_{34} = \mathbf{0}. \tag{2}$$

Remembering that $\mathbf{F}_{43} = -\mathbf{F}_{34}$, the in-plane components of $\mathbf{F}_{34}$ are the only two unknowns, and they are shared by these two equations.

The cross-product terms of these equations are

$$\begin{aligned}
\mathbf{R}_{G_3A} \times \left(-m_3\mathbf{A}_{G_3}\right) &= \left(59.4\hat{\mathbf{i}} + 67.6\hat{\mathbf{j}}\ \text{mm}\right) \times \left(70.2\hat{\mathbf{i}} + 233\hat{\mathbf{j}}\ \text{N}\right) = 18.6\hat{\mathbf{k}}\ \text{N}\cdot\text{m},\\
\mathbf{R}_{BA} \times \mathbf{F}_{43} &= \left(208.4\hat{\mathbf{i}} + 70.5\hat{\mathbf{j}}\ \text{mm}\right) \times \left(-F_{34}^x\hat{\mathbf{i}} - F_{34}^y\hat{\mathbf{j}}\ \text{N}\right)\\
&= \left(70.5F_{34}^x - 208.4F_{34}^y\right)\hat{\mathbf{k}}\ \text{N}\cdot\text{m},
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{R}_{G_4O_4} \times \left(-m_4\mathbf{A}_{G_4}\right) &= \left(81.2\hat{\mathbf{i}} + 38.7\hat{\mathbf{j}}\ \text{mm}\right) \times \left(313\hat{\mathbf{i}} + 415\hat{\mathbf{j}}\ \text{N}\right) = 21.6\hat{\mathbf{k}}\ \text{N}\cdot\text{m},\\
\mathbf{R}_{CO_4} \times \mathbf{F}_C &= \left(119.5\hat{\mathbf{i}} + 10.7\hat{\mathbf{j}}\ \text{mm}\right) \times \left(-800\hat{\mathbf{j}}\ \text{N}\right) = -95.6\hat{\mathbf{k}}\ \text{N}\cdot\text{m},\\
\mathbf{R}_{BO_4} \times \mathbf{F}_{34} &= \left(71.6\hat{\mathbf{i}} + 131.8\hat{\mathbf{j}}\ \text{mm}\right) \times \left(F_{34}^x\hat{\mathbf{i}} + F_{34}^y\hat{\mathbf{j}}\ \text{N}\right)\\
&= \left(-131.8F_{34}^x + 71.6F_{34}^y\right)\hat{\mathbf{k}}\ \text{N}\cdot\text{m}.
\end{aligned}$$

Then, substituting these values into Eqs. (1) and (2) gives

$$\sum \mathbf{M}_A = 18.6\hat{\mathbf{k}} + 1.43\hat{\mathbf{k}} + \left(70.5F_{34}^x - 208.4F_{34}^y\right)\hat{\mathbf{k}} = \mathbf{0}$$

and

$$\sum \mathbf{M}_{O_4} = 21.6\hat{\mathbf{k}} + 33.8\hat{\mathbf{k}} - 95.6\hat{\mathbf{k}} + \left(-131.8F_{34}^x + 71.6F_{34}^y\right)\hat{\mathbf{k}} = \mathbf{0}.$$

Rearranging terms gives two equations in two unknowns, that is,

$$70.5F_{34}^x - 208.4F_{34}^y = -20.0\ \text{N}\cdot\text{m}$$

and

$$-131.8F_{34}^x + 71.6F_{34}^y = 40.2\text{N}\cdot\text{m}.$$

Solving simultaneously, the two components of the force vector $\mathbf{F}_{34}$ are:

$$F_{34}^x = -0.310\ 16\ \text{kN} = -310.16\ \text{N} \quad \text{and} \quad F_{34}^y = -0.008\ 84\ \text{kN} = -8.84\ \text{N}.$$

Therefore, the force vector is

$$\mathbf{F}_{34} = -310.16\hat{\mathbf{i}} - 8.84\hat{\mathbf{j}}\ \text{N} = 310.29\ \text{N}\angle -178.4^\circ. \qquad \textit{Ans.}$$

Next, summing forces on link 4 yields the equation

$$\Sigma\mathbf{F}_{i4} = \mathbf{F}_{14} + \mathbf{F}_{34} + \mathbf{F}_C + (-m_4\mathbf{A}_{G_4}) = \mathbf{0},$$

and, upon solving, we find the force vector

$$\begin{aligned}\mathbf{F}_{14} &= -\mathbf{F}_{34} - \mathbf{F}_C - \left(-m_4\mathbf{A}_{G_4}\right)\\ &= -\left(-310.16\hat{\mathbf{i}} - 8.84\hat{\mathbf{j}}\ \text{N}\right) - \left(-800\hat{\mathbf{j}}\ \text{N}\right) - \left(313\hat{\mathbf{i}} + 415\hat{\mathbf{j}}\ \text{N}\right)\\ &= -2.84\hat{\mathbf{i}} + 393.84\hat{\mathbf{j}}\ \text{N} = 393.85\ \text{N}\angle 90.41^\circ. \qquad \textit{Ans.}\end{aligned}$$

Similarly, summing forces on link 3 yields

$$\Sigma\mathbf{F}_{13} = \mathbf{F}_{23} + \mathbf{F}_{43} + (-m_3\mathbf{A}_{G_3}) = \mathbf{0},$$

and solving this, the force vector is

$$\begin{aligned}\mathbf{F}_{23} &= -(-\mathbf{F}_{34}) - \left(-m_3\mathbf{A}_{G_3}\right)\\ &= -\left(310.16\hat{\mathbf{i}} + 8.84\hat{\mathbf{j}}\ \text{N}\right) - \left(-70.2\hat{\mathbf{i}} + 233\hat{\mathbf{j}}\ \text{N}\right)\\ &= -239.96\hat{\mathbf{i}} - 241.84\hat{\mathbf{j}}\ \text{N} = 340.69\ \text{N}\angle -134.78^\circ. \qquad \textit{Ans.}\end{aligned}$$

Now, for link 2, we have

$$\Sigma\mathbf{F}_{i2} = \mathbf{F}_{12} + \mathbf{F}_{32} + (-m_2\mathbf{A}_{G_2}) = \mathbf{0}.$$

Rearranging this gives

$$\mathbf{F}_{12} = -\mathbf{F}_{32} = \mathbf{F}_{23} = -239.96\hat{\mathbf{i}} - 241.84\hat{\mathbf{j}}\ \text{N} = 340.69\ \text{N}\angle -134.78^\circ. \qquad \textit{Ans.}$$

Summing moments on link 2 about O_2, gives

$$\sum \mathbf{M}_{O_2} = \mathbf{R}_{AO_2} \times \mathbf{F}_{32} + \mathbf{T}_{12} + \left(-I_{G_2}\boldsymbol{\alpha}_2\right) = \mathbf{0},$$

and solving for the driving torque gives

$$\mathbf{T}_{12} = -\mathbf{R}_{AO_2} \times \mathbf{F}_{32} = -\left(-25.4\hat{\mathbf{i}} + 54.4\hat{\mathbf{j}}\ \text{mm}\right) \times \left(239.96\hat{\mathbf{i}} + 241.84\hat{\mathbf{j}}\ \text{N}\right) = 19.2\hat{\mathbf{k}}\ \text{N}\cdot\text{m}.$$

Ans.

12.5 PRINCIPLE OF SUPERPOSITION

Linear systems are those in which effect is proportional to cause. This means that the response, or output, of a linear system is directly proportional to the drive, or input, to the system. An example of a linear system is a spring, where the deflection (output) is directly proportional to the force (input) exerted on the spring.

The *principle of superposition* may be used to solve problems involving linear systems by considering each of the inputs to the system separately. If the system is linear, the responses to each of these inputs can be summed or superimposed on each other to determine the total response of the system. Thus the principle of superposition states that, *for a linear system, the individual responses to several disturbances, or driving functions, can be superimposed on each other to obtain the total response of the system.*

The principle of superposition does not apply to nonlinear systems. Some examples of nonlinear systems, where superposition may not be used, are systems with static or Coulomb friction, systems with clearances or backlash, or systems with springs that change stiffness as they are deflected.

We have now reviewed all the principles necessary for making a complete dynamic force analysis of a planar mechanism. The steps in using the principle of superposition for making such an analysis are summarized as follows:

1. Perform a kinematic analysis of the mechanism. Locate the center of mass of each link, and find the acceleration of each mass center; also, find the angular acceleration of each link.
2. If the inertia forces are attached to all links simultaneously, along with other applied forces and moments, then often there are no two- or three-force members, and it may be difficult to find the lines of action of unknown constraint forces. Instead of doing this, it is sometimes convenient to ignore the masses and applied forces and moments on all but one or two links and to leave other links as two- or three-force members. By choosing in this manner, a solution may become possible for the constraint forces caused by the masses or applied forces and moments being considered, but without those caused by the masses and applied forces and moments being ignored.
3. Those masses and applied forces and moments considered in Step 2 can be ignored while a solution is obtained for additional constraint force components caused by some of the previously ignored masses or applied forces and moments. This process can be repeated until constraint force components caused by all masses and all applied forces and moments are found.
4. The results of Steps 2 and 3 can be vectorially added to obtain the resultant forces and moments on each link caused by the combined effects of all masses and all applied forces and moments.

This process of superposition is demonstrated in the following example.

EXAMPLE 12.5

Perform a complete dynamic force analysis of the four-bar linkage in the posture shown in Fig. 12.10, using the principle of superposition. The input crank is operating at constant angular velocity $\omega_2 = 60$ rad/s ccw, and the applied force is $\mathbf{F}_C = 40\hat{\mathbf{i}}$ lb. The weights of links 3 and 4 are $w_3 = 7.13$ lb and $w_4 = 3.42$ lb, respectively, and the mass moments of inertia of links 2, 3, and 4 are $I_{G_2} = 0.25\ \text{in} \cdot \text{lb} \cdot \text{s}^2$ $I_{G_3} = 0.625\ \text{in} \cdot \text{lbs}^2$, and $I_{G_4} = 0.037\ \text{in} \cdot \text{lb} \cdot \text{s}^2$, respectively.

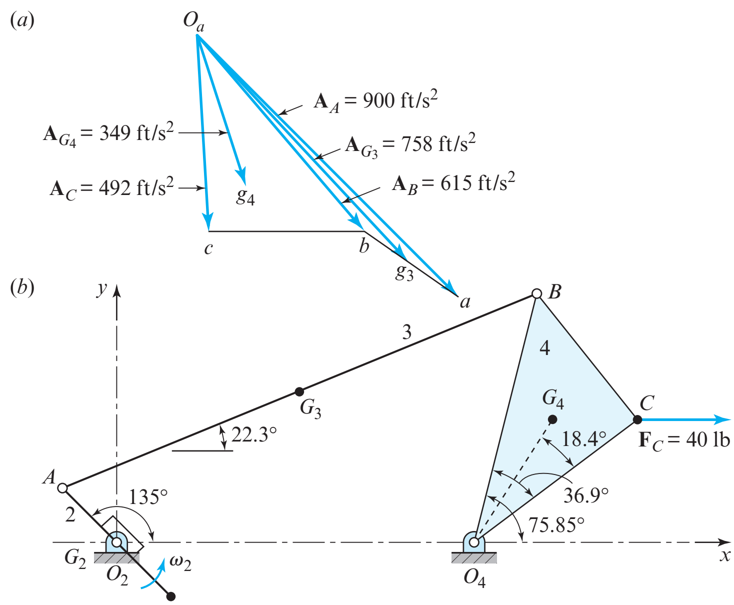


Figure 12.10 Example 14.5: (*a*) $R_{AO_2} = 3$ in, $R_{O_4O_2} = 14$ in, $R_{BA} = 20$ in, $R_{BO_4} = 10$ in, $R_{CO_4} = 8$ in, $R_{CB} = 6$ in, $R_{G_3A} = 10$ in, $R_{G_4O_4} = 5.69$ in; (*b*) acceleration polygon.

ASSUMPTIONS

Gravity and friction effects are negligible.

SOLUTION

STEP 1. All of the details of a complete kinematic analysis of the linkage are not included here, but the resulting acceleration polygon is shown in Fig. 12.10*a* (the numeric results are shown on the acceleration polygon). From the methods of Chap. 4, the angular accelerations of links 3 and 4 are

$$\alpha_3 = 148 \text{ rad/s}^2 \text{ ccw} \quad \text{and} \quad \alpha_4 = 604 \text{ rad/s}^2 \text{ cw}.$$

STEP 2. Since the center of mass of link 2 is located at bearing O_2, the force analysis is concerned primarily with links 3 and 4. Free-body diagrams of links 4 and 3 are shown in Figs. 12.11 and 12.12, respectively. Note that these diagrams are arranged in pseudo-equation form to emphasize the concept of superposition. Thus, in each figure, the forces in (*a*) plus those in (*b*) and (*c*) produce the results shown in (*d*). The action and reaction forces are also correlated; for example, $\mathbf{F}'_{34}$ in Fig. 12.11*a* is equal to $-\mathbf{F}'_{43}$ in Fig. 12.12*a* and so on. The following analysis is not difficult, but it is complex; therefore, it is important to read it slowly and to examine the figures carefully, detail by detail.

We start by considering the effects of the mass of link 4 alone, while ignoring the masses of links 2 and 3. Using Figs. 12.11*a* and 12.12*a*, we make the following calculations:

$$I_{G_4}\alpha_4 = \left(0.037\ \text{in}\cdot\text{lb}\cdot\text{s}^2\right)\left(604\ \text{rad/s}^2\right) = 22.3\ \text{in}\cdot\text{lb cw},$$

$$m_4 A_{G_4} = \frac{3.42\ \text{lb}}{32.2\ \text{ft/s}^2}\left(349\ \text{ft/s}^2\right) = 37.1\ \text{lb},$$

$$h_4 = \frac{I_{G_4}\alpha_4}{m_4 A_{G_4}} = \frac{22.3\ \text{in}\cdot\text{lb}}{37.1\ \text{lb}} = 0.602\ \text{in.}$$

Now, we position the inertia force $-m_4 A_{G_4} = 37.1$ lb on the free-body diagram of link 4 opposite in direction to $\mathbf{A}_{G_4}$ and offset from G_4 by the distance $h_4 = 0.602$ in. The direction of the offset is to the right of G_4 so that the inertia force $-m_4 A_{G_4}$ produces a counterclockwise inertia moment $-I_{G_4}\alpha_4$ about G_4, opposite to the clockwise sense of $\boldsymbol{\alpha}_4$. The line of action of $\mathbf{F}'_{34}$ is along link 3 for part (*a*) of our superposition approach, since link 3 is a two-force member in this step. The intersection of the inertia force $-m_4\mathbf{A}_{G_4}$ and the line of action of $\mathbf{F}'_{34}$ gives the point of concurrency and establishes the line of action of $\mathbf{F}'_{14}$. The force polygon for part (*a*) for link 4 can be constructed and the magnitudes of $\mathbf{F}'_{34}$ and $\mathbf{F}'_{14}$ can be determined. These values are shown in the legend to Fig. 12.11.

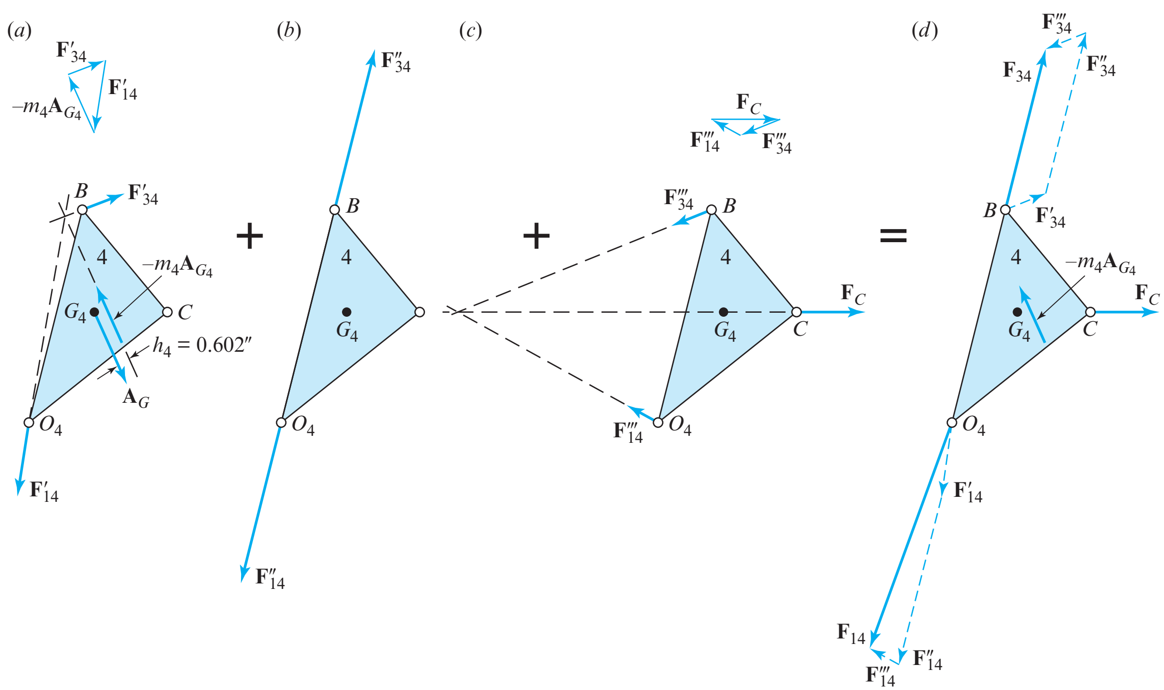


Figure 12.11 Free-body diagrams of link 4 showing superposition of forces: (*a*) $F'_{34} = 24.3$ lb and $F'_{14} = 44.3$ lb; (*b*) $F''_{34} = F''_{14} = 94.8$ lb; (*c*) $F'''_{34} = 25$ lb and $F'''_{14} = 19.3$ lb; (*d*) $F_{34} = 94.3$ lb and $F_{14} = 132$ lb.

Note that the forces $\mathbf{F}'_{43}$ and $\mathbf{F}'_{23}$ are now known, as shown in Fig. 12.12*a*.

STEP 3. Now, we consider the effects of the mass of link 3 alone. Using Fig. 12.12*b*, we make the following calculations:

$$I_{G_3}\alpha_3 = \left(0.625\ \text{in}\cdot\text{lb}\cdot\text{s}^2\right)\left(148\ \text{rad/s}^2\right) = 92.5\ \text{in}\cdot\text{lb ccw},$$

$$m_3A_{G_3} = \frac{7.13\ \text{lb}}{32.2\ \text{ft/s}^2}\left(758\ \text{ft/s}^2\right) = 168\ \text{lb},$$

$$h_3 = \frac{I_{G_3}\alpha_3}{m_3A_{G_3}} = \frac{92.5\ \text{in}\cdot\text{lb}}{168\ \text{lb}} = 0.550\ \text{in}.$$

Now, we position the inertia force $-m_3A_{G_3} = 168$ lb on the free-body diagram of link 3 opposite in direction to $\mathbf{A}_{G_3}$ and offset from G_3 by the distance $h_3 = 0.550$ in. The direction of the offset is to the left of G_3 so that the inertia force $-m_3\mathbf{A}_{G_3}$ produces a

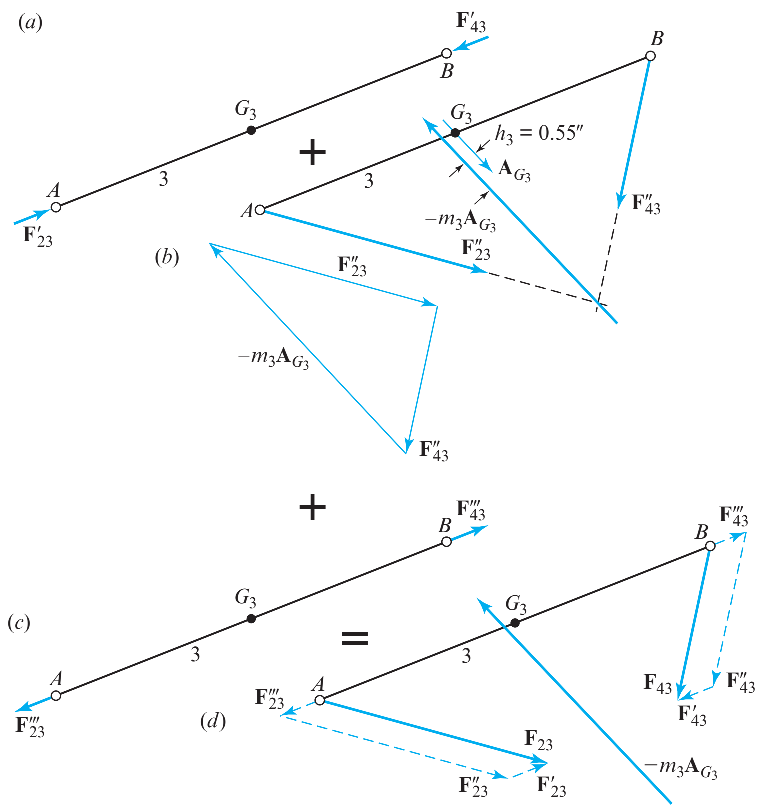


Figure 12.12 Free-body diagrams of link 3 showing superposition of forces: (*a*) $F'_{23} = F'_{43} = 24.3$ lb; (*b*) $F''_{23} = 145$ lb and $F''_{43} = 94.8$ lb; (*c*) $F'''_{23} = F'''_{43} = 25$ lb; (*d*) $F_{23} = 145$ lb and $F_{43} = 94.3$ lb.

clockwise inertia moment $-I_{G_3}\alpha_3$ about G_3, opposite to the counterclockwise sense of $\boldsymbol{\alpha}_3$. The line of action of $\mathbf{F}''_{43}$ is along link 4 for part (*b*) of our superposition approach, since link 4 is a two-force member in this step. The intersection of the line of action of $\mathbf{F}''_{43}$ and the inertia force $-m_3\mathbf{A}_{G_3}$ gives the point of concurrency. Thus, the line of action of $\mathbf{F}''_{23}$ is known, and the force polygon for link 3 can be constructed. The resulting magnitudes of $\mathbf{F}''_{43}$ and $\mathbf{F}''_{23}$ are included in the legend of Fig. 12.12.

Note that the forces $\mathbf{F}''_{34}$ and $\mathbf{F}''_{14}$ are now known, as shown in Fig. 12.11*b*. The results of the static force analysis with the applied force $F_c = 40$ lb as the specified loading, but with no inertia forces, as shown in Fig. 12.11*c*.

STEP 4. Recognizing that link 3 is again a two-force member, the force polygon in Fig. 12.11*c* determines the values of the forces acting on link 4. From these forces, the magnitudes and directions of the forces $\mathbf{F}''_{43}$ and $\mathbf{F}''_{23}$ acting on link 3 can be determined, as shown in Fig. 12.12*c*.

STEP 5. Now we perform the superposition of the results obtained in Steps 2, 3, and 4; this is accomplished by the vector additions shown in part (*d*) of each figure. The analysis is completed by taking the resultant force $\mathbf{F}_{23}$ from Fig. 12.12*d* and applying the negative value, $\mathbf{F}_{32}$, to link 2. This is shown in the free-body diagram of link 2, Fig. 12.13. There are no inertia force or inertia moment on link 2 because the center of mass is stationary and the angular velocity is constant. The offset distance, h_2, is measured as $h_2 = 1.56$ in Therefore, the external torque applied to link 2 is

$$T_{12} = h_2 F_{32} = (1.56 \text{ in})\,(145 \text{ lb}) = 226 \text{ in} \cdot \text{lb cw}.$$

Note that this torque is opposite in sense to the direction of rotation of link 2; this is not true for the entire cycle of operation, but it can occur at particular crank angles. The torque must sometimes be in the reverse direction to maintain the constant angular velocity of link 2.

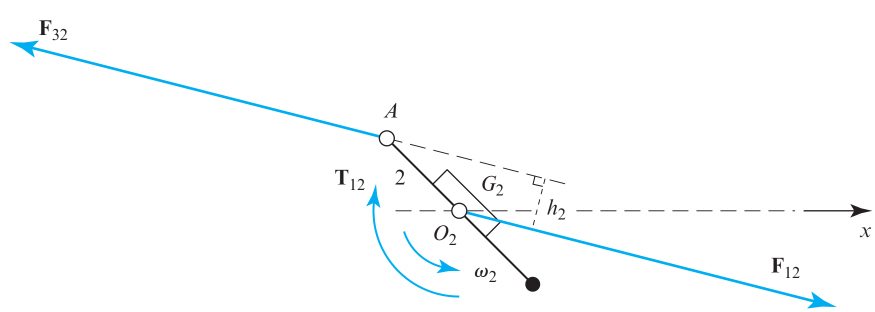


Figure 12.13 Free-body diagram of link 2 showing $F_{32} = F_{12} = 145$ lb and $T_{12} = 226$ in · lb cw.

12.6 PLANAR ROTATION ABOUT A FIXED CENTER

The previous sections have dealt with the case of dynamic forces for a rigid body having general planar motion. It is important to emphasize that the equations and methods of analysis investigated in these sections are general and apply to *all* problems with planar motion. It will be interesting now to study the application of these methods to a special case, that of a rigid body rotating about a fixed point [4].

Let us consider a rigid body constrained as shown in Fig. 12.14*a* to rotate with an angular velocity, $\boldsymbol{\omega}$, about some fixed point, O, not coincident with its center of mass, G. A system of forces (not shown) is applied to the body, causing it to undergo an angular acceleration, $\boldsymbol{\alpha}$. This motion of the body implies that the mass center, G, has normal and tangential components of acceleration, $\mathbf{A}^n_{GO}$ and $\mathbf{A}^t_{GO}$, whose magnitudes are $R_{GO}\omega^2$ and $R_{GO}\alpha$, respectively, as shown in Fig. 12.14*b*. Thus, if we resolve the resultant external force into normal and tangential components, these components must have magnitudes

$$F^n = mR_{GO}\omega^2 \quad \text{and} \quad F^t = mR_{GO}\alpha, \tag{a}$$

in accordance with Eq. (12.12). In addition, Eq. (12.13) states that an external moment must exist to create the angular acceleration and that the magnitude of this moment must be $M_G = I_G\alpha$.

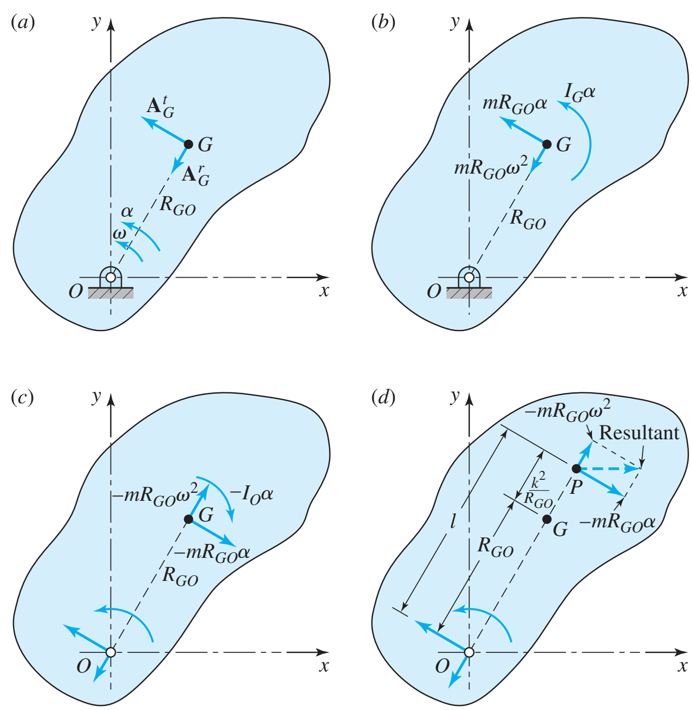


Figure 12.14

If we sum the moments of these forces about O, we have

$$\Sigma M_O = I_G\alpha + R_{GO}(mR_{GO}\alpha) = (I_G + mR_{GO}^2)\alpha. \tag{b}$$

Observe that the quantity in parentheses in Eq. (b) is identical in form with Eq. (12.11) and transfers the moment of inertia to an axis through O that is parallel to an axis through the center of mass. Therefore, Eq. (b) can be written in the form

$$\Sigma M_O = I_O\alpha.$$

Equations (12.15) and (12.16b) then become

$$\Sigma\mathbf{F} - m\mathbf{A}_G = \mathbf{0} \tag{12.19}$$

and

$$\Sigma\mathbf{M}_O - I_O\boldsymbol{\alpha} = \mathbf{0}, \tag{12.20}$$

by including the inertia force $-m\mathbf{A}_G$ and inertia moment $-I_O\boldsymbol{\alpha}$, as shown in Fig. 12.14c. We observe particularly that the system of forces does *not* reduce to a single couple because of the existence of the inertia force component, $-m\mathbf{R}_{GO}\omega^2$, which has no moment arm about point O. Thus, both Eqs. (12.19) and (12.20) are necessary.

A special case arises when the angular acceleration of the body is $\boldsymbol{\alpha} = \mathbf{0}$. Then, the external moment, $\Sigma\mathbf{M}_O$, is zero, and the only force is, from Fig. 12.14c, the inertia force, $-m\mathbf{R}_{GO}\omega^2$. A second special case exists under starting conditions, when $\omega = 0$ but α is not zero. Under these conditions, the only inertia force is $-mR_{GO}\boldsymbol{\alpha}$, and the system reduces to a single couple.

When a rigid body has only translation, the resultant inertia force and the resultant external force have the same line of action, which passes through the center of mass of the body. When a rigid body has both angular velocity and acceleration, again, the resultant inertia force and the resultant external force have the same line of action; however, in this case, this line does *not* pass through the center of mass. Let us now locate a point on this offset line of action.

The resultant inertia forces passes through some point, P, on line OG or its extension (Fig. 12.14d). This force can be resolved into two components: the component, $-m\mathbf{R}_{GO}\omega^2$, acting along OG; and the component, $-mR_{GO}\boldsymbol{\alpha}$, acting perpendicular to OG but not through point G. The distance from O to P, designated as l, can be determined by equating the moment of component $-mR_{GO}\boldsymbol{\alpha}$ through P (Fig. 12.14d) to the sum of the inertia moment and the moment of the inertia forces that act through G (Fig. 12.14c). Thus, taking moments about O in each figure, we have

$$(-mR_{GO}\alpha)l = -I_G\alpha + (-mR_{GO}\alpha)R_{GO}$$

or

$$l = \frac{I_G}{mR_{GO}} + R_{GO}. \tag{c}$$

Substituting Eq. (12.10) into Eq. (c) and eliminating mass gives

$$l = \frac{k^2}{R_{GO}} + R_{GO}. \tag{12.21}$$

Point P, located by Eq. (12.21), is called the *center of percussion.* As indicated, the resultant inertia force passes through P, and, consequently, the inertia force has zero moment about the center of percussion. If an external force is applied at P, perpendicular to OG, an angular acceleration, α, results, but the bearing reaction at O is zero except for the component caused by the inertia force, $-mR_{GO}\omega^2$. It is the usual practice in shock testing machines to apply the force at the center of percussion in order to eliminate the tangential bearing reaction that is otherwise caused by the externally applied shock load.

As another example, suppose we consider the impact of a baseball against a bat. If we crudely approximate the bat as a cylindric rod of length L, then its center of mass is located at approximately $R_G = L/2$, and its radius of gyration is approximately $k = L/\sqrt{12}$. Substituting these values into Eq. (12.21) gives the location of the center of percussion as $l = 2/3L$. Hitting the ball at that point produces no dynamic reaction force on the hands and is commonly referred to as the "sweet spot" in both baseball and tennis. Equation (12.21) shows that the location of the center of percussion is independent of the values of ω and α. If the axis of rotation is coincident with the center of mass, $R_{GO} = 0$ and Eq. (12.21) shows that $l = \infty$. Under these conditions there is no resultant inertia force but rather a resultant inertia couple, $-I_G\alpha$.

12.7 SHAKING FORCES AND MOMENTS

Of special interest to the designer are the forces transmitted to the frame or foundation of a machine owing to the inertia of the moving links. When these forces vary in magnitude or direction, they tend to shake or vibrate the machine (and the frame); consequently, such effects are called *shaking forces* and *shaking moments*.

If we consider some machine, say a four-bar linkage, for example, with links 2, 3, and 4 as the moving members and link 1 as the frame, then taking the entire group of moving parts as a system, not including the frame, and drawing a free-body diagram, we can immediately write

$$\sum \mathbf{F} = \mathbf{F}_{12} + \mathbf{F}_{14} + (-m_2\mathbf{A}_{G_2}) + (-m_3\mathbf{A}_{G_3}) + (-m_4\mathbf{A}_{G_4}) = \mathbf{0}.$$

Using $\mathbf{F}_s$ as a symbol for the resulting shaking force, we define this as equal to the resultant of all the reaction forces on ground link 1,

$$\mathbf{F}_s = \mathbf{F}_{21} + \mathbf{F}_{41}.$$

Therefore, from the previous equation, we have

$$\mathbf{F}_s = (-m_2\mathbf{A}_{G_2}) + (-m_3\mathbf{A}_{G_3}) + (-m_4\mathbf{A}_{G_4}). \tag{12.22}$$

Generalizing from this example, we can write the shaking force for any machine as

$$\mathbf{F}_s = \sum(-m_j\mathbf{A}_{G_j}). \tag{12.23}$$

This makes sense, since, if we consider a free-body diagram of the entire machine *including the frame,* all other applied and constraint forces have equal and opposite reaction forces, and these cancel within the free-body system. Only the inertia forces, having no reactions, are ultimately external to the system and remain unbalanced.* These are not balanced by reaction forces and produce unbalanced shaking effects between the frame and the surface on which it is mounted. These are the forces that require that the machine be fastened down to prevent it from moving.

A similar derivation can be made for unbalanced moments. Using the symbol $\mathbf{M}_s$ for the shaking moment, we take the summation of moments about the coordinate system origin and find

$$\mathbf{M}_s = \sum \left[\mathbf{R}_{G_j} \times (-m_j \mathbf{A}_{G_j})\right] + \sum (-I_{G_j} \boldsymbol{\alpha}_j). \tag{12.24}$$

12.8 COMPLEX-ALGEBRAIC APPROACH

In this section, we investigate the dynamic force analysis of a mechanism using the complex-algebraic approach. The general procedure that is adopted is as follows:

1. Perform a complete kinematic analysis.
2. Draw a free-body diagram of each link of the mechanism.
3. Write the dynamic force equations for each moving link.
4. Corresponding to each input angle, compute the following items:
   a. the axial and the transverse forces exerted on each moving link;
   b. the force components exerted on the ground bearings (here referred to as the *bearing loads*);
   c. the inertia moment; and
   d. the shaking moment on the foundation.

To demonstrate the complex-algebraic approach, we consider the four-bar linkage shown in Fig. 12.15.

For a specified angular velocity and acceleration of input link 2, the angular velocities and accelerations of coupler link 3 and output link 4 are determined following the methods of Chaps. 3 and 4. The accelerations of the mass centers of links 2, 3, and 4 are now expressed in coordinates attached to and aligned along the respective links.

The acceleration of the mass center of the input link 2 can be written as

$$\mathbf{A}_{G_2} = l_2(-\omega_2^2 + j\alpha_2)e^{j\theta_2} = (A_{G_2}^{x_2} + jA_{G_2}^{y_2})e^{j\theta_2}. \tag{12.25a}$$

Therefore, the x_2 and y_2 components of the acceleration, respectively, are

$$A_{G_2}^{x_2} = -l_2\omega_2^2 \quad \text{and} \quad A_{G_2}^{y_2} = -l_2\alpha_2. \tag{12.25b}$$

* The same arguments can be made about gravitational, magnetic, electric, or other non-contacting-force fields. However, because these are usually static forces, it is not customary to include them in the definition of shaking force.

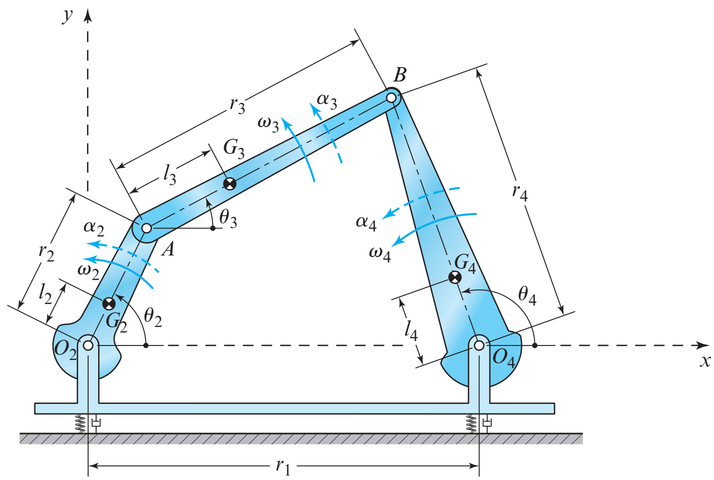


Figure 12.15 Four-bar linkage.

The acceleration of the mass center of the coupler link 3 can be written as

$$\mathbf{A}_{G_3} = r_2(-\omega_2^2 + j\alpha_2)e^{j\theta_2} + l_3(-\omega_3^2 + j\alpha_3)e^{j\theta_3}.$$

or as

$$\mathbf{A}_{G_3} = \left[r_2(-\omega_2^2 + j\alpha_2)e^{j\eta_{23}} + l_3(-\omega_3^2 + j\alpha_3)\right]e^{j\theta_3}, \tag{12.26}$$

where $\eta_{23} = \theta_2 - \theta_3$. The acceleration of the mass center of the coupler link can also be written as

$$\mathbf{A}_{G_3} = (A_{G_3}^{x_3} + jA_{G_3}^{y_3})e^{j\theta_3}. \tag{12.27}$$

Equating Eqs. (12.26) and (12.27), the x_3 and y_3 components of the acceleration are

$$A_{G_3}^{x_3} = r_2(-\omega_2^2 \cos\eta_{23} - \alpha_2 \sin\eta_{23}) - l_3\omega_3^2, \tag{12.28a}$$

$$A_{G_3}^{y_3} = r_2(-\omega_2^2 \sin\eta_{23} + \alpha_2 \cos\eta_{23}) - l_3\alpha_3. \tag{12.28b}$$

The acceleration of the mass center of the output link 4 can be written as

$$\mathbf{A}_{G_4} = l_4\left(-\omega_4^2 + j\alpha_4\right)e^{j\theta_4} = \left(A_{G_4}^{x_4} + jA_{G_4}^{y_4}\right)e^{j\theta_4}. \tag{12.29a}$$

Therefore, the x_4 and y_4 components of the acceleration, respectively, are

$$A_{G_4}^{x_4} = -l_4\omega_4^2 \quad \text{and} \quad A_{G_4}^{y_4} = l_4\alpha_4. \tag{12.29b}$$

The free-body diagrams of the four links are presented in Fig. 12.16.

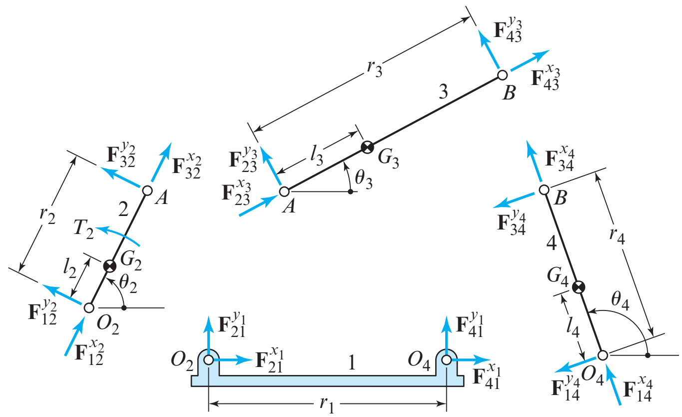


Figure 12.16 Free-body diagrams for the four-bar linkage.

For the input link, the sum of the forces gives

$$\left[(F_{12}^{x_2}+F_{32}^{x_2})+j(F_{12}^{y_2}+F_{32}^{y_2})\right]e^{j\theta_2}=m_2(A_{G_2}^{x_2}+jA_{G_2}^{y_2})e^{j\theta_2}, \tag{12.30a}$$

and the sum of the moments about bearing O_2 gives

$$r_2F_{32}^{y_2}+T_2=(I_{G_2}+m_2l_2^2)\alpha_2. \tag{12.30b}$$

For the coupler link, the sum of the forces gives

$$\left[(F_{23}^{x_3}+F_{43}^{x_3})+j(F_{23}^{y_3}+F_{43}^{y_3})\right]e^{j\theta_3}=m_3(A_{G_3}^{x_3}+jA_{G_3}^{y_3})e^{j\theta_3}, \tag{12.31a}$$

and the sum of the moments about the mass center, G_3, gives

$$F_{43}^{y_3}(r_3-l_3)-F_{23}^{y_3}l_3=I_{G_3}\alpha_3. \tag{12.31b}$$

For the output link, the sum of the forces gives

$$\left[(F_{14}^{x_4}+F_{34}^{x_4})+j(F_{14}^{y_4}+F_{34}^{y_4})\right]e^{j\theta_4}=m_4(A_{G_4}^{x_4}+jA_{G_4}^{y_4})e^{j\theta_4}, \tag{12.32a}$$

and the sum of the moments about bearing O_4 gives

$$r_4F_{34}^{y_4}=(I_{G_3}+m_4l_4^2)\alpha_4. \tag{12.32b}$$

For revolute joint A, the sum of the forces gives

$$(F_{32}^{x_2}+jF_{32}^{y_2})e^{j\theta_2}+(F_{23}^{x_3}+jF_{23}^{y_3})e^{j\theta_3}=0, \tag{12.33a}$$

and for revolute joint B, the sum of the forces gives

$$(F_{43}^{x_3}+jF_{43}^{y_3})e^{j\theta_3}+(F_{34}^{x_4}+jF_{34}^{y_4})e^{j\theta_4}=0. \tag{12.33b}$$

For bearing O_2, the sum of the forces gives

$$(F_{21}^{x_1} + jF_{21}^{y_1}) + (F_{12}^{x_2} + jF_{12}^{y_2})e^{j\theta_2} = 0, \tag{12.33c}$$

and for bearing O_4, the sum of the forces gives

$$(F_{41}^{x_1} + jF_{41}^{y_1}) + (F_{14}^{x_4} + jF_{14}^{y_4})e^{j\theta_4} = 0. \tag{12.33d}$$

To obtain solutions for the dynamic force analysis, we proceed as follows. From Eq. (12.32*b*), the transverse force acting at point B on output link 4 is

$$F_{34}^{y_4} = (I_{G_4} + m_4 l_4^2)\frac{\alpha_4}{r_4}. \tag{12.34}$$

Equating the real and imaginary parts of Eq. (12.30*a*), the x_2 and y_2 components of the forces, respectively, are

$$F_{12}^{x_2} + F_{32}^{x_2} = m_2 A_{G_2}^{x_2}, \tag{12.35a}$$

$$F_{12}^{y_2} + F_{32}^{y_2} = m_2 A_{G_2}^{y_2}. \tag{12.35b}$$

Equating the real and imaginary parts of Eq. (12.31*a*), the x_3 and y_3 components of the forces, respectively, are

$$F_{23}^{x_3} + F_{43}^{x_3} = m_3 A_{G_3}^{x_3}, \tag{12.36a}$$

$$F_{23}^{y_3} + F_{43}^{y_3} = m_3 A_{G_3}^{y_3}. \tag{12.36b}$$

Equating the real and imaginary parts of Eq. (12.32*a*), the x_4 and y_4 components of the forces, respectively, are

$$F_{14}^{x_4} + F_{34}^{x_4} = m_4 A_{G_4}^{x_4}, \tag{12.37a}$$

$$F_{14}^{y_4} + F_{34}^{y_4} = m_4 A_{G_4}^{y_4}. \tag{12.37b}$$

Substituting Eq. (12.36*b*) into Eq. (12.31*b*), the transverse force acting at revolute joint B on the coupler link is

$$F_{43}^{y_3} = \frac{1}{r_3}(I_{G_e}\alpha_3 + m_3 l_3 A_{G_3}^{y_3}). \tag{12.38a}$$

Then from Eq. (12.36*b*), the transverse force acting at revolute joint A on the coupler link is

$$F_{23}^{y_3} = m_3 A_{G_3}^{y_3} - F_{43}^{y_3}. \tag{12.38b}$$

Equation (12.33*b*) can be written as

$$(F_{43}^{x_3} + jF_{43}^{y_3})e^{j\eta_{34}} + (F_{34}^{x_4} + jF_{34}^{y_4}) = 0 \tag{12.39a}$$

or as

$$(F_{43}^{x3} + jF_{43}^{y3}) + (F_{34}^{x4} + jF_{34}^{y4})e^{-j\eta_{34}} = 0, \tag{12.39b}$$

where $\eta_{34} = \theta_3 - \theta_4$. Equating the imaginary parts of Eq. (12.39*a*), and equating the imaginary parts of Eq. (12.39*b*), respectively, gives

$$F_{43}^{x3} \sin\eta_{34} + F_{43}^{y3} \cos\eta_{34} + F_{34}^{y4} = 0, \tag{12.40a}$$

$$F_{43}^{y3} - (F_{34}^{x4} \sin\eta_{34} - F_{34}^{y4} \cos\eta_{34}) = 0. \tag{12.40b}$$

Then, from these two equations we have

$$F_{43}^{x3} = -(F_{43}^{y3} \cot\eta_{34} + F_{34}^{y4} \csc\eta_{34}), \tag{12.41a}$$

$$F_{34}^{x4} = F_{43}^{y3} \csc\eta_{34} + F_{34}^{y4} \cot\eta_{34}. \tag{12.41b}$$

Then, from Eqs. (12.36*a*) and (12.37*a*), the axial forces acting on links 3 and 4, respectively, are

$$F_{23}^{x3} = m_3 A_{G_3}^{x3} - F_{43}^{x3}, \tag{12.42a}$$

$$F_{14}^{x4} = m_4 A_{G_4}^{x4} - F_{34}^{x4}. \tag{12.42b}$$

Rewriting Eq. (12.33*a*) as

$$F_{32}^{x2} + jF_{32}^{y2} = -(F_{23}^{x3} + jF_{23}^{y3})e^{-j\eta_{23}}, \tag{14.43}$$

and equating the real and imaginary parts, respectively, gives

$$F_{32}^{x2} = -F_{23}^{x3} \cos\eta_{23} - F_{23}^{y3} \sin\eta_{23}, \tag{12.44a}$$

$$F_{32}^{y2} = F_{23}^{x3} \sin\eta_{23} - F_{23}^{y3} \cos\eta_{23}. \tag{12.44b}$$

The input torque (sometimes referred to as the *inertia torque*) can now be obtained by writing Eq. (12.30*b*) as

$$T_2 = -r_2 F_{32}^{y2} + (I_{G_2} + m_2 l_2^2)\alpha_2. \tag{12.45}$$

If the four-bar linkage is in *steady-state operation*, that is, if the input angular velocity is constant, then substituting $\alpha_2 = 0$ into Eq. (12.45) gives

$$T_2 = -r_2 F_{32}^{y2}. \tag{12.46}$$

From Eqs. (12.35) and (12.37) we have

$$F_{12}^{x2} = m_2 A_{G_2}^{x2} - F_{32}^{x2} \quad \text{and} \quad F_{14}^{x4} = m_4 A_{G_4}^{x4} - F_{34}^{x4}, \tag{12.47a}$$

$$F_{12}^{y2} = m_2 A_{G_2}^{y2} - F_{32}^{y2} \quad \text{and} \quad F_{14}^{y4} = m_4 A_{G_4}^{y4} - F_{34}^{y4}. \tag{12.47b}$$

If we write Eqs. (12.33*c*) and (12.33*d*), respectively, as

$$(F_{21}^{x_1}+jF_{21}^{y_1})=-(F_{12}^{x_2}+jF_{12}^{y_2})e^{j\theta_2},$$
$$(F_{41}^{x_1}+jF_{41}^{y_1})=-(F_{14}^{x_4}+jF_{14}^{y_4})e^{j\theta_4},$$

then the x and y components of the loads at bearing O_2, respectively, are

$$F_{21}^{x_1}=-F_{12}^{x_2}\cos\theta_2+F_{12}^{y_2}\sin\theta_2 \quad \text{and} \quad F_{21}^{y_1}=-F_{12}^{x_2}\sin\theta_2-F_{12}^{y_2}\cos\theta_2, \tag{12.48a}$$

and the x and y components of the loads at bearing O_4, respectively, are

$$F_{41}^{x_1}=-F_{14}^{x_4}\cos\theta_4+F_{14}^{y_4}\sin\theta_4 \quad \text{and} \quad F_{41}^{y_1}=-F_{14}^{x_4}\sin\theta_4-F_{14}^{y_4}\cos\theta_4. \tag{12.48b}$$

Finally, the dynamic shaking moment can be written as

$$M_s=r_1F_{41}^{y_1}.$$

Substituting Eq. (12.48*b*) into this equation gives

$$M_s=-r_1(F_{14}^{x_4}\sin\theta_4+F_{14}^{y_4}\cos\theta_4). \tag{12.49}$$

Let us consider the special case where the four-bar linkage is in static equilibrium because of the output torque, t_4, shown in Fig. 12.17, that is, the torque exerted on the load by the output crankshaft on bearing O_4. The free-body diagrams are as shown in Fig. 12.18.

From the dynamic force analysis, we can now determine the following as functions of the input angle:

1. The axial and the transverse force components exerted on the three moving links.
2. The bearing loads.
3. The inertia torque.
4. The shaking moment.

From Eqs. (12.30), the sum of the forces on link 2 and the sum of the moments about the bearing O_2 are

$$\left[(f_{12}^{x_2}+f_{32}^{x_2})+j(f_{12}^{y_2}+f_{32}^{y_2})\right]e^{j\theta_2}=0, \tag{12.50a}$$
$$r_2f_{32}^{y_2}+t_2=0. \tag{12.50b}$$

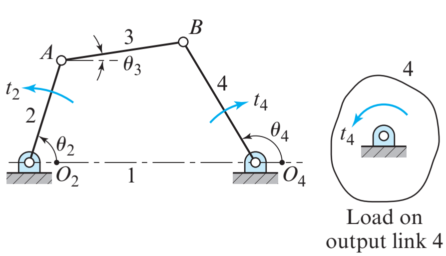


Figure 12.17 Output torque of the four-bar linkage.

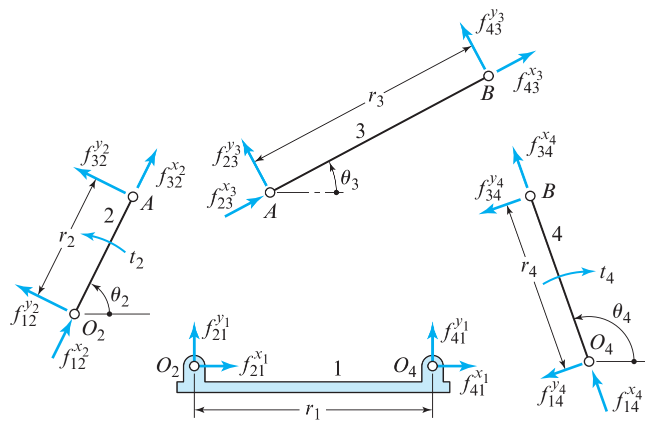


Figure 12.18 Free-body diagrams for the four-bar linkage.

From Eqs. (12.31), the sum of the forces on link 3 and the sum of the moments about the mass center G_3 are

$$\left[(f_{23}^{x_3} + f_{43}^{x_3}) + j(f_{23}^{y_3} + f_{43}^{y_3})\right] e^{j\theta_3} = 0, \tag{12.51a}$$

$$r_3 f_{43}^{y_3} = 0. \tag{12.51b}$$

From Eq. (12.32), the sum of the forces on link 4 and the sum of the moments about bearing O_4 are

$$\left[(f_{14}^{x_4} + f_{34}^{x_4}) + j(f_{14}^{y_4} + f_{34}^{y_4})\right] e^{j\theta_4} = 0, \tag{12.52a}$$

$$r_4 f_{34}^{y_4} + t_4 = 0. \tag{12.52b}$$

For revolute joint A, a sum of the forces gives

$$(f_{32}^{x_2} + jf_{32}^{y_2})e^{j\theta_2} + (f_{23}^{x_3} + jf_{23}^{y_3})e^{j\theta_3} = 0, \tag{12.53a}$$

and for revolute joint B, a sum of the forces gives

$$(f_{43}^{x_3} + jf_{43}^{y_3})e^{j\theta_3} + (f_{34}^{x_4} + jf_{34}^{y_4})e^{j\theta_4} = 0. \tag{12.53b}$$

For bearing O_2, a sum of the forces gives

$$(f_{21}^{x_1} + jf_{21}^{y_1}) + (f_{12}^{x_2} + jf_{12}^{y_2})e^{i\theta_2} = 0, \tag{12.54a}$$

and for bearing O_4, a sum of the forces gives

$$(f_{41}^{x_1} + jf_{41}^{y_1}) + (f_{14}^{x_4} + jf_{14}^{y_4})e^{i\theta_4} = 0. \tag{12.54b}$$

From Eqs. (12.51a), the axial and transverse forces at bearing A on the coupler link are

$$f_{23}^{x_3} = -f_{43}^{x_3} \quad \text{and} \quad f_{23}^{y_3} = -f_{43}^{y_3} = 0. \tag{12.55}$$

From Eq. (12.52b), the transverse force acting at bearing B on output link 4 is

$$f_{34}^{y_4} = -\frac{t_4}{r_4}. \tag{12.56}$$

Substituting Eq. (12.55) into Eq. (12.53a), and rearranging, gives

$$f_{32}^{x_2} + jf_{32}^{y_2} = -f_{23}^{x_3} e^{-j\eta_{23}}.$$

Equating the real and imaginary parts of this equation, respectively, gives

$$f_{32}^{x_2} = f_{43}^{x_3} \cos \eta_{23} \quad \text{and} \quad f_{32}^{y_2} = -f_{43}^{y_3} \sin \eta_{23}. \tag{12.57}$$

Equation (12.53b) can be written as

$$f_{43}^{x_3} e^{j\eta_{34}} + (f_{34}^{x_4} + jf_{34}^{y_4}) = 0.$$

Equating the real and imaginary parts of this equation, respectively, gives

$$f_{43}^{x_3} = -f_{23}^{x_3} = -\frac{t_4}{r_4} \frac{1}{\sin \eta_{34}} \quad \text{and} \quad f_{43}^{x_3} \cos \eta_{34} + f_{34}^{x_4} = 0. \tag{12.58}$$

Rearranging these two equations, the axial force on the output link can be written as

$$f_{34}^{x_4} = \frac{t_4}{r_4} \cot \eta_{34}. \tag{12.59}$$

From Eqs. (12.50a) and (12.52a) we have

$$f_{12}^{x_2} = -f_{32}^{x_2} \quad \text{and} \quad f_{12}^{y_2} = -f_{32}^{y_2}, \tag{12.60a}$$

$$f_{14}^{x_4} = -f_{34}^{x_4} \quad \text{and} \quad f_{14}^{y_4} = -f_{34}^{y_4}. \tag{12.60b}$$

Adding Eqs. (12.53a) and (12.54a) and with the aid of Eqs. (12.60a), we obtain

$$f_{21}^{x_1} + jf_{21}^{y_1} = f_{43}^{x_3} e^{j\theta_3}.$$

Equating the real and imaginary parts of this equation, respectively, gives

$$f_{21}^{x_1} = f_{43}^{x_3} \cos \theta_3 \quad \text{and} \quad f_{21}^{y_1} = f_{43}^{x_3} \sin \theta_3. \tag{12.61}$$

Adding Eqs. (12.53b) and (12.54b) and with the aid of Eqs. (12.60b), we obtain

$$f_{41}^{x_1} + jf_{41}^{y_1} = f_{43}^{x_3} e^{j\theta_3}.$$

Equating the real and imaginary parts of this equation, respectively, gives

$$f_{41}^{x_1} = f_{43}^{x_3} \cos \theta_3 \quad \text{and} \quad f_{41}^{y_1} = f_{43}^{x_3} \sin \theta_3. \tag{12.62}$$

Substituting Eq. (12.58) into Eq. (12.57), the transverse force acting at point A on the input link can be written as

$$f_{32}^{y_2} = \frac{t_4}{r_4}\frac{\sin\eta_{23}}{\sin\eta_{34}}.$$

Then, substituting this equation into (12.50b), and rearranging, the input torque can be written as

$$t_2 = -\frac{r_2}{r_4}\frac{\sin\eta_{23}}{\sin\eta_{34}}t_4. \tag{12.63}$$

Recall that the mechanical advantage of a four-bar linkage is defined as the size of the ratio of the output torque to the input torque (Secs. 1.10 and 3.20). Therefore, the mechanical advantage can be written from Eq. (12.63), as

$$MA = \frac{t_4}{t_2} = -\frac{r_4\sin\eta_{34}}{r_2\sin\eta_{23}}.$$

Note that this result is consistent with Eq. (3.41).

Finally, the shaking moment can be written with the aid of Eq. (12.62) as

$$m_s = r_1 f_{41}^{y_1} = r_1 f_{43}^{x_3}\sin\theta_3.$$

Substituting Eq. (12.58) into this equation, the shaking moment can be written as

$$m_s = \frac{r_1}{r_4}\frac{\sin\theta_3}{\sin\eta_{34}}t_4. \tag{12.64}$$

If friction in the mechanism is ignored, then the total load, which will be denoted here by an asterisk superscript, can be determined as the sum of the static load and the dynamic load; this is the principle of superposition. Therefore, the load torque is

$$T_4^* = t_4 + T_4. \tag{12.65}$$

The axial and transverse loads at bearing O_2, respectively, are

$$(F_{12}^{x_2})^* = -f_{32}^{x_2} + F_{12}^{x_2}, \quad (F_{12}^{y_2})^* = -f_{32}^{y_2} + F_{12}^{y_2}. \tag{12.66}$$

For the input link, the axial and transverse loads at revolute joint A, respectively, are

$$(F_{32}^{x_2})^* = f_{32}^{x_2} + F_{32}^{x_2}, \quad (F_{32}^{y_2})^* = f_{32}^{y_2} + F_{32}^{y_2}. \tag{12.67a}$$

For the coupler link, the axial and transverse loads at revolute joint A, respectively, are

$$(F_{23}^{x_3})^* = -f_{43}^{x_3} + F_{23}^{x_3}, \quad (F_{23}^{y_3})^* = F_{23}^{y_3}. \tag{12.67b}$$

For the coupler link, the axial and transverse loads at revolute joint B, respectively, are

$$(F_{43}^{x_3})^* = -f_{43}^{x_3} + F_{43}^{x_3}, \quad (F_{43}^{y_3})^* = F_{43}^{y_3}. \tag{12.68a}$$

For the output link, the axial and transverse loads at revolute joint B, respectively, are

$$(F_{34}^{x_4})^* = -f_{34}^{x_4} + F_{34}^{x_4}, \quad (F_{34}^{y_4})^* = f_{34}^{y_4} + F_{34}^{y_4}. \tag{12.68b}$$

The axial and transverse loads at bearing O_4, respectively, are

$$(F_{14}^{x_4})^* = -f_{34}^{x_4} + F_{14}^{x_4}, \quad (F_{14}^{y_4})^* = -f_{34}^{y_4} + F_{14}^{y_4}. \tag{12.69}$$

The x and y components of the load at bearing O_2, respectively, are

$$(F_{21}^{x_1})^* = f_{21}^{x_1} + F_{21}^{x_1} \quad \text{and} \quad (F_{21}^{y_1})^* = f_{21}^{y_1} + F_{21}^{y_1}. \tag{12.70a}$$

Therefore, the magnitude and the orientation, respectively, are

$$(F_{21})^* = \sqrt{(F_{21}^{x_1})^{*2} + (F_{21}^{y_1})^{*2}} \quad \text{and} \quad \psi_2 = \tan_2^{-1}\left(\frac{(F_{21}^{y_1})^*}{(F_{21}^{x_1})^*}\right). \tag{12.70b}$$

The x and y components of the load at bearing O_4, respectively, are

$$(F_{41}^{x_1})^* = f_{41}^{x_1} + F_{41}^{x_1} \quad \text{and} \quad (F_{41}^{y_1})^* = f_{41}^{y_1} + F_{41}^{y_1}. \tag{12.71a}$$

Therefore, the magnitude and orientation, respectively, are

$$(F_{41})^* = \sqrt{(F_{41}^{x_1})^{*2} + (F_{41}^{y_1})^{*2}} \quad \text{and} \quad \psi_4 = \tan_2^{-1}\left(\frac{(F_{41}^{y_1})^*}{(F_{41}^{x_1})^*}\right). \tag{12.71b}$$

The total driving torque to the input link 2 is

$$T_2^* = T_2 + t_2,$$

where the inertia input torque, T_2, and the static input torque, t_2, are given by Eqs. (12.45) and (12.63), respectively.

The total shaking moment is

$$M_s^* = M_s + m_s,$$

where the dynamic shaking moment, M_s, and the static shaking moment, m_s, are given by Eqs. (12.49) and (12.64), respectively.

For stable operation, we could vary the link lengths, l_1, l_2, and l_3, such that the power, $P = T_2^*\omega_2$, discussed in the following section, is maximized, and the shaking moment is minimized. For a better suspension system, we should try to minimize the vertical components of the loads at the two ground bearings, that is, $(F_{21}^{y_1})^*$ and $(F_{41}^{y_1})^*$.

12.9 EQUATION OF MOTION FROM POWER EQUATION

This section presents a method of obtaining the equation of motion of a mechanism based on energy considerations. We first write the power equation for the mechanism and then express this equation in terms of kinematic coefficients, as discussed in Chaps. 3 and 4.

The work-energy equation for a mechanism can be written as

$$W = \Delta E + \Delta U + W_f, \tag{12.72}$$

where

$W =$ the net work input to the mechanism, that is, the work input less the work output
$\Delta E =$ the change in the kinetic energy of the moving links
$\Delta U =$ the change in the potential energy stored in the mechanism; and
$W_f =$ the energy dissipated through damping and friction.

Differentiating Eq. (12.72) with respect to time gives

$$P = \frac{dW}{dt} = \frac{dE}{dt} + \frac{dU}{dt} + \frac{dW_f}{dt}, \tag{12.73}$$

where P is the *net power input* to the mechanism. Equation (12.73) is commonly referred to as the *power equation*. The net power is a scalar quantity and can be written as

$$P = Q\dot{\psi}, \tag{12.74}$$

where Q is the generalized input force (that is, the force or the torque acting on the input), and $\dot{\psi}$ is the generalized input velocity (that is, either the rectilinear or angular velocity of the input). If the product of these two signed quantities is positive (that is, if the signed quantities are acting in the same direction), then power is going into the system. If the product is negative (that is, if the two signed quantities are acting in opposite directions), then power is being removed from the system.

The kinetic energy of a link in the mechanism, say link j, can be written as

$$E_j = \frac{1}{2} m_j V_{G_j}^2 + \frac{1}{2} I_{G_j} \omega_j^2. \tag{12.75}$$

Recall from Sec. 3.12 that the velocity of the mass center of link, j, can be written in terms of first-order kinematic coefficients as

$$\mathbf{V}_{G_j} = (x'_{G_j}\hat{\mathbf{i}} + y'_{G_j}\hat{\mathbf{j}})\dot{\psi}, \tag{12.76a}$$

and the angular velocity of link j can be written as

$$\omega_j = \theta'_j \dot{\psi}. \tag{12.76b}$$

Substituting Eqs. (12.76) into Eq. (12.75), the kinetic energy of link j can be written as

$$E_j = \frac{1}{2} m_j \left(x'^2_{G_j} + y'^2_{G_j} \right) \dot{\psi}^2 + \frac{1}{2} I_{G_j} \theta'^2_j \dot{\psi}^2. \tag{12.77}$$

Differentiating this equation with respect to time, the time rate of change in the kinetic energy of link j can be written as

$$\frac{dE_j}{dt} = A_j \dot{\psi}\ddot{\psi} + B_j \dot{\psi}^3, \tag{12.78}$$

where

$$A_j = m_j(x'^2_{G_j} + y'^2_{G_j}) + I_{G_j}\theta'^2_j, \tag{12.79a}$$

$$B_j = m_j(x'_{G_j}x''_{G_j} + y'_{G_j}y''_{G_j}) + I_{G_j}\theta'_j\theta''_j. \tag{12.79b}$$

Comparing Eq. (12.79*a*) and Eq. (12.77), we see that the kinetic energy can be written as

$$E_j = \frac{1}{2}A_j\dot{\psi}^2. \tag{12.80}$$

From Eqs. (12.79*a*) and (12.79*b*), we also note the relationship between the coefficients, that is,

$$B_j = \frac{1}{2}\frac{dA_j}{d\psi}. \tag{12.81}$$

For a mechanism with n links and with link 1 fixed, the time rate of change in kinetic energy can be written, from Eq. (12.78), as

$$\frac{dE}{dt} = \sum_{j=2}^{n} A_j\dot{\psi}\ddot{\psi} + \sum_{j=2}^{n} B_j\dot{\psi}^3. \tag{12.82}$$

Next we consider the potential energy stored in link j of the mechanism:

1. *Due to gravity:* If we assume that the gravitational force acts in the negative y direction, then the potential energy due to gravity is

$$U_{gj} = m_j g y_{G_j}. \tag{12.83}$$

where m_j is the mass of the link, g is the gravitational constant, and y_{G_j} is the height of the mass center above a datum arbitrarily chosen at the origin, as shown

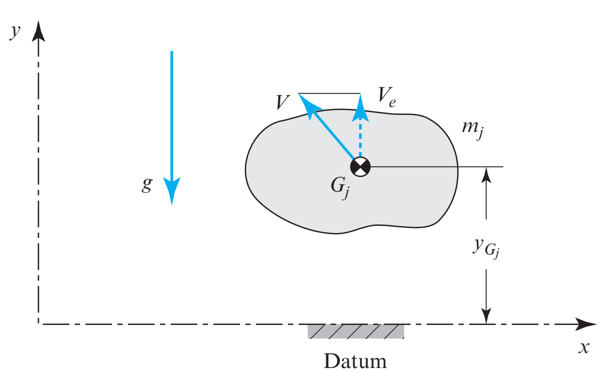


Figure 12.19 Potential energy due to elevation.

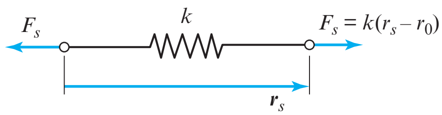


Figure 12.20 Vector across a rectilinear spring.

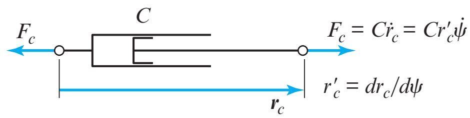


Figure 12.21 Vector across a damper.

in Fig. 12.19. Differentiating Eq. (12.83) with respect to time, the rate of change in the gravitational potential energy of link j can be written as

$$\frac{dU_{g_j}}{dt} = m_j g y'_{G_j}\dot{\psi}, \tag{12.84a}$$

where the first-order kinematic coefficient is

$$y'_{G_j} = \frac{dy_{G_j}}{d\psi}. \tag{12.84b}$$

Therefore, the time rate of change in the gravitational potential energy stored in the mechanism is

$$\frac{dU_g}{dt} = \sum_{j=2}^{n} m_j g y'_{G_j}\dot{\psi}. \tag{12.85}$$

2. *Due to a rectilinear spring:* If the mechanism contains a spring element, then the potential energy stored in the spring is

$$U_s = \frac{1}{2}k(r_s - r_0)^2, \tag{12.86}$$

where k is the spring rate, r_s is the actual length of the spring, and r_0 is the free length of the spring, as shown in Fig. 12.20. Differentiating Eq. (12.86) with respect to time, the time rate of change in the potential energy stored in the spring can be written as

$$\frac{dU_s}{dt} = k(r_s - r_0)r'_s\dot{\psi}, \tag{12.87a}$$

where the first-order kinematic coefficient of the spring is

$$r'_s = \frac{dr_s}{d\psi}. \tag{12.87b}$$

Next we consider the dissipative effects due to the following:

1. *A viscous damper:* The work done in overcoming the damping effect is

$$W_c = C\dot{r}_c\Delta r_c, \tag{12.88}$$

where C is the damping coefficient, and r_c is the length of the vector across the damper, as shown in Fig. 12.21. Differentiating Eq. (12.88) with respect to time, the time rate of change of the dissipative effect can be written

$$\frac{dW_c}{dt} = Cr_c'^2\dot{\psi}^2, \tag{12.89a}$$

where the first-order kinematic coefficient of the damper is

$$r_c' = \frac{dr_c}{d\psi}. \tag{12.89b}$$

2. *Coulomb friction:* The work done in overcoming Coulomb friction is

$$W_f = \mu N \Delta r_f, \tag{12.90}$$

where μ is the coefficient of friction, N is the normal force between the surfaces on the contacting links, and $\mathbf{r}_f$ is the vector along the surface to the point of contact shown as point C in Fig. 12.22. Differentiating Eq. (12.90) with respect to time, the time rate of change of the dissipative effect can be written as

$$\frac{dW_f}{dt} = \mu N |r_f'\dot{\psi}|, \tag{12.91a}$$

where the first-order kinematic coefficient is

$$r_f' = \frac{dr_f}{d\psi}. \tag{12.91b}$$

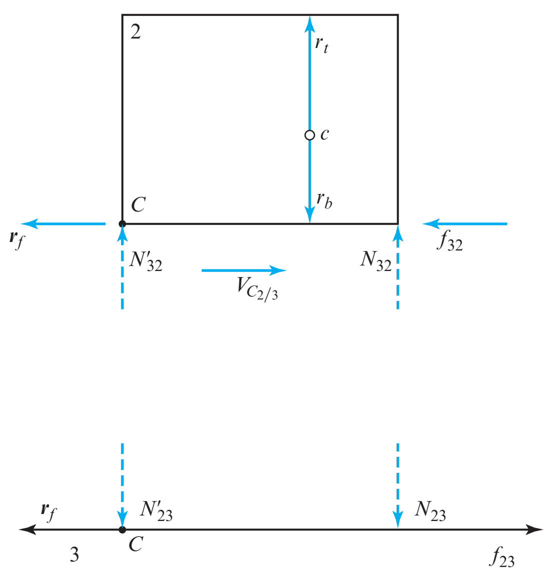


Figure 12.22 Vector to the point of contact.

Substituting Eqs. (12.74), (12.82), (12.85), (12.87*a*), (12.89*a*), and (12.91*a*) into Eq. (12.73), the net power can be written as

$$Q\dot{\psi} = \sum_{j=2}^{n} A_j \dot{\psi}\ddot{\psi} + \sum_{j=2}^{n} B_j \dot{\psi}^3 + \sum_{j=2}^{n} m_j g y'_{G_j} \dot{\psi}$$
$$+ k(r_s - r_0) r'_s \dot{\psi} + C r_c'^2 \dot{\psi}^2 + \mu N |r'_f \dot{\psi}|. \qquad (12.92)$$

We note that each term in this equation contains the generalized input velocity, $\dot{\psi}$. If we divide through by this common factor, then we obtain the *equation of motion* for the mechanism, that is,

$$Q = \sum_{j=2}^{n} A_j \ddot{\psi} + \sum_{j=2}^{n} B_j \dot{\psi}^2 + \sum_{j=2}^{n} m_j g y'_{G_j} + k(r_s - r_0) r'_s + C r_c'^2 \dot{\psi} + \mu N |r'_f|. \qquad (12.93)$$

This result is also valid when the generalized input velocity is zero, that is, when the input link is instantaneously stopped or when the mechanism is starting from rest. In this case, Eq. (12.93) reduces to

$$Q = \sum_{j=2}^{n} A_j \ddot{\psi} + \sum_{j=2}^{n} m_j g y'_{G_j} + k(r_s - r_0) r'_s + \mu N |r'_f|, \qquad (12.94)$$

and for this case there is no necessity to calculate $\sum_{j=2}^{n} B_j$ or the first-order kinematic coefficient of the viscous damper.

We will consider two cases for the input motion: (1) input rotation, that is, $\dot{\psi} = \dot{\theta}_i$, and (2) input translation, that is, $\dot{\psi} = \dot{r}_i$.

1. If the input is rotation, then Eq. (12.93) can be written as

$$T_i = \sum_{j=2}^{n} A_j \ddot{\theta}_i + \sum_{j=2}^{n} B_j \dot{\theta}_i^2 + \sum_{j=2}^{n} m_j g y'_{G_j} + k(r_s - r_0) r'_s + C r_c'^2 \dot{\theta}_i + \mu N \left| r'_f \right|, \qquad (12.95)$$

where T_i is the torque acting at the input. Note that the coefficient $\sum_{j=2}^{n} A_j$ must have units of moment of inertia and is referred to as the *equivalent mass moment of inertia* of the system and is denoted I_{EQ}. This moment of inertia, if thought of as concentrated on the input link alone, would have the same kinetic energy as the entire mechanism. From Eq. (12.80), the kinetic energy could then be written as

$$E = \frac{1}{2} I_{EQ} \dot{\theta}^2. \qquad (12.96)$$

The equation of motion, that is Eq. (12.95), could then be written as

$$T_i = I_{EQ}\ddot{\theta}_i + \frac{1}{2}\frac{dI_{EQ}}{d\theta_i}\dot{\theta}_i^2 + \sum_{j=2}^{n} m_j g y'_{G_j} + k\,(r_s - r_0)\,r'_s + Cr_c'^2\dot{\theta}_i + \mu N\left|r'_f\right|. \quad (12.97)$$

2. If the input is translation, then Eq. (12.93) can be written as

$$F_i = \sum_{j=2}^{n} A_j \ddot{r}_i + \sum_{j=2}^{n} B_j \dot{r}_i^2 + \sum_{j=2}^{n} m_j g y'_{G_j} + k(r_s - r_0)r'_s + Cr_c'^2\dot{r}_i + \mu N|r'_f|, \quad (12.98)$$

where F_i is the force acting at the input. For this case, the coefficient $\sum_{j=2}^{n} A_j$ has units of mass and is referred to as the *equivalent mass* of the system and denoted as m_{EQ}. This mass, if thought of as concentrated on the input link alone, would have the same kinetic energy as the entire mechanism. From Eq. (12.80), the kinetic energy could then be written as

$$E = \frac{1}{2} m_{EQ}\dot{r}_2. \quad (12.99)$$

The equation of motion, that is Eq. (12.98), could then be written as

$$F_i = m_{EQ}\ddot{r}_i + \frac{1}{2}\frac{dm_{EQ}}{dr_i}\dot{r}_i^2 + \sum_{j=2}^{n} m_j g y'_{G_j} + k\,(r_s - r_0)\,r'_s + Cr_c'^2\dot{r}_i + \mu N\left|r'_f\right|. \quad (12.100)$$

EXAMPLE 12.6

Determine the torque, T_2, that must be applied to input link 2 of the parallelogram four-bar linkage in the posture shown in Fig. 12.23. The angular velocity and angular acceleration of link 2 are $\omega_2 = 10$ rad/s ccw and $\alpha_2 = 10$ rad/s^2 cw, respectively. The masses of the links are $m_2 = 2.5$ kg, $m_3 = 5$ kg, and $m_4 = 2.5$ kg, and the mass moments of inertia of the links are $I_{G_2} = I_{G_4} = 0.25$ kg · m^2 and $I_{G_3} = 4.5$ kg · m^2, and the center of mass of each link is coincident with its geometric center. The free length of the spring is $r_0 = 0.6$ m, the spring rate is $k = 500$ N/m, and the damping constant is $C = 10$ N · s/m.

ASSUMPTIONS

Gravity acts vertically downward, and friction can be ignored.

SOLUTION

From the kinematic analysis of the four-bar linkage presented in Secs. 3.12 and 4.12, the first- and second-order kinematic coefficients of links 3 and 4 are

$$\theta'_3 = 0, \quad \theta'_4 = 1 \text{ rad/rad}, \quad \theta''_3 = 0, \quad \text{and} \quad \theta''_4 = 0. \quad (1)$$

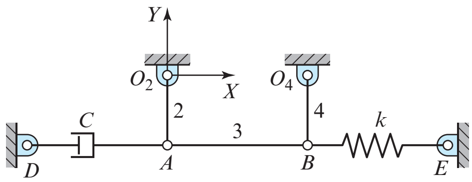


Figure 12.23
$r_1 = 0.8$ m, $r_2 = 0.4$ m, $r_3 = 0.8$ m, $r_4 = 0.4$ m, $r_s = R_{EB} = 0.8$ m, and $r_c = R_{DA} = 0.8$ m.

Recall that the angular velocities and angular accelerations of links 3 and 4 can be written as

$$\omega_3 = \theta_3'\omega_2 \quad \text{and} \quad \omega_4 = \theta_4'\omega_2$$
$$\alpha_3 = \theta_3''\omega_2^2 + \theta_3'\alpha_2 \quad \text{and} \quad \alpha_4 = \theta_4''\omega_2^2 + \theta_4'\alpha_2. \tag{2}$$

Substituting Eqs. (1) and the input angular velocity and angular acceleration into Eqs. (2) gives

$$\omega_3 = 0, \quad \omega_4 = 10 \text{ rad/s ccw}, \quad \alpha_3 = 0, \quad \text{and} \quad \alpha_4 = 10 \text{ rad/s}^2 \text{ cw}. \tag{3}$$

The x and y components of the position of mass center G_2 with respect to ground pivot O_2 are

$$x_{G_2} = \frac{1}{2}r_2\cos\theta_2 \quad \text{and} \quad y_{G_2} = \frac{1}{2}r_2\sin\theta_2. \tag{4}$$

Differentiating these equations twice with respect to θ_2 and substituting $\theta_2 = 270°$, the first- and second-order kinematic coefficients of the mass center of link 2 are

$$x_{G_2}' = 0.2 \text{ m/rad}, \quad y_{G_2}' = 0, \quad x_{G_2}'' = 0, \quad y_{G_2}'' = 0.2 \text{ m/rad}^2. \tag{5}$$

The x and y components of the position of mass center G_3 with respect to ground pivot O_2 can be written as

$$x_{G_3} = r_2\cos\theta_2 + \frac{1}{2}r_3\cos\theta_3 \quad \text{and} \quad y_{G_3} = r_2\sin\theta_2 + \frac{1}{2}r_3\sin\theta_3. \tag{6}$$

Differentiating these equations twice with respect to θ_2 and substituting $\theta_2 = 270°$ and $\theta_3 = 0°$, the first- and second-order kinematic coefficients of the mass center of link 3 are

$$x_{G_3}' = 0.4 \text{ m/rad}, \quad y_{G_3}' = 0, \quad x_{G_3}'' = 0, \quad y_{G_3}'' = 0.4 \text{ m/rad}^2. \tag{7}$$

The first- and second-order kinematic coefficients of the mass center of link 4 are the same as for the mass center of link 2, that is,

$$x_{G_4}' = x_{G_2}' = 0.2 \text{ m/rad}, \quad y_{G_4}' = y_{G_2}' = 0, \quad x_{G_4}'' = x_{G_2}'' = 0, \quad y_{G_4}'' = y_{G_2}'' = 0.2 \text{ m/rad}^2. \tag{8}$$

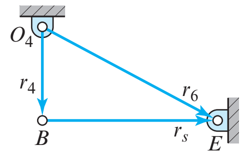


Figure 12.24 Vector loop for the spring.

To obtain the first-order kinematic coefficient for the spring, the vector loop-closure equation, shown in Fig. 12.24, can be written as

$$\mathbf{r}_4 + \mathbf{r}_s - \mathbf{r}_6 = \mathbf{0}. \tag{9}$$

The two scalar equations are

$$\begin{aligned} r_4 \cos\theta_4 + r_s \cos\theta_s - r_6 \cos\theta_6 &= 0, \\ r_4 \sin\theta_4 + r_s \sin\theta_s - r_6 \sin\theta_6 &= 0. \end{aligned} \tag{10}$$

Differentiating Eqs. (10) with respect to input θ_2 gives

$$-r_4 \sin\theta_4 \theta_4' - r_s \sin\theta_s \theta_s' + r_s' \cos\theta_s = 0, \tag{11a}$$

$$r_4 \cos\theta_4 \theta_4' + r_s \cos\theta_s \theta_s' + r_s' \sin\theta_s = 0. \tag{11b}$$

Substituting $\theta_s = 0°$, $\theta_4 = 270°$, and $\theta_4' = 1$ rad/rad into Eq. (11a), the first-order kinematic coefficient for the spring is

$$r_s' = -r_4\theta_4' = -0.4 \text{ m/rad}. \tag{12}$$

To obtain the first-order kinematic coefficient for the damper, the vector loop-closure equation, shown in Fig. 12.25, can be written as

$$\mathbf{r}_2 + \mathbf{r}_c - \mathbf{r}_8 = \mathbf{0}. \tag{13}$$

The two scalar equations are

$$r_2 \cos\theta_2 + r_c \cos\theta_c - r_8 \cos\theta_8 = 0, \tag{14a}$$

$$r_2 \sin\theta_2 + r_c \sin\theta_c - r_8 \sin\theta_8 = 0. \tag{14b}$$

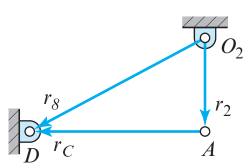


Figure 12.25 Vector loop for the viscous damper.

Differentiating Eqs. (14) with respect to input θ_2 gives

$$-r_2 \sin\theta_2 - r_c \sin\theta_c \theta_c' + r_c' \cos\theta_c = 0, \tag{15a}$$

$$r_2 \cos\theta_2 + r_c \cos\theta_c \theta_c' + r_c' \sin\theta_c = 0. \tag{15b}$$

Substituting $\theta_2 = 270°$ and $\theta_c = 180°$ into Eqs. (15a), the first-order kinematic coefficient for the damper is

$$r_c' = r_2 = 0.4 \text{ m/rad}. \tag{16}$$

The equivalent mass moment of inertia for the four-bar linkage is

$$I_{EQ} = \sum_{j=2}^{4} A_j.$$

From Eq. (12.79a), the equivalent mass moment of inertia can be written as

$$I_{EQ} = m_2(x_{G_2}'^2 + y_{G_2}'^2) + I_{G_2} + m_3(x_{G_3}'^2 + y_{G_3}'^2) + I_{G_3}\theta_3'^2 + m_4(x_{G_4}'^2 + y_{G_4}'^2) + I_{G_4}\theta_4'^2.$$

Substituting the known data and the kinematic coefficients into this equation gives

$$I_{EQ} = 0.1 + 0.25 + 0.8 + 0.1 + 0.25 = 1.5 \text{ kg} \cdot \text{m}^2. \tag{17}$$

From Eq. (12.79b), we can write

$$\sum_{j=2}^{4} B_j = m_2(x_{G_2}' x_{G_2}'' + y_{G_2}' y_{G_2}'') + m_3(x_{G_3}' x_{G_3}'' + y_{G_3}' y_{G_3}'') + I_{G_3}\theta_3'\theta_3''$$

$$+ m_4(x_{G_4}' x_{G_4}'' + y_{G_4}' y_{G_4}'') + I_{G_4}\theta_4'\theta_4''. \tag{18}$$

For the given posture, the kinematic coefficients are given by Eqs. (1), (5), (7), and (8). Substituting these kinematic coefficients into Eq. (18) gives

$$\sum_{j=2}^{n} B_j = 0. \tag{19}$$

Then, substituting the known data, the kinematic coefficients, and Eqs. (17) and (19) into Eq. (12.97), the equation of motion can be written

$$T_2 = \left(1.5 \text{ kg} \cdot \text{m}^2\right) \ddot{\theta}_2 + k\,(r_s - r_0)\,r_s' + C r_c'^2 \dot{\theta}_2.$$

Finally, substituting the known data and Eqs. (12) and (16) into this equation, the input torque is

$$T_2 = -15 \text{ N} \cdot \text{m} + (500 \text{ N/m})\,(0.8 \text{ m} - 0.6 \text{ m})\,(-0.4 \text{ m/rad})$$

$$+ (10 \text{ N} \cdot \text{s/m})\,(0.4 \text{ m/rad})^2\,(10 \text{ rad/s})$$

$$T_2 = -39 \text{ N} \cdot \text{m} = 39 \text{ N} \cdot \text{m cw}. \qquad \textit{Ans.}$$

Since the result is negative, the input torque acts in the direction opposite to the input angular velocity, that is, clockwise. Note that this is necessary to achieve the clockwise input angular acceleration specified. Also note that a clockwise input torque of $T_2 = 24\ \mathrm{N \cdot m}$ would yield a zero input acceleration at the instant under study.

12.10 MEASURING MASS MOMENT OF INERTIA

Sometimes the shapes of machine parts are very complex, and it is extremely tedious and time-consuming to calculate their mass moment(s) of inertia. Consider, for example, the problem of finding the mass moment of inertia of an automobile body about a vertical axis through its center of mass. For such problems, it is usually possible to determine the mass moment of inertia of a body by observing its dynamic behavior caused by a known rotational disturbance.

Many bodies, connecting rods, for example, are shaped so that their masses can be assumed to lie in a single plane. Once such bodies have been weighed and their mass centers located, they can be suspended like a pendulum and caused to oscillate. The mass moment of inertia of such a body can then be computed from an observation of its period or frequency of oscillation. For best experimental results, the part should be suspended with the pivot located close to, but not coincident with, the center of mass. It is not usually necessary to drill a hole to suspend the body; for example, a spoked wheel or a gear can be suspended on a knife edge at its rim.

When the body shown in Fig. 12.26*a* is displaced through angle θ, a gravity force mg acts at G. Summing moments about the pivot, O, gives

$$\sum M_o = -mg(r_G \sin\theta) - I_o\ddot{\theta} = 0. \qquad (a)$$

We intend that the pendulum be displaced only through a small angle, so that $\sin\theta$ can be replaced by θ. Equation (a) can then be written as

$$\ddot{\theta} + \frac{mgr_G}{I_o}\theta = 0. \qquad (b)$$

This second-order linear differential equation has the well-known solution

$$\theta = c_1 \sin\sqrt{\frac{mgr_G}{I_o}}t + c_2 \cos\sqrt{\frac{mgr_G}{I_o}}t, \qquad (c)$$

where c_1 and c_2 are constants of integration.

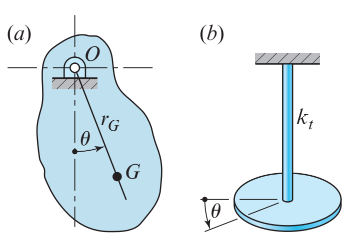


Figure 12.26 (a) Simple pendulum; (b) torsional pendulum.

We start the pendulum motion by displacing it through a small angle θ_0 and releasing it with no initial velocity from this posture. Thus, at time $t = 0$, we obtain $\theta = \theta_0$ and $\dot{\theta} = 0$. Substituting these conditions into Eq. (c) and its first time derivative enables us to evaluate the two constants. They are $c_1 = 0$ and $c_2 = \theta_0$. Therefore,

$$\theta = \theta_0 \cos \sqrt{\frac{mgr_G}{I_o}}t. \tag{12.101}$$

Since a cosine function repeats itself every 360° or 2π radians, the period of the motion is

$$\tau = 2\pi \sqrt{\frac{I_o}{mgr_G}}. \tag{d}$$

Therefore, the mass moment of inertia of the body about the pivot, O, is

$$I_O = mgr_G \left(\frac{\tau}{2\pi}\right)^2. \tag{12.102}$$

This equation indicates that the body must be weighed to get mg, the distance, r_G, must be measured, and then the pendulum must be suspended and oscillated so that the period, τ, can be observed; Eq. (12.102) can then be evaluated to give I_O about O. If the moment of inertia about the mass center is desired, it can be obtained by using the parallel-axis theorem, Eq. (12.11).

Figure 12.26b illustrates how the mass moment of inertia can be determined without actually weighing the body. The body is connected to a slender rod or wire at the mass center. The torsional stiffness, k_t, of the rod or wire is defined as the torque necessary to twist the rod through a unit angle. If the body of Fig. 12.26b is turned through a small angle, θ, and released, the equation of motion becomes

$$\ddot{\theta} + \frac{k_t}{I_G}\theta = 0. \tag{e}$$

This is similar to Eq. (b) and, with the same starting conditions, has the solution

$$\theta = \theta_o \cos \sqrt{\frac{k_t}{I_G}}t. \tag{12.103}$$

The period of oscillation is then

$$\tau = 2\pi \sqrt{\frac{I_G}{k_t}}, \tag{f}$$

and

$$I_G = k_t \left(\frac{\tau}{2\pi}\right)^2. \tag{12.104}$$

The torsional stiffness is often known or can be computed from a knowledge of the length and diameter of the rod or wire and its material. Then, the oscillation of the

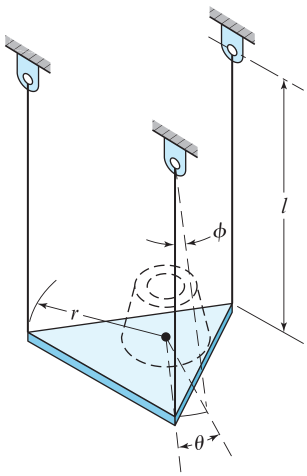


Figure 12.27 Trifilar pendulum.

body can be observed and Eq. (12.104) used to compute the mass moment of inertia, I_G. Alternatively, when the torsional stiffness, k_t, is unknown, a body with known mass moment of inertia, I_G, can be mounted and Eq. (12.104) can be used to determine k_t.

A *trifilar pendulum,* also called a *three-string torsional pendulum*, shown in Fig. 12.27, provides a very accurate method of measuring the mass moment of inertia. Three strings of equal length support a lightweight platform and are equally spaced about its center. A round platform serves just as well as the triangular one shown. The part whose mass moment of inertia is to be determined is carefully placed on the platform so that the center of mass of the object coincides with the platform center. The platform is then made to oscillate, and the number of oscillations is counted over a specified period of time [2, p. 129].

The notation for the three-string torsional pendulum analysis is as follows:

$m =$ mass of the part

$m_p =$ mass of the platform

$I_G =$ mass moment of inertia of the part

$I_p =$ mass moment of inertia of the platform

$r =$ platform radius

$\theta =$ platform angular displacement

$l =$ string length

$\phi =$ string angular displacement

$z =$ vertical axis through the center of the platform

We begin by writing the summation of moments about the z axis. This gives

$$\sum M^z = -r(m+m_p)g\sin\phi - (I_G+I_p)\ddot{\theta} = 0. \qquad (g)$$

Since we are assuming small displacements, the sine of an angle can be approximated by the angle itself. Therefore

$$\phi \approx \frac{r}{l}\theta, \tag{h}$$

and Eq. (g) becomes

$$\ddot{\theta} + \frac{(m+m_p)gr^2}{(I_G+I_p)l}\theta = 0. \tag{i}$$

This equation can be solved in the same manner as Eq. (b). The result is

$$I_G + I_p = \frac{(m+m_p)gr^2}{l}\left(\frac{\tau}{2\pi}\right)^2. \tag{12.105}$$

This equation can be used with an empty platform to find I_p. With m_p and I_p known, the equation can then be used to find I_G of the part being measured.

12.11 TRANSFORMATION OF INERTIA AXES

A brief review will confirm that all examples presented so far have been limited to planar motion. It is now time to extend our study to include spatial problems. The basic principles are not new, but the problems presented may seem more complex because of our difficulty in visualizing in three dimensions. In addition, our previous treatment of angular motion was not presented in enough detail to deal with three-dimensional rotations.

Up to this point, we have continually used what are called the *principal mass moments of inertia*. In a coordinate system with origin at the center of mass and with axes aligned in the principal axis directions, all *mass products of inertia* [Eqs. (12.8)] are zero. This choice of axes has been used to simplify the form of the equations from that required for other choices.

In Eq. (12.11), the parallel-axis formula, we demonstrated how translation of axes can be performed. However, up to now, we have said little of how inertia properties can be rotated to a coordinate system that is not parallel to the principal axes. This is the purpose of this section.

Let us assume that the principal axes are aligned along unit vector directions $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$, and that the rotated axes desired have directions denoted by $\hat{\mathbf{i}}'$, $\hat{\mathbf{j}}'$, and $\hat{\mathbf{k}}'$. Then, any point in the principal axis coordinate system is located by the position vector $\mathbf{R}$ and by $\mathbf{R}'$ in the rotated system. Since both are descriptions of the same point position, $\mathbf{R}' = \mathbf{R}$, and

$$R^{x'}\hat{\mathbf{i}}' + R^{y'}\hat{\mathbf{j}}' + R^{z'}\hat{\mathbf{k}}' = R^x\hat{\mathbf{i}} + R^y\hat{\mathbf{j}} + R^z\hat{\mathbf{k}}.$$

Now, by taking dot products of this equation with $\hat{\mathbf{i}}'$, $\hat{\mathbf{j}}'$, and $\hat{\mathbf{k}}'$, respectively, we find the transformation equations between the two sets of axes. They are

$$
\begin{aligned}
R^{x'} &= (\hat{\mathbf{i}}' \cdot \hat{\mathbf{i}})R^x + (\hat{\mathbf{i}}' \cdot \hat{\mathbf{j}})R^y + (\hat{\mathbf{i}}' \cdot \hat{\mathbf{k}})R^z,\\
R^{y'} &= (\hat{\mathbf{j}}' \cdot \hat{\mathbf{i}})R^x + (\hat{\mathbf{j}}' \cdot \hat{\mathbf{j}})R^y + (\hat{\mathbf{j}}' \cdot \hat{\mathbf{k}})R^z,\\
R^{z'} &= (\hat{\mathbf{k}}' \cdot \hat{\mathbf{i}})R^x + (\hat{\mathbf{k}}' \cdot \hat{\mathbf{j}})R^y + (\hat{\mathbf{k}}' \cdot \hat{\mathbf{k}})R^z.
\end{aligned} \tag{a}
$$

But, we may remember that the dot product of two unit vectors is equal to the cosine of the angle between them. For example, if the angle between $\hat{\mathbf{i}}'$ and $\hat{\mathbf{j}}$ is denoted by $\theta_{i'j}$, then

$$\hat{\mathbf{i}}' \cdot \hat{\mathbf{j}} = \cos\theta_{i'j}.$$

Thus, Eqs. (a) become

$$
\begin{aligned}
R^{x'} &= \cos\theta_{i'i}R^x + \cos\theta_{i'j}R^y + \cos\theta_{i'k}R^z,\\
R^{y'} &= \cos\theta_{j'i}R^x + \cos\theta_{j'j}R^y + \cos\theta_{j'k}R^z,\\
R^{z'} &= \cos\theta_{k'i}R^x + \cos\theta_{k'j}R^y + \cos\theta_{k'k}R^z.
\end{aligned} \tag{b}
$$

By substituting these into the definitions of mass moments of inertia in Eqs. (12.7), expanding, and then performing the integration, we get

$$
\begin{aligned}
I^{x'x'} &= \cos^2\theta_{i'i}I^{xx} + \cos^2\theta_{i'j}I^{yy} + \cos^2\theta_{i'k}I^{zz}\\
&\quad + 2\cos\theta_{i'i}\cos\theta_{i'j}I^{xy} + 2\cos\theta_{i'i}\cos\theta_{i'k}I^{xz} + 2\cos\theta_{i'j}\cos\theta_{i'k}I^{yz},\\
I^{y'y'} &= \cos^2\theta_{j'i}I^{xx} + \cos^2\theta_{j'j}I^{yy} + \cos^2\theta_{j'k}I^{zz}\\
&\quad + 2\cos\theta_{j'i}\cos\theta_{j'j}I^{xy} + 2\cos\theta_{j'i}\cos\theta_{j'k}I^{xz} + 2\cos\theta_{j'j}\cos\theta_{j'k}I^{yz},\\
I^{z'z'} &= \cos^2\theta_{k'i}I^{xx} + \cos^2\theta_{k'j}I^{yy} + \cos^2\theta_{k'k}I^{zz}\\
&\quad + 2\cos\theta_{k'i}\cos\theta_{k'j}I^{xy} + 2\cos\theta_{k'i}\cos\theta_{k'k}I^{xz} + 2\cos\theta_{k'j}\cos\theta_{k'k}I^{yz}.
\end{aligned} \tag{12.106}
$$

Similarly, for the mass products of inertia, starting from Eqs. (12.8), we get

$$
\begin{aligned}
I^{x'y'} &= -\cos\theta_{i'i}\cos\theta_{j'i}I^{xx} - \cos\theta_{i'j}\cos\theta_{j'j}I^{yy} - \cos\theta_{i'k}\cos\theta_{j'k}I^{zz}\\
&\quad + (\cos\theta_{i'i}\cos\theta_{j'j} + \cos\theta_{i'j}\cos\theta_{j'i})I^{xy}\\
&\quad + (\cos\theta_{i'j}\cos\theta_{j'k} + \cos\theta_{i'k}\cos\theta_{j'j})I^{yz}\\
&\quad + (\cos\theta_{i'k}\cos\theta_{j'i} + \cos\theta_{i'i}\cos\theta_{j'k})I^{zx}
\end{aligned} \tag{12.107a}
$$

$$
\begin{aligned}
I^{y'z'} &= -\cos\theta_{j'i}\cos\theta_{k'i}I^{xx} - \cos\theta_{j'j}\cos\theta_{k'j}I^{yy} - \cos\theta_{j'k}\cos\theta_{k'k}I^{zz}\\
&\quad + (\cos\theta_{j'i}\cos\theta_{k'j} + \cos\theta_{j'j}\cos\theta_{k'i})I^{xy}\\
&\quad + (\cos\theta_{j'j}\cos\theta_{k'k} + \cos\theta_{j'k}\cos\theta_{k'j})I^{yz}\\
&\quad + (\cos\theta_{j'k}\cos\theta_{k'i} + \cos\theta_{j'i}\cos\theta_{k'k})I^{zx}
\end{aligned} \tag{12.107b}
$$

$$
\begin{aligned}
I^{z'x'} = & -\cos\theta_{k'i}\cos\theta_{i'i}I^{xx} - \cos\theta_{k'j}\cos\theta_{i'j}I^{yy} - \cos\theta_{k'k}\cos\theta_{i'k}I^{zz} \\
& + (\cos\theta_{k'i}\cos\theta_{i'j} + \cos\theta_{k'j}\cos\theta_{i'i})I^{xy} \\
& + (\cos\theta_{k'j}\cos\theta_{i'k} + \cos\theta_{k'k}\cos\theta_{i'j})I^{yz} \\
& + (\cos\theta_{k'k}\cos\theta_{i'i} + \cos\theta_{k'i}\cos\theta_{i'k})I^{zx}.
\end{aligned}
\tag{12.107c}
$$

Once we have rotated away from the principal axes, or if we have started with other than the principal axes, then the situation is different. The parallel-axis formulae of Eq. (12.11) are no longer sufficient for the translation of axes. Let us assume that the mass moments and products of inertia are known in one coordinate system and are desired in another coordinate system, which is translated from the first by the equations

$$
\begin{aligned}
R^{x''} &= R^{x'} + d^{x'}, \\
R^{y''} &= R^{y'} + d^{y'}, \\
R^{z''} &= R^{z'} + d^{z'}.
\end{aligned}
\tag{12.108}
$$

Then, substituting into the definition of I^{xx} of Eq. (12.7), for example, and integrating:

$$
\begin{aligned}
I^{x''x''} &= \int [(R^{y''})^2 + (R^{z''})^2]dm \\
&= \int [(R^{y'} + d^{y'})^2 + (R^{z'} + d^{z'})^2]dm \\
&= \int [(R^{y'})^2 + (R^{z'})^2]dm + 2\left(\int R^{y'}\,dm\right)d^{y'} \\
&\quad + 2\left(\int R^{z'}\,dm\right)d^{z'} + (d^{y'})\int dm + (d^{z'})^2\int dm \\
&= I^{x'x'} + 2mR_G^{y'}d^{y'} + 2mR_G^{z'}d^{z'} + m(d^{y'})^2 + m(d^{z'})^2.
\end{aligned}
$$

Proceeding in this manner through each of Eqs. (12.7) and (12.8), we find

$$
\begin{aligned}
I^{x''x''} &= I^{x'x'} + 2mR_G^{y'}d^{y'} + 2mR_G^{z'}d^{z'} + m\left(d^{y'}\right)^2 + m\left(d^{z'}\right)^2, \\
I^{y''y''} &= I^{y'y'} + 2mR_G^{z'}d^{z'} + 2mR_G^{x'}d^{x'} + m\left(d^{z'}\right)^2 + m\left(d^{x'}\right)^2, \\
I^{z''z''} &= I^{z'z'} + 2mR_G^{x'}d^{x'} + 2mR_G^{y'}d^{y'} + m\left(d^{x'}\right)^2 + m\left(d^{y'}\right)^2, \\
I^{x''y''} &= I^{x'y'} + mR_G^{x'}d^{y'} + mR_G^{y'}d^{x'} + md^{x'}\,d^{y'}, \\
I^{y''z''} &= I^{y'z'} + mR_G^{y'}d^{z'} + mR_G^{z'}d^{y'} + md^{y'}\,d^{z'}, \\
I^{z''x''} &= I^{z'x'} + mR_G^{z'}d^{x'} + mR_G^{x'}d^{z'} + md^{z'}\,d^{x'}.
\end{aligned}
\tag{12.109}
$$

EXAMPLE 12.7

A circular steel disk, called a *swashplate*, is fastened at its center at an angle of 30° to a steel shaft, as shown in Fig. 12.28. Using 0.282 lb/in^3 as the density of steel, find the mass moments and products of inertia of the combined disk and shaft about the combined center of mass and aligned along the axes of the shaft.

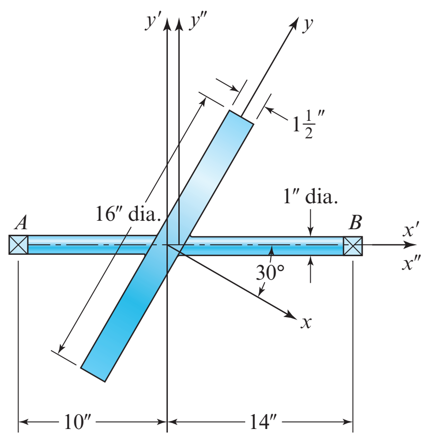


Figure 12.28 Shaft with swashplate.

SOLUTION

First, we calculate the mass of the disk and the principal mass moments of inertia of the disk using the equations of Table 5, Appendix A. These are

$$m = \frac{0.282\ \text{lb/in}^3}{386\ \text{in/s}^2}\pi(8\ \text{in})^2(1.5\ \text{in}) = 0.220\ \text{lb}\cdot\text{s}^2/\text{in},$$

$$I^{xx} = \frac{(0.220\ \text{lb}\cdot\text{s}^2/\text{in})(8\ \text{in})^2}{2} = 7.04\ \text{in}\cdot\text{lb}\cdot\text{s}^2,$$

$$I^{yy} = I^{zz} = \frac{(0.220\ \text{lb}\cdot\text{s}^2/\text{in})(8\ \text{in})^2}{4} = 3.52\ \text{in}\cdot\text{lb}\cdot\text{s}^2.$$

From Fig. 12.28 we now find the direction cosines between the coordinate systems. These are

$$\begin{array}{lll} \cos\theta_{i'i} = \ 0.866, & \cos\theta_{i'j} = 0.500, & \cos\theta_{i'k} = 0.000, \\ \cos\theta_{j'i} = -0.500, & \cos\theta_{j'j} = 0.866, & \cos\theta_{j'k} = 0.000, \\ \cos\theta_{k'i} = 0.000, & \cos\theta_{k'j} = 0.000, & \cos\theta_{k'k} = 1.000. \end{array}$$

Substituting these results into Eqs. (12.106) and (12.107) gives the mass moments and products of inertia of the disk aligned with the axes of the shaft. These are

$$I^{x'x'} = (0.866)^2(7.04\ \text{in}\cdot\text{lb}\cdot\text{s}^2) + (0.500)^2(3.52\ \text{in}\cdot\text{lb}\cdot\text{s}^2) = 6.16\ \text{in}\cdot\text{lb}\cdot\text{s}^2,$$

$$I^{y'y'} = (-0.500)^2(7.04\ \text{in}\cdot\text{lb}\cdot\text{s}^2) + (0.866)^2(3.52\ \text{in}\cdot\text{lb}\cdot\text{s}^2) = 4.40\ \text{in}\cdot\text{lb}\cdot\text{s}^2,$$

$$I^{z'z'} = I^{zz} = 3.52 \text{ in} \cdot \text{lb} \cdot \text{s}^2$$

$$I^{x'y'} = -(0.866)(-0.500)(7.04 \text{ in} \cdot \text{lb} \cdot \text{s}^2) - (0.500)(0.866)(3.52 \text{ in} \cdot \text{lb} \cdot \text{s}^2) = 1.52 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{y'z'} = I^{z'x'} = 0.$$

Next, we find the mass of the shaft and the principal mass moments of inertia of the shaft using the equations of Table 5, Appendix A. These are

$$m = \frac{0.282 \text{ lb/in}^3}{386 \text{ in/s}^2} \pi (0.5 \text{ in})^2 (24 \text{ in}) = 0.013\,8 \text{ lb} \cdot \text{s}^2/\text{in},$$

$$I^{xx} = \frac{(0.013\,8 \text{ lb} \cdot \text{s}^2/\text{in})(0.5 \text{ in})^2}{2} = 0.001\,72 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{yy} = I^{zz} = \frac{(0.013\,8 \text{ lb} \cdot \text{s}^2/\text{in})[3(0.5 \text{ in})^2 + (24 \text{ in})^2]}{12} = 0.662 \text{ in} \cdot \text{lb} \cdot \text{s}^2.$$

The mass products of inertia of both the disk and the shaft are zero because the axes used are principal axes.

Translating the values for the shaft to the same axes as the disk, Eq. (12.7) gives

$$I^{x'x'} = 0.001\,72 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{y'y'} = I^{z'z'} = (0.662 \text{ in} \cdot \text{lb} \cdot \text{s}^2) + (0.013\,8 \text{ lb} \cdot \text{s}^2/\text{in})(2 \text{ in})^2 = 0.717 \text{ in} \cdot \text{lb} \cdot \text{s}^2.$$

The values of the disk and the shaft are now both in the same coordinate system and are combined to give

$$I^{x'x'} = (6.16 \text{ in} \cdot \text{lb} \cdot \text{s}^2) + (0.001\,72 \text{ in} \cdot \text{lb} \cdot \text{s}^2) = 6.16 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{y'y'} = (4.40 \text{ in} \cdot \text{lb} \cdot \text{s}^2) + (0.717 \text{ in} \cdot \text{lb} \cdot \text{s}^2) = 5.12 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{z'z'} = (3.52 \text{ in} \cdot \text{lb} \cdot \text{s}^2) + (0.717 \text{ in} \cdot \text{lb} \cdot \text{s}^2) = 4.24 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{x'y'} = (1.52 \text{ in} \cdot \text{lb} \cdot \text{s}^2) + 0 = 1.52 \text{ in} \cdot \text{lb} \cdot \text{s}^2,$$

$$I^{y'z'} = I^{z'x'} = 0.$$

Finally, all values are translated to the combined center of mass using Eqs. (12.109):

$$m = (0.220 \text{ lb} \cdot \text{s}^2/\text{in}) + (0.013\,8 \text{ lb} \cdot \text{s}^2/\text{in}) = 0.234 \text{ lb} \cdot \text{s}^2/\text{in},$$

$$R_G^{x'} = \frac{(0.220 \text{ lb} \cdot \text{s}^2/\text{in})(0) + (0.013\,8 \text{ lb} \cdot \text{s}^2/\text{in})(2.0 \text{ in})}{0.235 \text{ lb} \cdot \text{s}^2/\text{in}} = 0.127 \text{ in},$$

$$R_G^{y'} = R_G^{z'} = 0,$$

$$d^{x'} = -0.127 \text{ in},$$

$$d^{y'} = d^{z'} = 0,$$

$$I^{x''x''} = 6.16 \text{ in} \cdot \text{lb} \cdot \text{s}^2, \qquad \textit{Ans.}$$

$$I^{y''y''} = (5.12\ \text{in} \cdot \text{lb} \cdot \text{s}^2) + 2(0.235\ \text{lb} \cdot \text{s}^2/\text{in})(0.118)(-0.118) + (0.234\ \text{in})(-0.118\ \text{in})^2$$
$$= 5.12\ \text{in} \cdot \text{lb} \cdot \text{s}^2, \qquad \textit{Ans.}$$
$$I^{z''z''} = (4.24\ \text{in} \cdot \text{lb} \cdot \text{s}^2) + 2(0.234\ \text{lb} \cdot \text{s}^2/\text{in})(0.118\ \text{in})(-0.118\ \text{in})$$
$$+ (0.234\ \text{in} \cdot \text{lb} \cdot \text{s}^2)(-0.118\ \text{in})^2$$
$$= 4.24\ \text{in} \cdot \text{lb} \cdot \text{s}^2, \qquad \textit{Ans.}$$
$$I^{x''y''} = 1.52\ \text{in} \cdot \text{lb} \cdot \text{s}^2, \qquad \textit{Ans.}$$
$$I^{y''z''} = I^{z''x''} = 0. \qquad \textit{Ans.}$$

12.12 EULER'S EQUATIONS OF MOTION

In Sec. 12.2, a detailed derivation found the form of Newton's law for a rigid body by integrating over all particles of the body. This demonstrated that the center of mass of the body was the single unique point of the body where Newton's law had the same form as that for a single particle. Repeating Eq. (12.12), this is

$$\sum \mathbf{F}_{ij} = m_j \mathbf{A}_{G_j}. \tag{12.110}$$

Taking much less care with derivation, we then gave the "equivalent" rotational form, Eq. (12.13), as

$$\sum \mathbf{M}_{Gij} = I_{G_j} \boldsymbol{\alpha}_j.$$

Although this equation is valid for all problems with only planar motion, this equation is oversimplified and is *not* always valid for problems with spatial motion. Our task now is to more carefully derive an appropriate equation that *is* valid for problems that include three-dimensional effects.

We start, as before, with Newton's law written for a single differential particle at location P on a rigid body. This can be written as

$$d\mathbf{F} = \mathbf{A}_P dm,$$

where $d\mathbf{F}$ is the net unbalanced force on the particle, $\mathbf{A}_P$ is the absolute acceleration of the particle, and dm is the mass of the particle. Next, we find the net unbalanced moment that this particle contributes to the body by taking the moment of its net unbalanced force about the center of mass of the body, point G:

$$\mathbf{R}_{PG} \times d\mathbf{F} = \mathbf{R}_{PG} \times \mathbf{A}_P dm$$
$$= \mathbf{R}_{PG} \times [\mathbf{A}_G + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PG}) + \boldsymbol{\alpha} \times \mathbf{R}_{PG}] dm.$$

On the left-hand side of this equation is the net unbalanced moment contribution of this single particle. If we rearrange the terms on the right-hand side, the equation becomes

$$d\mathbf{M} = \mathbf{R}_{PG} \times \mathbf{A}_G dm + \mathbf{R}_{PG} \times [\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{R}_{PG})] dm + \mathbf{R}_{PG} \times (\boldsymbol{\alpha} \times \mathbf{R}_{PG}) dm. \tag{a}$$

We now recall the following vector identity for the triple cross-product of three arbitrary vectors, say **A**, **B**, and **C**,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}.$$

Using this identity on the second term on the right-hand side of Eq. (a), we have

$$\begin{aligned} d\mathbf{M} = \mathbf{R}_{PG} \times \mathbf{A}_G dm + \mathbf{R}_{PG} \times (\boldsymbol{\omega} \cdot \mathbf{R}_{PG})\boldsymbol{\omega} dm \\ - \mathbf{R}_{PG} \times (\boldsymbol{\omega} \cdot \boldsymbol{\omega})\mathbf{R}_{PG} dm + \mathbf{R}_{PG} \times (\boldsymbol{\alpha} \times \mathbf{R}_{PG}) dm. \end{aligned} \tag{b}$$

Now, we hope to integrate this equation over all particles of the body. We will look at this term by term. The integral of the moment contributions of the individual particles gives the net externally applied unbalanced moment on the body:

$$\int d\mathbf{M} = \sum \mathbf{M}_{Gij}. \tag{c}$$

Recognizing that $\mathbf{A}_G$ is common for all particles, we see that the first term on the right-hand side of Eq. (b) integrates to

$$\int \mathbf{R}_{PG} \times \mathbf{A}_G dm = \int \mathbf{R}_{PG} dm \times \mathbf{A}_G = \mathbf{0}, \tag{d}$$

where we have taken advantage of the definition of the center of mass and noted that $\int \mathbf{R}_{PG} dm = m\mathbf{R}_{GG} = \mathbf{0}$.

The second term on the right-hand side of Eq. (b) can be expanded according to its components along the coordinate axes and integrated. The integrals will be recognized as the mass moments and products of inertia of the body. Thus,

$$\begin{aligned} \int \mathbf{R}_{PG} \times (\boldsymbol{\omega} \cdot \mathbf{R}_{PG})\boldsymbol{\omega} dm = {} & [-(I_G^{yy} - I_G^{zz})\omega^y\omega^z - I_G^{yz}(\omega^y\omega^y - \omega^z\omega^z) + (I_G^{xy}\omega^z - I_G^{xz}\omega^y)\omega^x]\hat{\mathbf{i}} \\ & + [-(I_G^{zz} - I_G^{xx})\omega^z\omega^x - I_G^{zx}(\omega^z\omega^z - \omega^x\omega^x) + (I_G^{yz}\omega^x - I_G^{yx}\omega^z)\omega^y]\hat{\mathbf{j}} \\ & + [-(I_G^{xx} - I_G^{yy})\omega^x\omega^y - I_G^{xy}(\omega^x\omega^x - \omega^y\omega^y) + (I_G^{zx}\omega^y - I_G^{zy}\omega^x)\omega^z]\hat{\mathbf{k}}. \end{aligned} \tag{e}$$

Since $\mathbf{R}_{PG} \times \mathbf{R}_{PG} = \mathbf{0}$ for every particle, the third term on the right-hand side of Eq. (b) integrates to

$$(\boldsymbol{\omega} \cdot \boldsymbol{\omega}) \int \mathbf{R}_{PG} \times \mathbf{R}_{PG} dm = \mathbf{0}. \tag{f}$$

The final term on the right-hand side of Eq. (b) is also expanded according to its components along the coordinate axes and integrated. Thus

$$\begin{aligned} \int \mathbf{R}_{PG} \times (\boldsymbol{\alpha} \times \mathbf{R}_{PG}) dm = {} & (I_G^{xx}\alpha^x - I_G^{xy}\alpha^y - I_G^{xz}\alpha^z)\hat{\mathbf{i}} \\ & + (I_G^{yx}\alpha^x - I_G^{yy}\alpha^y - I_G^{yz}\alpha^z)\hat{\mathbf{j}} \\ & + (I_G^{zx}\alpha^x - I_G^{zy}\alpha^y - I_G^{zz}\alpha^z)\hat{\mathbf{k}}. \end{aligned} \tag{g}$$

When we now substitute Eqs. (c) through (g) into Eq. (b) and equate components in the $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ directions, we obtain the following three equations:

$$\begin{aligned}
\sum M^x_{G_{ij}} &= I^{xx}_G\alpha^x - I^{xy}_G\alpha^y - I^{zx}_G\alpha^z \\
&\quad + (I^{xy}_G\omega^z - I^{zx}_G\omega^y)\omega^x - (I^{yy}_G - I^{zz}_G)\omega^y\omega^z - I^{yz}_G(\omega^y\omega^y - \omega^z\omega^z), \\
\sum M^y_{G_{ij}} &= -I^{xy}_G\alpha^x + I^{yy}_G\alpha^y - I^{yz}_G\alpha^z \\
&\quad + (I^{yz}_G\omega^x - I^{xy}_G\omega^z)\omega^y - (I^{zz}_G - I^{xx}_G)\omega^z\omega^x - I^{zx}_G(\omega^z\omega^z - \omega^x\omega^x), \\
\sum M^z_{G_{ij}} &= -I^{xz}_G\alpha^x - I^{yz}_G\alpha^y + I^{zz}_G\alpha^z \\
&\quad + (I^{zx}_G\omega^y - I^{yz}_G\omega^x)\omega^z - (I^{xx}_G - I^{yy}_G)\omega^x\omega^y - I^{xy}_G(\omega^x\omega^x - \omega^y\omega^y).
\end{aligned} \tag{12.111}$$

This set of equations is the most general case of the moment equation. It allows for angular velocities and angular accelerations in three dimensions. Its only restriction is that the summation of moments and the mass moments and products of inertia must be taken about the mass center of the body.

If we further require that the mass moments of inertia be measured around the principal axes of the body, then the mass products of inertia are zero. The above equations then reduce to

$$\begin{aligned}
\sum M^x_{G_{ij}} &= I^{xx}_G\alpha^x - (I^{yy}_G - I^{zz}_G)\omega^y\omega^z, \\
\sum M^y_{G_{ij}} &= I^{yy}_G\alpha^y - (I^{zz}_G - I^{xx}_G)\omega^z\omega^x, \\
\sum M^z_{G_{ij}} &= I^{zz}_G\alpha^z - (I^{xx}_G - I^{yy}_G)\omega^x\omega^y.
\end{aligned} \tag{12.112}$$

This important set of equations is called *Euler's equations of motion*. It should be emphasized that they govern any three-dimensional rotational motion of a rigid body, but that the principal axes of inertia must be used for the x-, y-, and z-component directions.*

EXAMPLE 12.8

Consider the swashplate that was analyzed in Example 12.7. The shaft is supported by radial bearings at A and B, as shown in Fig. 12.29. At the instant shown, the shaft is rotating at a speed of 1 500 rev/min cw, which is decreasing at a rate of 100 rev/min/s. Calculate the bearing reactions.

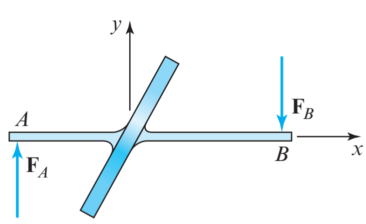


Figure 12.29 Shaft with swashplate.

* It should now be recognized that, when we restricted ourselves to motions in the xy plane, it was a special case of Eq. (12.112) that we were using in Eq. (12.13) and what followed.

SOLUTION

From the choice of axes shown in Fig. 12.29 and the data specified in the problem statement, the angular velocity and acceleration of the shaft, respectively, are

$$\boldsymbol{\omega} = -(1\,500 \text{ rev/min})(2\pi \text{ rad/rev})/(60 \text{ s/min})\hat{\mathbf{i}} = -157\hat{\mathbf{i}} \text{ rad/s},$$
$$\boldsymbol{\alpha} = +(100 \text{ rev/min/s})(2\pi \text{ rad/rev})/(60 \text{ s/min})\hat{\mathbf{i}} = +10.5\hat{\mathbf{i}} \text{ rad/s}^2.$$

The mass moments and products of inertia of the swashplate were found in Example 12.7.

Substituting these values into Eqs. (12.111) gives

$$\sum M^x = (6.16 \text{ in}\cdot\text{lb·s}^2)(10.5 \text{ rad/s}^2) = 64.7 \text{ in}\cdot\text{lb},$$
$$\sum M^y = -(1.52 \text{ in}\cdot\text{lb·s}^2)(10.5 \text{ rad/s}^2) = -16 \text{ in}\cdot\text{lb},$$
$$\sum M^z = -(1.52 \text{ in}\cdot\text{lb·s}^2)(-157 \text{ rad/s})^2 = -37\,500 \text{ in}\cdot\text{lb},$$
$$\sum \mathbf{M} = 64.7\hat{\mathbf{i}} - 16.0\hat{\mathbf{j}} - 37\,500\hat{\mathbf{k}} \text{ in}\cdot\text{lb}.$$

The $\sum M^x$ component is the result of the shaft torque causing the deceleration. The other two components are provided by forces at the bearings, A and B. From these we find, by taking moments about the mass center, that

$$\sum M^y = (10 \text{ in})(F_A^z) - (14 \text{ in})(F_B^z) = -16.0 \text{ in}\cdot\text{lb},$$
$$\sum M^z = -(10 \text{ in})(F_A^y) + (14 \text{ in})(F_B^y) = -37\,500 \text{ in}\cdot\text{lb}.$$

Remembering that $(F_A^y) + (F_B^y) = 0$ and $(F_A^z) + (F_B^z) = 0$ for equilibrium, we find that

$$F_A^z = -F_B^z = (-16.0 \text{ in}\cdot\text{lb})/(24 \text{ in}) = -0.667 \text{ lb} \approx 0,$$
$$F_A^y = -F_B^y = (37\,500 \text{ in}\cdot\text{lb})/(24 \text{ in}) = 1\,560 \text{ lb}.$$

Thus the two bearing reaction forces on the shaft are

$$\mathbf{F}_A = 1\,560\hat{\mathbf{j}} \text{ lb}, \qquad \textit{Ans.}$$
$$\mathbf{F}_B = -1\,560\hat{\mathbf{j}} \text{ lb}. \qquad \textit{Ans.}$$

Note that these bearing reactions are caused entirely by the z component of the inertia moment. This, in turn, is caused by the angular velocity of the shaft and not by its angular acceleration; the bearing reaction loads are still quite large even when the shaft is operating at constant speed.

Note also that this example exhibits only planar motion, but if it had been analyzed by the methods of Sec. 12.6, we would have found *no* net bearing reaction forces! This should be sufficient warning of the danger of ignoring the third dimension in dynamic force analysis, based solely on the justification that the *motion* is planar.

12.13 IMPULSE AND MOMENTUM

If we consider that both force and acceleration may be functions of time, we can multiply both sides of Eq. (12.110) by dt and integrate between two chosen times, t_1 and t_2. This results in

$$\sum \int_{t_1}^{t_2} \mathbf{F}_{ij}(t)dt = m_j \int_{t_1}^{t_2} \mathbf{A}_{G_j}(t)dt = m_j \mathbf{V}_{G_j}(t_2) - m_j \mathbf{V}_{G_j}(t_1), \qquad (a)$$

where $\mathbf{V}_{G_j}(t_2)$ and $\mathbf{V}_{G_j}(t_1)$ are the velocities of the centers of mass at times t_2 and t_1, respectively.

The product of the mass and the velocity of the center of mass of a moving body is called its *momentum*. Momentum is a vector quantity and is given the symbol $\mathbf{L}$. Thus, for body j, the momentum is

$$\mathbf{L}_j(t) = m_j \mathbf{V}_{G_j}(t). \qquad (12.113)$$

Using this definition, Eq. (a) becomes

$$\sum \int_{t_1}^{t_2} \mathbf{F}_{ij}(t)dt = \mathbf{L}_j(t_2) - \mathbf{L}_j(t_1) = \Delta \mathbf{L}_j. \qquad (12.114)$$

The integral of a force over an interval of time is called the *impulse* of the force. Therefore, Eq. (12.114) expresses the *principle of impulse and momentum*, which is that *the total impulse on a rigid body is equal to its change in momentum during the same time interval.* Conversely, *any change in momentum of a rigid body over a time interval is due to the total impulse of the external forces on that body.*

If we now make the time interval between t_1 and t_2 infinitesimal, divide Eq. (12.114) by this interval, and take the limit, it becomes

$$\sum \mathbf{F}_{ij} = \frac{d\mathbf{L}_j}{dt}. \qquad (12.115)$$

Therefore, *the resultant external force acting upon a rigid body is equal to the time rate of change of its momentum.* If there is no net external force acting on a body, then there can be no change in its momentum. Thus, with no net external force,

$$\frac{d\mathbf{L}_j}{dt} = \mathbf{0} \quad \text{or} \quad \mathbf{L}_j = \text{constant}, \qquad (12.116)$$

is a statement of the law of *conservation of momentum*.

12.14 ANGULAR IMPULSE AND ANGULAR MOMENTUM

When a body translates, its motion is described in terms of its velocity and acceleration, and we are interested in the forces that produce this motion. When a body rotates, we are interested in the moments, and the motion is described by angular velocity and angular acceleration. Similarly, in dealing with rotational motion, we must consider terms such